Supplementary Information

for "Debiased Counterfactual Estimation for Nonlinear Panel Factor Models: A Generative Deep Learning Approach"

Appendix A Theory and proofs

Appendix A.1 Adjustment information and prediction inputs

Sections 2–4 of the main text distinguish the unit-level information used for adjustment from the additional information used to predict outcomes in a particular period. We use the following notation for these objects.

Let A_i^X be the pre-treatment covariate average. Let A_i^Y and A_i^R contain the pre-treatment mean, SD, and $.10, .50, .90$ quantiles of outcomes and population linear-anchor residuals $Y_{is}(0) - X_{is}'\beta_0 - \lambda_i' f_s$, respectively. The unit-level adjustment state is

$$U_i = (\lambda_i', A_i^{X\prime}, A_i^{Y\prime}, A_i^{R\prime})'. \tag{1}$$

For predetermined or treatment-invariant covariates $X_{it}^\star$ admitted to prediction, the period-specific information is

$$S_{it} = \big(U_i', f_t', X_{it}^{\star\prime}, \lambda_i' f_t, |\lambda_i' f_t|, \widetilde{t}\big)', \tag{2}$$

where $\widetilde{t}$ is scaled time. The additional coordinates are known functions of the adjustment state, time factors, and prediction covariates.

The fitted unit-level state is

$$\hat{U}_i = (\hat{\lambda}_i', \bar{X}_i^{\mathrm{pre}\prime}, A_i^{Y,\mathrm{pre}\prime}, A_i^{R,\mathrm{pre}\prime})'. \tag{3}$$

The fitted residual summaries use the fold-specific linear anchor. Replacing the population coordinates in equation (2) by their fitted counterparts gives $\hat{S}_{it}$. Whole-unit folds produce treatment-probability predictions

$$\hat{e}_i = \widehat{\Pr}(D_i = 1 \mid \hat{U}_i). \tag{4}$$

Both branches use these scores and fold assignments. Appendix B specifies the coordinates and transformations used by each fitted network.

Write $e(U_i) = \Pr(D_i = 1 \mid U_i)$ and $\mu_{0t}(S_{it}) = \mathbb{E}\{Y_{it}(0) \mid S_{it}\}$. The joint unconfoundedness and compatibility restrictions are stated in Assumption 3; the overlap, moment, rank, and recovery conditions are in Assumptions 4–6. These are the formal versions of the corresponding assumptions in the main text.

Appendix A.2 Asymptotic regimes and targets

Units are the sampling clusters. Identification concerns the population law; recovery, inference, and finite-panel evaluation use the following regimes.

Let $\mathcal{H}_N$ denote the common design environment: the factor path, pre-specified post-period mixing weights, and exogenous constants in the assignment rule. All population expectations and cross-unit limits below may be interpreted conditionally on $\mathcal{H}_N$, and we suppress it from the notation. Conditional on this environment, the unit clusters must follow the law appearing in the displayed expectations and scores. In particular, an assignment intercept computed from the same realized cross-section is a generated common quantity

rather than an exogenous element of $\mathcal{H}_N$; its first-order effect must be included unless it is replaced by a fixed population-calibrated constant or shown negligible. Likewise, repeatedly drawing assignments until a treated-count event occurs changes the joint assignment law and is not represented by independent Bernoulli scores with propensity $e(U_i)$ unless that conditioning is derived explicitly.

O: oracle-state cluster regime.

$N \to \infty$ while the population state U_i and compatible period state S_{it} are treated as observed; T_0 and the number of post-treatment periods may be fixed. This regime justifies the form of an oracle AIPW score and its unit-cluster central limit theorem. It does not justify replacing U_i by an estimated loading when T_0 is fixed.

R: state-recovery regime.

$N_{\text{co}}, T_0 \to \infty$, the anchor rank is selected consistently, and $a_{NT} + b_{NT} \to 0$, so the rotation-aligned Stage-1 error $\rho_{NT} \to 0$. This regime supports feasible consistency when the downstream learners and numerical operations are also consistent. It does not by itself supply a $\sqrt{N}$ expansion.

J: joint-growth inference regime.

$N \to \infty$, $N_{\text{co}}/N \to \kappa_{\text{co}} \in (0,1)$, $N_{\text{tr}}/N \to p \in (0,1)$, and $T_0 \to \infty$ along one triangular array. The number of target post-treatment periods is fixed or uniformly bounded, and its pooling weights converge. In addition to the oracle score conditions, a feasible $\sqrt{N}$ result requires the total nuisance product and a generated-state oracle-equivalence remainder defined below to be $o_p(N^{-1/2})$.

F: realized finite-panel evaluation.

The generated or observed panel is held fixed and an estimate is compared with its realized treated-sample target. This describes the realized-target

point-error comparisons. It is not a population central-limit regime, and its score cannot be borrowed from a population-target theorem. Geithner's reported bootstrap intervals instead concern a company-population target conditional on the event calendar; the finite-cohort distinction is developed in the full-refit bootstrap section below.

For a fold k, let $\hat{\eta}^{(-k)}$ denote the complete Stage-1 artifact and let $\Psi_t(O_i;\theta,\eta)$ denote the ATT score evaluated with population state and nuisances. Define the generated-state oracle-equivalence remainder

$$\Delta_{\text{state},N} = \left| \frac{1}{N} \sum_k \sum_{i \in I_k} \left[\Psi_t\{O_i;\theta_0,\hat{\eta}^{(-k)}\} - \Psi_t(O_i;\theta_0,\eta_0) \right] \right|. \tag{5}$$

The remainder includes alignment, common factor estimation, loading estimation, and every derived coordinate of $\hat{U}_i$ and $\hat{S}_{it}$ after separating the ordinary outcome and propensity nuisance errors. An influence function for the canonical generated state remains to be derived. The feasible-estimator asymptotic-linear theorems therefore impose

$$\Delta_{\text{state},N} = o_p(N^{-1/2}). \tag{6}$$

If instead $\sqrt{N}\Delta_{\text{state},N}$ has a nondegenerate first-order term, that term must be derived and added to the score and variance before a feasible $\sqrt{N}$ claim can be made. Cross-fitting prevents direct unit leakage but does not, by itself, prove (6) for a common factor estimate.

The following conditions yield a rate for this remainder. Let $r_{S,N}^2 = N^{-1}\sum_i \|\hat{S}_i - S_i\|^2$ after fold-specific alignment, and let $\mathcal{M}(a) = \mathbb{E}[\Psi\{O;\theta_0,\eta(S + a(\hat{S} - S))\} \mid \mathcal{A}]$, where $\mathcal{A}$ contains the training folds. If, uniformly over folds, (i) the score map is twice Gateaux differentiable along the aligned state path, (ii) its first derivative vanishes, $\mathcal{M}'(0) = 0$, (iii) $|\mathcal{M}''(a)| \le C r_{S,N}^2$, and (iv) the centered cross-fitted score difference is $O_p(r_{S,N}/\sqrt{N})$, then

$$\Delta_{\text{state},N} = O_p\{r_{S,N}^2 + r_{S,N}/\sqrt{N}\}. \tag{7}$$

Thus $r_{S,N} = o_p(N^{-1/4})$ is a primitive sufficient rate only when state orthogonality and the stated stochastic bound have both been verified. Ordinary nuisance orthogonality can cancel state paths that are measurable with respect to the population state and independent training data, as shown below; it need not cancel an outcome-generated path. The current pipeline freezes the Stage-1 coordinates and cross-fits downstream scores, but its treated and held-out-control loadings use their own pre-treatment paths. Under unrestricted within-unit serial dependence this is not equivalent to independent generated-regressor cross-fitting. Its stitched training features also create indirect cross-fold dependence. Consequently conditions (ii)–(iv) are not established by the implementation, and (6) remains a high-level assumption rather than a proved consequence of the recovery rate.

If state orthogonality fails, a first-order expansion instead has the form

$$\sqrt{N}(\hat{\theta} - \theta_0) = N^{-1/2} \sum_{i=1}^{N} \{\psi_i + \Gamma_i\} + o_p(1), \tag{8}$$

provided the aligned Stage-1 estimator itself has an asymptotically linear representation that induces Γ_i. The variance is then $\mathrm{Var}(\psi_i + \Gamma_i)$, including their covariance. If no such representation is available, a generic norm-based sufficient alternative, under a bounded score derivative, is the much stronger $r_{S,N} = o_p(N^{-1/2})$. It is not the only alternative: suitable average cancellation and time-series bounds can suffice for individual components, but must be established for the executed estimator. Neither $\Gamma_i = 0$ nor the stronger rate is asserted for the canonical estimator.

Regime R, together with nuisance and state consistency, gives ATT and plug-in-QTE consistency. Regime J, the ATT product rate, (6), and an $o_p(N^{-1/2})$ numerical remainder give the conditional asymptotic-linear result for a population ATT.

Sequential asymptotics are not a fifth regime. One may first use regime R to study recovery as the pre-treatment information and control cross-section grow, and then use the regime-O score in an oracle limiting experiment. That iterated argument transfers to the

feasible same-sample statistic only if the recovery and score approximations are uniform enough to imply (6). Regime J states this requirement on a single triangular array. Thus neither an informal "T_0 first, N second" argument nor a fixed-T_0 oracle calculation substitutes for the feasible joint-growth gate.

Appendix A.3 Full conditions for identification, consistency, and conditional inference

The full hypotheses underlying the main text's identification, conditional-mean consistency, double-robustness, linear-limit, ATT-variance, and plug-in-QTE statements are collected here. The sampling targets and regimes are those in Appendix A.2; units, rather than unit–period rows, are the sampling clusters. The statements below retain the population, state-recovery, and joint-growth conditions used by their proofs.

Appendix A.3.1 Identification and recovery conditions

Assumption 1 (SUTVA)**.** *Unit i's outcome depends only on its own treatment status D_{it}.*

Assumption 2 (Anchored nonlinear factor structure)**.** *Untreated outcomes satisfy*

$$Y_{it}(0) = X_{it}'\beta_0 + \lambda_i' f_t + h_0(X_{it}, \lambda_i, f_t) + \varepsilon_{it}, \tag{9}$$

where $\lambda_i, f_t \in \mathbb{R}^r$ form a pervasive linear anchor that recovers the population adjustment state, and h_0 is a square-integrable, regular nonlinear correction on that same state. Assumption 6 governs recoverability. We require $\mathbb{E}[\varepsilon_{it} \mid X_{it}, \lambda_i, f_t] = 0$, but no parametric disturbance law. Observed outcomes satisfy

$$Y_{it} = Y_{it}(0) + \delta_{it} D_{it}, \tag{10}$$

with unit-time-specific treatment effects δ_{it}.

Assumption 3 (Common adjustment state and factor-based unconfoundedness)**.** *Use the fixed-dimensional adjustment state U_i and prediction inputs S_{it} defined in equations* (1)–(2)*. Conditional on U_i, assignment is independent jointly of the untreated outcome path and every predetermined or treatment-invariant covariate path admitted to prediction:*

$$\left(\{Y_{it}(0)\}_{t=1}^{T}, \{X_{it}^{\star}\}_{t=1}^{T}\right) \perp D_i \mid U_i. \tag{11}$$

With $e(U_i) = \Pr(D_i = 1 \mid U_i)$, prediction states S_{it} that contain U_i and otherwise use only $f_t, X_{it}^{\star}$ and declared functions of $(U_i, f_t, X_{it}^{\star})$ satisfy $Y_{it}(0) \perp D_i \mid S_{it}$ and $\Pr(D_i = 1 \mid S_{it}) = e(U_i)$. Both branches share this state and score; f_t is a common time index, not a unit-level assignment variable.

Assumption 4 (Overlap condition)**.** *For constants $0 < \underline{e} \leq \bar{e} < 1$,*

$$\underline{e} \leq e(U_i) \leq \bar{e}, \quad \textit{for all } i. \tag{12}$$

Clipping and capping are recorded finite-sample regularizations, not substitutes for overlap. The propensity-correctness branch and oracle-score inference below assume them asymptotically inactive. The SI derives the bias and frozen-map sampling expansion when stabilization remains binding.

Assumption 5 (Regularity conditions)**.** *Under the usual factor normalization, loadings have positive-definite second moment. The factor Gram matrix is nonsingular and converges to a positive-definite limit in recovery and joint-growth regimes. Independent unit paths have uniformly bounded $2 + \delta$ moments and satisfy the cluster laws of large numbers and central limit theorem along the joint-growth array. Post-period dependence within units is unrestricted; growing pre-period windows require the concentration conditions in the SI for loading and summary-coordinate recovery.*

Assumption 6 (Recoverability of the linear anchor)**.** *After partialling out $X_{it}'\beta_0$, the un-*

treated control matrix equals a pervasive, spectrally separated rank-r anchor plus a nonlinear component and an approximate-factor disturbance. The nonlinear component is asymptotically orthogonal to the covariate and factor tangent spaces and has operator norm $o\{\sqrt{N_{\mathrm{co}}T}\}$. The iterated IFE objective converges and the treated-pre-period rank rule selects r with probability approaching one.

Appendix A.3.2 Conditional-mean consistency

Assumption 7 (Consistency of propensity score estimation)**.** *On the common state in* (1)*, $\|\hat{e}(\hat{U}) - e(U)\|_\infty = o_p(1)$, with overlap and aligned state recovery as in Assumptions 4 and 6.*

Assumption 8 (Sufficiency of the critic function class)**.** *The* 1*-Lipschitz critic class distinguishes the weighted real and generated laws: its Wasserstein discrepancy vanishes only when they coincide.*

Theorem 1 (Conditional mean consistency under an approximate-extremum condition)**.** *In the recovery regime, impose Assumptions 1–6, 7, and 8. Require the Stage-3A mean-sufficiency condition and vanishing approximation error for its actual generator class relative to $\mu_{0t}(S_{it})$. Suppose the returned fit has $o_p(1)$ population-risk gap and a uniform law of large numbers holds as training unit clusters diverge. Require $M_N \to \infty$, aligned anchor error $\rho_{NT} \to 0$, and mean-truth compatibility of the complete objective or vanishing bias of incompatible penalties. Then the fixed-draw average with immutable buffers satisfies*

$$\mathbb{E}\Big[\big\{\bar{G}_\phi^{M_N}(\hat{S}_{it}) - \mu_{0t}(S_{it})\big\}^2\Big] \xrightarrow{p} 0. \tag{13}$$

Appendix A.3.3 Double robustness and the linear limit

Write $\mu_{0t}(S_{it}) = \mathbb{E}\{Y_{it}(0) \mid S_{it}\}$. Compatibility $\Pr(D_i = 1 \mid S_{it}) = e(U_i)$ makes the common weights balance the declared prediction coordinates.

Condition 1 (Consistency of the fitted outcome nuisance)**.** *The returned noise-average satisfies $\|\hat{\mu}_{0t} - \mu_{0t}\|_{L_2} = o_p(1)$ in the foldwise evaluation law, including state replacement and*

numerical error, under the allowed possibly misspecified propensity limit. Representability alone does not suffice.

Condition 2 (Correct specification of the propensity score model)**.** $\hat{e}(\hat{U}_i) \xrightarrow{p} e(U_i) = \Pr(D_i = 1 \mid U_i)$ *uniformly over* i*.*

Theorem 2 (Double robustness)**.** *In the recovery regime, under Assumptions 1–6, suppose the foldwise nuisance limits are functions of the population common state, the remaining nuisance has a stable limit in the overlap and moment envelopes, and the score averages obey a unit-cluster law of large numbers after generated-state replacement. With asymptotically inactive truncation for the population target and an* $o_p(1)$ *numerical remainder, the estimator* $\widehat{ATT}_t^{\mathrm{DR}}$ *in* (4) *satisfies:*

(a) If the fitted outcome nuisance is consistent (Condition 1), then $\widehat{ATT}_t^{\mathrm{DR}} \xrightarrow{p} ATT_t$*, even when the stable propensity limit is misspecified.*

(b) If the propensity score nuisance is consistent (Condition 2), then $\widehat{ATT}_t^{\mathrm{DR}} \xrightarrow{p} ATT_t$*, even if the outcome limit is misspecified, provided it has finite second moment.*

(c) If both models are correct and the joint-growth, product-rate, and oracle-equivalence conditions stated in the Supplementary Information hold, $\widehat{ATT}_t^{\mathrm{DR}}$ *is asymptotically linear with the oracle observed-state influence function.*

Proposition 1 (Population no-harm limit under a linear DGP)**.** *If* $h_0 \equiv 0$*, treatment is independent of the adjustment state, the population penalties are truth-compatible, and Theorem 1 holds, the noise-averaged DR-FGAN and LFM conditional-mean limits agree.*

Appendix A.3.4 Conditional ATT variance and plug-in QTE consistency

Proposition 2 (First-order variance estimation, conditional on oracle equivalence)**.** *Under the joint-growth asymptotic-linear representation,* L_2*-consistent estimated scores and a cluster law of large numbers for their squares give consistent score variance. Delete-one-unit*

variance additionally requires a mean-square approximation of centered deletion pseudovalues by the centered influence scores. Full re-estimation alone does not establish this deletion stability; frozen-nuisance deletion requires the same approximation.

Proposition 3 (Consistency of the plug-in QTE)**.** *In the recovery regime, suppose the common-state identification and overlap conditions hold, and the fold-specific QTE regression mean, aligned state, and conditional residual CDF are uniformly consistent on a neighborhood of the target support containing the aligned estimated states. The population residual CDF is uniformly continuous in its outcome and state arguments on a neighborhood containing the aligned estimated states. Suppose also that the population-CDF averages obey unit-cluster uniform laws of large numbers and numerical evaluation errors vanish. If the two marginal densities are continuous and bounded away from zero near a compact interior quantile set $\mathcal{A}$, then*

$$\sup_{\tau\in\mathcal{A}}\left|\widehat{QTE}_t^{\text{plug}}(\tau) - QTE_t(\tau)\right| = o_p(1).$$

The conclusion also holds for a fixed-period mixture formed before inversion when its marginal densities satisfy the same condition.

Implemented pipeline and inferential scope. The convergence premises concern the returned fits, including optimization, state replacement, and numerical evaluation. Section Appendix A.20 gives the exact stabilized-weight target and the frozen-map active-cap expansion; the latter is not an influence function for all estimated nuisances. Sections Appendix A.21 and Appendix A.22 state the selected-rank and complete-map dependence conditions. The released stitched pipeline has not been shown to satisfy those conditions or a general generated-state influence expansion. Sections Appendix A.23 and Appendix A.24 separate omitted-history information, total state recovery, and training-to-scoring prediction transfer. The conditional ATT variance statement therefore does not certify the released finite-panel interval; the QTE statement establishes plug-in consistency, not interval validity.

Appendix A.4 The Linear Factor Anchor as the Stage-1 Target

Stage 1 targets the latent state shared by the linear anchor and nonlinear correction. The conditional mean is estimated downstream.

Assumption 9 (Full recoverability conditions for the linear anchor)**.** *After partialling out $X_{it}'\beta_0$, write the untreated control matrix as*

$$\widetilde{Y}_{\mathcal{C}} = \Lambda_{\mathcal{C}} F' + \mathsf{H}_{\mathcal{C}} + \mathsf{E}_{\mathcal{C}}, \qquad [\mathsf{H}_{\mathcal{C}}]_{it} = h_0(X_{it}, \lambda_i, f_t). \tag{14}$$

The rank-r anchor is pervasive and separated: the nonzero singular values of $\Lambda_{\mathcal{C}} F'$ are of order $\sqrt{N_{\mathrm{co}} T}$ and the rth singular value is separated from the remainder. The nonlinear correction is defined after population projection so that, for every covariate panel X_k, it is asymptotically orthogonal to the covariate and factor tangent spaces,

$$\frac{\langle \mathsf{H}_{\mathcal{C}}, \mathsf{X}_k \rangle}{N_{\mathrm{co}} T} = o(1), \qquad \frac{\|T^{-1} \mathsf{H}_{\mathcal{C}} F\|_F}{\sqrt{N_{\mathrm{co}}}} = o_p(1),$$
$$\frac{\|N_{\mathrm{co}}^{-1} \mathsf{H}_{\mathcal{C}}' \Lambda_{\mathcal{C}}\|_F}{\sqrt{T}} = o_p(1).$$

and

$$a_{NT} := \frac{\|\mathsf{H}_{\mathcal{C}}\|_{\mathrm{op}}}{\sqrt{N_{\mathrm{co}} T}} \longrightarrow 0.$$

The disturbance matrix satisfies the moment and weak-dependence conditions of an approximate factor model (Bai 2003). The treated-pre-period rank rule selects r with probability approaching one.

Assumption 6 links the linear anchor to the nonlinear model. The leading linear component identifies (λ_i, f_t) up to rotation, while h_0 may remain important for individual predictions without introducing a competing pervasive factor space. Approximate-factor perturbation arguments then give consistency of the Stage-1 state; Section Appendix A.5 states the rate. Bai's linear IFE results apply to this anchor and its composite weak residual, not

to an arbitrary nonlinear g. The WGAN-GP subsequently fits a correction on the recovered inputs, subject to the mean-sufficiency and approximation conditions below.

Appendix A.5 Recovery of the latent adjustment state

The causal restriction concerns the population loading λ_i, whereas the estimator uses $\hat{\lambda}_i$. The anchored model connects the two through state recovery.

After removing $X_{it}'\beta_0$, let

$$\widetilde{\mathbf{Y}}_{\mathcal{C}} = \Lambda_{\mathcal{C}} F' + \mathsf{H}_{\mathcal{C}} + \mathsf{E}_{\mathcal{C}}$$

as in Assumption 6, and define

$$a_{NT} = \frac{\|\mathsf{H}_{\mathcal{C}}\|_{\mathrm{op}}}{\sqrt{N_{\mathrm{co}}T}}, \qquad b_{NT} = \|\hat{\beta} - \beta_0\|, \qquad \rho_{NT} = N_{\mathrm{co}}^{-1/2} + c_{T_0} + a_{NT} + b_{NT}.$$

All comparisons of factors below are made after a nonsingular population-aligning rotation R_{NT}. Here $c_{T_0} \to 0$ is a high-level pre-period projection and summary-coordinate rate. It includes projection of a non-training unit and perturbation of every declared pre-period anchor-residual summary, including the empirical quantiles. It equals $T_0^{-1/2}$ only if the nonlinear projection-bias component $T_0^{-1}\sum_{s\le T_0} f_s h_0(X_{is}, \lambda_i, f_s)$ is centered or otherwise negligible, and if a suitable time-series concentration inequality holds for the centered composite-residual average $T_0^{-1}\sum_{s\le T_0} f_s v_{is}$, where $v_{is} = h_0(X_{is}, \lambda_i, f_s) + \varepsilon_{is}$. It also requires a uniformly nonsingular pre-period factor Gram matrix and the local CDF regularity needed for the quantile summaries. Arbitrary dependence among a fixed vector of post-period outcomes is compatible with unit-cluster inference, but arbitrary dependence across a growing pre-period window does not imply $c_{T_0} \to 0$.

Proposition 4 (Linear-anchor recovery)**.** *In regime R, suppose Assumptions 2, 5, and 6 hold,* $\Pr(\hat{r} = r) \to 1$*, the canonical alternating least-squares/truncated-SVD objective converges to the separated rank-r population solution, and the composite idiosyncratic array obeys the approximate-factor moment and weak-dependence conditions. Suppose also that*

the pre-period projection concentration above holds at rate c_{T_0}. Then a sequence of nonsingular rotations R_{NT} exists such that

$$\left\{ T^{-1} \sum_{t=1}^{T} \|\hat{f}_t - R_{NT} f_t\|^2 \right\}^{1/2} = O_p\big(N_{\text{co}}^{-1/2} + T^{-1/2} + a_{NT} + b_{NT}\big)\,,$$
$$\left\{ N^{-1} \sum_{i=1}^{N} \|\hat{\lambda}_i - R_{NT}^{-\top} \lambda_i\|^2 \right\}^{1/2} = O_p(\rho_{NT}). \tag{15}$$

Consequently, if $a_{NT} + b_{NT} \to 0$ and $N_{\text{co}}, T_0 \to \infty$, any Lipschitz conditional mean or propensity nuisance evaluated on the aligned estimated state differs from the same nuisance evaluated on the population state by $O_p(\rho_{NT})$ in L_2.

Proof. The rank-r anchor has singular values of order $\sqrt{N_{\text{co}} T}$ and an asymptotically separated rth singular value. The composite perturbation has operator norm

$$O_p(\sqrt{N_{\text{co}}} + \sqrt{T}) + \|\mathsf{H}_{\mathcal{C}}\|_{\text{op}} + O_p\{\sqrt{N_{\text{co}} T}\, b_{NT}\}.$$

The variational characterization of the leading singular space therefore bounds its rotation-aligned error by the ratio of this perturbation to the anchor eigengap. This yields the first line of (15); it is the usual approximate-factor argument of Bai (2003), with the explicit a_{NT} term recording the nonlinear perturbation. Projecting a non-training unit's pre-treatment path on the recovered factor space adds c_{T_0} and gives the second line. This covers treated and scored-control loadings in the canonical artifact; it does not use their post-treatment outcomes. Lipschitz continuity and Cauchy–Schwarz then give the final statement. The linear IFE analysis of Bai (2009) supplies primitive conditions for the $h_0 \equiv 0$ benchmark; neither reference asserts recovery for an unrestricted nonlinear g. Consistency of the treated-pre-period leave-one-time-out rank rule is imposed through $\Pr(\hat{r} = r) \to 1$, not inferred from a finite candidate-grid minimum. □

Remark 1 (Population and estimated adjustment states)**.** *Assumption 3 directly imposes*

the joint treatment-assignment restriction on the population state U_i in equation (1)*; it is not imposed on an estimated loading alone. Proposition 4 permits plug-in nuisances based on the rotation-aligned $\hat{U}_i$ because all declared state coordinates converge. It does not assert the exact finite-sample relation $Y(0) \perp D \mid \hat{U}_i$. If a_{NT} fails to vanish because h_0 contains an omitted pervasive direction, the proposition is unavailable and factor-based identification requires an enriched Stage 1 state.*

State-path derivative. Fix a target period and suppress its index. Write $\mu = \mathbb{E}\{Y(0) \mid S\}$, $e = \Pr(D = 1 \mid U)$, $p = \Pr(D = 1)$, and $\varepsilon = Y(0) - \mu$. Maintain the common-state identification restriction and bounded overlap. Let a differentiable state path induce nuisance paths μ_a, e_a with $\mu_0 = \mu$, $e_0 = e$, and derivatives h_μ, h_e. Differentiation under the expectation of the ATT score in (22), holding the data law, p, and the target fixed, gives

$$\partial_a \mathbb{E}\psi(\mu_a, e_a)|_{a=0} = -\mathbb{E}\left[\frac{D-e}{p(1-e)}h_\mu\right] - \mathbb{E}\left[\frac{(1-D)\varepsilon}{p(1-e)^2}h_e\right]. \tag{16}$$

The control residual has mean zero at the truth, so differentiation of the normalized control denominator adds no first-order population term. Both terms vanish if, conditional on a genuinely independent training sample, the directions are measurable functions of the true compatible state. Thus ordinary nuisance orthogonality also removes these smooth state-coordinate perturbations. For directions depending on a scored unit's additional pre-treatment history, the joint assignment restriction can still cancel the first term, but the second requires a conditional residual-covariance bound. Freezing coordinates or cross-fitting downstream fits does not supply that bound.

Why a correct proxy-state fit is not enough. For a proxy V, conditional on independent training when applicable, put $m_d(V) = \mathbb{E}\{Y(0) \mid V, D = d\}$ and $e_V = \Pr(D = 1 \mid V)$, assuming proxy overlap. Even with both proxy nuisances correct, normalized AIPW has

limit $\theta_V = \mathbb{E}\{Y(1) - m_0(V) \mid D = 1\}$, so

$$\theta_V - ATT = \mathbb{E}\{m_1(V) - m_0(V) \mid D = 1\}. \tag{17}$$

This follows by conditioning the weighted control residual on V. Double robustness does not remove this residual-confounding term. Recovery and score-level error control are needed to connect the proxy target to ATT.

Appendix A.6 Nuisance regularity and approximation

Write $V_{it} = (X_{it}, \lambda_i, f_t)$ for the ATT generator's population inputs and $\mu_{0t}(S_{it}) = \mathbb{E}\{Y_{it}(0) \mid S_{it}\}$ for the mean required by the augmented score. These are different conditioning sets: S_{it} also contains the pre-treatment summaries used for adjustment.

Assumption 10 (ATT mean sufficiency on generator inputs)**.** *For the canonical generator's outcome-correct route, $\mu_{0t}(S_{it}) = m_{0t}(V_{it})$ almost surely. The actual generator class, including its GSC skip and projected loading–factor interaction, can approximate this mean in the foldwise evaluation law.*

The restriction $\mathbb{E}(\varepsilon_{it} \mid V_{it}) = 0$ in the structural equation does not imply $\mathbb{E}(\varepsilon_{it} \mid S_{it}) = 0$ when histories carry additional predictive information, for example through serial dependence. Assumption 10 supplies that bridge only where the canonical outcome learner is claimed to be correct. It is not required for the true-propensity branch of Theorem 2: a possibly misspecified mean on V_{it} remains measurable on S_{it}.

Let $\mathcal{G}_{\mathrm{NN}}(L, W, M)$ denote the implemented GSC-skip generator class, whose feed-forward correction uses LayerNorm and LeakyReLU, depth L, width W, and bounded weights M on covariates, projected loading–factor products, and noise. This is not an unrestricted network on all of S_{it}. A generic member is denoted $g = G_\phi$, with fitted generator $G_{\widehat{\phi}}$. Approximation is distinguished from estimation and optimization error.

Definition 1 (Approximation error)**.** *The approximation error on aligned population inputs relative to the score's full-state conditional mean is*

$$\epsilon_{\text{approx}} = \inf_{g \in \mathcal{G}_{\text{NN}}} \left\| \mathbb{E}_{\mathbf{z}} g(V_{it}, \mathbf{z}) - \mu_{0t}(S_{it}) \right\|_{L_2}. \tag{18}$$

Estimated-buffer replacement is handled separately. Under Assumption 10, the target equals $m_{0t}(V_{it})$, but vanishing approximation error through the projected interaction still requires verification; padding alone does not supply it.

Appendix A.6.1 Smoothness assumptions

Assumption 11 (Hölder smoothness)**.** *The true conditional mean function $\mu : [-1,1]^{d_\mu} \to \mathbb{R}$ and the generalized propensity score $e : [-1,1]^{d_e} \to (0,1)$ belong respectively to Hölder balls $\mathcal{W}^{\beta_\mu,\infty}([-1,1]^{d_\mu})$ and $\mathcal{W}^{\beta_p,\infty}([-1,1]^{d_e})$, i.e., weak derivatives D^α exist and are bounded for all orders $|\alpha| \leq \lfloor \beta \rfloor$, with the highest-order derivatives satisfying a Hölder continuity condition. Here d_μ and d_e are the effective input dimensions of the outcome model and the propensity score model, respectively. For the canonical ATT learner, μ denotes the same $\mu_{0t}(S_{it}) = m_{0t}(V_{it})$ under Assumption 10. Smoothness does not establish representation through the implemented projected interaction.*

Assumption 12 (Quantified overlap condition)**.** *For the common adjustment state, $e(U_i) \in [\underline{e}, \bar{e}]$ almost surely for fixed $0 < \underline{e} < \bar{e} < 1$. The oracle unit-cluster scores used below have uniformly bounded $2+\delta$ moments along the relevant array. Any clipping or capping rule is asymptotically inactive for the stated theorem; otherwise population-ATT centering requires a separate outcome-correctness or bias-cancellation argument, and a stabilization-dependent limiting estimand may arise.*

Oracle-state empirical-risk results provide candidate nuisance rates under their sampling and smoothness conditions (Farrell et al. 2021; Schmidt-Hieber 2020). Applying these rates to dependent panel rows, generated factor inputs, and the returned adversarial fit requires

learner-specific bounds, which we impose below.

Appendix A.7 Proof of Theorem 2

Throughout this appendix, expectations are taken under the observed-data law of the unit clusters, together with the auxiliary randomness from sample splitting and generator draws. For a fixed post-treatment period t, let U_i be the common unit adjustment state in equation (1), let S_{it} be the compatible prediction state in equation (2), let $e(U_i) = \Pr(D_i = 1 \mid U_i)$, and let $p = \Pr(D_i = 1) \in (0, 1)$. Write $\mu_{0t}(S_{it}) = \mathbb{E}\{Y_{it}(0) \mid S_{it}\}$ and $\varepsilon_{it} = Y_{it}(0) - \mu_{0t}(S_{it})$. The maintained restrictions give $\Pr(D_i = 1 \mid S_{it}) = e(U_i)$ and $\mathbb{E}(\varepsilon_{it} \mid S_{it}, D_i) = 0$. Decompose the generator output into its noise average and a noise residual, $\bar{Y}_{it}^{\text{gen}}(0) = \hat{\mu}_{it} + \zeta_{it}$, where $\hat{\mu}_{it} = \mathbb{E}_{\mathbf{z}}[G_{\hat{\phi}}(S_{it}, \mathbf{z})]$ and $\mathbb{E}(\zeta_{it} \mid \text{data}) = 0$. The population-ATT expansion assumes asymptotically inactive clipping and capping, as specified in Assumption 12. Units are independent or weakly dependent clusters under the regimes above, with moments bounded by Assumption 5.

Proposition 5 (Identification on the common state)**.** *Under Assumptions 1, 3, and 4, and the compatibility restriction* $\Pr(D_i = 1 \mid S_{it}) = e(U_i)$*,*

$$\mathbb{E}\{Y_{it}(0) \mid D_i = 1\} = \mathbb{E}\{\mu_{0t}(S_{it}) \mid D_i = 1\},$$
$$F_{0t}(y) = \mathbb{E}[\Pr\{Y_{it} \le y \mid S_{it}, D_i = 0\} \mid D_i = 1].$$

Consequently ATT_t *and the two marginal distributions defining* $QTE_t(\tau)$ *are identified. With fixed weights* ω_t*, the pooled law is identified by mixing the period CDFs before inversion. This proposition is a population result for* U_i, S_{it}*; replacement by their estimated analogues uses Proposition 4 and does not hold exactly in a finite panel.*

Proof. Conditional independence gives $\mathcal{L}\{Y_{it}(0) \mid S_{it}, D_i = 1\} = \mathcal{L}\{Y_{it}(0) \mid S_{it}, D_i = 0\}$ on the overlap support. For controls $Y_{it} = Y_{it}(0)$. Iterated expectations yield both displays; the

ATT follows after subtracting the identified counterfactual mean from the observed treated mean. A finite weighted mixture preserves identification. □

Starting from the definition of the estimator, for every post-treatment period t,

$$\begin{aligned} ATT_t - \widehat{ATT}_t^{\mathrm{DR}} = (ATT_t - ATT_{t,N}^{\mathrm{real}}) \\ + \underbrace{\frac{1}{N_{\mathrm{tr}}} \sum_{i \in \mathcal{T}} \big(\bar{Y}_{it}^{\mathrm{gen}}(0) - Y_{it}(0)\big)}_{A_t} \\ + \underbrace{\sum_{i \in \mathcal{C}} \tilde{w}_{it} \big(Y_{it} - \bar{Y}_{it}^{\mathrm{gen}}(0)\big)}_{B_t}. \end{aligned} \tag{19}$$

since $Y_{it} = Y_{it}(1)$ for treated post-treatment observations and $Y_{it} = Y_{it}(0)$ for controls. The realized target $ATT_{t,N}^{\mathrm{real}}$ is the average individual treatment effect among the sampled treated units. The unit-cluster law of large numbers gives $ATT_{t,N}^{\mathrm{real}} - ATT_t = o_p(1)$, so consistency follows from $A_t + B_t = o_p(1)$. The population–sample difference cannot be discarded in a $\sqrt{N}$ expansion; its treated-sample variation is included in the population influence score below.

Lemma 1 (Weight stabilization)**.** *Suppose Assumptions 3, 4, and 5 hold and* $\hat{e}(\hat{U}_i) \xrightarrow{p} e^*(U_i)$ *uniformly over* i *for a limit* e^* *bounded away from zero and one. Let* h_{it} *be a fixed measurable unit-cluster function with finite second moment satisfying the weighted unit-cluster laws of large numbers used below, or a generated array for which replacement by such a function and the same laws have been established. Then*

$$\sum_{i \in \mathcal{C}} \tilde{w}_{it}\, h_{it} \xrightarrow{p} \frac{\mathbb{E}\big[(1 - D_i)\, \rho^*(U_i)\, h_{it}\big]}{\mathbb{E}\big[(1 - D_i)\, \rho^*(U_i)\big]}, \qquad \rho^* = \frac{e^*}{1 - e^*}.$$

If in addition $e^* = e$ *and* $\mathbb{E}(h_{it} \mid U_i, D_i = 0) = \mathbb{E}(h_{it} \mid U_i, D_i = 1)$*, the limit equals* $\mathbb{E}(h_{it} \mid D_i = 1)$*. The latter condition holds for functions of* S_{it} *and* $Y_{it}(0)$ *under Assumption 3 and the state-compatibility restriction.*

Proof. Write $\sum_{i\in\mathcal{C}} \tilde{w}_{it} h_{it} = S_1/S_0$ with $S_1 = N^{-1}\sum_{i=1}^{N}(1-D_i)\hat{\rho}(\hat{U}_i)h_{it}$ and $S_0 = N^{-1}\sum_{i=1}^{N}(1-D_i)\hat{\rho}(\hat{U}_i)$, where $\hat{\rho} = \hat{e}/(1-\hat{e})$. Uniform convergence and overlap give $\sup_i |\hat{\rho}(\hat{U}_i) - \rho^*(U_i)| = o_p(1)$ with ρ^* bounded, so replacing $\hat{\rho}$ by ρ^* changes S_0 and S_1 by $o_p(1)$. The unit-cluster law of large numbers yields the stated ratio limit. When $e^* = e$, iterated expectations and the displayed conditional-mean restriction give

$$\begin{aligned}\mathbb{E}\left[\frac{(1-D)e(U)}{1-e(U)}h\right] &= \mathbb{E}[e(U)\mathbb{E}(h \mid U, D=0)] \\ &= \mathbb{E}[e(U)\mathbb{E}(h \mid U, D=1)] = \mathbb{E}(Dh).\end{aligned}$$

The denominator is p, so the ratio is $\mathbb{E}(Dh)/p = \mathbb{E}(h \mid D=1)$. □

Lemma 2 (Noise averaging)**.** *Under independently randomized evaluation draws conditional on the data and trained fits, with uniformly bounded conditional generator-output variances,* $N_{\text{tr}}^{-1}\sum_{i\in\mathcal{T}}\zeta_{it} = o_p(1)$ *and* $\sum_{i\in\mathcal{C}}\tilde{w}_{it}\,\zeta_{it} = o_p(1)$.

Proof. Conditional on the data and the trained generator, the noise residuals ζ_{it} are mean zero, independent across i, and have finite variance, since the generator is a fixed Lipschitz map of the Gaussian draws at evaluation time. Chebyshev's inequality gives the first claim. For the second, the inactive truncation and Assumption 4 imply $\tilde{w}_{it} \le C/N_{\text{co}}$ for a finite constant C, so the weighted sum has conditional variance at most $C^2 N_{\text{co}}^{-1}\max_i \text{Var}(\zeta_{it})$, which vanishes. □

Proof of Theorem 2. Part (a). Condition 1 directly assumes consistency of the actual fitted mean, $r_g = \|\hat{\mu} - \mu_{0t}\|_{L_2} = o_p(1)$, under the allowed propensity limit. Consistency is imposed on the fitted mean, rather than on its function class. Decompose

$$A_t = \frac{1}{N_{\text{tr}}}\sum_{i\in\mathcal{T}}\left\{\hat{\mu}_{it} - \mu_{0t}(S_{it})\right\} - \frac{1}{N_{\text{tr}}}\sum_{i\in\mathcal{T}}\varepsilon_{it} + \frac{1}{N_{\text{tr}}}\sum_{i\in\mathcal{T}}\zeta_{it}.$$

The first average is $O_p(r_g) = o_p(1)$ by Cauchy–Schwarz and overlap, which transfers the population L_2 rate to the treated population. The second is $o_p(1)$ by the unit-cluster law

of large numbers because $\mathbb{E}(\varepsilon_{it} \mid S_{it}, D_i) = 0$, and the third vanishes by Lemma 2. Hence $A_t = o_p(1)$. For the correction term, write

$$B_t = \sum_{i \in \mathcal{C}} \tilde{w}_{it} \{\mu_{0t}(S_{it}) - \hat{\mu}_{it}\} + \sum_{i \in \mathcal{C}} \tilde{w}_{it}\, \varepsilon_{it} - \sum_{i \in \mathcal{C}} \tilde{w}_{it}\, \zeta_{it}.$$

Because $\sum_i \tilde{w}_{it} = 1$ and $\tilde{w}_{it} \leq C/N_{\text{co}}$, Jensen's inequality and the assumed evaluation law of large numbers bound the first sum by $O_p(r_g) = o_p(1)$. For the second, replace the generated weights by their stable population-state limit under the theorem's replacement and score-law-of-large-numbers conditions. The limiting control weight is a bounded function of U_i, so its weighted residual expectation is zero by $\mathbb{E}(\varepsilon_{it} \mid S_{it}, D_i = 0) = 0$. The weighted sample average is therefore $o_p(1)$. The third sum vanishes by Lemma 2, subject to the stated draw and numerical conditions. Thus $A_t + B_t = o_p(1)$; adding the population–sample term in (19) proves consistency. The propensity limit need not be the true propensity, but boundedness, stability, and the fitted-mean consistency premise are required.

Part (b). Suppose the propensity model is correctly specified in the sense of Condition 2, and suppose the generator's noise average converges in L_2 to a finite-second-moment limit μ^*, not necessarily equal to μ_{0t}. Replacing $\hat{\mu}$ by μ^* in A_t and B_t changes each by $o_p(1)$ by the same Cauchy–Schwarz and bounded-weight arguments as in part (a). By the law of large numbers,

$$A_t \xrightarrow{p} \mathbb{E}\big[\mu^*(S_{it}) - Y_{it}(0) \mid D_i = 1\big],$$

and by Lemma 1 applied with $e^* = e$ and $h_{it} = Y_{it}(0) - \mu^*(S_{it})$, using $Y_{it} = Y_{it}(0)$ for controls,

$$B_t \xrightarrow{p} \mathbb{E}\big[Y_{it}(0) - \mu^*(S_{it}) \mid D_i = 1\big].$$

The two limits are negatives of each other, so $A_t + B_t \xrightarrow{p} 0$ and $\widehat{ATT}_t^{\text{DR}} \xrightarrow{p} ATT_t$ by (19). Thus inverse probability weights transport the control-group mean of the outcome-model error to its treated-group mean, where it cancels against A_t exactly; no additional mean-

zero restriction on that error is required.

Part (c). Suppose both models are correctly specified and the joint-growth gates of Theorem 3 hold. Proposition 6 then gives

$$\widehat{ATT}_t^{\mathrm{DR}} - ATT_t = \frac{1}{N}\sum_{i=1}^{N} \psi_t(Z_i; ATT_t, \mu_{0t}, e, p) + o_p(N^{-1/2}),$$

with ψ_t the oracle observed-state influence function in (22). The cluster central limit theorem gives asymptotic normality with variance V_t conditionally on the generated-state and numerical gates. □

The feasible-estimator result requires unit-fold construction of the Stage-1 artifact, propensity, and outcome nuisance. Sample reuse requires an explicit empirical-process and oracle-equivalence bound.

Appendix A.8 ATT rates and conditional inference

Conditional $\sqrt{N}$ ATT inference combines the oracle-state orthogonal expansion with a bound on the generated-state remainder.

Appendix A.8.1 Convergence rates for nuisance functions

The central-limit scaling is always the number of unit clusters N. A unit–period row count may enter a learner-specific risk bound only after its within-unit dependence conditions are verified; it is never substituted mechanically for the independent sample size. Regime O permits fixed periods only because the state is treated as observed. Feasible Stage-1 recovery uses regimes R or J and therefore requires $T_0 \to \infty$.

Lemma 3 (Convergence rates for nuisance estimates)**.** *Under Assumptions 2, 5, 6, and 11–12, suppose the oracle-state conditional-mean and propensity learners attain rates* $r_{\mu,N}^{\mathrm{orc}}$ *and* $r_{e,N}^{\mathrm{orc}}$*. Suppose, additionally, that replacing every fitted and evaluated state by its aligned oracle counterpart changes each complete fitted nuisance map by* $O_p(\rho_{NT})$ *in evaluation* L_2*.*

If both population nuisances are Lipschitz in the aligned state with bounded constants, the triangle inequality gives the conditional bounds

$$r_g := \|\hat{\mu}_{\mathrm{NN}} - \mu_{0t}(S_{it})\|_{L_2} = O_p\big(r^{\mathrm{orc}}_{\mu,N} + \rho_{NT}\big), \tag{20}$$

$$r_e := \|\hat{e} - e\|_{L_2} = O_p\big(r^{\mathrm{orc}}_{e,N} + \rho_{NT}\big). \tag{21}$$

The oracle rates and complete fitted-map stability premise require learner-specific results. Lipschitz continuity of the population regression alone does not control refitting on estimated inputs. For the canonical ATT generator, the mean rate also requires Assumption 10 and approximation by its input architecture. Both bounds contain ρ_{NT} because the propensity and outcome nuisances use the same estimated state. The actual-objective bound in Section Appendix A.16 retains a reconstruction-history covariance term, which can enter under a square root rather than as an additive $O_p(\rho_{NT})$. The score-level condition (6) remains a separate requirement.

Appendix A.8.2 Neyman orthogonality of AIPW and the product rate

The unstabilized population score underlying $\widehat{ATT}_t^{\mathrm{DR}}$ in (4) possesses Neyman orthogonality (Chernozhukov et al. 2018) at the true nuisances. Transferring this property to the implemented statistic requires the inactive-stabilization and generated-state conditions stated above. Binding stabilization has the different derivatives in Section Appendix A.20. Let U_i denote the common unit adjustment state and S_{it} the compatible period prediction state. Write $\mu_{0t}(S_{it}) = \mathbb{E}\{Y_{it}(0) \mid S_{it}\}$, $e(U_i) = \Pr(D_i = 1 \mid U_i)$, and $p = \Pr(D_i = 1)$. The equality $\Pr(D_i = 1 \mid S_{it}) = e(U_i)$ is part of the state contract rather than inferred from predictive performance. The ATT influence function in this oracle expansion is

$$\begin{aligned}\psi_t(\mathbf{Z}_i; \theta_t, \mu_{0t}, e, p) = \frac{D_i}{p}\big\{Y_{it} - \mu_{0t}(S_{it}) - \theta_t\big\} \\ - \frac{(1 - D_i)e(U_i)}{p\{1 - e(U_i)\}}\big\{Y_{it} - \mu_{0t}(S_{it})\big\},\end{aligned} \tag{22}$$

where $\theta_t = ATT_t$. At the truth this score has mean zero. Indeed, the first term averages $Y_{it}(1) - \mu_{0t}(S_{it}) - \theta_t$ over treated units, while the second term has conditional mean zero among controls. The term $-D_i\theta_t/p$ is essential: it incorporates sampling variation in the treated-group normalization and cannot be replaced by an unweighted $-\theta_t$ term.

For square-integrable directions h and k, the Gateaux derivatives of the population moment at the truth are

$$\partial_\mu \mathbb{E}\psi_t[h] = -\frac{1}{p}\mathbb{E}[D_i h(S_{it})] + \frac{1}{p}\mathbb{E}\left[\frac{(1-D_i)e(U_i)}{1-e(U_i)}h(S_{it})\right] = 0, \tag{23}$$

$$\partial_e \mathbb{E}\psi_t[k] = -\frac{1}{p}\mathbb{E}\left[\frac{(1-D_i)k(U_i)}{\{1-e(U_i)\}^2}\{Y_{it} - \mu_{0t}(S_{it})\}\right] = 0. \tag{24}$$

The first equality uses the balancing identity implied jointly by $e(U_i)$ and $\Pr(D_i = 1 \mid S_{it}) = e(U_i)$, and the second uses the control residual's conditional mean zero. The derivative with respect to p also vanishes because the numerator of (22) has mean zero at the truth. Thus the score is Neyman-orthogonal in $\eta = (\mu_{0t}, e, p)$.

The resulting nuisance-bias bound is the product of the two estimation errors.

Proposition 6 (Oracle expansion and feasible remainder for ATT)**.** *Under Assumptions 11–12 and the conditions of Theorem 1, let r_g and r_e be as defined in Lemma 3. Suppose the complete downstream nuisance fits are cross-fitted by unit, their foldwise limits remain uniformly inside the overlap and moment envelopes, and clipping and capping are asymptotically inactive. For the oracle-state centered-score bound, require the untreated residual's second moment conditional on all admitted scored-unit information to be uniformly bounded. With a numerical remainder $\zeta_{\mathrm{ATT},N} = o_p(1)$, the feasible Hájek estimator satisfies*

$$\begin{aligned}\widehat{ATT}_t^{\mathrm{DR}} - ATT_t = \frac{1}{N}\sum_{i=1}^{N}\psi_t(\mathbf{Z}_i; ATT_t, \mu_{0t}, e, p) \\ + O_p(r_g r_e) + O_p(\Delta_{\mathrm{state},N}) + O_p(\zeta_{\mathrm{ATT},N}) + o_p(N^{-1/2}).\end{aligned} \tag{25}$$

Proof. First hold the population state fixed and condition on a training fold. For generic

limits $(\bar{\mu}, \bar{e})$, put $\bar{\rho} = \bar{e}/(1-\bar{e})$, $\bar{q} = \mathbb{E}\{(1-D)\bar{\rho}(U)\}$, and $\delta = \bar{\mu} - \mu_0$. The population bias of the normalized-odds ATT functional is

$$B(\bar{\mu}, \bar{e}) = -\frac{\mathbb{E}(D\delta)}{p} + \frac{\mathbb{E}\{(1-D)\bar{\rho}\delta\}}{\bar{q}}. \tag{26}$$

It is zero when $\bar{\mu} = \mu_0$. It is also zero when $\bar{e} = e$, because $\bar{q} = p$ and $\mathbb{E}[(1-D)e(U)\{1-e(U)\}^{-1}h(S)] = \mathbb{E}[Dh(S)]$. For general consistent limits, add and subtract the same expression with $\rho = e/(1-e)$ and use overlap. Since $\|\bar{\rho} - \rho\|_2 \le C\|\bar{e} - e\|_2$ and $|\bar{q} - p| \le C\|\bar{e} - e\|_2$, Cauchy–Schwarz applied to (26) gives $|B(\bar{\mu}, \bar{e})| \le C\|\delta\|_2\|\bar{e} - e\|_2$.

In the oracle-state comparison with independently trained foldwise nuisances, unit-fold cross-fitting makes each scored unit conditionally independent of the downstream training sample. The conditional variance of the centered score difference is therefore $O_p\{(r_g + r_e)^2/N\}$; foldwise Cauchy–Schwarz gives the displayed $O_p(r_g r_e)$ conditional bias. A first-order ratio expansion of $N^{-1}\sum D_i$ and $N^{-1}\sum(1-D_i)\hat{\rho}_i$ around p yields exactly the oracle Hájek score in (22), including $-D_i ATT_t/p$. These steps prove the oracle-state expansion with remainder $O_p(r_g r_e) + o_p(N^{-1/2})$.

Replacing the population state by the fold-specific aligned artifact adds $\Delta_{\text{state},N}$ by definition. Approximation of the generator mean by its fixed draw bank, score-solver and optimization tolerances, and any finite-precision normalization add $\zeta_{\text{ATT},N}$. A truncation rule that remains binding is not covered by the numerical remainder: the executed score differs from the oracle score, and standard population-ATT centering then requires a separate correctness or bias-cancellation argument. The proposition excludes that case by imposing asymptotic inactivity. Combining the terms proves (25). □

Remark 2 (Empirical processes and sample reuse)**.** *In the oracle-state argument, unit-fold cross-fitting controls the empirical-process dependence between downstream nuisance fits and scored observations. The Stage-1 artifact is also fold-specific, but a scored unit's loading uses its own pre-period path, and global rank selection may use treated pre-period outcomes. These*

operations require the generated-state condition in addition to downstream cross-fitting. In particular, the canonical common-state table is stitched from the out-of-fold state row of each unit. For a control $i \in I_k$ and a training unit $j \in I_\ell$, $\ell \neq k$, the Stage-1 fit excluding I_ℓ can use i's control outcomes when constructing j's state. The propensity fit excluding I_k then uses this state of j. Hence direct training-ID exclusion does not eliminate the path from i's outcomes through another fold's state to i's fitted propensity. GAN training weights inherit the same stitched propensity artifact. This is an indirect dependence, not use of treated post-treatment responses in nuisance fitting. Its negligibility requires a multistage stability bound or a genuinely nested splitting argument; neither has been established for this implementation.

Theorem 3 (Conditional feasible-estimator $\sqrt{N}$ limit for ATT)**.** *Consider regime J and the population target* `ATT-POP-T`*. Under the conditions of Proposition 6, suppose the oracle-state learner rates satisfy*

$$r_g r_e = o_p(N^{-1/2}), \tag{27}$$

and impose the high-level gates $\Delta_{\text{state},N} = o_p(N^{-1/2})$ and $\zeta_{\text{ATT},N} = o_p(N^{-1/2})$. If the oracle unit scores satisfy a cluster central limit theorem and the estimated scores are L_2 consistent, then

$$\sqrt{N}\big(\widehat{ATT}_t^{\text{DR}} - ATT_t\big) \xrightarrow{d} \mathcal{N}(0, V_t), \tag{28}$$

where $V_t = \text{Var}\{\psi_t(\mathbf{Z}; ATT_t, \mu_{0t}, e, p)\}$. This is a conditional oracle-equivalence theorem. It is not a fixed-T_0 result and does not establish an efficiency bound for the model with a latent estimated state.

A direct cluster influence-function estimator evaluates (22) with out-of-fold nuisances and $\hat{p} = N_{\text{tr}}/N$:

$$\hat{\psi}_{it} = \frac{D_i}{\hat{p}}\{Y_{it} - \hat{\mu}_{0t}(\hat{S}_{it}) - \widehat{ATT}_t^{\text{DR}}\} - N(1 - D_i)\tilde{w}_i\{Y_{it} - \hat{\mu}_{0t}(\hat{S}_{it})\}. \tag{29}$$

Here $\tilde{w}_i$ is the globally normalized control-odds weight used in the point estimate (zero

for a treated unit), as in the implemented score. Under the both-consistent and inactive-stabilization conditions, $N\tilde{w}_i$ has the oracle limit $e(U_i)/[p\{1-e(U_i)\}]$ for controls. Its unit-cluster variance estimator is

$$\widehat{V}_t = \frac{1}{N}\sum_{i=1}^{N}\left(\hat{\psi}_{it} - \bar{\hat{\psi}}_t\right)^2, \qquad \widehat{\mathrm{Var}}(\widehat{ATT}_t^{\mathrm{DR}}) = \frac{\widehat{V}_t}{N}. \tag{30}$$

For a statistic pooling post-treatment periods, $\hat{\psi}_{it}$ is replaced by the same weighted period average used to define the estimand. A delete-one-unit jackknife targets the same first-order cluster variance only with the additional pseudovalue-stability condition in Proposition 7. Full re-estimation alone does not establish this condition; frozen-nuisance deletion requires equivalence at the centered-pseudovalue level.

Remark 3 (Scope of the oracle score)**.** *Equation* (22) *is the standard ATT influence function in the nonparametric model where the adjustment state is observed (Hahn 1998; Chernozhukov et al. 2018). Theorem 3 uses it only after imposing feasible-to-oracle equivalence. Proposition 4 gives state consistency but does not alone prove that equivalence at the* $N^{-1/2}$ *scale. Because the tangent space and first-stage contribution for the latent-state model have not been derived, the paper makes no semiparametric-efficiency claim for the feasible estimator.*

Point robustness versus inference robustness. For frozen nuisance functions, write $r = Y - \bar{\mu}$, $q = \bar{e}/(1-\bar{e})$, $s = \mathbb{E}\{(1-D)q\}$, $a = \mathbb{E}(Dr)/p$, and $b = \mathbb{E}\{(1-D)qr\}/s$, so the normalized functional is $a - b$. Its two-ratio influence function is

$$\psi_{\mathrm{ratio}} = \frac{D}{p}(r-a) - \frac{(1-D)q}{s}(r-b).$$

The score implemented in (29) has limiting form $D\{r-(a-b)\}/p - (1-D)qr/s$, which differs by $b\{D/p - (1-D)q/s\}$. Under a consistent outcome mean, $b = 0$ and the difference disappears. If only the propensity is consistent, this equality need not hold, and fitted-

nuisance first-order terms may also remain. Thus the point-consistency result does not establish a doubly robust standard error. Fixed evaluation draws may contribute algorithmic variation; the fixed $M = 50$ implementation is not itself an $M_N \to \infty$ sequence. Neither the numerical gate nor generated-state oracle equivalence is certified by computing the empirical score variance.

Appendix A.8.3 High-level rate decomposition for the conditional-mean nuisance

Theorem 3 requires a rate for the fixed-draw WGAN-GP conditional mean supplied to Stage 4. The canonical generator uses a fixed Stage-1 loading and factor buffer, a GSC skip connection, a spectrally normalized critic with gradient penalty, and a weighted objective containing adversarial, reconstruction, correction-size, and treated-pre-period calibration terms. Calibration and checkpoint selection use disjoint treated units. The rate decomposition must account for this complete objective and the returned nonconvex fit.

Here $m_0 = \mu_{0t}(S_{it})$; writing it as a function of V_{it} uses Assumption 10. Let $m_N^\dagger$ be the conditional mean of a population minimizer of the complete, tuning-dependent objective over the actual generator class, and let $r_{\text{fit},N}$ denote the oracle-state L_2 error of the returned fit relative to $m_N^\dagger$. Let $r_{\text{state},N}$ bound the L_2 change in the complete fitted map when both training and evaluation inputs are replaced by aligned oracle states. The triangle inequality gives

$$\|\hat{m}^M - m_0\|_2 \le r_{\text{fit},N} + \|m_N^\dagger - m_0\|_2 + r_{\text{state},N} + O_p(M_N^{-1/2}). \tag{31}$$

The second term includes omitted-state and architectural approximation error as well as objective bias. Its convergence to zero requires mean sufficiency, vanishing approximation error, and mean-truth compatibility of the complete objective or vanishing tuning bias. A squared-error rate for a stand-alone regression network does not establish these requirements or $r_{\text{fit},N}$ for the combined adversarial objective. Nor does a Lipschitz evaluation map alone establish $r_{\text{state},N} = O_p(\rho_{NT})$ after refitting. Section Appendix A.16 supplies the architecture-

specific objective bound and distinguishes refitting from fixed-map perturbations. Spectral normalization and gradient penalty regularize the critic; convergence of the returned fit remains a separate premise. Theorem 3 imposes a rate on the total left side of (31).

The last term records the fixed evaluation-draw average. One valid asymptotic sequence lets $M_N \to \infty$ sufficiently fast for the required product and numerical gates. With fixed M, unit-specific independent draw error instead enters the larger algorithmic influence function in Proposition 12; it cannot be deleted while retaining the oracle-mean score. That proposition requires full-generator convergence and the stated history and stability conditions, which have not been verified for the complete implementation. Finally, the stated `ATT-POP-T` theorem requires clipping and capping to be asymptotically inactive. If they remain binding, population-ATT centering requires a separate outcome-correctness or bias-cancellation argument, and a stabilization-dependent limiting estimand may arise. These qualifications concern both inference and any consistency branch relying on the propensity; outcome-model correctness may still deliver population-ATT centering under separately stated conditions.

Appendix A.9 Conditional variance procedures

Proposition 7 (Conditional score variance and deletion stability)**.** *Suppose the joint-growth conditions of Theorem 3 hold and the estimator has the stated unit-cluster asymptotic-linear representation. Then the empirical variance of the corresponding estimated unit scores is consistent. For delete-one-unit variance, write* $\hat{\theta}_t = \widehat{ATT}_t^{\mathrm{DR}}$*, let* $\hat{\theta}_t^{(-i)}$ *denote the estimate with unit* i *deleted, and define*

$$v_{it}^J = N\hat{\theta}_t - (N-1)\hat{\theta}_t^{(-i)}, \qquad \bar{v}_t^J = N^{-1}\sum_i v_{it}^J.$$

Let ψ_{it} be the oracle unit score in (22) *and $\bar{\psi}_t = N^{-1}\sum_i \psi_{it}$. In addition to the oracle second-moment law of large numbers, a sufficient deletion-stability condition is*

$$\frac{1}{N}\sum_{i=1}^{N}\left[(v_{it}^{J} - \bar{v}_t^{J}) - (\psi_{it} - \bar{\psi}_t)\right]^2 = o_p(1). \tag{32}$$

Under this additional condition,

$$N\widehat{\mathrm{Var}}_{\mathrm{jack}}(\hat{\theta}_t) \xrightarrow{p} V_t, \qquad V_t = \mathrm{Var}(\psi_{it}).$$

This estimates the first-order cluster variance; estimator asymptotic linearity or full nuisance re-estimation alone does not imply (32)*.*

Proof. The delete-one variance equals

$$\frac{N-1}{N}\sum_i \left(\hat{\theta}_t^{(-i)} - \bar{\hat{\theta}}_t^{(-\cdot)}\right)^2 = \frac{1}{N(N-1)}\sum_i (v_{it}^{J} - \bar{v}_t^{J})^2.$$

By Cauchy–Schwarz, (32) and the oracle second-moment law of large numbers imply $N^{-1}\sum_i (v_{it}^{J} - \bar{v}_t^{J})^2 \xrightarrow{p} V_t$. The displayed identity gives the conclusion. □

For a frozen-nuisance deletion to inherit a valid full-reestimation jackknife, its centered pseudovalues must satisfy

$$\frac{1}{N}\sum_i \left[(v_{it}^{J,\mathrm{frozen}} - \bar{v}_t^{J,\mathrm{frozen}}) - (v_{it}^{J,\mathrm{full}} - \bar{v}_t^{J,\mathrm{full}})\right]^2 = o_p(1). \tag{33}$$

Uniform $o_p(N^{-1/2})$ agreement between the deleted point estimates is not sufficient by itself, because pseudovalues multiply deletion differences by $N-1$. Unit deletion preserves within-unit dependence, but neither orthogonality nor full re-estimation proves these deletion-stability conditions. They remain imposed rather than verified for the reported algorithm.

Appendix A.10 Distributional consistency and rates for the ETC-GAN plug-in QTE

Consistency of the unaugmented ETC-GAN plug-in QTE follows from residual-distribution learning and CDF inversion. Let U_i be the common unit adjustment state and let S_{it} be the compatible population predictive state in equations (1)–(2). Let $\mu^Q(S_{it})$ be the QTE branch's regression mean and let $F_\varepsilon(\cdot \mid s)$ denote the conditional CDF of $\varepsilon_{it}(0) = Y_{it}(0) - \mu^Q(S_{it})$. Factor states are identified only up to rotation; throughout, $\hat{U}_i$ and $\hat{S}_{it}$ denote fold-specific estimated states after population alignment. The canonical ETC-GAN training sample combines controls in every period with training-treated units in pre-treatment periods. We require the conditional residual law given $S = s$ to be the same across these observed-untreated source cells and the target post-treatment cell, and require target-state support to lie in their overlap. Including event time in S permits time-varying laws but does not create support where none is observed. The odds entering the control part of the criterion are functions of the common $\hat{U}_i$; treated pre-period rows have unit mass. Under the shared-law condition, mixing these two observed-untreated sources changes the state measure but not the conditional residual target. Transfer from training risk to target risk also requires the measure-domination condition below.

For two conditional CDFs define the treated-population Cramér metric

$$d^2_{\mathrm{C},1}(F, G) = \mathbb{E}\left[\int_{\mathbb{R}} \{F(r \mid S) - G(r \mid S)\}^2 \, dr \,\middle|\, D = 1\right]. \tag{34}$$

For a real-valued distribution F, write $ES(F, y) = \mathbb{E}_F|X - y| - \frac{1}{2}\mathbb{E}_F|X - X'|$. If $Y \sim G$, then $\mathbb{E}_G ES(F, Y) - \mathbb{E}_G ES(G, Y) = \int \{F(r) - G(r)\}^2 dr$. Conditional integration over the treated-state measure gives (34); the score is strictly proper on laws with finite first moment (Gneiting and Raftery 2007; Székely and Rizzo 2013). The ordinary energy term used by ETC-GAN is its two-draw sample representation. The tail-transformed energy and finite-grid pinball terms are also minimized at the truth when correctly evaluated, while the global

feature critic is an auxiliary joint-distribution discrepancy. The proper-score component identifies the conditional law.

Assumption 13 (ETC-GAN distributional sieve)**.** *This assumption describes an expanding mathematical sequence. Section Appendix A.18 separately bounds the implemented finite-grid transport, including its endpoint atoms and approximation floor. Let $\mathcal{S}_1$ be a compact neighborhood of the treated-state support. Density and dimension conditions below concern intrinsic continuous state coordinates, after removing fixed time labels and deterministic derived coordinates such as the factor signal and its absolute value. Discrete time cells are handled separately; a density is not assumed in the redundant concatenated input vector.*

1. *Factor-based unconfoundedness, the shared residual-law condition above, and overlap hold for U_i, with $e(U_i) \in [\underline{e}, \bar{e}] \subset (0,1)$ and $\Pr(D_i = 1 \mid S_{it}) = e(U_i)$. The weighted observed-untreated training-state measure dominates each target treated-state measure with a uniformly bounded Radon–Nikodym derivative. Support inclusion alone is not this quantitative condition; if the training weight on a target time cell vanishes, the condition needs separate justification or a cell-specific criterion. The treated predictive-state distribution has a density bounded away from zero on each continuous component of $\mathcal{S}_1$, whose support satisfies an interior-cone condition. Finite discrete components, such as event time, are handled cell by cell.*

2. *The monotone transported-quantile classes $\mathcal{F}_N$ have maximum quantile-grid spacing $h_N \to 0$ and endpoint levels tending to zero and one, and*

$$a_N := \inf_{F \in \mathcal{F}_N} \|F - F_\varepsilon\|_{\infty,\mathcal{S}_1} \longrightarrow 0, \quad \|H\|_{\infty,\mathcal{S}_1} := \sup_{s \in \mathcal{S}_1, r \in \mathbb{R}} |H(r \mid s)|.$$

Uniform tail tightness turns the interior-grid approximation into an approximation on $\mathbb{R}$. The target and sieve CDFs have uniformly bounded first moments and are equi-Lipschitz in (r, s). These are mathematical sieve conditions, not properties implied by a fixed clamped quantile grid: endpoint atoms would require a separate approximation

remainder or continuous endpoint treatment. The fixed shift and log-spacing bounds in the reported code satisfy the approximation condition only if the target lies in their closure; an alternative formal sequence may let those bounds expand slowly. A fixed bounded class that excludes the truth has $a_N \not\to 0$ and is not covered.

3. *Cross-fitted propensity, QTE-branch regression mean, and aligned state estimates obey*

$$b_N := \|\hat{e}(\hat{U}) - e(U)\|_\infty + \|\hat{\mu}^Q - \mu^Q\|_\infty + \|\hat{S} - S\|_\infty = o_p(1).$$

The weighted empirical proper-score and adversarial criteria obey a unit-cluster uniform law of large numbers on $\mathcal{F}_N$. All row-level resampling bounds account for dependence within a unit rather than counting unit–period rows as independent. Uniform convergence of the aligned state here is stronger than the average-L_2 anchor rate alone; it must be supplied separately. Neither this law of large numbers nor downstream cross-fitting establishes a root-N evaluation bound for a random CDF built from a shared estimated state.

4. *Let $\mathfrak{R}_N(F)$ be the complete population criterion, including pinball, ordinary and tail energy, global feature discrepancy, mean preservation, and transport regularization, evaluated on the population state and residual law. The ordinary energy weight is bounded away from zero. Every term treated as truth-compatible must be minimized at F_ε under the maintained law; this is not automatic for a fixed finite-draw mean penalty or transport regularizer. Let $\alpha_N \to 0$ bound the square root of the positive complete-criterion excess risk of a suitable sieve approximant, and let $\kappa_N \to 0$ bound the metric-scale bias of non-truth-compatible penalties. Specifically, for some $F_N \in \mathcal{F}_N$ and fixed $c > 0$,*

$$[\mathfrak{R}_N(F_N) - \mathfrak{R}_N(F_\varepsilon)]_+ \le \alpha_N^2, \qquad \mathfrak{R}_N(F) - \mathfrak{R}_N(F_\varepsilon) \ge c\, d_{\mathrm{C},1}^2(F, F_\varepsilon) - \kappa_N^2 \quad (F \in \mathcal{F}_N).$$

The lower bound follows from the positive proper-score weight, training-to-target measure domination, and a uniform bound on any negative penalty excess. The complete-risk approximation α_N is a separate requirement: uniform CDF approximation a_N on an unbounded outcome domain does not by itself give a same-order Cramér or complete-risk bound. The returned adversarial-phase checkpoint obeys the metric-scale excess-risk condition

$$\mathfrak{R}_N(\widehat{F}_\varepsilon) - \inf_{F \in \mathcal{F}_N} \mathfrak{R}_N(F) = O_p(\upsilon_N^2), \qquad \upsilon_N = o(1). \tag{35}$$

This υ_N includes empirical-process, optimization, finite-draw, finite-batch, validation-selection, and estimated-input criterion errors on the Cramér scale. In particular, the shared-state cross-pair diagonal in the implemented critic introduces an $O(B^{-1})$ risk discrepancy relative to the independent-joint-law criterion in (164)*. This discrepancy must vanish within the excess-risk gate, for example along an appropriate $B_N \to \infty$ sequence; fixed reported batches do not establish that limit. A unit-cluster law of large numbers alone does not supply a particular rate for these errors.*

5. *The spectrally normalized critic classes are complexity controlled. Their population feature-energy discrepancies are nonnegative and vanish when the real and generated joint laws agree. The adversarial term therefore cannot remove the identification supplied by the positive-weight proper score. Propensity clipping/capping is asymptotically inactive for the treated-population target.*

Assumption 13 follows the sieve approach of Chen (2007): approximation error falls while complexity grows slowly enough for uniform convergence. Early stopping and local numerical solutions are allowed when their excess risk vanishes. The approximation–sampling decomposition follows the adversarial oracle approach of Liang (2021); our excess-risk condition separately retains optimization and implementation errors. The reported fixed transport penalty and bounds remain finite-sample choices; a consistency sequence must satisfy the

stated approximation and risk conditions.

Theorem 4 (Weighted residual-distribution consistency and rate)**.** *In regime R, under Assumption 13, let* $\widehat{F}_\varepsilon$ *be the conditional residual CDF induced by the ETC-GAN quantile transport and define*

$$r_N = \alpha_N + \kappa_N + b_N + h_N + \upsilon_N. \tag{36}$$

Then

$$d_{\mathrm{C},1}(\widehat{F}_\varepsilon, F_\varepsilon) = O_p(r_N). \tag{37}$$

Let d_s *denote the continuous dimension of the conditioning state. The integrated Cramér rate also implies*

$$\sup_{s\in\mathcal{S}_1, r\in\mathbb{R}} |\widehat{F}_\varepsilon(r \mid s) - F_\varepsilon(r \mid s)| = O_p\left(r_N^{\frac{2}{d_s+3}}\right) = o_p(1). \tag{38}$$

Under factor-based unconfoundedness the conditional target equals the untreated conditional residual law on the overlap region. Odds weighting determines the treated-state measure in (34)*; it does not create double robustness.*

Proof. For the control component and any integrable loss $\ell(S, \varepsilon(0))$, iterated expectations, factor-based unconfoundedness, and overlap give

$$\frac{\mathbb{E}[(1-D)e(U)\{1-e(U)\}^{-1}\ell]}{\mathbb{E}[(1-D)e(U)\{1-e(U)\}^{-1}]} = \mathbb{E}[\ell \mid D=1].$$

The directly observed treated-pre component has the same conditional residual law by assumption, so adding it preserves the conditional target. For the ordinary energy score evaluated under the target treated-state measure, strict propriety gives the identity

$$R_{\mathrm{E}}(F) - R_{\mathrm{E}}(F_\varepsilon) = d^2_{\mathrm{C},1}(F, F_\varepsilon),$$

so F_ε is its unique minimizer. Training-to-target measure domination and the explicit

complete-risk lower bound in Assumption 13 transfer this identification to the training criterion. The approximate-extremum and approximant bounds then give

$$c\, d_{\mathrm{C},1}^2(\widehat{F}_\varepsilon, F_\varepsilon) \le \alpha_N^2 + \kappa_N^2 + O_p(\upsilon_N^2),$$

which is bounded by $O_p(r_N^2)$. This proves (37); see Chen (2007) and the adversarial oracle formulation of Liang (2021). The contributions of b_N or h_N, including estimated-input and grid errors, must satisfy the stated metric-scale risk bound.

For the uniform statement, let $H_N(s,r) = \widehat{F}_\varepsilon(r \mid s) - F_\varepsilon(r \mid s)$ and $q_N = \|H_N\|_{\infty,\mathcal{S}_1}$. Equi-Lipschitzness implies that if $|H_N(s_0,r_0)| = q_N$, then $|H_N| \ge q_N/2$ on a ball of radius proportional to q_N around (s_0,r_0). The lower state-density and interior-cone conditions make that ball have probability–Lebesgue volume of order $q_N^{d_s+1}$. Hence $d_{\mathrm{C},1}^2 \ge C q_N^{d_s+3}$, so $q_N \le C d_{\mathrm{C},1}^{2/(d_s+3)}$. Equation (38) follows from (37). □

For a fixed post-treatment period, write

$$F_{0t}(y) = \Pr\{Y_{it}(0) \le y \mid D_i = 1\} = \mathbb{E}\big[F_\varepsilon\{y - \mu^Q(S_{it}) \mid S_{it}\} \mid D_i = 1\big]$$

and estimate it by

$$\widehat{F}_{0t}^{\mathrm{plug}}(y) = \frac{1}{N_{\mathrm{tr}}} \sum_{i \in \mathcal{T}} \widehat{F}_\varepsilon^{(-k(i))}\{y - \hat{\mu}_{it}^Q \mid \hat{S}_{it}\}.$$

There is no augmented-CDF residual or logistic fluctuation in this estimator.

Proposition 8 (Plug-in counterfactual-CDF consistency)**.** *Suppose Theorem 4 and the state/mean conditions in Assumption 13 hold uniformly over a fixed number of folds and target periods. Let*

$$\epsilon_N = \max_k \|\widehat{F}_\varepsilon^{(-k)} - F_\varepsilon\|_{\infty,\mathcal{S}_1} = o_p(1), \qquad \ell_N = \max_t \sup_y \left| \frac{1}{N_{\mathrm{tr}}} \sum_{i \in \mathcal{T}} m_{t,y}(S_{it}) - F_{0t}(y) \right| = o_p(1),$$

where the latter is a unit-cluster uniform law of large numbers for the fixed population

conditional-CDF functions $m_{t,y}(s) = F_\varepsilon\{y - \mu^Q(s) \mid s\}$. *If numerical evaluation contributes a uniform error* $\eta_N^{\text{eval}} = o_p(1)$, *then*

$$\max_t \sup_y |\widehat{F}_{0t}^{\text{plug}}(y) - F_{0t}(y)| \le \epsilon_N + O_p(b_N) + \ell_N + \eta_N^{\text{eval}} = o_p(1). \tag{39}$$

This conclusion does not require conditional independence of the fitted maps and their evaluation units: the first two terms are controlled uniformly before taking the sample average. It does require the stated uniform learner and aligned-state consistency, not merely an average state-recovery rate.

Proof. For each scored unit, add and subtract the true residual CDF evaluated at its estimated mean and state, and then the population conditional CDF $m_{t,y}(S_{it})$. Uniform conditional-CDF error bounds the first difference by ϵ_N regardless of sample reuse. Equi-Lipschitzness and the uniform aligned-state/mean condition bound the second by $O_p(b_N)$. Averaging the remaining fixed population functions gives ℓ_N, and evaluation error adds η_N^{eval}. Taking suprema proves the bound. Own-history loading estimates, common factors, and rank selection need not be declared independent to use this deterministic decomposition; their error is part of the uniform state/learner premises. □

Corollary 1 (Period-specific and pooled marginal-QTE consistency)**.** *Under Proposition 8, let* $\mathcal{A} = [\underline{\tau}, \bar{\tau}] \subset (0,1)$. *Suppose the observed treated empirical CDF obeys a uniform law of large numbers, and* F_{0t} *and* F_{1t} *are continuously differentiable with densities uniformly bounded below by* $\underline{f} > 0$ *in neighborhoods of the quantiles indexed by* $\mathcal{A}$. *Then*

$$\sup_{\tau \in \mathcal{A}} |\widehat{QTE}_t^{\text{plug}}(\tau) - QTE_t(\tau)| = o_p(1). \tag{40}$$

For a fixed number of post-treatment periods, the same conclusion holds for equation (168) *when the mixed marginals satisfy the same local density condition; period CDFs are mixed using* ω *before inversion, and estimated weights add* $\|\hat{\omega} - \omega\|_1 = o_p(1)$. *This does not concern*

an average of period-specific QTEs, which is a different estimand. If the unobserved realized treated empirical counterfactual law also obeys the corresponding cluster uniform law of large numbers, its quantiles converge to their population counterparts, so the plug-in estimator is also consistent for the realized finite-panel QTE along this sequence. This consistency equivalence does not equate their first-order scores or furnish an interval for a fixed realized cohort.

Proof. Proposition 8 and the observed-marginal uniform law of large numbers give uniform consistency of both marginal CDFs. The inverse-CDF map is locally Lipschitz with constant at most $\underline{f}^{-1}$ on $\mathcal{A}$ (van der Vaart 1998, Chapter 21); applying it to both marginals and differencing proves (40). A finite weighted mixture preserves the CDF bound, and its own positive-density condition permits inversion. The realized-target statement follows by a triangle inequality with its population counterpart. □

The plug-in estimator is consistent under the stated recovery, learner, and numerical conditions. A root-N plug-in limit requires additional control of the fitted-CDF contribution, as formalized in the conditional bootstrap results below. Propensity weighting alone does not make the estimator doubly robust or orthogonal.

Scope of the current QTE analysis. Geithner reports pointwise bootstrap intervals under the company-population sampling and full-refit regularity conditions below. Archived targeted-QTE theory concerns a different estimator and supplies no justification for these plug-in intervals or for a fixed realized cohort.

Appendix A.11 Exposure shift, fitted linear predictions, and residual correction

This section specializes the theory to the paired ATT benchmark and its covariate-shift extension. Write $L_i = (1.5\lambda_{i1}, 1.2\lambda_{i2})'$ and

$$Y_{it}(0) = 0.7Z_i + 0.5W_{it} + L_i' f_t + h(Z_i, W_{it}) - a(Z_i) + \varepsilon_{it},$$
$$h(z,w) = 3\tanh(1.2z)\tanh\{1.2(z+w)\}.$$

Here $a(z) = \mathbb{E}_{Q_0} h(z, W)$, $Q_0 = U[-2,2]$, and $Q_\delta = (1/2-\delta)U[-2,0] + (1/2+\delta)U[0,2]$ is the post-period exposure law in both arms, with $\delta \in [0, 1/2]$. The shared design uses $\delta = 0.15$ or 0. The bounded, mean-zero loadings are independent of (Z, D, W); $Z \sim U[-1.5, 1.5]$, $e(Z) = \text{expit}(c + Z)$, and independent errors are $N(0, 0.3^2)$, independent of all other primitives. Loading components are independent with $\mathbb{E}(LL') = \text{diag}(2.25, 1.44)$; exposures are independent across units and periods under their specified period laws. Treatment adds $\tau = 2$. Expectations $\mathbb{E}_d$ condition on $D = d$.

Define the control-pre projection and its omitted response by

$$b = \frac{\mathbb{E}_{0,\text{pre}}[W\{h(Z,W) - a(Z)\}]}{4/3}, \qquad \beta_0 = (0.7, 0.5 + b)',$$
$$\ell_{it} = (Z_i, W_{it})\beta_0 + L_i' f_t, \qquad H_{it} = h(Z_i, W_{it}) - a(Z_i) - bW_{it}. \tag{41}$$

These are analytical population objects, not inputs supplied to a learner. Put $K(z) = \mathbb{E}_{U[0,2]} h(z, W) - \mathbb{E}_{U[-2,0]} h(z, W)$. Then

$$\mathbb{E}(H \mid Z = z, \text{post}) = \delta K(z) - 2b\delta, \qquad A_\delta = \mathbb{E}_{1,\text{post}} H = \delta\{\mathbb{E}_1 K - 2b\}. \tag{42}$$

An unweighted control-residual subtraction instead leaves

$$B_\delta = \delta(\mathbb{E}_1 K - \mathbb{E}_0 K) = \frac{\delta \operatorname{Cov}\{K(Z), e(Z)\}}{p(1-p)}, \qquad p = \Pr(D = 1).$$

Thus the raw LFM error and a treated-minus-control residual error are different objects.

Exact finite-panel decomposition. In a fold, let $\widehat{F}_-$ and $\widehat{F}_+$ denote the estimated pre and post factor matrices, and put $P = \widehat{F}_+ \widehat{F}_-^+$. The own-pre-history loading regression gives $\widehat{m}_{L,i,+} = X_{i,+}\widehat{\beta} + P(Y_{i,-} - X_{i,-}\widehat{\beta})$ exactly. Consequently,

$$\begin{aligned} \widehat{\tau}_L - \tau = \operatorname{avg}_{D_i=1,+} \big[& H_{i,+} + \varepsilon_{i,+} - P(H_{i,-} + \varepsilon_{i,-}) \\ & - (X_{i,+} - PX_{i,-})(\widehat{\beta} - \beta_0) + (F_+ - PF_-)L_i \big]. \end{aligned} \tag{43}$$

This identity applies at the selected rank. It separates omitted response, own-history projection, coefficient error, and factor extrapolation.

Proposition 9 (Fitted LFM bias in the centered shift design)**.** *Embed the shared design in a sequence with the number of training controls $n \to \infty$, $T_0 \to \infty$ through complete 12-period factor cycles, and a fixed post window $J = 12$. Keep finitely many unit folds with growing training and scoring groups, allocated independently of outcomes conditional on treatment status. Use rank two and the implemented pooled-OLS initialization followed by coefficient least squares and rank-two SVD, with any stopping iteration bounded by a fixed finite $K_{\max}$. Use exact arithmetic, or errors vanishing in the normalized norms below. Then the treated-post linear predictions converge to ℓ in average squared error, and*

$$\widehat{\tau}_L - \tau \xrightarrow{p} A_\delta.$$

For the specified response and assignment rule, $A_\delta > 0$ for $\delta > 0$ and $A_0 = 0$.

Proof. Let $M_0 = LF'$, $R = H + \varepsilon$, and $T = T_0 + J$. In pre periods, $\mathbb{E}(H \mid Z) = 0$,

$\mathbb{E}_{0,\text{pre}}WH = 0$, and $\mathbb{E}W = 0$. Independence, complete factor cycles, and $J/T \to 0$ give

$$\frac{X'X}{nT} \to_p \operatorname{diag}(\mathbb{E}_0 Z^2, 4/3), \qquad \frac{X'R}{nT} \to_p 0, \qquad \frac{X'M_0}{nT} \to_p 0,$$

where products with X vectorize panel cells. Conditional on Z, D, pre entries of R are independent, centered, uniformly sub-Gaussian. Their operator norm is $O_p(\sqrt{n}+\sqrt{T_0})$; the fixed post block has Frobenius norm $O_p(\sqrt{nJ})$. Hence $\|R\|_{\text{op}}/\sqrt{nT} = o_p(1)$ and $\|R\|_F/\sqrt{nT} = O_p(1)$.

The displayed moments make the pooled-OLS initializer consistent for β_0. If an iterate is consistent, put $E_j = R - X(\widehat{\beta}^{(j)} - \beta_0)$. Then $\|E_j\|_{\text{op}} = o_p(\sqrt{nT})$. Rank-two SVD of $M_0 + E_j$ returns M_j satisfying $\|M_j - M_0\|_{\text{op}} \le 2\|E_j\|_{\text{op}}$ and $\|M_j - M_0\|_F \le 4\|E_j\|_{\text{op}}$. The next coefficient error equals

$$(X'X/nT)^{-1}\{X'R/(nT) + X'(M_0 - M_j)/(nT)\} = o_p(1)$$

by Cauchy–Schwarz. Induction and a finite union bound cover any stopping iteration at most $K_{\max}$; global convergence of alternating optimization is not needed here.

The pre factor Gram tends to I_2 and $L'L/n$ tends to $\operatorname{diag}(2.25, 1.44)$. The two signal singular values are therefore of order $\sqrt{nT}$. Singular-subspace perturbation and the SVD normal equations give, after alignment to $F(F'F/T)^{-1/2}$ and using $F'F/T = I_2 + o(1)$, $\|\widehat{F} - F\|_F/\sqrt{T} = o_p(1)$ and $\|\widehat{L} - L\|_F/\sqrt{n} = o_p(1)$. For each fixed post period, $\mathbb{E}_0(LR_t) = 0$, so $L'R_t/n = o_p(1)$. Replacing L by $\widehat{L}$ changes this by at most $\|\widehat{L} - L\|_F\|R_t\|/n = o_p(1)$. The equation $\widehat{f}_t = (\widehat{L}'\widehat{L})^{-1}\widehat{L}'(Y_{C,t} - X_{C,t}\widehat{\beta})$ thus gives $\widehat{f}_t \to_p f_t$.

For a scored treated unit, pre-period loading least squares has a stable Gram matrix. Its true-factor residual term $F_-'R_{i,-}/T_0$ has conditional second moment $O(T_0^{-1})$. Replacing F_- by $\widehat{F}_-$ adds a term whose average squared norm is bounded by $\|\widehat{F}_- - F_-\|_F^2/T_0$ times the average normalized residual second moment, and hence vanishes. Coefficient and signal errors vanish by the preceding steps. This proves average squared loading consistency and

then treated-post prediction consistency. Cross-sectional averaging leaves $\mathbb{E}_{1,\text{post}}H = A_\delta$.

Finally, K is odd and, for $z > 0$,

$$K(z) = \frac{3\tanh(1.2z)}{2}\int_0^2 [\tanh\{1.2(z+w)\} - \tanh\{1.2(z-w)\}]\,dw > 0.$$

Symmetry and increasing e imply $\mathbb{E}_1 K > 0$. Likewise $k(z) = \mathbb{E}_{Q_0}[Wh(z,W)]$ is odd and positive for $z > 0$. The control density is proportional to $1 - e(z)$, so $\mathbb{E}_0 k < 0$; therefore $b < 0$. Equation (42) proves the sign. □

The result covers the code's finite iteration cap, not its rank selector. Extending it to selected rank requires $\Pr(\widehat{r}_k = 2\ \forall k) \to 1$ or forecast stability at other selected ranks. Higher ranks can have the same limiting pre prediction risk, so a positive validation-risk gap is not established here. Official GSC additionally uses fixed effects and its own rank search; Proposition 9 is not a theorem for that separate implementation. The asymptotic embedding $J/T \to 0$ explains a mechanism and sign, not the exact bias at the released $T_0 = 36$.

What residual correction removes. For an honest candidate mean $m = \ell + \alpha H$, let w be control weights depending on pre information with $\mathbb{E}_0 w = 1$, independent of post W, ε conditional on Z, L, D. Raw ATT bias is $(1-\alpha)A_\delta$; normalized residual correction leaves

$$(1-\alpha)\delta\{\mathbb{E}_1 K - \mathbb{E}_0[wK]\}. \tag{44}$$

To prove this, take the conditional post expectation in (42) and subtract its weighted control counterpart; the constant $-2b\delta$ cancels. Correct normalized odds make $\mathbb{E}_0[wK] = \mathbb{E}_1 K$. A correct mean ($\alpha = 1$) also removes the bias. This cancellation is supplied by the DR score, including when its mean learner is linear; a flexible mean reduces the remaining imbalance's coefficient. It is not an adversarial-specific cancellation.

Appendix A.12 Adversarial moments and target error

Within a fold, fix the generator, state maps, and weight function. The ATT critic compares weighted controls pooled over all training periods, not only target-post observations. Let ν_{tr} be that state law on the critic's actual coordinates $V = (X, \lambda, f)$, and let P_{tr} and $Q_{g,\mathrm{tr}}$ be the real and generated joint laws of (V, Y) with this same state marginal. Their conditional means are μ_{tr} and m_g. Let ν_1 and ν_{0w} be the treated-post and normalized weighted-control-post state laws, with untreated means μ_1, μ_{0w}. These means need not agree after history-based weighting.

Proposition 10 (Target-weighted critic control)**.** *For normalized DR ATT, set $\Lambda = \nu_1 - \nu_{0w}$, suppose $\nu_1, \nu_{0w} \ll \nu_{\mathrm{tr}}$, and write $a_\Lambda = d\Lambda/d\nu_{\mathrm{tr}}$. Assume the following integrals exist. Define*

$$T_\Lambda = \int (\mu_1 - \mu_{\mathrm{tr}})\, d\nu_1 - \int (\mu_{0w} - \mu_{\mathrm{tr}})\, d\nu_{0w}.$$

The population ATT bias is exactly

$$B_g = T_\Lambda + (P_{\mathrm{tr}} - Q_{g,\mathrm{tr}})\{a_\Lambda(V)Y\}.$$

Let $\mathcal{F}$ be a symmetric critic class and $D_{\mathcal{F}}(P, Q) = \sup_{f \in \mathcal{F}} |(P - Q)f|$. If some $c > 0, f \in \mathcal{F}$ satisfies $\mathbb{E}_P|a_\Lambda Y - cf| + \mathbb{E}_Q|a_\Lambda Y - cf| \le \epsilon_{\mathrm{app}}$, then

$$|B_g| \le |T_\Lambda| + cD_{\mathcal{F}}(P_{\mathrm{tr}}, Q_{g,\mathrm{tr}}) + \epsilon_{\mathrm{app}}. \tag{45}$$

For plug-in ATT, use $\Lambda = \nu_1$ and omit the second term in T_Λ.

Proof. Subtract the target ATT from the two residual means, obtaining $B_g = \int(\mu_1 - m_g)d\nu_1 - \int(\mu_{0w} - m_g)d\nu_{0w}$. Adding and subtracting μ_{tr} gives $T_\Lambda + \int(\mu_{\mathrm{tr}} - m_g)d\Lambda$. Iterated expectations identify the last integral with the joint-law moment in the statement. Add and subtract cf and apply the triangle inequality to obtain (45). □

The common conditional-law restriction makes $T_\Lambda = 0$. Otherwise matching the weighted training law can match the wrong target law. In the implementation, weights use broader own-history summaries than the critic's coordinates, and training includes pre outcomes used in those summaries. Thus neither a small discrepancy nor a rich critic alone establishes $T_\Lambda = 0$.

For a fitted critic $\widehat{f}$, write $\widehat{d}(f) = (\widehat{P} - \widehat{Q}_g)f$, $\epsilon_{\text{crit}} = \sup_f \widehat{d}(f) - \widehat{d}(\widehat{f})$, and let Δ_g bound the uniform empirical-to-population discrepancy. Then $D_{\mathcal{F}} \le \widehat{d}(\widehat{f}) + \epsilon_{\text{crit}} + \Delta_g$. The latter two terms require cluster generalization and optimization control. Gradient penalty changes the maximization objective; spectral normalization does not prove witness coverage or maximization. A logged minibatch loss is therefore not an upper certificate. Proposition 17 gives the corresponding penalized-critic bound, including the target witness's gradient penalty; Section Appendix A.17 also accounts for generator changes and the spectral-normalization map used in evaluation. For comparison, Cauchy–Schwarz also gives $|B_g - T_\Lambda| \le \|a_\Lambda\|_2 \|\mu_{\text{tr}} - m_g\|_2$. Adversarial moments offer a distinct route to target control, not a general accuracy ordering over a squared-error or noADV learner.

Appendix A.13 History-dependent nuisances and the inference remainder

Fix one post period, or aggregate the post outcome and prediction with the same fixed weights. Let $\mathcal{G} = \sigma(U, S, H_{\text{pre}})$ contain all admitted own-history information. Assume $\mathbb{E}[D \mid \mathcal{G}, Y(0)] = e(U)$ and $\mathbb{E}[\varepsilon \mid U, S] = 0$, where $\varepsilon = Y(0) - \mu_0(S)$. The first restriction concerns the joint admitted history and outcome, not just their separate marginals. Freeze nuisance maps trained independently of a fresh scored unit. They may use $\mathcal{G}$. Put

$$d_m = \widehat{m} - \mu_0, \quad d_e = \widehat{e} - e, \quad v = \frac{d_e}{1 - \widehat{e}}, \quad q = p + \mathbb{E}v, \quad c_m = \frac{\mathbb{E}(e d_m)}{p}.$$

Proposition 11 (Exact normalized history-bias identity)**.** *With true and fitted propensities*

bounded away from zero and one, finite second moments, and q uniformly bounded away from zero, the population normalized-odds ATT functional satisfies

$$\Theta(\widehat{m},\widehat{e}) - ATT = \frac{\mathbb{E}\{v(d_m - c_m)\} - \mathbb{E}(v\varepsilon)}{q}. \tag{46}$$

Its absolute bias is bounded by $C\|d_e\|_2\|d_m\|_2 + |\mathbb{E}(v\varepsilon)|/q$.

Proof. Use $(1-e)\widehat{\rho} = e + v$ and the assignment restriction for group moments. The mean restriction gives $\mathbb{E}(e\varepsilon) = \mathbb{E}[e(U)\mathbb{E}(\varepsilon \mid U, S)] = 0$. The treated residual mean is $ATT - c_m$; the weighted control denominator is q and its numerator is $\mathbb{E}(v\varepsilon) - pc_m - \mathbb{E}(vd_m)$. Subtracting the two ratios proves the identity. Overlap and Cauchy–Schwarz bound its first term. □

Full-history mean sufficiency, $\mathbb{E}(\varepsilon \mid \mathcal{G}) = 0$, eliminates the additional history term. Correctness only relative to a smaller state need not do so. For root-N inference it is sufficient to show $\mathbb{E}(v\varepsilon) = o_p(N^{-1/2})$; a product-rate assertion alone does not bound it.

An explicit interface for stitched states. For each fold k, freeze its complete fitted maps and let $\mathbb{E}^*$ integrate those maps over an independent new unit from the same population law for every fold. This is not joint conditioning on all training folds. Write $\mathbb{P}_k$ for its scored average, $\pi_k = n_k/N$, $\rho = e/(1-e)$, and set

$$u = \sum_k \pi_k \mathbb{E}^*(ed_{mk}), \quad v_* = \sum_k \pi_k \mathbb{E}^* v_k, \quad z = \sum_k \pi_k \mathbb{E}^*(v_k d_{mk}), \quad h_* = \sum_k \pi_k \mathbb{E}^*(v_k \varepsilon).$$

The following are explicit empirical-stability requirements:

$$\begin{aligned}
R_A &= \sum_k \pi_k(\mathbb{P}_k - \mathbb{E}^*)Dd_{mk} = o_p(N^{-1/2}),\\
R_Q &= \sum_k \pi_k(\mathbb{P}_k - \mathbb{E}^*)(1-D)(\widehat{\rho}_k - \rho) = o_p(N^{-1/2}),\\
R_B &= \sum_k \pi_k(\mathbb{P}_k - \mathbb{E}^*)(1-D)\{(\widehat{\rho}_k - \rho)\varepsilon - \widehat{\rho}_k d_{mk}\} = o_p(N^{-1/2}).
\end{aligned} \tag{47}$$

If the nuisance errors vanish in L_2, conditional residual second moments are bounded, denominators stay positive, stabilization is asymptotically inactive, and additional numerical errors are $o_p(N^{-1/2})$, then

$$\widehat{\theta} - \theta = \mathbb{P}_N \psi + \frac{z - h_* - u v_*/p}{p + v_*} + o_p(N^{-1/2}). \tag{48}$$

Here ψ is (22), and its oracle cluster CLT and moment conditions are retained. To verify the expansion, let A_N, B_N, P_N, Q_N be the oracle treated numerator, control numerator, treated denominator, and odds denominator. Equation (47) gives $\widehat{A}_N = A_N - u + o_p(N^{-1/2})$, $\widehat{Q}_N = Q_N + v_* + o_p(N^{-1/2})$, and $\widehat{B}_N = B_N + h_* - u - z + o_p(N^{-1/2})$. The difference from the oracle two-ratio estimator is

$$-\frac{u}{P_N} + \frac{u + z - h_*}{Q_N + v_*} + \frac{B_N v_*}{Q_N(Q_N + v_*)} + o_p(N^{-1/2}).$$

Use $P_N, Q_N = p + O_p(N^{-1/2})$, $B_N = O_p(N^{-1/2})$, and $u, v_*, h_* = o_p(1)$ to obtain (48). Thus the product rate, $h_* = o_p(N^{-1/2})$, and the three stability conditions suffice. Honest full-pipeline fold training with bounded conditional residual variance verifies the centered-score conditions by conditional variance bounds and finitely many folds when scored units retain this population law. Treatment-stratified folds additionally require control of their centering difference from the common fresh-unit law; within-stratum independence alone is insufficient. The actual stitched-state and global rank-selection operations still need their own stability argument.

Appendix A.14 Fixed generator draws in ATT inference

Proposition 12 (Conditional fixed-M algorithmic influence function)**.** *Suppose a stable full generator limit $g_0(V, z)$ exists, nonrandom conditional on the common environment. Let $\mathcal{H}$ contain the full admitted own history and U, V, S, with $\mathbb{E}(D \mid \mathcal{H}) = e(U)$ and $\mathbb{E}_z g_0(V, z) = \mu_0(S)$. For fixed M, use independent unit-specific evaluation marks, independent of the full*

unit records (including treatment and history) and of fitting and selection. The propensity map does not use these marks. Let

$$\zeta_i^M = M^{-1}\sum_{m=1}^{M}\{g_0(V_i, z_{im}) - \mu_0(S_i)\}, \qquad q_{0i} = \mu_0(S_i) + \zeta_i^M.$$

Require L_2 convergence of the returned evaluated prediction to q_0 at rate $r_{m,M} = o_p(1)$, propensity rate $r_e = o_p(1)$ with $r_{m,M}r_e = o_p(N^{-1/2})$, and the history, empirical-stability, inactive-stabilization, and numerical conditions of (48) for residual $\varepsilon - \zeta^M$. In particular require $h_* = o_p(N^{-1/2})$, or the stronger condition $\mathbb{E}(\varepsilon \mid \mathcal{G}) = 0$, and a bounded conditional second moment for $\varepsilon - \zeta^M$ given the admitted unit information before auxiliary marks. Assume the augmented unit scores satisfy a CLT and their estimated scores are L_2 consistent with a second-moment law of large numbers. Then the influence function is

$$\psi_i^M = \psi_i - \frac{D_i - e_i}{p(1-e_i)}\zeta_i^M, \qquad V_M = V + \frac{1}{M}\mathbb{E}\left[\frac{e(U)\sigma_g^2(V)}{p^2\{1-e(U)\}}\right], \tag{49}$$

where $\sigma_g^2(V) = \mathrm{Var}_z\{g_0(V,z)\}$ holds V fixed. The empirical variance of the evaluated-mean score estimates V_M.

Proof. Replace the reference prediction by q_0 in the two-ratio expansion. Because the propensity excludes evaluation marks and $\mathbb{E}_z\zeta^M = 0$, $\mathbb{E}[v(\varepsilon - \zeta^M)] = \mathbb{E}(v\varepsilon)$. The same history and product-rate argument therefore applies to $\widehat{m}^M - q_0$. Substituting q_0 in the oracle score subtracts $\{D-e\}\zeta^M/[p(1-e)]$, giving the first identity. Conditional independence and zero mean make this added term uncorrelated with ψ. Since $\mathbb{E}[(D-e)^2 \mid \mathcal{H}] = e(1-e)$ and U, V are $\mathcal{H}$-measurable, its variance is the nonnegative second term in (49). Score L_2 consistency and the augmented-cluster law of large numbers prove the variance assertion. □

For a pooled target, aggregate ζ_{it}^M with the target's period weights and retain its within-unit covariance. The code uses a distinct post draw bank for each unit and evaluates its influence score using the resulting mean, which is compatible with this form. The theorem

concerns repeated sampling including independent auxiliary marks; recording a PRNG seed does not establish generator convergence or stitched-state stability. Fixed-M randomization adds variance, not a general efficiency advantage over a deterministic conditional mean.

Appendix A.15 From average distributional risk to marginal QTE

Marginal estimation need not require uniform accuracy at every state. Conditional on a fixed common-factor path, let n treated units be independent draws. For each unit, let F_i be its true conditional untreated CDF and G_i an arbitrary fitted valid CDF. For a fixed post window, each is the same predetermined weighted period mixture. The G_i may depend on all data. Define

$$\bar{F}_n = n^{-1}\sum_i F_i, \quad \bar{G}_n = n^{-1}\sum_i G_i, \quad F_0 = \mathbb{E}F_i, \quad \mathcal{R}_n = n^{-1}\sum_i \int_{\mathbb{R}} (G_i - F_i)^2\, dy.$$

The oracle $\bar{F}_n$ is a conditional mixture for realized states, not the unobserved empirical distribution of realized counterfactuals.

Proposition 13 (Average-risk marginal-CDF bound)**.** *Suppose $\mathcal{R}_n < \infty$ and the population marginal F_0 is L-Lipschitz. With $\ell_n = \|\bar{F}_n - F_0\|_\infty$,*

$$\|\bar{G}_n - F_0\|_\infty \le \ell_n + (3L\mathcal{R}_n)^{1/3}. \tag{50}$$

Independent target units give $\ell_n = O_p(n^{-1/2})$. Thus $\mathcal{R}_n = O_p(r_n^2)$ implies marginal-CDF error $O_p(r_n^{2/3} + n^{-1/2})$, plus any uniform numerical evaluation error. No continuity of G_i, or uniform-in-state learner rate, is required.

Proof. Jensen and Tonelli give $\int(\bar{G}_n - \bar{F}_n)^2 \le \mathcal{R}_n$. If $q = \bar{G}_n(x) - F_0(x) > \ell_n$, monotonicity and the Lipschitz bound give $\bar{G}_n(y) - \bar{F}_n(y) \ge q - \ell_n - L(y - x)$ for $y \ge x$. Integrating its squared nonnegative part up to $x + (q - \ell_n)/L$ yields $\mathcal{R}_n \ge (q - \ell_n)^3/(3L)$. The negative gap uses $y \le x$; take suprema to obtain the result. Introduce independent draws $Y_i^* \mid S_i \sim F_i$.

Their empirical CDF has conditional expectation $\bar{F}_n$ and marginal law F_0. Conditional Jensen and the expected empirical-CDF supremum bound give $\mathbb{E}\ell_n \le Cn^{-1/2}$, hence the asserted stochastic order. A finite period mixture is covered by first drawing its period index with the declared weights; within-unit time independence is unnecessary. □

If F_0, F_1 have densities at least $c_0, c_1 > 0$ on neighborhoods of the quantiles in $\mathcal{A} \Subset (0,1)$, generalized-inverse bracketing gives, for sufficiently small CDF errors,

$$\sup_{\tau\in\mathcal{A}} |\widehat{QTE}(\tau) - QTE(\tau)| \le \frac{\|\widehat{F}_0 - F_0\|_\infty}{c_0} + \frac{\|\widehat{F}_1 - F_1\|_\infty}{c_1}. \tag{51}$$

Indeed $F^{-1}(\tau - e) \le \widehat{F}^{-1}(\tau) \le F^{-1}(\tau + e)$ when $\|\widehat{F} - F\|_\infty \le e$, and the inverse is locally $1/c$-Lipschitz. Small density near a tail amplifies distribution error. Endpoint atoms in the fitted law do not invalidate this argument.

Learner, mean, and state contributions. For a single period, write $F_i(y) = F_\varepsilon(y - \mu_i \mid S_i)$ and $G_i(y) = \widehat{F}_\varepsilon(y - \widehat{\mu}_i \mid \widehat{S}_i)$. Suppose the true residual family has density bounded by L_ε and is L_S-Lipschitz in state in CDF L_2 norm on the relevant neighborhood. Define

$$A_n^2 = n^{-1}\sum_i \|\widehat{F}_\varepsilon(\cdot \mid \widehat{S}_i) - F_\varepsilon(\cdot \mid \widehat{S}_i)\|_2^2, \quad a_n^2 = n^{-1}\sum_i \|\widehat{S}_i - S_i\|^2, \quad b_n^2 = n^{-1}\sum_i |\widehat{\mu}_i - \mu_i|^2.$$

Minkowski gives $\sqrt{\mathcal{R}_n} \le A_n + L_S a_n + \sqrt{L_\varepsilon}\, b_n$. For the last term, the CDF translation identity and bounded density give $\int\{F(y-a) - F(y-b)\}^2 dy \le \|F(\cdot - a) - F(\cdot - b)\|_\infty \|F(\cdot - a) - F(\cdot - b)\|_1 \le L_\varepsilon (a-b)^2$. For a pooled target apply this decomposition periodwise and average with the fixed period weights, using Jensen for the mixture risk. The conditional energy-score identity in the preceding subsection controls the squared-CDF learner loss. Transferring population training risk to the actual scored-cohort A_n still requires honest full-pipeline evaluation, a uniform law for the complete fitted map, or a stability bound. Downstream fold labels alone do not supply that step. This route applies to ETC-GAN and its proper-score noADV counterpart. It establishes point consistency and an error bound, not root-N

QTE inference or a first-order interval for a fixed realized cohort.

What the Cramér critic can additionally test. For real and generated joint laws P, Q with the target state marginal, let $y = \mu(S) + R$ and consider the two-point feature $\phi_a(S, R) = v$ if $y \le a$ and $-v$ otherwise, with $\|v\| = 1$. Writing $p_a = P(y \le a)$ and $q_a = Q(y \le a)$, direct enumeration in the energy distance gives $\mathcal{D}_E(\phi_a \# P, \phi_a \# Q) = 4(p_a - q_a)^2$. If the critic class approximates every such witness with $\mathbb{E}_P\|\phi - \phi_a\| + \mathbb{E}_Q\|\phi - \phi_a\| \le \zeta$, the triangle inequality for distances changes its energy discrepancy by at most 4ζ. Consequently,

$$\|F_P - F_Q\|_\infty \le \frac{1}{2}\sqrt{\sup_{\phi \in \Phi} \mathcal{D}_E(\phi \# P, \phi \# Q) + 4\zeta}.$$

This is a target-CDF witness bound. A different training-state marginal requires target reweighting or an additional transport bound. The outcome threshold must also be measurable in the critic's actual coordinates. The fitted QTE mean uses information not directly included in the residual critic, so this measurability is a separate requirement; Section Appendix A.17 retains it in the implemented-critic bound. The fixed neural critic and finite alternating updates do not establish witness coverage or maximization. The adversarial component therefore adds adaptive joint-distribution tests, while the proper-score risk bound remains available without it.

Appendix A.16 Deep-network approximation and ATT prediction error

The correction network enlarges the response class beyond a linear prediction surface, while the augmented score controls residual imbalance. This section relates these two steps to the network and training criterion in `R/wgan_training.py`. We condition on a fold's fitted upstream coordinates. Approximation, optimization, selection, and target-law transfer enter separately; the resulting bounds do not require an assumption that the fitted checkpoint is

a global optimum.

Appendix A.16.1 The exact generator and objective

Write V for all state information supplied to a frozen fitted map and $b(V)$ for its fitted LFM skip. The implemented class is

$$G_\phi(V, z) = b(V) + \delta_\phi\big(X, (P_u\widehat{\lambda}) \odot (P_t\widehat{f}), z\big), \qquad z \sim N(0, I).$$

Here δ_ϕ has three Linear–LayerNorm–LeakyReLU blocks and a linear output in the default specification. The upstream coordinates are fixed buffers; the two projections and downstream layers are trained. In the bounds below, g denotes a candidate G_ϕ, $\widehat{g} = G_{\widehat{\phi}}$ the returned generator, and f a frozen critic D_ψ. The symbol V collects the fitted inputs; S continues to denote the population adjustment state where that distinction matters. Define $m_g(V) = \mathbb{E}_z g(V, z)$ and $v_g(V) = \mathrm{Var}_z\{g(V, z)\}$.

The reconstruction loss is a squared error of a *single random draw*, not the squared error of m_g. Let ν be the normalized weighted control-state law, pooling all training periods, and write $Y = \mu + \varepsilon$ for a declared structural untreated mean. The state may include generated own-history features. With

$$u_g^2 = \mathbb{E}_\nu(m_g - \mu)^2 + \mathbb{E}_\nu v_g, \qquad c_g = \mathbb{E}_\nu\{(m_g - \mu)\varepsilon\},$$

independent fresh generator noise gives the exact identities

$$\mathbb{E}_\nu\mathbb{E}_z(Y - g)^2 = u_g^2 + \mathbb{E}_\nu\varepsilon^2 - 2c_g, \tag{52}$$

$$\mathbb{E}_\nu\mathbb{E}_z(g - b)^2 = u_g^2 + \|\mu - b\|_{L_2(\nu)}^2 + 2\mathbb{E}_\nu\{(\mu - b)(m_g - \mu)\}. \tag{53}$$

Neither identity requires $c_g = 0$. That equality follows if $\mathbb{E}(\varepsilon \mid V) = 0$, but it need not hold merely because $\mathbb{E}(\varepsilon \mid S) = 0$ for an oracle state S contained in V. An own-pre-history

feature can contain the same residual that appears in a pooled pre-period training outcome.

For the actual default $\lambda_{\text{var}} = 0$, the expectation over fresh calibration draws, conditional on its observed calibration panel, is exactly the noise-averaged calibration loss $C(g)$:

$$C(g) = \frac{1}{T_0} \sum_{t \le T_0} \left[\left\{ \frac{1}{n_c} \sum_{i \in C_c} (m_g(V_{it}) - Y_{it}) \right\}^2 + \frac{1}{M_c n_c^2} \sum_{i \in C_c} v_g(V_{it}) \right]. \tag{54}$$

The generator noises are independent across draws and calibration cells, which proves the variance term. The optional squared variance-calibration term, when enabled, is another nonnegative term in C; its expectation is not simply a population variance mismatch. Calibration is an average of period-level treated-pre mean errors. It is not a post-period ATT validation score.

For a frozen critic f, the objective corresponding to a generator update is

$$J_f(g) = a\,\mathbb{E}_\nu \mathbb{E}_z (Y - g)^2 + d\,\mathbb{E}_\nu \mathbb{E}_z (g - b)^2 + t\,C(g) - \mathbb{E}_\nu \mathbb{E}_z f(V, g), \tag{55}$$

where $a = \lambda_{\text{reg}}$, $d = \lambda_\delta$, and $t = \lambda_{\text{cal}}$, with default values $a = 5$, $d = 0.5$, and $t = 1$. The critic's input contains $(Y, X, \widehat{\lambda}, \widehat{f})$ separately; it is not the generator's projected interaction input. The gradient penalty is in the *critic* update. Adam weight decay is an update rule and is not an additional term in the logged generator objective; no claim below identifies it with an exactly optimized penalized risk.

Appendix A.16.2 An oracle inequality for the implemented objective

Let $\mathcal{G}$ be any fixed parameter-constrained subclass of this architecture, containing the released $\widehat{g}$ and a comparator $g_\circ$. A random class is also allowed for the deterministic statement below. Write $\widehat{J}_f$ for a *common* empirical evaluator of (55), with its normalized weights and

noise convention held fixed across candidate generators. Suppose

$$\sup_{g\in\mathcal{G}} |\widehat{J}_f(g) - J_f(g)| \le \Delta, \qquad \widehat{J}_f(\widehat{g}) \le \inf_{g\in\mathcal{G}} \widehat{J}_f(g) + \epsilon_{\text{alg}}.$$

These inequalities define events or quantities to establish; the saved epoch losses do not establish them. Those losses use changing critics, fresh noises, and possibly changing mini-batches.

Proposition 14 (Implemented-objective ATT network bound)**.** *Assume $a > 0$, $d, t \ge 0$, finite moments in the displayed expressions, and that $f(V,\cdot)$ is uniformly L_f-Lipschitz. Put*

$$B = \|\mu - b\|_{L_2(\nu)}, \qquad a_\circ = \{\mathbb{E}_\nu \mathbb{E}_z (g_\circ - \mu)^2\}^{1/2}, \qquad h_{\text{rec}} = (c_{\widehat{g}} - c_{g_\circ})_+.$$

Then

$$\|m_{\widehat{g}} - \mu\|_{L_2(\nu)} \le u_{\widehat{g}} \le a_\circ + \frac{2dB + L_f}{a+d} + \left\{ \frac{tC(g_\circ) + 2\Delta + \epsilon_{\text{alg}} + 2ah_{\text{rec}}}{a+d} \right\}^{1/2}. \tag{56}$$

For a strict noADV update, remove the adversarial term and set $L_f = 0$. This is an upper bound, not an ordering of the two trained estimators.

Proof. The two empirical inequalities imply $J_f(\widehat{g}) \le J_f(g_\circ) + 2\Delta + \epsilon_{\text{alg}}$. Apply (52)–(53) and cancel the common irreducible variance and skip error B^2. The difference of reconstruction cross terms is bounded by $2ah_{\text{rec}}$. Coupling $g_\circ$ and $\widehat{g}$ with the same z and applying Cauchy–Schwarz gives

$$|\mathbb{E}\{(\mu - b)(m_{g_\circ} - m_{\widehat{g}})\}| \le B(a_\circ + u_{\widehat{g}}), \quad \mathbb{E}\{f(V,\widehat{g}) - f(V,g_\circ)\} \le L_f(u_{\widehat{g}} + a_\circ).$$

Discard the nonnegative $tC(\widehat{g})$ and obtain, with $A = a + d$, $q = 2dB + L_f$, and $R =$

$tC(g_\circ) + 2\Delta + \epsilon_{\text{alg}} + 2ah_{\text{rec}}$,

$$Au_{\widehat{g}}^2 \le Aa_\circ^2 + q(u_{\widehat{g}} + a_\circ) + R.$$

Solving this quadratic gives $u_{\widehat{g}} \le q/(2A) + \sqrt{(a_\circ + q/(2A))^2 + R/A} \le a_\circ + q/A + \sqrt{R/A}$, as claimed. □

The terms separate distinct sources of prediction error. The comparator error $a_\circ$ includes *generator variance*; a deterministic comparator is appropriate to this conditional-mean problem when the architecture can supply one. The fixed skip penalty can pull the network toward a misspecified linear mean. The adversarial term can also change the population optimum. Finally, pre-calibration may conflict with the target law, and own-history features can make reconstruction correlated with the structural residual. None of these terms disappears solely because the network is deep.

The objective-mismatch allowance can alternatively be left sharp. Put $\Omega_f(g) = d\mathbb{E}(g - b)^2 + tC(g) - \mathbb{E}f(V, g)$. Directly from the same proof,

$$\|m_{\widehat{g}} - \mu\|_2^2 \le a_\circ^2 + \frac{[\Omega_f(g_\circ) - \Omega_f(\widehat{g})]_+ + 2\Delta + \epsilon_{\text{alg}}}{a} + 2h_{\text{rec}}. \tag{57}$$

The simpler bound (56) replaces this comparison by checkable norm quantities, at the price of conservatism.

For the visited checkpoint set $\mathcal{P} \subseteq \mathcal{G}$, define

$$\epsilon_{\text{visit}} = \inf_{g \in \mathcal{P}} \widehat{J}_f(g) - \inf_{g \in \mathcal{G}} \widehat{J}_f(g), \qquad \epsilon_{\text{choice}} = \widehat{J}_f(\widehat{g}) - \inf_{g \in \mathcal{P}} \widehat{J}_f(g).$$

Both are nonnegative, and their sum is ϵ_{alg}. The first includes optimization and finite training-budget failure. The second is generally not zero: the code selects by treated-pre mean-prediction MSE, not by $\widehat{J}_f$. The code does not save a common-critic, common-noise objective evaluation for all checkpoints, so neither gap is currently certified.

A fixed-penalty counterexample. Even with a deterministic generator class containing every real constant, no adversarial or calibration term, and $Y = \mu + \varepsilon$ with mean-zero noise, the population objective $a\mathbb{E}(Y-g)^2 + d(g-b)^2$ has minimizer $g^* = (a\mu + db)/(a+d)$. Its mean error is $d|\mu - b|/(a+d) > 0$ when $d > 0$ and $b \neq \mu$. Thus representation of the truth and infinite data alone do not prove consistency of this fixed-penalty training objective. This simplified subproblem isolates one obstruction; it is not a claim that the complete calibrated adversarial fit has that exact bias.

Appendix A.16.3 Checkpoint selection and target-law transfer

Let P_s denote the treated-pre selection law. For a candidate g and an independent M_c-draw bank define $\overline{g}_{M_c} = M_c^{-1} \sum_{j=1}^{M_c} g(V, z_j)$. Write $Y = \mu_s(V) + \varepsilon_s$ with $\mathbb{E}(\varepsilon_s \mid V) = 0$. Its fresh-unit selection risk is

$$R_s(g) = \mathbb{E}_s(Y - \overline{g}_{M_c})^2 = \mathbb{E}_s\varepsilon_s^2 + \|m_g - \mu_s\|_{L_2(P_s)}^2 + \frac{\mathbb{E}_s v_g}{M_c}. \tag{58}$$

Here μ_s is the conditional mean of the full admitted selection information, not automatically the structural mean μ. On the event $\sup_{g\in\mathcal{P}} |\widehat{R}_s(g) - R_s(g)| \leq \Delta_s$, the exact saved-score minimizer satisfies

$$\|m_{\widehat{g}} - \mu_s\|_{L_2(P_s)}^2 + \frac{\mathbb{E}_s v_{\widehat{g}}}{M_c} \leq \inf_{g\in\mathcal{P}} \left\{ \|m_g - \mu_s\|_{L_2(P_s)}^2 + \frac{\mathbb{E}_s v_g}{M_c} \right\} + 2\Delta_s. \tag{59}$$

To prove it, sandwich the empirical minimum between population risks using the uniform deviation event, and subtract the common noise variance. This is an oracle inequality over *visited* checkpoints, not over all network parameters. It remains a deterministic implication if the visited path is data dependent. The deviation event still requires proof.

Suppose the post target state law P_t satisfies $dP_t/dP_s \leq C_{ts}$, and put $\eta_{ts} = \|\mu_t - \mu_s\|_{L_2(P_t)}$

for the target/reference conditional-mean discrepancy. Then

$$\|m_g - \mu_t\|_{L_2(P_t)} \le \sqrt{C_{ts}}\|m_g - \mu_s\|_{L_2(P_s)} + \eta_{ts}. \tag{60}$$

This is the triangle inequality followed by integration of the squared first term against the density ratio. If a sufficient common state makes the conditional mean invariant, $\eta_{ts} = 0$. If history distributions alter the conditional mean, or the post state lacks pre support, small selection risk does not itself control post ATT risk. The same transfer statement applies to the training law ν in (56). It is particularly important not to equate all-period weighted-control risk with treated-post risk without this step.

Appendix A.16.4 Generated coordinates and the finite architecture

The current class has an explicit, finite-parameter continuity bound. For LayerNorm with affine coefficient vector γ_j and numerical constant $\epsilon_j > 0$, its Euclidean Lipschitz constant is at most $\|\gamma_j\|_\infty/\sqrt{\epsilon_j}$. To see this, first apply the orthogonal centering projection. The derivative of $x \mapsto x/(\|x\|^2/p + \epsilon_j)^{1/2}$ has operator norm at most $1/\sqrt{\epsilon_j}$; affine scaling then gives the claim. LeakyReLU with negative slope 0.2 is 1-Lipschitz. Therefore the downstream network has valid constant

$$L_{\delta,\phi} \le \|A_{\text{out}}\|_{\text{op}} \prod_j \frac{\|\gamma_j\|_\infty \|A_j\|_{\text{op}}}{\sqrt{\epsilon_j}}.$$

On a domain where $\|\lambda\|, \|\widehat{\lambda}\| \le B_\lambda$ and $\|f\|, \|\widehat{f}\| \le B_f$,

$$\|(P_u\widehat{\lambda}) \odot (P_t\widehat{f}) - (P_u\lambda) \odot (P_t f)\|_2 \le \|P_u\|_{\text{op}}\|P_t\|_{\text{op}}\{B_f\|\widehat{\lambda} - \lambda\|_2 + B_\lambda\|\widehat{f} - f\|_2\}.$$

This follows by adding and subtracting $(P_u\lambda)\odot(P_t\widehat{f})$ and using $\|a\odot b\|_2 \le \|a\|_2\|b\|_2$. The skip contributes the separate difference $|\widehat{b} - b|$. Consequently, if an oracle-coordinate comparator has error a_G and its fitted-coordinate perturbation has L_2 norm at most r_A, then $a_\circ \le a_G + r_A$

in Proposition 14, where r_A is bounded by the displayed network constant times coordinate error plus skip error. This is a statement for that comparator, not a uniform small constant for every trained network.

If, under the weighted reference law ν, oracle-state means satisfy $\mathbb{E}_\nu(\varepsilon \mid S) = 0$ and the same network evaluated at oracle coordinates is S-measurable after noise averaging, then

$$|c_g| \le \|m_g(\widehat{S}) - m_g(S)\|_2 \|\varepsilon\|_2.$$

It follows by subtracting the oracle-coordinate mean inside the covariance and applying Cauchy–Schwarz. Thus state perturbation also controls the reconstruction-history covariance, but its generic contribution to (56) is under a square root. An additive $L\rho_{NT}$ bound alone requires a stronger conditional-mean or orthogonality premise; it does not follow from coordinate Lipschitzness.

Proposition 15 (A constructive approximation subclass of the actual architecture)**.** *Fix a compact input set $\mathcal{X}$, a depth-L ordinary feedforward LeakyReLU network F with the same negative slope* 0.2*, at most p neurons in each hidden layer, and a linear output. If $d_c \ge 2p+2$, then, for every $\eta > 0$, an L-block Linear–LayerNorm–LeakyReLU network of width d_c has output F_η satisfying $\sup_{x\in\mathcal{X}} |F_\eta(x) - F(x)| \le \eta$. In particular, the actual three-block width-128 correction class has in its uniform closure every three-hidden-layer LeakyReLU network of width at most 63. It also contains, in uniform closure, every ordinary three-hidden-layer ReLU network of width at most 31.*

Proof. For a target hidden-layer preactivation vector $a \in \mathbb{R}^p$, make the actual affine layer output $(a, -a, A, -A, 0, \dots, 0)$. Its coordinate mean is exactly zero. Write $s_A^2 = 2A^2/d_c + \epsilon$, where $\epsilon > 0$ is this LayerNorm's numerical constant. Give the first $2p$ normalized coordinates affine scale s_A and zero affine shift, and give the remaining coordinates zero scale and shift.

The positive signal coordinates before activation are exactly

$$a_j \left(1 + \frac{2\|a\|_2^2}{d_c s_A^2}\right)^{-1/2},$$

and the negative coordinates have the opposite signs. On a bounded set of a these converge uniformly to a_j and $-a_j$ as $A \to \infty$; the error is $O(A^{-2})$ by $0 \le 1 - (1+x)^{-1/2} \le x/2$ for $x \ge 0$. Activation preserves uniform convergence because LeakyReLU is 1-Lipschitz. At the next block, use only the positive signal outputs to form the target network's next affine preactivations and construct their positive and negative pairs and two anchors in the same way. Continuity and compactness keep all finitely many intermediate signals uniformly bounded for sufficiently large A. Induction over the fixed L blocks, followed by the target linear output using the positive signals, proves uniform convergence. For fixed target weights and depth the resulting error is $O(A^{-2})$, with a constant depending on that target network and compact set.

For the ReLU claim, let σ_α be LeakyReLU with $\alpha = 0.2$. The exact scalar identity

$$\max(x, 0) = \frac{\sigma_\alpha(x) + \alpha\sigma_\alpha(-x)}{1 - \alpha^2}$$

expresses each ReLU signal using two LeakyReLU neurons. Layerwise substitution represents any width-31 ReLU network by a width-62 LeakyReLU network at the same number of hidden layers. The first construction needs $2 \cdot 62 + 2 = 126 \le 128$ coordinates. □

The construction establishes approximation capacity for the implemented LayerNorm architecture. A statistical or optimization rate requires additional control of its parameter norms and fitting procedure. The large anchor biases and LayerNorm scales grow with A; a parameter-norm constraint or weight-decay training path can restrict how close the construction is attained. In particular, for the DGP correction $H(Z, W)$ on its compact domain, the architecture's best deterministic approximation error is at most the best error of the stated ordinary-network subclass, in the limit over finite parameter bounds. The

construction does not prove a numerical small error for that fixed-width subclass or that Adam finds the construction.

The covariates $X = (Z, W)$ enter directly; projected factor/noise columns can be ignored by zeroing their first-layer weights. This therefore applies to corrections depending only on (Z, W), the form of the shared DGP's omitted mean. Generic rates for growing sparse ReLU networks cannot simply be substituted for the fixed implemented network. Moreover the elementwise projected product is not an injective representation of arbitrary (λ, f): if $f = 0$, all loading vectors yield the same interaction. The class therefore does not automatically approximate every continuous $h(X, \lambda, f)$ from the broad identifying model.

Appendix A.16.5 Independent units, not independent panel cells

An empirical-process bound can be supplied without pretending the nT cells are independent. Here is a deliberately elementary version. Let $\mathcal{H}$ be a fixed finite collection of K candidate losses on unit histories O_i that are i.i.d. conditional on the admitted common environment, and let $q(O_i) \geq 0$ be a fixed weighting function. All expectations and variances in this paragraph are conditional on that environment. Suppose $\mathbb{E}q = Q \geq q_0 > 0$, $\mathrm{Var}(q) \leq v_q$, $\sup_h \mathrm{Var}(qh) \leq v_h$, and $\sup_h |\mathbb{E}(qh)|/Q \leq B_h$. For $\widehat{R}_h = \mathbb{P}_n(qh)/\mathbb{P}_n q$ and $R_h = \mathbb{E}(qh)/Q$, Chebyshev's inequality and a union bound give, with probability at least $1 - \alpha$,

$$\sup_h |\widehat{R}_h - R_h| \leq \frac{2}{q_0} \left\{ \sqrt{\frac{2K v_h}{n\alpha}} + B_h \sqrt{\frac{2v_q}{n\alpha}} \right\}, \tag{61}$$

provided $\sqrt{2v_q/(n\alpha)} \leq q_0/2$. Indeed, the numerator and denominator deviations are bounded by the two respective square roots with joint failure probability at most α; the empirical denominator is then at least $q_0/2$. The exact ratio difference is bounded by the numerator deviation plus B_h times the denominator deviation, divided by that lower bound. Each $h(O_i)$ may average arbitrarily dependent periods and noise banks. A deterministic loss-class cover extends the result with its covering error. A data-dependent visited path is

not automatically such a fixed class; an independent selection sample or a suitable uniform stability/class argument is needed. Nor is the conditionally frozen upstream pipeline in this lemma automatically the stitched pipeline of the code. Accordingly this lemma identifies the correct sampling unit and one provable route to Δ, not a claimed rate for the released fit. In particular, a square-root excess-risk bound from this elementary lemma is not by itself a strict $o(n^{-1/4})$ nuisance rate.

Appendix A.16.6 Connection to the current DGP and the DR score

Proposition 9 establishes a persistent raw LFM ATT bias A_δ in the centered-shift design under its rank-two asymptotic embedding. For $m = \ell + \alpha H$, residual correction changes this to $(1-\alpha)\delta\{\mathbb{E}_1 K - \mathbb{E}_0(wK)\}$ by (44). The present bound states what must be added before replacing this illustrative αH by the actual trained network: approximation error of the actual architecture, generated-coordinate and skip error, learning and selection gaps, objective mismatch, history effects, and transfer to the treated-post law. It does not prove that α is positive in every fit, or that the network beats LFM in finite samples.

Under Proposition 11, let r_m and r_e denote the full-history L_2 mean and propensity errors, and retain the normalized denominator q from (46). Then

$$|\Theta(\widehat{m}, \widehat{e}) - ATT| \le C\, r_e r_m + \frac{|\mathbb{E}\{v\varepsilon\}|}{q}, \qquad v = \frac{\widehat{e} - e}{1 - \widehat{e}}.$$

Thus deep learning has a specific possible role: reducing nonlinear mean approximation error, which reduces sensitivity to propensity error. The final residual cancellation is a property of the DR score, not a GAN-only identification result. The score's history covariance is distinct from h_{rec} in the training-risk inequality. Clipping/capping and the complete cross-fold stability conditions remain separate requirements. Finally, the released post prediction is a finite M-draw average, not m_g. For independent fresh evaluation draws its additional integrated squared error is $\mathbb{E}v_g/M$; root-N treatment of a fixed M belongs to the algorithmic

influence-function calculation in Proposition 12. Increasing N alone does not remove this unit-level auxiliary variation.

Code correspondence. The architecture and default coefficients are defined by `ATTGANConfig` and `Generator` in `R/wgan_training.py`. The identities (52)–(54) correspond to `generator_loss_components` and the calibration block of `_fit_one_fold`. The selection risk corresponds to `_selection_score`; `_predict_post_mean` uses the separate finite evaluation bank. The normalized residual correction is implemented in `R/dr_att.py`. The default architecture has three width-128 blocks, calibration uses $M_c = 4$, and post prediction uses $M = 50$; individual run contracts record any overrides. The legacy learner under `gscgan/src` is not the implementation analyzed here. The replication archive records function locations, source hashes, and the verified statement-to-code mapping in `audit/deep_theory_code_alignment_20261017/att_mapping.json`.

Appendix A.17 Adaptive joint-distribution constraints

The adversarial term tests discrepancies in the joint outcome–state law. Its role differs from the approximation capacity of the generator and from orthogonality of the ATT score. The following results identify target-relevant discrepancies that a critic can detect and retain the approximation, optimization, and sampling errors needed to connect that discrepancy to causal estimation.

Appendix A.17.1 Fixed moments and target-relevant witnesses

Proposition 16 (Fixed moments versus a target witness)**.** *Let $\mathcal{A} = \{a_1, \ldots, a_J\}$ be a finite joint support of (V, Y), partitioned into state strata. Form the matrix A whose rows are the state-stratum indicators and a finite set of specified moments $\phi_1, \ldots, \phi_K$, evaluated on $\mathcal{A}$. Let $h = (h(a_1), \ldots, h(a_J))'$. If h is outside the row space of A, there exist probability laws P, Q with identical state marginals and identical values of all K moments, but $(P - Q)h \neq 0$.*

If a symmetric critic class contains h/c for some $c > 0$, it detects these laws: $D_{\mathcal{F}}(P, Q) \geq |(P - Q)h|/c > 0$. More generally the lower bound becomes $\{|(P - Q)h| - \epsilon_h\}/c$ if a critic approximates h/c with $\mathbb{E}_P|h - cf| + \mathbb{E}_Q|h - cf| \leq \epsilon_h$.

Proof. Let $v = h - \Pi_A h$, where Π_A is Euclidean projection on the row space of A. Then $v \neq 0$, $Av = 0$, and $v'h = \|v\|_2^2$. Take any strictly positive probability vector p_0 and choose $0 < \epsilon < \min_{j:v_j \neq 0} p_{0j}/|v_j|$. Set $p = p_0 + \epsilon v$ and $q = p_0 - \epsilon v$. The stratum-indicator rows sum to the constant vector, so $\sum_j v_j = 0$. Thus p, q are nonnegative probability vectors; $Av = 0$ gives both the common state marginal and the matched moments. Nevertheless $(p-q)'h = 2\epsilon\|v\|_2^2 > 0$. For the critic conclusion, write $(P-Q)h = c(P-Q)f + (P-Q)(h-cf)$ and bound the last term by ϵ_h. □

An ATT-relevant example. Let $V \in \{-1, 1\}$ have equal probabilities. Under P, let $Y \in \{-1, 1\}$ be independent of V, again with equal probabilities. Under Q, set $Y = V$. Both state marginals and both outcome marginals are exactly equal, hence all separate moments of V and Y agree. Set $\nu_1(V = 1) = 3/4$, $\nu_{0w}(V = 1) = 1/4$, and $\nu_{\mathrm{tr}}(V = 1) = 1/2$. Then $a_\Lambda = d(\nu_1 - \nu_{0w})/d\nu_{\mathrm{tr}} = V$. The real conditional mean is zero and the generated mean is V. Under a common real conditional law across training and target groups, the normalized DR bias is therefore

$$B = (P - Q)(VY) = -1.$$

Separate marginal calibration does not detect it, whereas the joint witness $h(V, Y) = VY$ does. On these four support points, $h(V, Y) = 1 - |V - Y|$; the latter is a continuous piecewise-linear function, representable by a small leaky-ReLU network up to scaling. This observation shows why joint critics can be useful, but is not a proof that the trained spectrally normalized critic finds the witness. Correct normalized odds would make $\nu_{0w} = \nu_1$ and eliminate this particular ATT bias without any adversarial term.

A distribution-relevant example. At a single state, let $P(Y=-1)=P(Y=1)=1/2$ and let $Q(Y=-\sqrt{2})=Q(Y=\sqrt{2})=1/4$, $Q(Y=0)=1/2$. Both means equal zero and both second moments equal one, while $F_P(-1/2)=1/2$ and $F_Q(-1/2)=1/4$. A threshold witness consequently detects an outcome-distribution error missed by mean–variance matching. The general proposition covers any finite prescribed moment list whenever the target witness is not in its span. It does not cover a loss that already identifies the full conditional law. In particular, the conditional energy score retained by the ETC-GAN noADV implementation is a proper distributional score; the example gives no identification advantage over that score.

Appendix A.17.2 Target control under the penalized critic objective

Write g and f for candidates from the generator G_ϕ and critic D_ψ classes. Fix g and let $d_g(f)=(P-Q_g)f$. Let $\widehat{d}_g(f)$ be a specified empirical contrast, not merely the last training minibatch. Work with a symmetric class $\mathcal{F}$ and a nonnegative, sign-symmetric empirical penalty $\widehat{R}_g(f)=\widehat{R}_g(-f)$. For ATT this penalty can include the actual gradient penalty $\lambda_{\mathrm{gp}}\widehat{\mathrm{GP}}_g(f)$. Spectral normalization instead restricts the function class; it is not an additional additive penalty. Define

$$\widehat{J}_g(f)=\widehat{d}_g(f)-\widehat{R}_g(f), \qquad \epsilon_{\mathrm{opt}}(g,\widehat{f})=\sup_{f\in\mathcal{F}}\widehat{J}_g(f)-\widehat{J}_g(\widehat{f}), \quad \Delta_g=\sup_{f\in\mathcal{F}}|d_g(f)-\widehat{d}_g(f)|.$$

These quantities are nonnegative except for the attained objective $\widehat{J}_g(\widehat{f})$. The evaluated objective alone does not determine the two error terms.

Proposition 17 (Target-specific penalized-critic certificate)**.** *Use the ATT bias decomposition in Proposition 10, $B_g=T_\Lambda+(P-Q_g)h$, where $h=a_\Lambda(V)Y$ and T_Λ records the training-to-target conditional-law difference. If $f_h\in\mathcal{F}$, $c>0$ obey $\mathbb{E}_P|h-cf_h|+\mathbb{E}_{Q_g}|h-cf_h|\le\epsilon_h$, then*

$$|B_g|\le|T_\Lambda|+\epsilon_h+c\{\widehat{J}_g(\widehat{f})+\epsilon_{\mathrm{opt}}+\widehat{R}_g(f_h)+\Delta_g\}. \tag{62}$$

The penalty paid is that of a target witness, not an assumed uniform bound on the penalties of every critic.

Proof. The approximation condition implies $|B_g| \le |T_\Lambda| + \epsilon_h + c|d_g(f_h)|$. Choose $s \in \{-1, 1\}$ so that $d_g(sf_h) = |d_g(f_h)|$. By symmetry $sf_h \in \mathcal{F}$. Uniform empirical control and the definition of the optimization gap give

$$d_g(sf_h) \le \widehat{d}_g(sf_h) + \Delta_g = \widehat{J}_g(sf_h) + \widehat{R}_g(f_h) + \Delta_g \le \widehat{J}_g(\widehat{f}) + \epsilon_{\text{opt}} + \widehat{R}_g(f_h) + \Delta_g.$$

Substitution proves the claim. Since the left side of this display is nonnegative, the complete brace is necessarily nonnegative when its assumed bounds hold; an isolated negative logged critic value is not a negative error certificate. □

In the ATT implementation,

$$\widehat{\mathrm{GP}}_g(f) = \sum_j \bar{w}_j (\|\nabla f(\alpha_j O_j + (1-\alpha_j)\widetilde{O}_{g,j})\|_2 - 1)^2, \qquad O = (Y, X, \widehat{\lambda}, \widehat{f}_t).$$

Thus a target witness with small gradients is not necessarily unpenalized: the penalty is centered at gradient norm one, not at zero. Taking $\widehat{R} = \lambda_{\text{gp}}\widehat{\mathrm{GP}}$ gives $\widehat{J} = \widehat{d} - \lambda_{\text{gp}}\widehat{\mathrm{GP}}$, the displayed critic loss with its sign reversed. The optimizer additionally uses weight decay. One may either leave its effect in ϵ_{opt}, or define a parameterized penalty including $\eta\|\theta\|^2/2$ and pay that same additional witness penalty. Neither convention asserts global optimization by Adam. The last minibatch value, finite critic steps, and gradient clipping alone provide no upper bound on ϵ_{opt}. Moreover, spectral normalization updates its power-iteration buffers during training-mode forward calls. Real, fake, and interpolated inputs can therefore be evaluated by slightly different numerical maps. The theorem uses a fixed evaluation snapshot and a common reference contrast. Comparing the logs to that contrast requires an additional map-drift bound; it cannot be inferred to be zero from spectral normalization. Approximate power iteration also does not itself certify an exact unit operator norm for every layer.

Proposition 18 (The stale-generator optimization term)**.** *For a common reference sample and critic class, define* $\Omega(g, g^-) = \sup_f |\widehat{J}_g(f) - \widehat{J}_{g^-}(f)|$*. For any critic* $\widehat{f}$*,*

$$\epsilon_{\text{opt}}(g, \widehat{f}) \le \epsilon_{\text{opt}}(g^-, \widehat{f}) + 2\Omega(g, g^-).$$

Proof. For each f, $\widehat{J}_g(f) \le \widehat{J}_{g^-}(f) + \Omega$ and $\widehat{J}_g(\widehat{f}) \ge \widehat{J}_{g^-}(\widehat{f}) - \Omega$. Subtract the second inequality from the first and take the supremum. □

The ATT code trains a critic, updates the generator, and then saves both after validation. Therefore even a certified critic for the pre-update generator would require the drift term for the saved generator. In addition, each stage draws fresh random noise; converting its minibatch losses to a common reference objective requires sampling error terms. The gradient-penalty part of Ω is not controlled by a scalar Lipschitz bound alone, since it also involves gradients.

What a cluster generalization assumption would supply. For clarity, suppose $U_1, \dots, U_n$ are independent unit clusters, weights $a_i \ge 0$ are fixed with $\sum_i a_i = 1$, and a deterministic class of cluster contrasts $\mathcal{A}$ is bounded by B_A. Allow arbitrary time dependence inside each U_i. If a sup-norm ϵ-net has cardinality K, weighted Hoeffding and a union bound give, with probability at least $1 - \delta$,

$$\sup_{A \in \mathcal{A}} \left| \sum_i a_i \{A(U_i) - \mathbb{E}A(U_i)\} \right| \le 2\epsilon + B_A \sqrt{2 \sum_i a_i^2 \log(2K/\delta)}. \tag{63}$$

To verify the bound, each centered net contrast has weighted summand range of length at most $2a_i B_A$, hence its two-sided tail at t is at most $2\exp\{-t^2/(2B_A^2 \sum_i a_i^2)\}$. Union bound over K elements and substitute the displayed t. Replacing a contrast by a net element changes its empirical and population averages by at most ϵ each. The effective sample size here is $(\sum_i a_i^2)^{-1}$, not nT. For fitted generators or state maps the class must cover their joint selection, or the maps must be fitted on independent data. History-dependent weights

and stitched estimated states in the actual pipeline do not automatically satisfy this simple independent-cluster setup. Equation (63) is a sufficient interface, not a verified rate for the present implementation.

Appendix A.17.3 The implemented Cramér statistic and target CDFs

For feature maps with $\|\phi(o)\| \le 1$, write

$$D_\phi(P,Q) = 2\mathbb{E}\|\phi(O) - \phi(\widetilde{O}')\| - \mathbb{E}\|\phi(O) - \phi(O')\| - \mathbb{E}\|\phi(\widetilde{O}) - \phi(\widetilde{O}')\|,$$

where primed draws are independent and have the indicated marginals. The canonical QTE critic uses spectral normalization and normalizes its output vector. It has no gradient penalty. Its real and generated rows share the same sampled state, whereas its cross-distance average includes the diagonal pairs. This produces a finite-batch discrepancy from the independent-product energy distance.

Proposition 19 (Same-state cross-diagonal term)**.** *Let (A_i, B_i), $i = 1, \dots, b$, be independent pairs, $b \ge 2$, with common marginal laws equal to the real and generated feature laws and $\|A_i\|, \|B_i\| \le 1$. Dependence within each pair is allowed. Let $\widehat{D}_b$ use all b^2 cross distances and exclude the diagonals of both within-sample averages, exactly as in the code. Then*

$$\mathbb{E}\widehat{D}_b = D_\phi(P,Q) + \frac{2}{b}\{\mathbb{E}\|A_1 - B_1\| - \mathbb{E}\|A_1 - B_2\|\}, \qquad |\mathbb{E}\widehat{D}_b - D_\phi(P,Q)| \le \frac{4}{b}.$$

Proof. There are b diagonal cross pairs and $b(b-1)$ off-diagonal cross pairs. The expected cross average is therefore $b^{-1}\mathbb{E}\|A_1 - B_1\| + (1 - b^{-1})\mathbb{E}\|A_1 - B_2\|$. Both within-sample diagonal-excluded averages have the independent-pair expectations in D_ϕ. Subtracting proves the identity. Each cross expectation lies in $[0, 2]$, proving the bound. □

Conditioned on the observed training rows and fixed fitted maps, the code's with-replacement row sampling supplies the independent pairs in this proposition. It does

not remove dependence among the original unit-period data, which requires a separate generalization term. At $b = 1$ the within terms vanish by convention in the code; the displayed identity is restricted to $b \geq 2$.

For an adapted fitted critic, take bounds uniform over the chosen critic/generator class. Let Δ_{data} bound the energy discrepancy between population and empirical-row laws, and let Δ_{MC} bound deviation of the evaluated batch statistic from its conditional mean. If real and fake forward calls do not use the same frozen feature-map snapshot, also let Δ_{map} bound their energy-statistic difference from the common-map statistic. For example, average feature errors e_P, e_Q yield $\Delta_{\text{map}} \leq 4(e_P + e_Q)$, by the triangle inequality applied to the cross and the two diagonal-excluded within averages. Write L_{eval} for the reported value and require $\widehat{D}_b(\widehat{\phi}, g) \leq L_{\text{eval}} + \Delta_{\text{map}}$, where $\widehat{D}_b$ and its optimization gap below always refer to a common fixed feature-map snapshot. Thus L_{eval} may be a training log only with its map-drift allowance, not by definition. Then a valid uniform comparison is

$$\sup_{\phi \in \Phi} D_\phi(P, Q_g) \leq L_{\text{eval}} + \epsilon^E_{\text{opt}} + \Delta_{\text{data}} + \Delta_{\text{MC}} + \Delta_{\text{map}} + 4/b,$$

where $\epsilon^E_{\text{opt}} = \sup_\phi \widehat{D}_b(\phi, g) - \widehat{D}_b(\widehat{\phi}, g)$. The proof is just to upper-bound each population discrepancy by its empirical counterpart plus these uniform error terms and then take the supremum.

To connect this bound to outcomes, fix the fitted mean map and let $Y = \widehat{\mu}(S) + R$ in the joint residual–state laws. Suppose that, for every threshold a, a feature critic approximates $\phi_a(S, R) = v\mathbf{1}\{Y \leq a\} - v\mathbf{1}\{Y > a\}$, $\|v\| = 1$, with total $L^1(P) + L^1(Q)$ feature error at most ζ. Direct enumeration gives $D_{\phi_a} = 4(F_P(a) - F_Q(a))^2$. Replacing ϕ_a by its approximating feature map changes D by at most 4ζ: the cross term contributes at most 2ζ and the two within terms together at most 2ζ. If η_{target} bounds the sum of the two Kolmogorov distances

from training to target outcome laws (real and generated), then

$$\|F_{P,\text{target}} - F_{Q,\text{target}}\|_\infty \le \eta_{\text{target}} + \frac{1}{2}\sqrt{L_{\text{eval}} + \epsilon_{\text{opt}}^E + \Delta_{\text{data}} + \Delta_{\text{MC}} + \Delta_{\text{map}} + 4/b + 4\zeta}. \quad (64)$$

This follows from the preceding energy bound followed by the triangle inequality for the two training-to-target CDF differences. The full radicand is nonnegative whenever the assumed upper bounds hold. A trained finite-dimensional continuous critic need not approximate all discontinuous thresholds in the required L^1 norm; a margin or density condition and sufficient capacity would be needed to turn that approximation premise into a quantitative rate. Residual thresholds must also encode the fitted mean when the target is the outcome CDF rather than the residual CDF. In particular, the mean network uses summaries of raw pre outcomes, whereas the distribution critic uses summaries of pre residuals and does not directly receive the fitted mean or the mean-model fold identifier. Measurability of that mean in the critic's coordinates is not automatic. One must prove the required witness approximation on the actual coordinates or retain its error; silently adjoining the fitted mean would change the critic. In the current QTE code, disabling parameter gradients during the generator step does not switch the critic to evaluation mode; spectral-normalization buffers can still update. Thus $\Delta_{\text{map}} = 0$ should be reserved for evaluation of a fixed common snapshot, not asserted for the logs.

Appendix A.17.4 Interpretation and code correspondence

For ATT, the adversarial term can penalize joint prediction errors whose target-weighted moments matter after nonlinear response and exposure shift. It does not itself cancel propensity error, repair history confounding, or establish state recovery. Its training law is weighted controls over all periods; it is not automatically the target-post law. The term T_Λ must therefore remain unless the corresponding conditional-law transport restriction is justified.

For QTE, the adaptive critic can test discrepancies beyond a small preselected collec-

tion of outcome moments. The actual noADV learner already retains pinball and proper conditional-energy losses, so the present results establish neither an identification advantage nor a strict risk improvement over noADV. The generator combines those losses with mean and transport regularization; none of the critic bounds proves that this full objective is minimized at the true law.

Applying these inequalities to a fitted checkpoint requires three additional controls. First, target-witness coverage must be checked for the actual state and critic class, including state-coordinate and target-law differences. Second, one needs an honest uniform generalization or full-pipeline stability argument, not an independent-row approximation. Third, evaluation at the saved generator requires critic optimization and sampling-error bounds. ATT saves a possibly stale critic with the selected generator; QTE returns only the selected generator and does not save the critic associated with that checkpoint. A logged training gap therefore cannot replace these quantities. These qualifications concern the mapping from training to target error; they do not alter the estimators or numerical comparisons reported in the paper.

Code correspondence. For ATT, `weighted_wasserstein_terms`, `compute_gradient_penalty`, and `_make_critic_input` in `R/wgan_training.py` define the weighted contrast, full-input gradient penalty, and critic coordinates. The update order in `_fit_one_fold` gives the generator-drift term in Proposition 18. For QTE, `GlobalCramerFeatureCritic` and `feature_energy_distance` in `gscgan/simulations/end_to_end_single_global_cramer_gan_qte_experiment.py` define the normalized features and same-state cross-distance statistic. `train_empirical_transport_model` retains the conditional energy and pinball terms when the adversarial coefficient is zero. The archive `audit/deep_theory_code_alignment_20261017/adversarial_mapping.json` records the exact function locations and source hashes.

Appendix A.18 Deep residual transport and marginal QTE error

ETC-GAN estimates the counterfactual distribution by a state-dependent monotone transport of empirical residual quantiles. This subsection connects its constrained network and checkpoint rule to marginal QTE error, without DR-CDF targeting. Statements are conditional on the common environment in Section Appendix A.2 and the declared period-mixture weights. Sampling independence, when required, concerns unit profiles. Code correspondence appears at the end of the subsection.

Appendix A.18.1 The monotone transport architecture

Let $0 < \alpha = u_1 < \cdots < u_K = \beta < 1$ denote the knot levels. The code fixes $K = 19$, $\alpha = .005$, $\beta = .995$, with largest interior gap .10. Let $\widehat{b}_j$ be the weighted empirical residual quantiles. With $d_0 = .001$, $A = 1.25$, and $B = .55$, the network generates

$$\begin{aligned} \widehat{q}_{\theta,1}(s) &= \widehat{b}_1 + a_\theta(s), & a_\theta(s) &= A\tanh\{v_{\theta,1}(s)\}, \\ \widehat{q}_{\theta,j+1}(s) - \widehat{q}_{\theta,j}(s) &= \widehat{d}_j \exp\{\ell_{\theta,j}(s)\}, & & \\ \widehat{d}_j &= \max(\widehat{b}_{j+1} - \widehat{b}_j, d_0), & \ell_{\theta,j}(s) &= B\tanh\{v_{\theta,j+1}(s)\}. \end{aligned} \tag{65}$$

Here v_θ is a two-hidden-layer SiLU network. Denote by $I(q)(u)$ the linear interpolation of an ordered knot vector q, made constant below α and above β. The residual generator's ideal real-arithmetic version is $G_\theta(s, U) = I(\widehat{q}_\theta(s))(U)$ for $U \sim U[0, 1]$. Here θ indexes the QTE residual generator, distinct from the ATT generator's parameters ϕ. Its implemented conditional residual CDF is denoted $F_{\varepsilon,\theta}$; numerical interpolation error is accounted for below.

Proposition 20 (Noncrossing, endpoint atoms, and numerical interpolation)**.** *Every finite-parameter network in* (65) *has strictly increasing knots. Its ideal transport is nondecreasing in U and defines a valid conditional distribution, with endpoint masses α and $1-\beta$. In exact real arithmetic, replacing each interpolation denominator $h_j = u_{j+1} - u_j$ by $h_j + \epsilon$, as in the*

code with $\epsilon = 10^{-8}$, preserves monotonicity and changes a draw by at most

$$a_{\text{num}}(\widehat{b}) = \max_{j<K} \frac{e^B \widehat{d}_j \epsilon}{h_j + \epsilon}. \tag{66}$$

This denominator creates very small upward jumps at interior knots; the terminal atom is at the implemented terminal draw rather than exactly at $\widehat{q}_{\theta,K}$. Floating-point rounding is an additional numerical error, not included in (66).

Proof. Each increment is at least $d_0 e^{-B} > 0$. Linear interpolation is nondecreasing, and clamping U places the stated probability at each endpoint. On an interval, the implemented fraction has positive slope. Its left limit at the upper knot is the knot value minus $(q_{j+1} - q_j)\epsilon/(h_j + \epsilon)$; the next interval starts at that knot value, so the jump is upward. Clamping at the final interval gives the terminal value stated above. The difference from the ideal interpolation is nonnegative and at most that same increment times $\epsilon/(h_j + \epsilon)$, bounded by (66). □

Appendix A.18.2 A distributional bridge that allows fitted atoms

For a CDF F, write $Q_F(u) = \inf\{y : F(y) \geq u\}$. The following inequality is useful because a density restriction on the fitted distribution would contradict the actual endpoint clamping.

Proposition 21 (CDF loss controlled by monotone transport loss)**.** *Let F be L-Lipschitz and let G be any CDF with finite second moment, as also assumed for F. Then*

$$\int_{\mathbb{R}} \{G(y) - F(y)\}^2 \, dy \leq L \int_0^1 \{Q_G(u) - Q_F(u)\}^2 \, du. \tag{67}$$

The right side is L times the squared monotone-coupling Wasserstein distance. No density or continuity of G is needed.

Proof. Fix y and let $a = F(y)$ and $b = G(y)$. For almost every u between a and b, y lies between $Q_F(u)$ and $Q_G(u)$. Since F is continuous and L-Lipschitz, $|Q_F(u) - y| \geq |u - a|/L$

for those u. Consequently

$$|G(y) - F(y)|^2 \le 2L \int_{\min(a,b)}^{\max(a,b)} |Q_F(u) - y|\, du.$$

Tonelli's theorem bounds the integral over y by

$$2L \int_0^1 \int_{\min(Q_F(u),Q_G(u))}^{\max(Q_F(u),Q_G(u))} |Q_F(u) - y|\, dy\, du = L \int_0^1 |Q_G(u) - Q_F(u)|^2\, du.$$

This proves the claim. Generalized-inverse identities and the crossing argument hold outside null sets in u, including when G has atoms. □

Appendix A.18.3 Architecture, grid, tails, and empirical-base error

Let $q_s(u)$ denote the true residual quantile function. Fix a reference ordered base $b = (b_j)$, for example the population analogue of the pooled weighted residual base, and put $d_j = \max(b_{j+1} - b_j, d_0)$. Define the projected state-specific parameters

$$\begin{aligned}
a^*(s) &= \mathrm{clip}(q_s(\alpha) - b_1, -A, A),\\
d_j^*(s) &= \mathrm{clip}\{q_s(u_{j+1}) - q_s(u_j), e^{-B} d_j, e^{B} d_j\},\\
\ell_j^*(s) &= \log\{d_j^*(s)/d_j\}.
\end{aligned}$$

These are a comparison construction, not an oracle used in training. For a parameter vector θ define

$$\begin{aligned}
C(s) &= \mathrm{dist}\{q_s(\alpha) - b_1, [-A, A]\} + \sum_{j<K} \mathrm{dist}\{q_s(u_{j+1}) - q_s(u_j), [e^{-B} d_j, e^{B} d_j]\},\\
A_\theta(s) &= |a_\theta(s) - a^*(s)| + e^{B} \sum_{j<K} d_j |\ell_{\theta,j}(s) - \ell_j^*(s)|,\\
B_b &= |\widehat{b}_1 - b_1| + e^{B} \sum_{j<K} |\widehat{d}_j - d_j|.
\end{aligned}$$

C measures range and spacing-floor restrictions; A_θ measures the actual SiLU network's approximation to admissible parameter maps; B_b measures estimation of the weighted empirical base. The inequality $|\widehat{d}_j - d_j| \le |\widehat{b}_{j+1} - b_{j+1}| + |\widehat{b}_j - b_j|$ follows because the maximum with d_0 is 1-Lipschitz. No independent-observation quantile rate is presumed for this pooled, generated-residual base.

For any state measure ν, let

$$T_\nu^2 = \int \left[\int_0^\alpha |q_s(u) - q_s(\alpha)|^2 du + \int_\beta^1 |q_s(u) - q_s(\beta)|^2 du \right] d\nu(s),$$
$$H_\nu^2 = \int \sum_{j<K} \int_{u_j}^{u_{j+1}} |q_s(u) - I((q_s(u_k))_{k=1}^K)(u)|^2 du \, d\nu(s).$$

They are tail truncation and interior interpolation errors.

Proposition 22 (Finite architecture-error accounting)**.** *If the true residual CDF at each relevant state has density bounded by L_ε, and the displayed terms are finite, then the implemented conditional CDF $F_{\varepsilon,\theta}$ satisfies*

$$\left[\int \int (F_{\varepsilon,\theta} - F_\varepsilon)^2 dy \, d\nu \right]^{1/2} \le \sqrt{L_\varepsilon} \left\{ T_\nu + H_\nu + \|C\|_{L_2(\nu)} + \|A_\theta\|_{L_2(\nu)} + B_b + a_{\text{num}} \right\}. \quad (68)$$

If q_s is twice differentiable on the interior knot intervals with $|q_s''| \le M(s)$, then $H_\nu^2 \le \|M\|_{L_2(\nu)}^2 \sum_{j<K} h_j^5/120$.

Proof. Construct reference knots starting at $b_1 + a^*(s)$ with increments $d_j^*(s)$. Telescoping and the definition of projection bound their maximum deviation from the true knots by $C(s)$. The mean-value inequality $|e^x - e^z| \le e^B |x - z|$ for $x, z \in [-B, B]$ bounds the network's deviation from these reference knots, using base b, by $A_\theta(s)$. Replacing b with $\widehat{b}$ changes every network knot by at most B_b. Interpolation and endpoint clamping are nonexpansive in the maximum knot norm. Minkowski in $L_2(\nu \otimes du)$ therefore bounds the draw error by the terms in braces; (66) adds the implemented denominator error. Proposition 21 applied conditionally and integrated proves the claim. Finally, the linear-interpolation remainder

is at most $M(s)(u-u_j)(u_{j+1}-u)/2$. Squaring and integrating gives $M(s)^2 h_j^5/120$ on each interval. □

For fixed knots and fixed endpoints, H_ν and T_ν need not vanish. Even with a perfect optimizer, a conditional law whose required spacings or lower endpoint lie outside the specified ranges can have $C > 0$. In particular, the fixed clamped law is not generally identical to an unbounded Gaussian-tailed residual law. Increasing a network's width alone does not remove these restrictions.

Appendix A.18.4 Validation-selected transport

The saved checkpoint minimizes a held-out, tail-weighted *pinball* score over eligible visited checkpoints. It does not minimize held-out energy score or the full generator objective. Let $\omega \in [0,1)$ be the validation tail weight, $K_L = \#\{j : u_j \le .30\}$, and

$$\gamma_j = (1-\omega)/K + \omega\, \mathbf{1}\{u_j \le .30\}/K_L.$$

Thus $\gamma_j > 0$ and $\sum_j \gamma_j = 1$. At the default $\omega = .5$, $K_L = 9$. Let $V(\theta)$ be the population version of this score under a reference validation-state law ν_v, using true residuals and states, and let V^* be its value at the true conditional knot quantiles. The empirical $\widehat{V}$ is the score the code evaluates, using estimated states and out-of-fold residuals. Denote the finite set of eligible checkpoints by $\mathcal{C}$ and set

$$\Delta_v = \sup_{\theta\in\mathcal{C}} |\widehat{V}(\theta) - V(\theta)|, \qquad \epsilon_{\text{path}} = \inf_{\theta\in\mathcal{C}} V(\theta) - \inf_{\theta\in\Theta} V(\theta) \ge 0.$$

Θ is the actual fixed-base neural transport class containing $\mathcal{C}$. In the adversarial branch, $\mathcal{C}$ excludes warmup checkpoints. The nonadversarial branch can retain its initial checkpoint. Finite iterations, fixed penalty coefficients, and optimizing a loss different from the selection loss all enter ϵ_{path}; no optimizer-convergence claim is made for this quantity.

For interpolation weights, define

$$\lambda_1 = \alpha + h_1/2, \qquad \lambda_K = 1 - \beta + h_{K-1}/2,$$
$$\lambda_j = (h_{j-1} + h_j)/2 \quad (1 < j < K), \qquad C_\gamma = \max_j \sqrt{\lambda_j/\gamma_j}.$$

These weights sum to one and account for endpoint clamping.

Proposition 23 (Validation-selected transport oracle inequality)**.** *Suppose the reference residual density is bounded above by L_ε and below by $c > 0$ on each segment between its true knot quantile and every candidate knot being compared. Let the target evaluation-state law ν_* satisfy $d\nu_*/d\nu_v \le \kappa$ and have the same true residual conditional law. For any comparator $\theta_0 \in \Theta$, define $D_v = \|C + A_{\theta_0} + B_b\|_{L_2(\nu_v)}$. If $\widehat{\theta}$ minimizes $\widehat{V}$ on $\mathcal{C}$, then*

$$\left[\int\int (G_{\widehat{\theta}} - F_\varepsilon)^2 dy\, d\nu_*\right]^{1/2} \le \sqrt{L_\varepsilon}\left[T_{\nu_*} + H_{\nu_*} + a_{\mathrm{num}} + C_\gamma\sqrt{\frac{\kappa}{c}\{L_\varepsilon D_v^2 + 2\epsilon_{\mathrm{path}} + 4\Delta_v\}}\right]. \tag{69}$$

The same inequality applies to noADV, with its own eligible path.

Proof. The pinball excess-loss identity is

$$E\{\rho_u(R-q) - \rho_u(R - q_s(u)) \mid s\} = \int_{q_s(u)}^{q} \{F_\varepsilon(z \mid s) - u\}\, dz.$$

The density bounds place this expression between $c(q - q_s(u))^2/2$ and $L_\varepsilon(q - q_s(u))^2/2$. Empirical minimization and the definition of Δ_v give

$$V(\widehat{\theta}) - V^* \le \inf_\Theta (V - V^*) + \epsilon_{\mathrm{path}} + 2\Delta_v \le L_\varepsilon D_v^2/2 + \epsilon_{\mathrm{path}} + 2\Delta_v.$$

The last inequality follows from the maximum knot-error bound in the preceding proof for θ_0. Therefore

$$E_{\nu_*}\sum_j \gamma_j |\widehat{q}_{\widehat{\theta},j}(S) - q_S(u_j)|^2 \le \frac{\kappa}{c}\{L_\varepsilon D_v^2 + 2\epsilon_{\mathrm{path}} + 4\Delta_v\}.$$

For knot errors e_j, convexity of the square gives $\int_0^1 |I(e)(u)|^2 du \le \sum_j \lambda_j e_j^2 \le C_\gamma^2 \sum_j \gamma_j e_j^2$. Minkowski adds the true-quantile grid and tail errors and (66). Proposition 21 completes the proof. □

The overlap assumption in this theorem is about *validation to target states and their conditional residual law.* Normalized or capped training odds do not establish it. If target and validation residual CDFs differ, their L_2 difference must be added by Minkowski before using the theorem. The lower density constant can be small at the extreme .005 and .995 knots; the numerical bound can accordingly be weak even when central quantiles are accurately estimated.

When target-state domination fails, its remainder cannot be set to zero. More explicitly, write the Lebesgue decomposition $\nu_* = r\nu_v + \nu_\perp$, let $e(s) = \|G_{\widehat{\theta}}(\cdot \mid s) - F_\varepsilon(\cdot \mid s)\|_2$, and fix any finite κ. Then

$$A_* \le \sqrt{\kappa}\, A_v + T_{\text{transfer}}, \qquad T_{\text{transfer}}^2 = \int (r-\kappa)_+ e(s)^2 d\nu_v(s) + \int e(s)^2 d\nu_\perp(s),$$

where $A_*^2 = \int e^2 d\nu_*$ and $A_v^2 = \int e^2 d\nu_v$. The inequality follows by integrating $r \le \kappa + (r-\kappa)_+$ and taking square roots. Apply the selection theorem with $\nu_* = \nu_v$ to bound A_v, and retain T_{transfer}. This identifies the extra assumption needed when post-treatment exposure/state support extends beyond validation support. It is not an estimate of that remainder, and training-odds truncation does not bound it. A mismatch of the conditional residual laws adds the separate L_2 law-discrepancy term.

Validation generalization. The decomposition of Δ_v can be made explicit. Let $\widetilde{V}$ evaluate the same candidate functions on validation units' true states and true residuals. The pinball loss is 1-Lipschitz in its residual-minus-quantile argument, so

$$\Delta_v \le \sup_{\theta\in\mathcal{C}} |\widetilde{V}(\theta) - V(\theta)| + \frac{1}{n_v}\sum_{i=1}^{n_v}\sum_t \pi_t \left[|\widehat{\mu}_{it} - \mu_{it}| + \sup_{\theta\in\mathcal{C}} \sum_j \gamma_j |q_{\theta,j}(\widehat{S}_{it}) - q_{\theta,j}(S_{it})| \right]. \quad (70)$$

Here π_t are the actual validation-period weights. If the entire candidate path and upstream fitted maps were independent of validation unit clusters and the unit-aggregated ideal losses were in $[0, H]$, Hoeffding and a union bound would give the first term at most $H\sqrt{\log(2|\mathcal{C}|/\delta)/(2n_v)}$ with probability $1-\delta$. This assertion concerns n_v units, not n_vT independent observations. Without a bounded-loss envelope one needs an appropriate tail inequality instead. The current global factor fit, generated pre-residual summaries, and stitched mean fits do not automatically satisfy the independence premise. A complete-pipeline uniform law or a stability bound is an alternative, but has not been established here.

Appendix A.18.5 Proper scoring and the finite-batch objective

For conditionally independent $R \sim F$, $Z, Z' \sim G$, finite first moments imply

$$E|Z-R| - \tfrac{1}{2}E|Z-Z'| - \tfrac{1}{2}E|R-R'| = \int (G-F)^2 dy, \tag{71}$$

where R' is an independent copy of R. To prove this, use $E|A-B| = \int \{F_A(1-F_B)+F_B(1-F_A)\}dy$ for independent real variables and subtract. Applying the identity conditionally on state proves strict propriety of the conditional energy component. It is present in both ETC-GAN and noADV; distribution identification therefore does not belong exclusively to the adversarial term.

The full finite-batch training objective is not just this proper score. For example, with batch size b and four independent generated draws per sampled state, the mean penalty has expectation

$$E\left[\left\{b^{-1}\sum_{k=1}^{b}\tfrac{1}{4}\sum_{m=1}^{4} Z_{km}\right\}^2\right] = (EZ)^2 + \frac{\mathrm{Var}_S\{E(Z \mid S)\} + \tfrac{1}{4}E_S \mathrm{Var}(Z \mid S)}{b}. \tag{72}$$

This follows by the variance decomposition for independent, with-replacement batch rows,

conditional on the fixed empirical training distribution. Thus even a zero-mean true residual law can receive a positive variance penalty. Transport regularization is also nonzero for a nontrivial true transport. The code then selects a checkpoint using a different, pinball-based criterion. Proper scoring of one summand is not proof that the truth minimizes the complete training/selection procedure; this is why the path gap remains in (69). The learned critic's additional target-witness role is analyzed separately.

Appendix A.18.6 From the transport bound to the implemented marginal QTE

For the scored cohort, set

$$F_i(y) = F_\varepsilon(y - \mu_i \mid S_i), \qquad G_i(y) = \widehat{F}_\varepsilon(y - \widehat{\mu}_i \mid \widehat{S}_i), \qquad \mathcal{R}_n = n^{-1} \sum_i \int (G_i - F_i)^2 dy.$$

Period mixtures use the same fixed period weights on both sides. States must be aligned in a common coordinate system; changing rank without an alignment or embedding theorem does not define a small state error. In addition, the true residual family must have a specified extension to a neighborhood containing the estimated states. Estimated coordinates can lie off the population-state support, where an ordinary conditional law is otherwise defined only almost surely and $F_\varepsilon(\cdot \mid \widehat{S}_i)$ has no identified meaning. Assume this extension is L_S-Lipschitz in state in CDF L_2 norm, with density bounded by L_ε throughout that neighborhood. Write $r_S^2 = n^{-1} \sum_i \|\widehat{S}_i - S_i\|^2$ and $r_\mu^2 = n^{-1} \sum_i |\widehat{\mu}_i - \mu_i|^2$. Let A_* be the population residual-CDF error on the evaluation-state law at the *estimated* coordinates, bounded by (69) when its premises hold. If

$$\xi_{\text{eval}} = \left[\left\{ n^{-1} \sum_i \|\widehat{F}_\varepsilon(\cdot \mid \widehat{S}_i) - F_\varepsilon(\cdot \mid \widehat{S}_i)\|_2^2 \right\}^{1/2} - A_* \right]_+,$$

then the exact empirical-to-population interface yields

$$\sqrt{\mathcal{R}_n} \le A_* + \xi_{\text{eval}} + L_S r_S + \sqrt{L_\varepsilon}\, r_\mu. \tag{73}$$

Indeed add and subtract the true residual CDF at $\widehat{S}_i$ and use Minkowski. For the translation term, $\|F(\cdot-a)-F(\cdot-b)\|_2^2 \le \|F(\cdot-a)-F(\cdot-b)\|_\infty \|F(\cdot-a)-F(\cdot-b)\|_1 \le L_\varepsilon (a-b)^2$. This extends the learner–mean–state decomposition in Section Appendix A.15 with a transport and checkpoint-selection bound for its learner term. Honest cohort evaluation, full-class uniform control, or suitable stability must bound ξ_{eval}. The same is true of Δ_v in (70). A different target conditional residual law adds its discrepancy separately; removing DR-CDF does not remove this transport requirement.

Let $\bar{F}_n = n^{-1}\sum_i F_i$, $F_0 = EF_i$, and $\ell_n = \|\bar{F}_n - F_0\|_\infty$. If F_0 is L_0-Lipschitz, Proposition 13 gives

$$\|\bar{G}_n - F_0\|_\infty \le \ell_n + (3L_0\mathcal{R}_n)^{1/3}. \tag{74}$$

The proof is given in Section Appendix A.15. For independent target units following a common conditional law, $\ell_n = O_p(n^{-1/2})$ concerns the population-mixture target, not the realized finite-cohort counterfactual ECDF.

Finite draws. Conditional on fitted quantities, independent auxiliary uniforms give $K_s = nJM$ independent, not identically distributed, draws in the uniformly pooled post-window implementation. Their average CDF is $\bar{G}_n$. For its simulated ECDF $\widehat{F}_{0,M}$, symmetrization, the $K_s + 1$ threshold patterns, and bounded differences give

$$\|\widehat{F}_{0,M} - \bar{G}_n\|_\infty \le 2\sqrt{\frac{2\log\{2(K_s+1)\}}{K_s}} + \sqrt{\frac{\log(1/\delta)}{2K_s}} \tag{75}$$

with probability at least $1-\delta$. To see this, symmetrization bounds the expected supremum by twice the Rademacher supremum; conditional on draw values, the threshold class has at most $K_s + 1$ vectors and their negatives have at most $2(K_s + 1)$, so the finite-class Rademacher bound gives the first term. Changing one draw changes the supremum by at most $1/K_s$, and McDiarmid gives the second. This bound only describes auxiliary Monte Carlo error, not information from nJ independent observed outcomes. For period-specific

QTE use $K_s = nM$. Code rounding, and NumPy's linearly interpolated empirical quantile convention rather than a generalized-inverse quantile, require their own numerical allowances. The latter discrepancy is bounded by the largest adjacent ordered-draw gap covering the evaluated levels; it need not be called zero in a finite sample.

Finally, if the *true marginal* F_0, F_1 have densities bounded below by $c_0, c_1 > 0$ near the evaluated quantiles, then, for errors small enough to remain in those neighborhoods,

$$\sup_{\tau \in \mathcal{A}} |\widehat{QTE}(\tau) - QTE(\tau)| \leq \frac{\ell_n + (3L_0\mathcal{R}_n)^{1/3} + \eta_M}{c_0} + \frac{\|\widehat{F}_1 - F_1\|_\infty}{c_1} + \eta_{\text{quantile}}, \tag{76}$$

where η_M includes (75) and any CDF evaluation error, and η_{quantile} includes the empirical-quantile convention and rounding. Generalized-inverse bracketing $F^{-1}(\tau - e) \leq \widehat{F}^{-1}(\tau) \leq F^{-1}(\tau + e)$ and the local inverse-Lipschitz property prove (76). No lower density is required of the fitted atomic distribution.

Appendix A.18.7 The finite-cohort target scored in the simulation table

Table 2 uses quantiles of the realized, held-out untreated outcomes, not quantiles of the population mixture F_0. For n treated units, J post periods, and fixed nonnegative period weights π_t summing to one, define

$$F_{0,n}^{\text{real}}(y) = \frac{1}{n}\sum_{i=1}^{n}\sum_{t=1}^{J} \pi_t \mathbf{1}\{Y_{it}(0) \leq y\}, \qquad \bar{F}_n(y) = \frac{1}{n}\sum_{i=1}^{n}\sum_{t=1}^{J} \pi_t F_{it}(y).$$

Here F_{it} is the true conditional untreated CDF under the declared oracle information, and the code uses $\pi_t = 1/J$ for pooling. Let $F_i = \sum_t \pi_t F_{it}$ and $G_i = \sum_t \pi_t G_{it}$ in the definition of $\mathcal{R}_n$. If all true F_{it} have density at most L, then $\bar{F}_n$ is L-Lipschitz. The same triangular-gap proof used for (74), now taking $\bar{F}_n$ itself as reference, gives

$$\|\bar{G}_n - \bar{F}_n\|_\infty \leq (3L\mathcal{R}_n)^{1/3}.$$

Consequently, with $\xi_{\text{real}} = \|F_{0,n}^{\text{real}} - \bar{F}_n\|_\infty$,

$$\|\widehat{F}_{0,M} - F_{0,n}^{\text{real}}\|_\infty \le (3L\mathcal{R}_n)^{1/3} + \xi_{\text{real}} + \eta_M =: e_n. \tag{77}$$

This is a pathwise interface: ξ_{real} separates realization noise in the scored oracle outcomes from conditional-distribution learning error. It is not a nuisance term that training can legitimately use. Those counterfactual outcomes are available only to simulation scoring, not to the estimator in an application.

Since $F_{0,n}^{\text{real}}$ is discrete, a positive-density inverse bound must not be assigned to it. Instead let $Q_{0,n}^{\text{real}}$ be its generalized inverse. For $e_n < \min(\tau, 1-\tau)$, exact inverse bracketing yields

$$Q_{0,n}^{\text{real}}(\tau - e_n) \le \widehat{F}_{0,M}^{-1}(\tau) \le Q_{0,n}^{\text{real}}(\tau + e_n). \tag{78}$$

For example, if $x < Q_{0,n}^{\text{real}}(\tau - e_n)$, then $F_{0,n}^{\text{real}}(x) < \tau - e_n$ and $\widehat{F}_{0,M}(x) < \tau$, proving the lower bracket; at $x = Q_{0,n}^{\text{real}}(\tau + e_n)$ the corresponding inequalities give $\widehat{F}_{0,M}(x) \ge \tau$, proving the upper one.

The code uses the *same* observed-treated sample and quantile convention in its estimated and realized-target QTE. Its treated quantile therefore cancels exactly in their difference. For generalized-inverse quantiles,

$$\begin{aligned}|\widehat{QTE}_n(\tau) - QTE_n^{\text{real}}(\tau)| \le \max\{&Q_{0,n}^{\text{real}}(\tau + e_n) - Q_{0,n}^{\text{real}}(\tau),\\ &Q_{0,n}^{\text{real}}(\tau) - Q_{0,n}^{\text{real}}(\tau - e_n)\}.\end{aligned} \tag{79}$$

For NumPy's interpolated empirical quantiles, add the absolute convention discrepancy for the generated-counterfactual sample and for the realized-counterfactual sample, plus rounding. There is no additional treated-sample convention term because it cancels. These finite-sample brackets can be wide near a large ordered-outcome gap; no root-N quantile rate follows from the brackets alone.

One sufficient stochastic bound for the realization term also respects unit clustering. Conditional on oracle states and the common-factor path, suppose the n untreated outcome *profiles* are independent, while time dependence within each profile is arbitrary, and that F_{it} are their conditional marginals. Then, with conditional probability at least $1-\delta$,

$$\xi_{\text{real}} \leq 2\sqrt{\frac{2\log\{2(nJ+1)\}}{n}} + \sqrt{\frac{\log(1/\delta)}{2n}}.$$

To prove this, apply symmetrization to the unit-level functions $H_i(y) = \sum_t \pi_t \mathbf{1}\{Y_{it}(0) \leq y\}$, which lie in $[0,1]$. On a realized sample the vectors $(H_1(y), \ldots, H_n(y))$ have at most $nJ+1$ threshold patterns, each with Euclidean norm at most $\sqrt{n}$. The finite-class Rademacher bound and the factor two from symmetrization give the first term. Replacing one whole profile changes the supremum by at most $1/n$, so McDiarmid gives the second. The effective sampling size is n, not nJ. This conditional-independence premise is not supplied by conditioning on data-dependent fitted states or by downstream fold labels. If it fails, retain ξ_{real} rather than substituting this stochastic bound. The result connects the theory to the table's scoring target without reusing hidden counterfactual outcomes in fitting.

Inferential scope. These propositions supply a finite-error chain from the exact deep transport to marginal QTE. They establish neither a vanishing fixed-architecture floor nor root-N QTE inference. For example, even if a growing implementation and the missing generalization interfaces gave $\mathcal{R}_n = O_p(r_n^2)$, (74) yields an upper bound of order $r_n^{2/3} + n^{-1/2}$, not a first-order expansion. A root-N assertion would require a separate expansion, undersmoothing or bias control, full generated-state dependence analysis, and a valid variance or resampling theorem. The functional benefit established here is a state-dependent, noncrossing counterfactual distribution with an explicit error decomposition. Proper-score noADV shares that identification route; a strict adversarial accuracy advantage is not implied by these proofs.

Code correspondence. The functions below are in `gscgan/simulations/end_to_end_single_global_cramer_gan_qte_experiment.py`. The versioned source mapping is `audit/deep_theory_code_alignment_20261017/qte_mapping.json`.

Transport and empirical base.

`weighted_quantiles` and `EmpiricalTransportGenerator.quantiles/sample` implement the weighted base, bounded shift, positive spacings, and interpolation in Propositions 20 and 22.

Generated state and validation.

`build_distribution_arrays` constructs the generated state, training rows, and weights. `tail_aware_validation` and `train_empirical_transport_model` determine the eligible checkpoints in Proposition 23. The module's noADV training-length default differs from its adversarial default; matched-budget experiments are defined by their runner overrides and saved contracts, not inferred from these defaults.

Scoring and target.

`conditional_energy_score` supplies the proper-score summand. `sample_quantile_generator` and `evaluate_counterfactual_samples` implement the auxiliary draws, pooled empirical quantiles, and realized-panel scoring in Section Appendix A.18.7. Counterfactual truth enters evaluation only.

Appendix A.19 History, cross-fold dependence, and ATT inference

All statements in this section concern a population ATT with independent unit clusters, conditional on an admitted common environment. A unit may contain arbitrarily dependent periods. A fixed post-period average is treated as one outcome. Neither panel rows nor simulation replications replace the number of independent units N in a sampling limit.

Appendix A.19.1 History-sensitive bias

Let $p = \mathbb{P}(D = 1)$, $\varepsilon = Y(0) - m_0(S)$, and $\mathcal{H} = \sigma(U, S, H_{\text{pre}})$. Suppose

$$\mathbb{E}[D \mid \mathcal{H}, Y(0)] = e(U), \qquad \mathbb{E}(\varepsilon \mid U, S) = 0.$$

Freeze fitted maps $\widehat{m}, \widehat{e}$ before evaluating an independent new unit; they may use its admitted history. Write $d_m = \widehat{m} - m_0$, $d_e = \widehat{e} - e$, $v = d_e/(1 - \widehat{e})$, $q = p + \mathbb{E}v$, and $c_m = \mathbb{E}(ed_m)/p$. Both true and fitted propensities are bounded away from zero and one, and q is bounded away from zero. Proposition 11 gives the exact normalized-odds bias B in (46). Its history term admits the following quantitative bound.

Proposition 24 (History predictability is a separate inference rate)**.** *Let $r_e = \|d_e\|_2$, $r_m = \|d_m\|_2$, and $\eta_H = \|\mathbb{E}(\varepsilon \mid \mathcal{H})\|_2$ under the fresh-unit law. Then, for a constant depending only on overlap and the denominator,*

$$|B| \le Cr_e r_m + Cr_e \eta_H. \tag{80}$$

In particular the history part is negligible on a root-N scale if $r_e \eta_H = o_p(N^{-1/2})$. Full-history mean sufficiency makes $\eta_H = 0$ exactly; sufficiency only for (U, S) does not.

Proof. The fitted maps and v are $\mathcal{H}$-measurable, so iterated expectations give $\mathbb{E}(v\varepsilon) = \mathbb{E}\{v\mathbb{E}(\varepsilon \mid \mathcal{H})\}$. Cauchy–Schwarz and $\|v\|_2 \le Cr_e$ bound this by $Cr_e\eta_H$. Also $|c_m| \le \|e\|_2 r_m/p$, hence $\|d_m - c_m\|_2 \le Cr_m$. Apply the same inequality to the other term of (46) and divide by q. □

If the residual is a stationary Gaussian AR(1) process, $\varepsilon_t = \phi\varepsilon_{t-1} + \nu_t$, and the admitted history contains the residual path through T_0 and no future innovations, then

$$\mathbb{E}(\varepsilon_{T_0+j} \mid \mathcal{H}) = \phi^j \varepsilon_{T_0}, \qquad \eta_H = \sigma_\varepsilon |\phi|^j.$$

This follows by iterating the recursion and using independence of future innovations. Extending the pre-period window does not make this quantity vanish at a fixed adjacent post period. The shared simulation instead uses independent post innovations; its primitive history term is zero at the full sufficient state. Dependence caused by reusing other units in fitted maps remains a different issue.

Appendix A.19.2 Indirect cross-fold dependence

For fold k, let P_k be the scored-unit empirical average, $\pi_k = n_k/N$, and $\mathbb{E}_k^*$ integrate a frozen map over a new unit under the same population law for every fold. The fold index denotes different fitted maps, not a different population treated fraction. Define three error functions

$$
\begin{aligned}
f_{Ak} &= Dd_{mk}, \\
f_{Qk} &= (1-D)(\widehat{\rho}_k - \rho), \\
f_{Bk} &= (1-D)\{(\widehat{\rho}_k - \rho)\varepsilon - \widehat{\rho}_k d_{mk}\}, \qquad \rho = e/(1-e).
\end{aligned}
$$

Their empirical remainders are $R_j = \sum_k \pi_k (P_k - \mathbb{E}_k^*) f_{jk}$, $j = A, Q, B$. These are the three centered scores in (47); their control is distinct from direct training-fold exclusion.

Proposition 25 (Honest-surrogate transfer)**.** *For each j, k, let f_{jk}^H be a hypothetical honest counterpart trained without any scored-fold records, including all upstream feature construction and rank selection. Suppose, conditional on its training information, the centered scored average has variance at most r_{Nk}^2/n_k and mean zero, where r_{Nk} is measurable with respect to that training information and $r_N = \max_k r_{Nk} = o_p(1)$. For treatment-stratified folds this premise additionally requires control of the difference between stratum-wise centering and the common population centering; it does not follow from within-stratum independence alone. Let K be fixed with n_k/N bounded away from zero. If*

$$
s_N = \max_{j \in \{A,Q,B\}} \sum_k \pi_k (P_k + \mathbb{E}_k^*) |f_{jk} - f_{jk}^H|,
$$

then $\max_j |R_j| = O_p(r_N/\sqrt{N}) + O_p(s_N)$. *Consequently* $s_N = o_p(N^{-1/2})$ *transfers the honest centered-score bound to the actual stitched implementation.*

Proof. For each fold, conditional Chebyshev bounds the honest centered average by $O_p(r_N/\sqrt{n_k})$. A finite sum gives $O_p(r_N/\sqrt{N})$; independence across fitted folds is not used. For the difference, the deterministic triangle inequality gives

$$|(P_k - \mathbb{E}_k^*)(f_{jk} - f_{jk}^H)| \le (P_k + \mathbb{E}_k^*)|f_{jk} - f_{jk}^H|.$$

Multiply by π_k, sum, and take the finite maximum. □

A primitive, conservative version is available. If complete-map replacement of one scored record by an external fixed dummy changes each score-error function uniformly by at most β_N (including its generated inputs), and replacing every record in a scored fold produces an honest map, a telescoping argument gives $s_N \le 2 \max_k n_k \beta_N$. Thus $\beta_N = o(N^{-3/2})$ is sufficient here, not necessary. No such bound has been established for the nonsmooth rank selection, boosting, Adam trajectory, or early-stopping operations. The proposition isolates the complete-map stability condition needed in addition to downstream cross-fitting. Section Appendix A.22 replaces this conservative whole-fold bound with an exact bias/covariance decomposition and a leave-one/leave-two transfer. Its uniform sufficient rate is $\beta_N = o(N^{-1/2})$, together with consistent score-error maps; the rate is not established for the implemented learner.

Appendix A.19.3 Rate assembly without counting state error twice

For clarity, let a_{mN} and a_{eN} denote oracle-coordinate learner errors under the appropriate evaluation law, with all approximation, optimization, selection, domain-transfer and fixed-penalty terms retained. If a coupling argument for the actual fitted maps gives

$$r_m \le a_{mN} + L_m \rho_{NT}, \qquad r_e \le a_{eN} + L_e \rho_{NT},$$

then their product is bounded by

$$a_{mN}a_{eN} + (L_e a_{mN} + L_m a_{eN})\rho_{NT} + L_m L_e \rho_{NT}^2. \tag{81}$$

The Lipschitz constants may depend on N and must be included in rates. In particular, fixed positive architectural or penalty errors cannot simply be dropped when asserting both nuisance consistency or efficient inference. Sections Appendix A.23 and Appendix A.24 refine two components for the current DGP. The first bounds omitted-history predictability by anchor prediction error; the second controls the difference between full-panel training and own-pre-period scoring coordinates. They provide interfaces for the complete-map rates above, not additional biases to add after those rates already include history and state replacement.

Theorem 5 (Conditional inference upgrade)**.** *Assume the assignment, overlap and moment restrictions above, a unit-cluster CLT for the oracle normalized ATT score, and fold-maximal nuisance errors* $r_m = \max_k r_{mk} = o_p(1)$ *and* $r_e = \max_k r_{ek} = o_p(1)$*. Assume the three empirical remainders* $R_A, R_Q, R_B = o_p(N^{-1/2})$*, positive denominators, asymptotically inactive clipping/capping, and numerical score errors* $o_p(N^{-1/2})$*. For finitely many folds suppose*

$$r_m r_e = o_p(N^{-1/2}), \qquad \max_k r_{ek}\eta_{Hk} = o_p(N^{-1/2}).$$

Then the estimator has the oracle first-order expansion and root-N normal limit. An empirical score variance is consistent if, additionally, the estimated scores converge in L_2 *and their squared values satisfy the required unit-cluster law of large numbers.*

Proof. Put $u = \sum_k \pi_k \mathbb{E}_k^*(ed_{mk})$, $v_* = \sum_k \pi_k \mathbb{E}_k^* v_k$, $z = \sum_k \pi_k \mathbb{E}_k^*(v_k d_{mk})$, and $h = \sum_k \pi_k \mathbb{E}_k^*(v_k \varepsilon)$. Let A_N, B_N, P_N, Q_N be the oracle treated numerator, control numerator, treated fraction and control-odds denominator. Direct expansion of the three estimated

sample moments gives

$$\widehat{A}_N = A_N - u - R_A, \quad \widehat{Q}_N = Q_N + v_* + R_Q, \quad \widehat{B}_N = B_N + h - u - z + R_B.$$

Subtracting the oracle two-ratio statistic, using $P_N, Q_N = p + O_p(N^{-1/2})$, $B_N = O_p(N^{-1/2})$, and $u, v_* = o_p(1)$, gives the bias remainder

$$\frac{z - h - uv_*/p}{p + v_*} + o_p(N^{-1/2}).$$

Cauchy–Schwarz bounds z and uv_* by $Cr_m r_e$; Proposition 24 bounds h by the stated history rate. The remainder therefore vanishes on the root-N scale. Apply the oracle cluster CLT and, for studentization, the assumed score L_2 convergence and second-moment law of large numbers. □

This formulation uses errors of the complete fitted maps relative to true states and explicitly controls their reuse. An additional undifferentiated state remainder is unnecessary *if all of these full-map conditions are proved*; they are not yet proved. If one instead establishes learner rates only with oracle inputs, the generated-state remainder in (6) is still required. Ordinary AIPW nuisance orthogonality is not itself a proof of either route.

Appendix A.19.4 The variance target with fixed generator draws

Proposition 12 supplies the fixed-M counterpart of Theorem 5. The released ATT configuration averages $M = 50$ post-period draws, distinct from its four-draw calibration bank. Under the proposition's stable-generator, independent-mark, full-history, product-rate, and complete-map stability conditions, the reference is $m_0 + \zeta^M$, rather than the integrated mean m_0. The added influence term $-(D-e)\zeta^M/\{p(1-e)\}$ has mean zero conditional on the full

unit record, and its covariance with the ordinary score is zero. Its variance is

$$\frac{1}{M}\mathbb{E}\left[\frac{e(U)\,\mathrm{Var}_z\{g_0(V,z)\}}{p^2\{1-e(U)\}}\right].$$

For a pooled target the variance retains the within-unit covariance of the pooled draw error. A fixed evaluation bank is part of the estimator; saving its seed does not prove convergence to the stable-generator limit. The additional variance is algorithmic variation, not an efficiency gain over a deterministic conditional mean.

Appendix A.19.5 Conditions for feasible inference

The conditional inference theorem separates the following requirements.

1. **Stage-1 recovery and selected rank.** The linear anchor, orthogonality of the nonlinear perturbation, eigengap and own-history loading concentration must hold. Section Appendix A.21 proves transfer under selection consistency or a common first-order ATT score across selected ranks, and gives a validation-risk margin condition. Those premises remain unproved for the executed criterion; agreement of pre-period prediction risk is not enough.

2. **Full-history centering.** The shared DGP's independent errors support it at the true sufficient state. General applications require a substantive history condition, a quantitative bound such as (80), or a derived extra influence term. Section Appendix A.23 gives a sharper directional bound when the true propensity is measurable in the outcome prediction information. Its current-DGP anchor bound does not assert exact sufficiency at fixed T_0.

3. **Stitched-map stability.** In `common_adjustment.py`, another fold's state can use the current control's outcomes before the propensity is fitted. Frozen buffers and direct training-ID exclusion do not remove that path. Proposition 25 gives a precise

sufficient interface, not a claim that the implemented algorithm already satisfies it. Section Appendix A.22 gives the sharper single- and double-deletion bound and traces the fixed-rank return path in the executed source.

4. **Optimization and tuning bias.** The ATT and QTE bounds above retain fixed objectives and checkpoint choices. No finite iteration budget has been proved to make the gaps vanish. Section Appendix A.24 separates oracle-input support transfer and same-network sensitivity from learning with outcome-generated inputs. The latter still needs a target-risk argument; uniform own-pre loadings alone would not remove self-inclusion of pre-period training outcomes.

5. **Weight regime.** A cap that remains binding changes the normalized-odds score. General population-ATT centering cannot be inferred from propensity consistency alone in that regime. Section Appendix A.20 derives its exact bias, frozen-map sampling expansion, and nuisance derivatives. The shared DGP has strict population margins, making inactive stabilization a consequence of sample-uniform propensity consistency, not an unconditional property of the fitted runs.

6. **Variance.** The current score calculation preserves unit clustering, but does not estimate a general generated-state influence contribution. The smooth stacked-Z helper requires an externally justified first-stage influence representation and keeps its cross covariance; it does not supply that representation for the present pipeline.

7. **QTE.** The plug-in transport results establish finite error control and conditional consistency when all terms vanish. They do not establish a root-N plug-in expansion, orthogonality, double robustness, or confidence-band coverage.

Code correspondence. Source map: `audit/deep_theory_code_alignment_20261017/inference_mapping.json`.

State and history.

`gscgan/src/gsc_nuisance.py` selects rank on treated-pre data and fits own-pre loadings; `gscgan/src/common_adjustment.py` stitches histories before cross-fitted propensity estimation. Proposition 25 addresses this indirect reuse.

Evaluated prediction and normalized score.

`_fixed_draw_bank` and `_predict_post_mean` in `R/wgan_training.py` separate calibration and evaluation marks. In `R/dr_att.py`, `compute_att_aipw` computes the normalized residual correction; `compute_att_unit_cluster_if` aggregates its score by unit.

Generated-state variance.

`gscgan/src/generated_state_inference.py` combines supplied smooth first-stage and ATT influences, retaining their cross covariance. The first-stage representation remains unproved for selected-rank, stitched neural training.

Appendix A.20 Clipping, global capping, and the ATT inference target

This section distinguishes an exact population identity, a sampling theorem for frozen maps, and sufficient conditions for transferring the uncapped theory to estimated maps. None is a proof of the complete stitched learner's rates. All sampling statements use independent unit clusters conditional on an admitted common environment; one outcome can be a fixed weighted average of post periods. Folds, periods, and simulation replications are not additional independent units.

Appendix A.20.1 The precise implemented transformation

For raw fitted probabilities $\widehat{e}_i$, put

$$h_i = \frac{[\widehat{e}_i]_\ell^u}{1-[\widehat{e}_i]_\ell^u}, \qquad \bar{h}_0 = \frac{1}{N_0}\sum_{D_i=0} h_i, \qquad \widehat{t} = \kappa \bar{h}_0, \quad [x]_\ell^u = \min\{u, \max(\ell, x)\}.$$

The shared policy uses $(\ell, u, \kappa) = (0.05, 0.95, 5)$ in the shared-design ATT release. Algebraically its final control weights are

$$w_i = \frac{\min\{h_i, \widehat{t}\}}{\sum_{D_j=0} \min\{h_j, \widehat{t}\}}, \qquad D_i = 0. \tag{82}$$

Indeed, its first normalization gives $h_i/(N_0\bar{h}_0)$; capping that value at κ/N_0 and renormalizing cancels the common $N_0\bar{h}_0$ factor. Thus the threshold on the odds scale is estimated from all controls, not fixed at a population value and not fitted separately within folds. The cap is imposed *before* the last normalization: the final maximum weight need not be at most κ/N_0.

Appendix A.20.2 An exact identity for arbitrary stable weights

Let $\mathcal{H}$ contain the admitted state and own pre-treatment history. Assume consistency and $\mathbb{E}[D \mid \mathcal{H}, Y(0)] = e(U)$, where $e(U)$ is $\mathcal{H}$-measurable and bounded away from zero and one. Set $p = \mathbb{P}(D = 1)$. Freeze the fitted maps before integrating over an independent fresh unit. Let $a \geq 0$ and m be $\mathcal{H}$-measurable, with $q_a = \mathbb{E}\{(1-e)a\} > 0$ and all following moments finite. The weight a may already include fixed population clipping and capping. Define

$$\Theta(m, a) = \frac{\mathbb{E}\{D(Y-m)\}}{p} - \frac{\mathbb{E}\{(1-D)a(Y-m)\}}{q_a}.$$

Proposition 26 (Stable-weight population bias and surviving branches)**.** *Let $\mu_0(S)$ obey $\mathbb{E}\{Y(0) - \mu_0(S) \mid U, S\} = 0$, and write $\varepsilon = Y(0) - \mu_0(S)$, $d_m = m - \mu_0(S)$, $c_m = \mathbb{E}(ed_m)/p$, and $v_a = (1-e)a - e$. Then $q_a = p + \mathbb{E}v_a$ and*

$$\Theta(m, a) - ATT = \frac{\mathbb{E}\{v_a(d_m - c_m)\} - \mathbb{E}(v_a\varepsilon)}{q_a}. \tag{83}$$

In particular, a correct full-history conditional mean $m = \mu_H := \mathbb{E}\{Y(0) \mid \mathcal{H}\}$ makes the bias zero for any such a. Correct weights up to a positive constant, $a = ce/(1-e)$, also make the bias zero for any admitted m. Conversely, if centering is required for every bounded

perturbation of μ_H, it requires $a = c\,e/(1-e)$ almost surely.

Proof. The assignment restriction converts treated and weighted control expectations to e and $(1-e)a$. Since $\mathbb{E}(e\varepsilon) = 0$, their residual means are respectively $ATT - c_m$ and $\{\mathbb{E}(v_a\varepsilon) - pc_m - \mathbb{E}(v_a d_m)\}/q_a$. Subtract to obtain (83). Alternatively, with $\delta = m - \mu_H$, the same assignment restriction gives the especially direct identity

$$\Theta(m,a) - ATT = \mathbb{E}\left[\left\{\frac{(1-e)a}{q_a} - \frac{e}{p}\right\}\delta\right]. \tag{84}$$

This proves both sufficient branches. If the integral is zero for every bounded δ, the bracket is zero almost surely (test its sign, or indicators of its positive and negative sets). Overlap then gives $a = (q_a/p)e/(1-e)$. This proves necessity as stated. □

Consistency of the *raw* propensity does not restore the second branch when its limit is transformed into $a = \min\{[e]_\ell^u/(1-[e]_\ell^u), \kappa\mathbb{E}_0([e]_\ell^u/(1-[e]_\ell^u))\}$. The necessary proportionality generally fails under binding stabilization. Special outcome-error cancellation can still occur; its exact requirement is (84), not propensity consistency alone. The stabilized residual functional generally depends on the limiting outcome map; it is not automatically the ATT of a single trimmed population. Moreover, correctness for a smaller state is not full-history correctness: when $m = \mu_0(S)$ the bias in (83) can remain $-\mathbb{E}(v_a\varepsilon)/q_a$. For example, let $e = 0.4$ be constant, let an admitted history variable H be equally likely -1 or 1, let $Y(0) = H$, and set $\mu_0 = 0$, $m = 0$, and $a = 2 + H$. Assignment is independent of H, yet the ATT bias is $-1/2$. Using $m = H$ or constant odds removes it.

Equation (83) is conditional on frozen maps evaluated on a fresh unit. It is not an assertion that conditioning jointly on every fitted fold makes the observed, stitched sample independent. A law of large numbers for the complete fitted maps is still needed to identify the probability limit of the actual estimator with this functional.

Appendix A.20.3 An active-cap sampling expansion for frozen maps

Here $m(\mathcal{H})$ and $h(\mathcal{H})$ are fixed functions, with $0 < h_{\min} \le h \le h_{\max} < \infty$. They can be trained on an independent external sample and then treated as fixed, provided the scored units remain conditionally iid under the asserted law. Define

$$\begin{aligned} s &= 1-p, & b_0 &= \mathbb{E}(h \mid D=0), & t &= \kappa b_0, \\ a_t &= \min(h,t), & q &= \mathbb{E}\{(1-D)a_t\}, & R &= Y-m, \\ A &= \mathbb{E}(R \mid D=1), & C &= \frac{\mathbb{E}\{(1-D)a_t R\}}{q}, & \theta &= A-C. \end{aligned}$$

The estimator is the treated residual average minus the weighted control average using (82). Its center here is $\theta = \Theta(m, a_t)$; Proposition 26 determines whether this equals ATT.

Proposition 27 (Both denominators and the estimated cap threshold)**.** *Suppose* $0 < p < 1$, $\mathbb{E}R^2 < \infty$, $\kappa > 0$ *is fixed, and* $\mathbb{P}(h = t \mid D = 0) = 0$. *Under the conditional iid sampling law,*

$$\sqrt{N}(\widehat{\theta} - \theta) = N^{-1/2} \sum_{i=1}^{N} \phi_{\text{cap}}(O_i) + o_p(1), \tag{85}$$

where

$$\begin{aligned} T(O) &= \frac{\kappa(1-D)}{s}(h - b_0), \\ \Gamma &= \frac{\mathbb{E}\{(1-D)\mathbf{1}(h > t)(R-C)\}}{q}, \\ \phi_{\text{cap}}(O) &= \frac{D}{p}(R-A) - \frac{(1-D)a_t}{q}(R-C) - \Gamma T(O). \end{aligned} \tag{86}$$

This influence function has mean zero and finite variance, giving the usual normal limit (possibly degenerate). It includes the estimation of N_0/N*, both residual-mean denominators, and the global cap threshold.*

Proof. The ratio expansion of $\widehat{t} = \kappa P_N\{(1-D)h\}/P_N(1-D)$ gives $\widehat{t} - t = P_N T + o_p(N^{-1/2})$. For $v = 1, R$ define $B_v(t) = \mathbb{E}\{(1-D)a_t v\}$. The no-atom condition and dominated con-

vergence imply $B_v'(t) = \mathbb{E}\{(1-D)\mathbf{1}(h > t)v\}$. This derivative also gives the empirical expansion at $\widehat{t}$. Indeed the pointwise remainder in the minimum function is bounded by $|\widehat{t}-t|\mathbf{1}\{|h-t| \le |\widehat{t}-t|\}$; after multiplying by $(1-D)|v|$, its empirical average is $o_p(N^{-1/2})$. To see the last claim, first bound $|\widehat{t}-t|$ by $M/\sqrt{N}$ with arbitrarily high probability, then dominate the shrinking band by a fixed band of width δ. Its empirical mean converges to its expectation; that expectation tends to zero as $\delta \downarrow 0$. The derivative's empirical average converges by the law of large numbers. Thus

$$P_N\{(1-D)a_{\widehat{t}}v\} = B_v(t) + (P_N - P)\{(1-D)a_t v\} + B_v'(t)P_N T + o_p(N^{-1/2}).$$

The quotient derivative for B_R/B_1 is Γ, and its fixed-t empirical part is $(1-D)a_t(R-C)/q$. The treated ratio contributes $D(R-A)/p$. Subtracting proves (86); the boundedness of h, a_t and the residual second moment give its finite variance. □

If h has mass at t, differentiating the minimum at that mass may give a directional, nonlinear limit. The proposition does not cover that case. Clipping can create atoms at its endpoint odds, so the no-atom hypothesis must be checked rather than inferred from a continuous raw propensity.

Why the released score is not the general active-cap influence. With the same fixed maps, the population counterpart of `compute_att_unit_cluster_if` is

$$\widetilde{\phi} = \frac{D}{p}(R-\theta) - \frac{(1-D)a_t}{q}R.$$

Although it is centered, direct subtraction gives

$$\widetilde{\phi} - \phi_{\text{cap}} = C\left\{\frac{D}{p} - \frac{(1-D)a_t}{q}\right\} + \Gamma T. \tag{87}$$

Consequently empirical zero mean is not enough to justify its variance for a general two-ratio, active-cap statistic. A correct full-history mean makes $C = \Gamma = 0$; otherwise either term can be nonzero. This observation does not change the released estimator or retrofit a variance estimate.

Estimated nuisances remain a separate first-order problem. At a fixed population law, bounded directions $\dot{m}, \dot{h}$ and the same no-atom condition give

$$\begin{aligned}
\dot{\Theta}_m &= -\frac{\mathbb{E}(D\dot{m})}{p} + \frac{\mathbb{E}\{(1-D)a_t\dot{m}\}}{q}, \\
\dot{a}_t &= \mathbf{1}(h < t)\dot{h} + \mathbf{1}(h > t)\kappa\mathbb{E}(\dot{h} \mid D = 0), \\
\dot{\Theta}_h &= -\frac{\mathbb{E}\{(1-D)(R-C)\dot{a}_t\}}{q}.
\end{aligned} \tag{88}$$

These follow by dominated differentiation and the quotient rule; they are derivatives in the *post-clipping odds* map h. Differentiating a raw propensity through clipping requires its own endpoint condition. When $m = \mu_H$, $\dot{\Theta}_h = 0$, including the threshold part, but $\dot{\Theta}_m$ generally does not vanish under imbalanced stabilized weights. Therefore outcome-model correctness can establish consistency without establishing the usual orthogonal, product-rate inference result. An estimated outcome map then needs a negligible first-order linear functional, a sufficiently fast rate, or a justified additional influence term. If both maps are estimated, their joint first-order behavior and remainder must be controlled, not inferred from Proposition 27.

The actual algorithm first stitches own-history state summaries and then cross-fits its propensity on those summaries. Another fold's training state can depend on outcomes of a currently scored unit. Whole-unit downstream exclusion and immutable buffers do not establish the external training premise used above. Treatment-stratified folds also require their own centering argument. Full-map convergence, empirical stability, and any generated-state influence remain required for the actual procedure.

Appendix A.20.4 A sufficient inactive-stabilization bridge

Proposition 28 (Strict margins and sample-uniform propensity consistency)**.** *Suppose iid units have true propensity e_i, $0 < p < 1$, and for constants $\eta_e, \eta_\rho > 0$,*

$$\ell + \eta_e \le e_i \le u - \eta_e \quad \textit{almost surely}, \qquad \operatorname{ess\,sup} \frac{e}{1-e} \le \kappa \frac{p}{1-p} - \eta_\rho.$$

If the actually evaluated raw predictions, however constructed, satisfy $\max_{i \le N} |\widehat{e}_i - e_i| = o_p(1)$, then with probability tending to one no observation is clipped and no control weight is capped. Holding the same fitted outcome predictions fixed, the stabilized and unstabilized evaluation scores are then exactly equal; in particular their difference is $o_p(N^{-1/2})$.

Proof. Sample-uniform consistency and the first margin put every predicted probability strictly inside $[\ell, u]$ with probability tending to one. On this event the odds map is uniformly Lipschitz on that interval, so $\max_i |\widehat{\rho}_i - \rho_i| = o_p(1)$. The control ratio law of large numbers and $\mathbb{E}\{(1-D)\rho\} = p$ give $N_0^{-1} \sum_{D_i=0} \widehat{\rho}_i \to_p p/(1-p)$. The second margin therefore separates the largest fitted control odds from κ times their empirical mean with probability tending to one. Equation (82) is then ordinary normalized odds. For the same outcome predictions, exact equality outside an event of vanishing probability implies the asserted stochastic order without requiring a rate on that probability. □

This sufficient condition is stronger than L_2 propensity consistency. An L_2 rate alone does not rule out a few clipped or capped observations; weaker conditions can suffice if their weighted score contribution is shown directly to be $o_p(N^{-1/2})$. This theorem does not claim that fixed-budget boosting on selected-rank, stitched states is sample-uniformly consistent. It also does not equate a separately retrained uncapped procedure to the released procedure: shared weights enter outcome training as well as final evaluation. Outcome-map convergence and complete-map stability remain separate requirements for inference on the full implemented estimator.

For the current shared simulation, $Z \sim U[-1.5, 1.5]$ and $e(Z) = \text{expit}(c + Z)$ with $p = 0.35$. Its population-calibrated intercept has the exact expression

$$c = \log \frac{\exp(3p) - 1}{\exp(1.5) - \exp(3p - 1.5)} \simeq -0.7272164888033855.$$

It gives $e \in [0.0973329242, 0.6841227153]$, strictly inside $[0.05, 0.95]$, and

$$\sup \rho = 2.1657863620 < 5\frac{0.35}{0.65} = 2.6923076923, \qquad \frac{\sup \rho}{\mathbb{E}_0 \rho} = 4.0221746722 < 5.$$

Thus the true distribution has a cap margin about 0.5265213303 on the odds scale. These are population algebraic checks, not learned-weight diagnostics. Binding stabilization in finite fitted panels does not disprove the conditional inactive limit, and the strict population margin does not prove that the fitted pipeline reaches that limit.

Executable correspondence and scope. The source-hash and line map is `audit/inference_closure_20261018/weights_code_mapping.json`. The canonical weight transformation is in `gscgan/src/common_adjustment.py`, lines 996–1011; its ATT adapter passes the same global arrays to every fold (lines 1205–1267). `R/wgan_training.py`, lines 446–466 and 1026–1032, reconstructs the identical final OOF weights. `R/dr_att.py`, lines 31–63, contains the canonical two-residual-mean computation, while lines 107–119 compute the empirically centered score discussed in (87). The differently shaped legacy `gscgan/src/dr_att.py` routine is not substituted for this path. The shared20 execution driver, the archived `simulations/run_att_table2_rankcheck.py`, lines 487–513, finalizes the same global weights and calls `simulations/run_att_exploration.py`, lines 147–163, to summarize the pooled residual means. That point-estimation wrapper is algebraically equivalent; it does not itself call the canonical variance routine. The canonical inference coverage runner rejects any binding clipping or cap at `simulations/run_att_if_coverage.py`, lines 533–544. That finite-run guard is not an active-cap theorem and cannot supply missing

nuisance or stitched-map rate arguments.

The canonical source bytes above were compared with the immutable shared20 source archive, not only with the current checkout. The accompanying small algebra tests check the exact transformation, population bias, nuisance derivatives, and the distributional derivative in (86). They do not train a nuisance, rerun a simulation ledger, or establish empirical coverage.

Appendix A.21 Finite-grid rank selection and ATT inference

This section closes the logical transfer from a fixed-rank result to a data-selected rank. It does not assert the unproved premises needed to apply that transfer to the released estimator. All probabilities concern the joint unit-cluster array and algorithmic randomization, conditional on an admitted common environment as in Section Appendix A.2. The population ATT is denoted by θ_0; N counts units, not periods or simulation replications.

Appendix A.21.1 The criterion actually evaluated

Let $I_1, \dots, I_K$ be the outer unit folds, $\mathcal{T}$ all treated units, and $\mathcal{C}_{-k}$ the controls outside I_k. For each candidate $r \in \mathcal{R}$, the code fits $\widehat{\beta}_{kr}$ and $\widehat{F}_{kr}$ using all observed periods of $\mathcal{C}_{-k}$. For every treated unit and pre date s, it computes

$$\widehat{\lambda}_{i,kr}^{(-s)} = \widehat{F}_{kr,-s}^{+}(Y_{i,-s} - X_{i,-s}\widehat{\beta}_{kr}), \qquad \widehat{L}_{kN}(r) = \frac{1}{|\mathcal{T}|T_0} \sum_{i\in\mathcal{T}} \sum_{s=1}^{T_0} \{Y_{is} - X_{is}'\widehat{\beta}_{kr} - \widehat{f}_{kr,s}'\widehat{\lambda}_{i,kr}^{(-s)}\}^2.$$

Here $-s$ denotes the other pre dates only, and $+$ denotes the Moore–Penrose inverse. The control fit is held fixed while the treated loading projection leaves out each pre date. Neither that control fit nor its factor row at date s is re-estimated with date s removed. The executed common rank is

$$\widehat{r} = \operatorname*{argmin}_{r\in\mathcal{R}} \big(\widehat{L}_N(r), r\big), \qquad \widehat{L}_N(r) = K^{-1} \sum_{k=1}^{K} \widehat{L}_{kN}(r), \tag{89}$$

where minimization is lexicographic: an exact numerical tie favors the smaller rank. Every final Stage-1 fold is then re-fitted at this one rank. The released grid is $\{1, 2, 3, 4\}$ and $K = 3$.

Thus a treated unit in I_k contributes pre outcomes directly to the selection criterion, including its own fold's criterion. A control in I_k can affect $\widehat{L}_{\ell N}$ through the control fit whenever $\ell \neq k$. The selected rank is therefore not measurable only with respect to the training records outside I_k. No treated post outcome enters this selection, but exclusion of those outcomes is not independence of selection from the scored unit. Nor does leaving out one time point make the other pre dates independent of it under arbitrary serial dependence. The common-state constructor subsequently requires the same dimension in every fold, stitches each unit's out-of-fold state row, and then fits the propensity on those stitched rows. Rank selection and the indirect stitched-state dependence in Section Appendix A.19 are distinct issues.

Appendix A.21.2 Two valid ways to transfer a fixed-rank result

For each $r \in \mathcal{R}$, define $\widehat{\theta}_N(r)$ as the *complete* ATT branch at that common rank, including state construction, propensity, outcome training, stabilization, checkpoint selection, evaluation draws and normalization. Couple all branches using the same data, folds and specified auxiliary randomness, so the executed statistic satisfies $\widehat{\theta}_N = \widehat{\theta}_N(\widehat{r})$. This coupling is an analytical comparison, not an instruction to re-run the experiment. A fixed-rank oracle comparison must use these same non-rank operations. If the algorithm can fail, each statement below additionally requires a common successful-definition event with probability approaching one; finite-run gate passage alone does not establish that event.

Proposition 29 (Selection-consistency transfer)**.** *Suppose* $r_0 \in \mathcal{R}$ *and* $\Pr(\widehat{r} \neq r_0) \to 0$. *For every deterministic scale* $a_N > 0$,

$$a_N\{\widehat{\theta}_N(\widehat{r}) - \widehat{\theta}_N(r_0)\} = o_p(1).$$

In particular, a root-N asymptotic-linear representation and weak limit for the complete fixed-r_0 branch transfer to the selected branch. If the coupled fixed-r_0 variance estimate is consistent, its selected counterpart is consistent as well, and the corresponding studentized limit transfers whenever its limiting variance is positive. No $o(N^{-1/2})$ bound on $\Pr(\widehat{r} \neq r_0)$ is needed.

Proof. On $\{\widehat{r} = r_0\}$ the coupled statistics are exactly equal. Consequently, for every $\epsilon > 0$,

$$\Pr\Big(a_N|\widehat{\theta}_N(\widehat{r}) - \widehat{\theta}_N(r_0)| > \epsilon\Big) \leq \Pr(\widehat{r} \neq r_0) \longrightarrow 0.$$

The same argument applies to the variance estimate and studentized statistic. Add the fixed-rank expansion and apply Slutsky's theorem. No bound on the size of the error on the exceptional event is used. □

This is a probability and distribution result, not a statement about expected bias, mean squared error, or uniform validity over a family of data laws. For example, a difference equal to N with probability $1/\log(N+1)$ and zero otherwise vanishes in probability at every deterministic rescaling in the displayed sense, while its expectation diverges. Moment transfer needs additional tail control; uniform inference needs uniformly small selection-error probabilities and uniform fixed-rank approximations. The argument removes only the rank-selection step. It does not prove the fixed-rank ATT expansion, history centering, stitched-map stability, or the treatment-assignment law.

Theorem 6 (Uniform score transfer without a unique selected rank)**.** *Let $\mathcal{A}_N \subseteq \mathcal{R}$ be a deterministic, nonempty set conditional on the common environment, with $\Pr(\widehat{r} \in \mathcal{A}_N) \to 1$. Suppose there is one common unit-score array ψ_{Ni} such that*

$$\max_{r \in \mathcal{A}_N} \left| \sqrt{N}\{\widehat{\theta}_N(r) - \theta_0\} - N^{-1/2} \sum_{i=1}^{N} \psi_{Ni} \right| = o_p(1). \tag{90}$$

If $N^{-1/2}\sum_i \psi_{Ni} \Rightarrow \mathcal{N}(0, V)$, $0 < V < \infty$, then the selected branch has that same expansion

and normal limit. If also $\max_{r\in\mathcal{A}_N}|\widehat{V}_N(r)-V|=o_p(1)$*, its studentized statistic converges to* $\mathcal{N}(0,1)$*. The selector may depend on all scored data.*

Proof. For every $\epsilon>0$, the probability that the selected expansion remainder exceeds ϵ is at most

$$\Pr(\widehat{r}\notin\mathcal{A}_N)+\Pr\left(\max_{r\in\mathcal{A}_N}\left|\sqrt{N}\{\widehat{\theta}_N(r)-\theta_0\}-N^{-1/2}\sum_i\psi_{Ni}\right|>\epsilon\right).$$

Both terms vanish. The identical maximum argument transfers variance consistency, and Slutsky's theorem proves studentization. □

For a fixed finite set of admitted ranks, separate $o_p(1)$ remainder bounds relative to the *same* score imply the maximum bound by a finite union bound. Independence of the branches is unnecessary. Equivalently one can establish $\max_{r\in\mathcal{A}_N}\sqrt{N}|\widehat{\theta}_N(r)-\widehat{\theta}_N^{\mathrm{ref}}|=o_p(1)$ relative to one valid reference statistic. State dimensions may differ: the comparison is between scalar ATT statistics or their scores, not between unaligned loading vectors of different lengths. The fixed-M version must use the common *augmented* score in Proposition 12. Equal integrated generator means do not ensure equal fixed-draw score limits across ranks.

Separate normal limits with the same marginal variance are insufficient. For a simple mathematical counterexample, let $Z_N\sim\mathcal{N}(0,1)$, $\widehat{\theta}_N(1)=Z_N/\sqrt{N}$ and $\widehat{\theta}_N(2)=-Z_N/\sqrt{N}$. Let $\widehat{L}_N(1)=1-N^{-1}\tanh Z_N$ and $\widehat{L}_N(2)=1+N^{-1}\tanh Z_N$ for $N\geq 2$. Both losses converge uniformly to the tied risk 1, and each candidate has an exact standard-normal root-N limit. Minimizing the losses gives $\sqrt{N}\widehat{\theta}_N(\widehat{r})=|Z_N|$, not a centered normal limit. This example shows why separate marginal normal limits cannot replace the common-score premise of Theorem 6 under potentially score-dependent selection; it is not a model for the executed CV losses.

Appendix A.21.3 What a validation-risk argument can establish

Let $L_N(r)$ be a deterministic benchmark risk conditional on the common environment, for the *executed* criterion, and put $\epsilon_N=\max_{r\in\mathcal{R}}|\widehat{L}_N(r)-L_N(r)|$. Any asserted bound on ϵ_N

must cover estimated control factors, own-history projections, overlapping folds, pre-time dependence and numerical fitting errors. A finite candidate list alone supplies none of these bounds. It only permits finitely many already valid bounds to be combined.

Proposition 30 (Finite-grid risk and margin bounds)**.** *Suppose a selector is a γ_N-approximate minimizer of $\widehat{L}_N$, with $\gamma_N \geq 0$. Then, pathwise,*

$$L_N(\widehat{r}) - \min_{r \in \mathcal{R}} L_N(r) \leq 2\epsilon_N + \gamma_N. \tag{91}$$

If r_0 uniquely minimizes L_N with gap $\Delta_N = \min_{r \neq r_0}\{L_N(r) - L_N(r_0)\} > 0$, then

$$\Pr(\widehat{r} \neq r_0) \leq \Pr(2\epsilon_N + \gamma_N \geq \Delta_N).$$

Thus $\epsilon_N = o_p(\Delta_N)$ and $\gamma_N = o_p(\Delta_N)$ imply selection consistency, even when $\Delta_N \to 0$. More generally, for deterministic $b_N \geq 0$, set $\mathcal{A}_N(b_N) = \{r : L_N(r) - \min_s L_N(s) \leq b_N\}$. If $\Pr(2\epsilon_N + \gamma_N \leq b_N) \to 1$, then $\Pr\{\widehat{r} \in \mathcal{A}_N(b_N)\} \to 1$. Combined with (90) on that set, this gives selected-rank ATT inference without selecting one rank consistently.

Proof. For a risk minimizer r_*,

$$L_N(\widehat{r}) \leq \widehat{L}_N(\widehat{r}) + \epsilon_N \leq \widehat{L}_N(r_*) + \gamma_N + \epsilon_N \leq L_N(r_*) + \gamma_N + 2\epsilon_N.$$

If $2\epsilon_N + \gamma_N < \Delta_N$, any wrong rank contradicts this inequality. The near-minimizer assertion is the same inequality with threshold b_N. □

The actual selector has $\gamma_N = 0$ for its computed finite scores; optimization and rounding differences from a theoretical risk belong in ϵ_N. Neither the gap nor its relation to that error is proved for the reported estimator. In particular, a structural anchor rank is not automatically the unique minimizer of a prediction risk. Overfitted ranks may have the same limiting pre prediction risk as the anchor rank, and a vanishing gap requires a correspondingly

stronger error bound. The smaller-rank rule resolves *exact sample ties*, not unequal random scores converging to the same population risk. The boundary gate, which checks whether the selected upper rank improves on its neighboring candidate beyond a fixed numerical tolerance, is not a population-risk margin condition and is not a rank-consistency test.

Appendix A.21.4 Prediction-risk equivalence is not ATT-score equivalence

The CV loss measures linear-anchor prediction on treated pre periods. It does not directly measure the post-period conditional mean, the propensity, the weighted ATT moment, or the complete neural fit returned after training and checkpoint selection. The distinction is visible in an exact finite-sample identity. Write $\mathbb{P}_N$ for the unit average, $p_N = \mathbb{P}_N D$, $q_N(r) = \mathbb{P}_N\{(1-D)\widehat{\rho}_i(r)\}$, and, for positive denominators,

$$c_i(r) = \frac{D_i}{p_N} - \frac{(1-D_i)\widehat{\rho}_i(r)}{q_N(r)}, \qquad \widehat{\theta}_N(r) = \mathbb{P}_N[c_i(r)\{Y_i - \widehat{m}_i(r)\}].$$

Here Y_i and $\widehat{m}_i(r)$ may be a common fixed post-period average, and $\widehat{\rho}_i(r)$ is the actual nonnegative raw weight whose normalization is used in the statistic, including any stabilization. For two ranks r, s,

$$\begin{aligned} \widehat{\theta}_N(r) - \widehat{\theta}_N(s) = & -\mathbb{P}_N[c_i(r)\{\widehat{m}_i(r) - \widehat{m}_i(s)\}] \\ & + \mathbb{P}_N[\{c_i(r) - c_i(s)\}\{Y_i - \widehat{m}_i(s)\}]. \end{aligned} \tag{92}$$

This follows by adding and subtracting $\mathbb{P}_N[c_i(r)\{Y_i - \widehat{m}_i(s)\}]$. Writing $\|a\|_N^2 = \mathbb{P}_N a_i^2$, Cauchy–Schwarz gives

$$|\widehat{\theta}_N(r) - \widehat{\theta}_N(s)| \le \|c(r)\|_N \|\widehat{m}(r) - \widehat{m}(s)\|_N + \|c(r) - c(s)\|_N \|Y - \widehat{m}(s)\|_N.$$

A uniform $o_p(N^{-1/2})$ bound on the right is a conservative sufficient transfer condition. Orthogonal-score cancellation may yield weaker sufficient rates, but requires a proof for

the complete fitted maps, such as the hypotheses of Theorem 5 uniformly over the admitted ranks. Pre MSPE alone does not bound either term in (92). Even identical pre predictions can coexist with different post predictions; even identical post means can coexist with different weights. For variance transfer one additionally needs common estimated-score L_2 convergence and the appropriate second-moment law of large numbers, not merely agreement of point estimates.

Established result and remaining gate. The propositions prove the selection-transfer and validation-margin implications, including the fact that a selection-error probability tending to zero is sufficient for root-N *distributional* transfer. They do not establish that the actual treated-pre CV selects the structural rank, that tied or nearly tied ranks yield a common ATT score, or that the fixed-rank estimator satisfies all inference gates. The observed selection of rank two in all forty released tasks is a finite-design diagnostic, not a proof of any of these asymptotic premises. Neither the fixed-rank mechanism result nor a successful boundary gate supplies those missing implications.

Code correspondence and frozen execution. The detailed line and SHA-256 map is in `audit/inference_closure_20261018/rank_code_audit.json`. The canonical criterion is in `gscgan/src/gsc_nuisance.py`, and the mean-fold selection and fixed-rank re-fit are in `gscgan/simulations/qte_canonical_pipeline.py`. `R/gsc_nuisance.py` is a compatibility wrapper; its explicitly legacy BIC function is not the released rank selector. The frozen ATT wrapper delegates to `simulations/run_att_table2_rankcheck.py` inside the archived source, whose observer calls the same canonical selector with the registered Stage-1 iteration cap and tolerance and additionally stops on any unconverged candidate or final fit. These source bytes and all archived rank records are unchanged.

Appendix A.22 A sharper interface for complete-map dependence

This subsection strengthens the sufficient interface in Proposition 25; it does not verify that interface for the released ATT estimator. All conditional expectations below are under independent unit clusters, conditional on a common environment $\mathcal{C}$. Random seeds and a fold allocation independent of the unit records may be included in $\mathcal{C}$. Conditioning on all fitted nuisances is *not* permitted as a way to obtain independence. The results are finite-sample identities and inequalities; asymptotic conclusions apply along sequences satisfying their displayed rates.

The actual return paths. Fix a control i in scoring fold k. Even at a prespecified rank, the outcome path of i is used to fit the factors and coefficients for another fold $\ell \neq k$. A unit j in fold ℓ receives an own-pre loading and residual summaries formed using that fit. Its stitched row $\widehat{U}_j$ therefore can depend on $Y_{i,+}$, although $\widehat{U}_i$ itself does not use $Y_{i,+}$ at fixed rank. The propensity learner for fold k is trained on these already-stitched rows. Thus the allowed computational path is

$$Y_{i,+} \longrightarrow (\widehat{F}_{-\ell}, \widehat{\beta}_{-\ell}) \longrightarrow \widehat{U}_j \longrightarrow \widehat{e}_{-k} \longrightarrow \widehat{e}_i \longrightarrow \widehat{w}_i.$$

This is a path of possible dependence, not a claim that every numerical perturbation changes every downstream value. Freezing arrays after their construction and recording disjoint direct training IDs do not remove it.

The same shared propensity artifact supplies weights of the controls used in fold k's adversarial and reconstruction objectives. Those weights may depend on i through stitched features, propensity fits for other folds, and the global weight policy. Consequently the path can also return to $\widehat{m}_{-k}$ through training losses and checkpoint selection. The generator in this ATT implementation uses the fold-k factor/loading state, not the stitched $\widehat{S}$ table as its direct neural input; the stitched-table return path to that generator is through the

shared weights. The final no-cap normalization is an ordinary ratio operation, not by itself evidence of outcome leakage. A common multiplicative scale also cancels in the implemented normalized weighted minibatch losses. Binding caps can change relative weights; neither observation removes the preceding feature and propensity paths.

There is an additional route not requiring the propensity learner. The rank-selection driver averages candidate validation scores across all outer folds and uses its minimizer for every final fold refit. A control's outcomes enter other folds' candidate factor fits, and every treated unit's pre history enters the candidate validation scores. These records can therefore affect the common selected rank, including the rank used to score that same unit. Fixing the rank removes this selection route but not the stitched-feature route.

Notation for a complete-map remainder. Consider any one of R_A, R_Q, R_B in (47). With $k(i)$ the scoring fold, put $f_i = f_{k(i)}$ and

$$X_i = f_i(O_i) - P^* f_i, \qquad R = N^{-1} \sum_{i=1}^{N} X_i.$$

Here P^* integrates the frozen *complete* fitted map over an independent new unit with the common population law. All upstream state, rank, propensity, optimization, and evaluation-mark operations belong to that map. This notation requires an evaluation extension to a new unit; merely having an array of in-sample predictions is insufficient. All quantities in the following propositions are square integrable. Write $\mathcal{F}_{-I} = \sigma(\mathcal{C}, O_j : j \notin I)$.

Proposition 31 (Exact conditional-bias and covariance decomposition)**.** *Define*

$$b_i = \mathbb{E}(X_i \mid \mathcal{F}_{-i}), \quad Z_i = X_i - b_i, \quad \bar{b} = N^{-1} \sum_i b_i, \quad \Delta_j Z_i = Z_i - \mathbb{E}(Z_i \mid \mathcal{F}_{-j}).$$

For $i \neq j$,

$$\mathbb{E}(Z_i Z_j \mid \mathcal{C}) = \mathbb{E}(\Delta_j Z_i \, \Delta_i Z_j \mid \mathcal{C}). \tag{93}$$

Set $v_N^2 = N^{-1}\sum_i \mathbb{E}(Z_i^2 \mid \mathcal{C})$, $a_{ij}^2 = \mathbb{E}\{(\Delta_j Z_i)^2 \mid \mathcal{C}\}$, *and* $c_N = N^{-1}\sum_{i\neq j} a_{ij}a_{ji}$. *Then*

$$\mathbb{E}\{(R-\bar{b})^2 \mid \mathcal{C}\} \leq \frac{v_N^2 + c_N}{N}. \tag{94}$$

Thus $v_N^2 + c_N = o_p(1)$ *and* $\bar{b} = o_p(N^{-1/2})$ *imply* $R = o_p(N^{-1/2})$. *The bias condition concerns the* random average $\bar{b}$, *not only its expectation.*

Proof. By construction $\mathbb{E}(Z_i \mid \mathcal{F}_{-i}) = 0$. Write $Z_i^{(-j)} = \mathbb{E}(Z_i \mid \mathcal{F}_{-j})$. The products $Z_i Z_j^{(-i)}$ and $Z_i^{(-j)} Z_j$ have conditional expectation zero by conditioning on $\mathcal{F}_{-i}$ and $\mathcal{F}_{-j}$, respectively. Given $\mathcal{F}_{-\{i,j\}}$, the two projected variables $Z_i^{(-j)}$ and $Z_j^{(-i)}$ are functions of the independent records O_i and O_j, respectively. Their conditional means are zero by the tower property, for example $\mathbb{E}(Z_i^{(-j)} \mid \mathcal{F}_{-\{i,j\}}) = \mathbb{E}(Z_i \mid \mathcal{F}_{-\{i,j\}}) = 0$. Their product therefore also has conditional expectation zero. Expanding the right side of (93) proves the identity. Expanding the square of $N^{-1}\sum_i Z_i$ and applying conditional Cauchy–Schwarz to each off-diagonal product proves (94). Conditional Chebyshev, truncating the conditional bound by one when necessary, proves the stochastic assertion. □

This proposition separates two issues that a statement of ordinary nuisance consistency does not settle: self-use bias $\bar{b}$ and covariance of conditionally centered errors. Its projections are theoretical objects, not estimable certificates from one released run. A complete-map deletion construction gives a more operational sufficient condition for both issues.

Proposition 32 (Leave-one/leave-two transfer)**.** *For each* i, *suppose there is a map* f_i^{-i} *measurable with respect to* $\mathcal{F}_{-i}$; *for each* $i \neq j$, *suppose there is a map* f_i^{-ij} *measurable with respect to* $\mathcal{F}_{-\{i,j\}}$. *The maps may be produced by deleting or externally replacing complete unit records and rebuilding every downstream object. Define*

$$\xi_i = f_i^{-i}(O_i) - P^* f_i^{-i}, \qquad \xi_i^{-j} = f_i^{-ij}(O_i) - P^* f_i^{-ij}, \qquad d_i = X_i - \xi_i,$$
$$r_N^2 = N^{-1}\sum_i \mathbb{E}(\xi_i^2 \mid \mathcal{C}), \qquad \alpha_{ij}^2 = \mathbb{E}\{(\xi_i - \xi_i^{-j})^2 \mid \mathcal{C}\}, \qquad q_N = N^{-1}\sum_{i\neq j} \alpha_{ij}\alpha_{ji}.$$

Then

$$\mathbb{E}\left[\left\{N^{-1}\sum_i \xi_i\right\}^2 \middle| \mathcal{C}\right] \le \frac{r_N^2 + q_N}{N}, \qquad R = N^{-1}\sum_i \xi_i + N^{-1}\sum_i d_i. \tag{95}$$

In particular $r_N^2 + q_N = o_p(1)$ *and* $N^{-1}\sum_i d_i = o_p(N^{-1/2})$ *suffice for* $R = o_p(N^{-1/2})$. *A conservative sufficient condition for the last requirement is*

$$N^{-1}\sum_i \{\mathbb{E}(d_i^2 \mid \mathcal{C})\}^{1/2} = o_p(N^{-1/2}).$$

Proof. Measurability and the fresh-unit definition give $\mathbb{E}(\xi_i \mid \mathcal{F}_{-i}) = 0$ and $\mathbb{E}(\xi_i^{-j} \mid \mathcal{F}_{-i}) = 0$. Also ξ_j^{-i} is $\mathcal{F}_{-i}$-measurable, so $\mathbb{E}(\xi_i \xi_j^{-i} \mid \mathcal{C}) = 0$. The analogous cross product vanishes with i, j reversed. The two double-deletion scores are conditionally independent and centered given $\mathcal{F}_{-\{i,j\}}$, hence $\mathbb{E}(\xi_i^{-j}\xi_j^{-i} \mid \mathcal{C}) = 0$. It follows exactly that

$$\mathbb{E}(\xi_i \xi_j \mid \mathcal{C}) = \mathbb{E}\{(\xi_i - \xi_i^{-j})(\xi_j - \xi_j^{-i}) \mid \mathcal{C}\}.$$

Expand the square and use conditional Cauchy–Schwarz to obtain the bound. The decomposition is algebraic. Conditional Minkowski bounds the L_2 norm of the average defect by the displayed average of norms; conditional Chebyshev proves both sufficient assertions. □

A substantially weaker uniform-stability corollary. Suppose, for a deterministic bound β_N, all actual-to-one-deletion and one-to-two-deletion complete *score-error maps* obey

$$\sup_o |f_i(o) - f_i^{-i}(o)| \le \beta_N, \qquad \sup_o |f_i^{-i}(o) - f_i^{-ij}(o)| \le \beta_N \quad \text{almost surely.}$$

Then $|d_i| \le 2\beta_N$, $\alpha_{ij} \le 2\beta_N$, and $q_N \le 4(N-1)\beta_N^2$. Consequently, for $t > 0$,

$$\Pr\left\{|R| > 2\beta_N + t\sqrt{\frac{r_N^2 + 4(N-1)\beta_N^2}{N}} \;\middle|\; \mathcal{C}\right\} \le t^{-2}.$$

In particular $r_N = o_p(1)$ and $\beta_N = o(N^{-1/2})$ suffice. The hypothetical stability rate $\beta_N = O(N^{-1})$ is adequate here. This avoids the accumulation of $O(N)$ individual deletions in the whole-fold triangle bound, which required the much stronger $o(N^{-3/2})$ rate. The new condition still concerns complete score maps, includes the own-deletion defect, and applies to one- *and* two-deletion data sets. A propensity-parameter bound alone does not establish it. Unbounded outcomes may make this uniform version inappropriate; the L_2 and average-pair version of Proposition 32 does not require uniform boundedness.

Why both checks are necessary. For independent Rademacher records O_i and $a_N \to 0$, the map $f_i(o) = a_N O_i o$ has fresh-unit L_2 norm a_N, but $X_i = a_N$ and $R = a_N$. Choosing $a_N = N^{-1/4}$ invalidates the desired root-N remainder despite consistency. Conversely the own-excluded map $f_i^{-i}(o) = a_N o \prod_{j \neq i} O_j$ gives $\xi_i = a_N \prod_j O_j$ for every i. Every score is conditionally centered, but their average has variance a_N^2, not a_N^2/N. Deleting another record and replacing it by zero yields $\alpha_{ij} = a_N$, and the pair bound is exact. These examples are logical counterexamples to omitted conditions, not models of the fitted GAN or evidence about its observed error size.

Own-history bias is different. An honest evaluation map may still use the scored unit's pre history. Its loading and residual summaries need not be independent of its post residual. For frozen maps, that issue is the population term $h = \mathbb{E}^*(v\varepsilon)$ in (46), bounded by $Cr_e\eta_H$ in Proposition 24. It is not $\bar{b}$, which concerns using a unit to fit the map against which that same unit is evaluated. The fresh-unit integrations already retain allowed own history. Independent primitive post innovations in the shared DGP can make the full-state history term zero; they do not force zero dependence after a scored control's post outcome has travelled through other folds' fits. For the fixed-M interpretation, augment the record by independent unit-specific evaluation marks and use its stated reference score; coupling those marks across deletions is a proof device, not an alteration of the released draw count.

An honest outer-fold comparison, not the released algorithm. A fully honest alternative can be defined precisely. For scoring fold I_k, perform *all* rank selection, factor estimation, feature alignment, propensity training, training-weight construction, calibration, and checkpoint selection using only I_k^c. Any inner folds must also be wholly inside I_k^c. Apply the resulting frozen history-to-state maps separately to each scored unit's admitted own history; do not rebuild training features or training weights using scored-unit records. Use the scored units only to evaluate their final nuisance predictions and the final ATT sample ratios. Different outer folds may select different ranks; they require compatible prediction targets, not a rank selected from a pooled outer-validation table.

For outcome-independent folds with fixed K and n_k/N bounded away from zero, suppose the resulting fold error map satisfies $P^* f_k^2 = o_p(1)$. Conditional on I_k^c and independent algorithmic randomness, the scored records are independent draws from the common law, so the centered fold average has mean zero and conditional variance at most $P^* f_k^2/n_k$. Conditional Chebyshev and a finite sum show $R = o_p(N^{-1/2})$ without independence between trained folds. This is a proof of the empirical-remainder statement for the alternative algorithm, not of nuisance convergence, history sufficiency, optimization rates, or weight-regime validity. Combining it with the other independently verified conditions of Theorem 5 would give that theorem's conclusion for the alternative. Treatment-stratified allocation instead requires stratum-wise integration and control of the centering difference; the common-law argument cannot simply be reused unchanged.

What is and is not closed. The exact bias/covariance identities, deletion bound, and honest-algorithm comparison are proved above. No required $\bar{b}$, q_N, or complete-map deletion rate has been proved for the selected-rank, boosting, Adam, and checkpoint pipeline. A finite perturbation can falsify exact exclusion, but cannot establish an asymptotic stability rate. Even a proof resolving the rank-selection branch would leave the fixed-rank stitched-feature branch. Therefore these results sharpen the missing proof obligations; they do not certify

the released ATT confidence interval or its current score-only variance estimate.

Source correspondence and execution version. The audit record `audit/inference_closure_20261018/dependence_mapping.json` contains exact source hashes and line ranges. The five core source files traced here are byte-identical between canonical commit `ca795ff3a6ec077b0204dd91d52b43dde13f0592` and the archived execution snapshot identified by commit `7f00c009cf32d0a335474e3896052a54c5ea2c88`. The released driver assigns folds by original, pre-sorting unit IDs and a prescribed random stream, not outcomes or treatment labels. It does not use the treatment-stratified default in `R/wgan_training.py`. The common-law argument assumes fixed row IDs and fold randomization exogenous to the unit array; an offset seed alone does not prove this. Under that interpretation no stratified-centering adjustment is needed, while the fitted-map dependence remains.

Appendix A.23 Outcome information, propensity information, and history error

The outcome generator and propensity learner need not use the same compression of the pre-treatment record. This section isolates the part of that difference that enters the normalized ATT score. The population identities concern frozen maps evaluated on an independent fresh unit, conditional on the common environment in Section Appendix A.2. They apply directly to a fully external training construction. Transferring them to the released stitched implementation also requires its complete-map empirical conditions in Sections Appendix A.19 and Appendix A.22; freezing a fitted map does not establish those conditions for its in-sample evaluation. For an externally trained map, the fresh-unit expectations also condition on that training realization, including the fitted network parameters.

Appendix A.23.1 A directional history remainder

Let $Y(0)$ denote one post outcome or a fixed weighted post-period average, D denote unit treatment, and $p = \mathbb{P}(D = 1)$. Let $\mathcal{V}$ be the information determining the fitted integrated outcome mean, and let $\mathcal{J}$ contain both $\mathcal{V}$ and the admitted features used by the propensity learner. Throughout this subsection suppose

$$\mathbb{E}\{D \mid \mathcal{J}, Y(0)\} = e, \qquad e \text{ is } \mathcal{V}\text{-measurable}, \qquad 0 < e < 1. \tag{96}$$

Define

$$m_{\mathcal{V}} = \mathbb{E}\{Y(0) \mid \mathcal{V}\}, \qquad m_{\mathcal{J}} = \mathbb{E}\{Y(0) \mid \mathcal{J}\}, \qquad \delta_H = m_{\mathcal{J}} - m_{\mathcal{V}}.$$

The fitted mean $\widehat{m}$ is $\mathcal{V}$-measurable. A nonnegative weight a may be any $\mathcal{J}$-measurable function, including a fixed population stabilization. Use the functional $\Theta(\widehat{m}, a)$ of Section Appendix A.20 and put

$$d_m = \widehat{m} - m_{\mathcal{V}}, \qquad v_a = (1-e)a - e, \qquad b_a = \mathbb{E}v_a, \qquad q_a = p + b_a,$$
$$\bar{v}_a = \mathbb{E}(v_a \mid \mathcal{V}), \qquad v_{a\perp} = v_a - \bar{v}_a.$$

All norms and expectations below use this fresh-unit law, and the displayed variables are square-integrable whenever a norm bound is invoked.

Proposition 33 (Directional decomposition of stabilized ATT drift)**.** *If* (96) *holds and* $q_a > 0$*, then*

$$\Theta(\widehat{m}, a) - ATT = \frac{\mathbb{E}\{(\bar{v}_a - eb_a/p)d_m\} - \mathbb{E}(v_{a\perp}\delta_H)}{q_a}. \tag{97}$$

Consequently, with $r_{\|,a} = \|\bar{v}_a - eb_a/p\|_2$*,* $r_{\perp,a} = \|v_{a\perp}\|_2$*, and* $\eta_H = \|\delta_H\|_2$*,*

$$|\Theta(\widehat{m}, a) - ATT| \le q_a^{-1}\{r_{\|,a}\|d_m\|_2 + r_{\perp,a}\eta_H\}. \tag{98}$$

Proof. Write $\epsilon_V = Y(0) - m_{\mathcal{V}}$ and $u = \mathbb{E}(ed_m)$. Since e is $\mathcal{V}$-measurable, $\mathbb{E}(e\epsilon_V) = 0$. The

treated residual numerator is $p\,ATT - u$. The weighted control numerator is

$$\mathbb{E}\{(e + v_a)(\epsilon_V - d_m)\} = \mathbb{E}(v_a\epsilon_V) - u - \mathbb{E}(v_a d_m).$$

Subtracting the two normalized residual means gives $\{\mathbb{E}(v_a d_m) - \mathbb{E}(v_a\epsilon_V) - ub_a/p\}/q_a$. Conditional projection yields $\mathbb{E}(v_a d_m) = \mathbb{E}(\bar{v}_a d_m)$ and $\mathbb{E}(v_a\epsilon_V) = \mathbb{E}(v_a\delta_H)$. Moreover, $\mathbb{E}(\delta_H \mid \mathcal{V}) = 0$, so $\mathbb{E}(\bar{v}_a\delta_H) = 0$. These identities prove (97); Cauchy–Schwarz proves (98). □

For un-capped fitted odds $a = \widehat{e}/(1 - \widehat{e})$,

$$v_a = \frac{\widehat{e} - e}{1 - \widehat{e}}.$$

If fitted probabilities are bounded away from one, conditional-projection contraction gives $r_{\parallel,a} + r_{\perp,a} \le C\|\widehat{e} - e\|_2$. This recovers Proposition 24. Equation (98) is sharper: only the component of the weight error varying within $\mathcal{V}$ interacts with the extra history information. In particular $v_{a\perp} = 0$ if the weight is $\mathcal{V}$-measurable, even if $m_{\mathcal{J}} \neq m_{\mathcal{V}}$.

For finitely many fitted folds, let $\mathbb{E}_k^*$ integrate each frozen map over a fresh unit, and let $\pi_k = n_k/N$. With one global control normalization, set $b = \sum_k \pi_k \mathbb{E}_k^* v_{ak}$. The numerator of (97) becomes

$$\sum_k \pi_k \mathbb{E}_k^*\{(\bar{v}_{ak} - eb/p)d_{mk}\} - \sum_k \pi_k \mathbb{E}_k^*(v_{ak\perp}\delta_{Hk}). \tag{99}$$

The same global b appears in every term. This follows by applying the preceding numerator calculation to the mixture before taking the two ratios. For an externally trained construction, centered empirical errors are controlled one fold at a time; independence of all fitted folds is not required. For the stitched construction, the complete-map conditions remain necessary to connect this population calculation to the sample statistic.

Appendix A.23.2 The additive skip supplies a comparison mean

For the implemented generator, $\mathcal{V}$ must contain more than the direct inputs of its residual network. Conditional on the fitted weights, its integrated prediction at period t has the form

$$\begin{aligned} \widehat{m}_t &= s_t + \mathbb{E}_z \delta_\theta(X_t, K_t \widehat{\lambda}_{\text{pre}}, z), \\ s_t &= X_t' \widehat{\beta} + \widehat{f}_t' \widehat{\lambda}_{\text{pre}}, \qquad K_t = \operatorname{diag}(B \widehat{f}_t) A. \end{aligned} \tag{100}$$

where A and B are the learned loading and factor projections. Thus $\mathcal{V}$ contains X_t, the projected interaction, and the additive skip s_t. For a pooled outcome it contains these quantities at all target dates. This measurability statement does not assert that the residual network can represent every function of $\mathcal{V}$.

Proposition 34 (A projection bound for omitted history information)**.** *Suppose* $Y(0) = \mu_{\text{full}} + \epsilon$ *and* $\mathbb{E}(\epsilon \mid \mathcal{J}) = 0$*. For any square-integrable,* $\mathcal{V}$*-measurable comparison* q_V*,*

$$\|\mu_{\text{full}} - q_V\|_2^2 = \|\mu_{\text{full}} - m_{\mathcal{J}}\|_2^2 + \|m_{\mathcal{J}} - m_{\mathcal{V}}\|_2^2 + \|m_{\mathcal{V}} - q_V\|_2^2, \tag{101}$$

$$\eta_H \leq \|\mu_{\text{full}} - q_V\|_2. \tag{102}$$

Proof. The conditional means of $Y(0)$ equal those of μ_{full} in $\mathcal{J}$ and $\mathcal{V}$. Decompose $\mu_{\text{full}} - q_V$ into the three differences on the right of (101). The first is orthogonal to every square-integrable $\mathcal{J}$-measurable variable. The second has conditional mean zero given $\mathcal{V}$ and is therefore orthogonal to the third. Squaring and integrating proves the identity and the bound. □

In the shared simulation, write

$$c(Z, W) = 0.7Z + 0.5W + h(Z, W) - a(Z), \qquad \mu_{\text{full},t} = c(Z, W_t) + f_t' \Lambda,$$

where Λ denotes the anchor-scaled loading. The independent post innovations satisfy the proposition's centering condition when scored features use only the unit's pre-treatment

history and treatment-invariant covariates. The comparison

$$q_{V,t} = c(Z, W_t) + s_t - X_t'\widehat{\beta} = c(Z, W_t) + \widehat{f}_t'\widehat{\lambda}_{\text{pre}}$$

is $\mathcal{V}$-measurable, so

$$\eta_{Ht} \leq \|f_t'\Lambda - \widehat{f}_t'\widehat{\lambda}_{\text{pre}}\|_2. \tag{103}$$

The population function c defines a comparison, not an oracle input to the learner. The bound does not require K_t to recover the full loading vector. For target weights ω_t, the analogous pooled bound is

$$\eta_{H,\text{pool}} \leq \left\| \sum_t \omega_t (f_t'\Lambda - \widehat{f}_t'\widehat{\lambda}_{\text{pre}}) \right\|_2. \tag{104}$$

Let $a_{mN} = \|\widehat{m} - q_V\|_2$ and $r_{AN} = \|\mu_{\text{full}} - q_V\|_2$. The same projection gives $\|\widehat{m} - m_{\mathcal{V}}\|_2 \leq a_{mN} + r_{AN}$. For denominators bounded away from zero, (98) therefore yields

$$|\Theta(\widehat{m}, a) - ATT| \leq C\{r_{\|,a}(a_{mN} + r_{AN}) + r_{\perp,a} r_{AN}\}. \tag{105}$$

Thus state recovery can control this information-set difference without adding the propensity summaries to the outcome network. Learning the comparison mean and transferring between training and scoring state laws remain separate components; Section Appendix A.24 treats that transfer. A complete-map outcome-error bound already incorporating these terms should not be augmented by counting them again as separate biases.

Appendix A.23.3 What the shared design supplies

The current assignment is $e(Z)$, and Z is retained in X_t and in the pre-X means used by the propensity learner. For any externally constructed, treatment-label-free history compression retaining Z,

$$\mathbb{E}\{D \mid \mathcal{V}, Y(0)\} = e(Z).$$

Indeed, conditioning first on all admitted histories and $Y(0)$ gives $e(Z)$ by independence of the assignment uniform; conditioning again on the smaller information set preserves it. Consequently the proxy-conditional mean identifies ATT in this particular design even at fixed T_0. This does not extend the same conclusion to a design with residual latent confounding, nor equate proxy identification to latent-oracle inference.

The complete pre-treatment factor cycles make the history-error bound explicit. In ideal aligned coordinates, $F'_{\mathrm{pre}}F_{\mathrm{pre}} = T_0 I_2$ and $F'_{\mathrm{pre}}\mathbf{1} = 0$. Let b_0 be the population control-pre projection coefficient of $h(Z,W) - a(Z)$ on W, and set $\beta_0 = (0.7, 0.5 + b_0)'$. Then

$$\widetilde{\lambda}_{\mathrm{pre}} - \Lambda = T_0^{-1} F'_{\mathrm{pre}} \xi, \qquad \xi_s = h(Z, W_s) - a(Z) - b_0 W_s + \varepsilon_s.$$

Conditional on Z, the ξ_s are independent and centered, with variance $\sigma_\xi^2(Z) = \mathrm{Var}\{h(Z,W) - a(Z) - b_0 W \mid Z\} + 0.09$.

Proposition 35 (Oracle-coordinate contribution of pre-state uncertainty)**.** *For the shared design and deterministic target weights, let $\bar{f} = \sum_t \omega_t f_t$. Then*

$$\begin{aligned} \mathbb{E}\big[\{f_t'(\widetilde{\lambda}_{\mathrm{pre}} - \Lambda)\}^2\big] &= \frac{\|f_t\|^2}{T_0}\mathbb{E}\sigma_\xi^2(Z), \\ \mathbb{E}\big[\{\bar{f}'(\widetilde{\lambda}_{\mathrm{pre}} - \Lambda)\}^2\big] &= \frac{\|\bar{f}\|^2}{T_0}\mathbb{E}\sigma_\xi^2(Z). \end{aligned} \tag{106}$$

For the twelve equally weighted post dates in the shared design,

$$\bar{f} = \sqrt{2}\left(\frac{2+\sqrt{3}}{6}, -\frac{1}{6}\right)', \qquad \|\bar{f}\|^2 = \frac{4+2\sqrt{3}}{9} > 0. \tag{107}$$

Proof. Conditional on Z, the projected loading error has mean zero and covariance $\sigma_\xi^2(Z)(F'_{\mathrm{pre}}F_{\mathrm{pre}})^{-1}$. Multiplication by f_t or $\bar{f}$, followed by integration over Z, proves (106). The post path repeats phases $1, \ldots, 6$ twice. Summing their sine and cosine coordinates gives $\sum_{j=1}^{6} \sin(\pi j/6) = 2 + \sqrt{3}$ and $\sum_{j=1}^{6} \cos(\pi j/6) = -1$, proving (107). □

The pooled post factor does not cancel. Equations (103)–(106) give an $O(T_0^{-1/2})$ history bound in ideal coordinates, not a zero error at fixed T_0. For estimated coordinates of the correct dimension, with a full-column-rank pre-factor matrix, the exact aligned identities are

$$\begin{aligned}\widehat{\lambda}_{\text{pre}} - \Lambda &= B_{\text{pre}}^{\dagger}\{\xi - X_{\text{pre}}(\widehat{\beta} - \beta_0) - (B_{\text{pre}} - F_{\text{pre}})\Lambda\} + \nu, \\ \widehat{f}_t'\widehat{\lambda}_{\text{pre}} - f_t'\Lambda &= (\widehat{f}_t - f_t)'\Lambda + \widehat{f}_t'(\widehat{\lambda}_{\text{pre}} - \Lambda),\end{aligned} \tag{108}$$

where B_{pre} is the aligned estimated pre-factor matrix and ν records numerical projection error. Substitution proves the first identity; direct subtraction proves the second. Their rate consequences retain rank, alignment, pre-Gram conditioning, coefficient error, and pointwise post-factor error. An average pre-factor rate alone does not control a particular target post row.

For example, if $r_{\perp,a} = O_p(N^{-\alpha})$ and $r_{AN} = O_p(T_0^{-1/2})$ with other anchor errors negligible, then $r_{\perp,a} r_{AN} = o_p(N^{-1/2})$ follows from $T_0/N^{1-2\alpha} \to \infty$. These are sufficient joint-growth conditions, not rates established by a fixed training budget.

Appendix A.23.4 Clipping and capping preserve a directional bound

Treat $\widehat{e}$ as the clipped fitted propensity and write $\widehat{\rho} = \widehat{e}/(1-\widehat{e})$, $v = (\widehat{e}-e)/(1-\widehat{e})$. Clipping is nonexpansive toward a true e inside the clipping interval. For a fixed odds threshold γ, define $a_\gamma = \min(\widehat{\rho}, \gamma)$ and $v_\gamma = (1-e)a_\gamma - e$.

Proposition 36 (Capping contracts within-information weight variation)**.** *Under* (96)*, for every fixed* γ*,*

$$v_\gamma = \min\{v, (1-e)\gamma - e\}, \tag{109}$$

$$\|v_\gamma - \mathbb{E}(v_\gamma \mid \mathcal{V})\|_2 \le \|v - \mathbb{E}(v \mid \mathcal{V})\|_2. \tag{110}$$

If I *is an interval satisfying* $\inf I > \operatorname{ess\,sup} e/(1-e)$*, then* $|v_\gamma| \le |v|$ *for all* $\gamma \in I$*.*

Proof. Multiplying the minimum by $1 - e > 0$ and subtracting e gives (109). Conditional on $\mathcal{V}$, its threshold is fixed because e is $\mathcal{V}$-measurable. Let v' be an independent copy under that conditional law. For the 1-Lipschitz map $f(x) = \min\{x, (1-e)\gamma - e\}$,

$$\mathrm{Var}\{f(v) \mid \mathcal{V}\} = \tfrac{1}{2}\mathbb{E}[\{f(v) - f(v')\}^2 \mid \mathcal{V}] \leq \tfrac{1}{2}\mathbb{E}\{(v - v')^2 \mid \mathcal{V}\} = \mathrm{Var}(v \mid \mathcal{V}).$$

Integration proves (110). On I, the threshold in (109) is positive, so truncating at that threshold cannot increase $|v|$. □

The directional history term in (97) is therefore bounded by $\|v - \mathbb{E}(v \mid \mathcal{V})\|_2 \eta_H$ uniformly over fixed thresholds. This does not remove the parallel term: if the true odds are capped, v_γ can be a nonzero $\mathcal{V}$-measurable distortion even when $v = 0$.

The implemented threshold is instead the global random quantity $\widehat{\gamma} = \kappa N_0^{-1} \sum_{D_i=0} \widehat{\rho}_i$. Proposition 36 is a population bound; replacing γ by $\widehat{\gamma}$ in a sampling expansion also requires uniform control of the centered score errors and positive denominators. One sufficient route is Proposition 28. A weaker route must show $\mathbb{P}(\widehat{\gamma} \in I) \to 1$ and control the empirical remainders uniformly over $\gamma \in I$ at the required root-N rate. The displayed contraction supplies its population history bound but not that empirical premise. The frozen-map active-cap expansion in Proposition 27 concerns a different regime.

Appendix A.23.5 Fixed short panels and the remaining inference conditions

Mean correctness on $\mathcal{V}$ alone does not make the richer-history term negligible. As an analytic example, let $\mathcal{V}$ be constant, $H \in \{-1, 1\}$ have equal probabilities, $\mathcal{J} = \sigma(H)$, D be independent Bernoulli with probability $1/2$, and $Y(0) = H$. For $\widehat{m} = m_{\mathcal{V}} = 0$ and $\widehat{e} = 1/2 + \delta_N H$, with $|\delta_N| < 1/2$, normalized fitted odds give

$$\Theta(\widehat{m}, \widehat{\rho}) - ATT = -\frac{4\delta_N}{1 + 4\delta_N^2}. \tag{111}$$

Indeed, $\mathbb{E}v = 4\delta_N^2/(1-4\delta_N^2)$ and $\mathbb{E}(vH) = 2\delta_N/(1-4\delta_N^2)$; substitute these into (97). If $\delta_N = N^{-1/4}$, the propensity is consistent and the narrower conditional mean is exact, but the drift is first order in δ_N. This example establishes the need for a history condition; it is not a result for a fitted simulation panel.

In the shared design, the full-history posterior offers another view of the distinction. With known c and F_{pre}, put $R_{\text{pre}} = Y_{\text{pre}} - c(Z, W_{\text{pre}})$. The bounded-uniform loading prior and Gaussian innovations imply a posterior density proportional to

$$\mathbf{1}\{\lambda \text{ belongs to the loading rectangle}\} \exp\{-\|R_{\text{pre}} - F_{\text{pre}}\lambda\|^2/(2 \cdot 0.09)\}.$$

It depends on the history through $F'_{\text{pre}}R_{\text{pre}}$, whereas the implemented linear-residual projection also contains $F'_{\text{pre}}\{h(Z, W_{\text{pre}}) - a(Z) - b_0 W_{\text{pre}}\}$. No identity in the implementation makes the generator's compression a sufficient statistic for this posterior at fixed T_0.

Remark 4 (An additional propensity influence term)**.** *Suppose, as a separate smooth-model setting, that $e_\vartheta(\mathcal{J})$ is differentiable, $e_{\vartheta_0} = e$, and the outcome limit is $m_{\mathcal{V}}$. Differentiating* (97) *at ϑ_0 gives*

$$L = -\frac{1}{p}\mathbb{E}\left\{\frac{\partial_\vartheta e_{\vartheta_0}}{1-e}\delta_H\right\}. \tag{112}$$

To verify the formula, along direction a write $v = s\, a'\partial_\vartheta e_{\vartheta_0}/(1-e) + o(s)$ in L_2; the numerator at $\widehat{m} = m_{\mathcal{V}}$ is $-\mathbb{E}(v\delta_H)$, and its denominator tends to p. If each balanced-fold training estimator has a common asymptotically linear expansion with influence $\ell(O)$, averaging the external training fits counts each observation $K-1$ times and yields $N^{-1}\sum_i \ell(O_i) + o_p(N^{-1/2})$. With suitable smooth remainders, the final score then includes $L'\ell(O)$ in addition to the proxy-reference score. Such a fitting expansion has not been established for the selected-tree GBM and generated inputs; this remark does not supply its standard error.

For a joint-growth inference argument, the directional rates above can replace the conservative history bound when the remaining premises of Theorem 5 hold. These include a common limiting reference score, control of the complete fitted-map empirical remainders,

the appropriate stabilization regime, and valid variance estimation. Fixed evaluation draws require the marked-score conditions of Proposition 12; they are not removed by a state projection bound. A fixed finite-tree propensity ensemble also does not automatically become consistent for the continuously varying $e(Z)$: an asymptotic learning argument must specify its complexity or approximation target rather than infer convergence from additional simulation seeds.

Code correspondence. The versioned source and line map is

`audit/interface_integration_20261018/code_mapping.json`

In `R/wgan_training.py`, `Generator.forward` (lines 205–219) constructs the bilinear interaction, neural residual, and additive skip in (100); `_control_weights_from_ps` (lines 446–466) implements the stabilization in Section Appendix A.20. In the separately versioned external-fold construction, archived at

`audit/interface_integration_20261018/source/att_honest_outer.py`

`own_pre_features` (lines 120–139) forms the richer propensity inputs. The routines `evaluate_scoring_state` and `_prediction_from_frozen_fit` (lines 269–276 and 302–324) evaluate the frozen map on each scored unit's pre-history. This construction illustrates the external-training premise; it does not replace the stitched procedure underlying the reported tables. The shared-design formulas are in `simulations/att_table2_anchor_dgp.py`, and the pre/post phase construction is in `simulations/att_anchor_shift_dgp.py`, lines 122–129.

Appendix A.24 Training-state and target-law transfer

The ATT learner is trained on control loadings estimated from their full observed panels, whereas treated and held-out loadings use pre-treatment outcomes. This section quantifies

the resulting difference and separates it from exposure-law transfer. The projection identities hold for any fixed returned factor fit, including that used for the reported estimates. Fresh-unit conditional statements additionally require an external training map. Complete outer-fold isolation supplies that exclusion in the separately versioned implementation; it does not retroactively remove the return paths of the released stitched estimator in Section Appendix A.22. That estimator still requires the complete-map conditions stated there.

Appendix A.24.1 Full-panel and pre-treatment loading projections

Fix the returned factor matrix $B = (B_0', B_+')'$ and coefficient $\widehat{\beta}$, in a common coordinate system. The two blocks contain T_0 pre-treatment and T_+ post-treatment dates. Set

$$A_0 = B_0'B_0, \quad A_+ = B_+'B_+, \quad A = A_0 + A_+, \qquad q_j = Y_j - X_j\widehat{\beta}, \quad j \in \{0, +\}.$$

Assume A_0 is invertible. Define the pre and full least-squares loadings $\ell_0 = A_0^{-1}B_0'q_0$ and $\ell_F = A^{-1}(B_0'q_0 + B_+'q_+)$.

Proposition 37 (Exact loading update)**.** *For every realized panel and fixed returned fit,*

$$\ell_F - \ell_0 = A^{-1}B_+'(q_+ - B_+\ell_0). \tag{113}$$

If $\lambda_{\min}(A_0) \geq \kappa T_0$ *and* $\max_{t>T_0} \|b_t\| \leq L_B$*, then*

$$\|\ell_F - \ell_0\| \leq \frac{L_B\sqrt{T_+}}{\kappa T_0}\|q_+ - B_+\ell_0\|. \tag{114}$$

Consequently a fresh-unit conditional residual bound $\|q_+ - B_+\ell_0\|_2 = O_p(\sqrt{T_+})$*, together with* $L_B = O_p(1)$ *and a positive lower bound on* κ*, gives* $\|\ell_F - \ell_0\|_2 = O_p(T_+/T_0)$*. The analogous empirical RMS assertion follows by averaging the squared deterministic bound over training controls.*

Proof. Since $A\ell_0 = B_0'q_0 + A_+\ell_0$, subtraction from the normal equation for ℓ_F gives (113).

Also $A \succeq A_0$ implies $\|A^{-1}\|_{\mathrm{op}} \leq (\kappa T_0)^{-1}$, while $\|B_+\|_{\mathrm{op}} \leq \|B_+\|_F \leq L_B\sqrt{T_+}$. Apply these inequalities to the update and then take the stated norm. No independence between a training control and the factors fitted from it is used in the deterministic statement. The conditional fresh-unit assertion requires its own residual moment bound. □

The SVD loading in the iterated IFE fit equals the full least-squares projection at each returned iterate, since its factors satisfy $B'B = TI$. This fact does not require a global optimum. If numerical projections have errors ν_F, ν_0, add $\nu_F - \nu_0$ to (113); a stopping tolerance for the IFE objective does not itself bound these projection errors.

The post-block leverage matrix $K_+ = B_+A^{-1}B_+'$ gives the equivalent residual identity

$$q_+ - B_+\ell_F = (I - K_+)(q_+ - B_+\ell_0). \tag{115}$$

Its nonzero eigenvalues are $\lambda_j/(1+\lambda_j)$, where λ_j are the nonzero eigenvalues of $B_+A_0^{-1}B_+'$. This follows by writing $C = B_+A_0^{-1/2}$ and $K_+ = C(I + C'C)^{-1}C'$. Thus full-loading post residuals have no larger Euclidean norm than pre-loading post residuals for the fixed fit. This in-sample projection comparison is not a comparison of ATT accuracy.

The shared simulation. Use $\lambda = (1.5\lambda_1, 1.2\lambda_2)'$ for the scaled anchor loading in this subsection, and let b_0 be the population control-pre coefficient in the projection of $h(Z,W) - a(Z)$ on W. At the oracle anchor,

$$\beta = (.7, .5 + b_0)', \qquad Y_t(0) = X_t'\beta + f_t'\lambda + \xi_t, \qquad \xi_t = h(Z, W_t) - a(Z) - b_0W_t + \varepsilon_t.$$

Conditional on Z, the innovations ξ_t are independent across dates. Their pre mean is zero; denote their post mean by $d(Z)$ and their pre/post variances by $\sigma_0^2(Z), \sigma_+^2(Z)$. In the shift design,

$$d(z) = .15\{\mu_+(z) - \mu_-(z)\} - .3b_0, \quad \mu_\pm(z) = \mathbb{E}\{h(z,W) \mid W \sim U(I_\pm)\}, \tag{116}$$

where $I_+ = (0, 2)$ and $I_- = (-2, 0)$. The difference $\mu_+ - \mu_-$ is zero at $z = 0$ and positive for $z > 0$, because $h(z, w)$ is then strictly increasing in w. Hence subtracting a constant cannot make d identically zero. In the no-shift design $d = 0$ and $\sigma_+^2 = \sigma_0^2$.

The pre factors cover complete twelve-date sine/cosine cycles. The twelve post dates repeat phases one through six twice. They therefore satisfy

$$F_0'F_0 = T_0 I_2, \quad F_+'F_+ = 12 I_2, \quad F_0'\mathbf{1} = 0, \qquad s = F_+'\mathbf{1} = \begin{pmatrix} 2\sqrt{2}(2+\sqrt{3}) \\ -2\sqrt{2} \end{pmatrix}. \tag{117}$$

Proposition 38 (Current-design projection discrepancy)**.** *At the oracle factors and β, let $\Delta = \ell_F - \ell_0$. Then*

$$\mathbb{E}(\Delta \mid Z, \lambda) = \frac{s\, d(Z)}{T_0 + 12}, \tag{118}$$

$$\mathrm{Var}(\Delta \mid Z, \lambda) = \frac{12\sigma_+^2(Z) + (144/T_0)\sigma_0^2(Z)}{(T_0+12)^2} I_2. \tag{119}$$

In particular the post-average linear-skip discrepancy has conditional mean

$$\mathbb{E}\{(s/12)'\Delta \mid Z\} = \frac{64 + 32\sqrt{3}}{12(T_0+12)} d(Z). \tag{120}$$

Proof. Substitute $q_j = F_j\lambda + \xi_j$ into (113) and use (117):

$$\Delta = \frac{1}{T_0 + 12}\left(F_+'\xi_+ - \frac{12}{T_0}F_0'\xi_0\right).$$

Conditional block independence, the zero pre mean and post mean $d(Z)\mathbf{1}$ give (118). The two conditional covariance matrices are $\sigma_0^2 I$ and $\sigma_+^2 I$; premultiplying by the displayed factor matrices gives (119). Finally $\|s\|^2 = 64 + 32\sqrt{3}$, proving (120). The loading cancels from the discrepancy, so the proof allows its actual bounded-uniform distribution. □

The oracle discrepancy has mean and SD of order T_0^{-1} with twelve post dates. In contrast,

$$\mathrm{Var}(\ell_0 - \lambda \mid Z) = \sigma_0^2(Z) I_2 / T_0.$$

The T_0^{-1} coupling rate is not a T_0^{-1} latent-recovery rate. Nor is (120) the bias of the fitted DR estimator: it isolates how a shifted nonlinear component enters an oracle full-panel loading. For estimated factors and coefficients, (113) remains exact, but the oracle conditional covariance formula cannot be reused by conditioning on factors estimated from those same training controls.

Appendix A.24.2 Sensitivity of the fitted generator

For fixed fitted ATT parameters ϕ, write the implemented generator as

$$G_\phi(X, \ell, b, z) = X'\widehat{\beta} + b'\ell + \delta_\phi\{X, (P_u\ell) \odot (P_t b), z\}.$$

The residual network contains affine–LayerNorm–LeakyReLU blocks and an affine output. The following bound concerns evaluation of this same network, not training a second network on different coordinates.

Proposition 39 (Fixed-network coordinate sensitivity)**.** *Let W_l and γ_l denote a block's weight matrix and LayerNorm gain, let $\epsilon_l > 0$ be its normalization constant, and let W_o be the output weight. A Lipschitz bound for the fitted residual map is*

$$L_\delta = \|W_o\|_{\mathrm{op}} \prod_l \frac{\|W_l\|_{\mathrm{op}} \|\gamma_l\|_\infty}{\sqrt{\epsilon_l}}. \tag{121}$$

At fixed X, b, z,

$$|G_\phi(X, \ell, b, z) - G_\phi(X, \ell', b, z)| \le L_{\phi,b} \|\ell - \ell'\|, \quad L_{\phi,b} = \|b\| + L_\delta \|\,\mathrm{diag}(P_t b) P_u\|_{\mathrm{op}}. \tag{122}$$

The same bounds hold for integrated means and finite-draw means coupled with the same generator-noise draws.

Proof. Put $C_d = I - \mathbf{1}\mathbf{1}'/d$ and $r(x) = \{\|C_d x\|^2/d + \epsilon\}^{1/2}$. The derivative of $C_d x/r(x)$ is

$$C_d/r(x) - (C_d x)(C_d x)'/\{d r(x)^3\}.$$

Its eigenvalues are zero on the constant direction, $1/r(x)$ on centered directions orthogonal to $C_d x$, and $\epsilon/r(x)^3$ on its radial direction. The operator norm is at most $\epsilon^{-1/2}$. Multiplication by the diagonal gain gives $\|\gamma\|_\infty/\sqrt{\epsilon}$. LeakyReLU with slope .2 is one-Lipschitz; composition proves (121). Subtract the two skips and note that the interaction difference is $\mathrm{diag}(P_t b)P_u(\ell - \ell')$ to obtain (122). Integrating, or averaging the same bound over paired draws, proves the last assertion. □

This constant is finite for fixed finite parameters but may grow along a statistical sequence. The critic's spectral normalization does not bound the generator's weights. Gradient clipping bounds update gradients, not the resulting parameter norms. Thus an asymptotic use retains $L_{\delta,N}$. Paired finite-draw sensitivity also does not remove the separate Monte Carlo error of a fixed $M = 50$ evaluation average.

Let $\ell = \lambda + d_\ell$ and $b = f + d_f$ in consistently aligned coordinates. Evaluate the same learned residual at oracle coordinates and use the oracle linear skip to define G^o_ϕ. Direct subtraction gives

$$\begin{aligned} |G_\phi - G^o_\phi| \le{}& |X'(\widehat{\beta} - \beta)| + |f' d_\ell| + |d'_f \lambda| + |d'_f d_\ell| \\ &+ L_\delta \|P_u\|_{\mathrm{op}} \|P_t\|_{\mathrm{op}} \{\|d_\ell\|\|f\| + \|\lambda\|\|d_f\| + \|d_\ell\|\|d_f\|\}. \end{aligned} \tag{123}$$

The interaction terms follow by expanding $(P_u\ell)\odot(P_t b) - (P_u\lambda)\odot(P_t f)$ and using $\|a \odot b\| \le \|a\|\|b\|$. Taking an L_2 norm requires control of the displayed products, for example bounded conditional factor errors or suitable fourth moments. This is a same-fitted-map coupling; no

refitting-stability term has been removed.

Appendix A.24.3 Oracle support transfer in the shared design

Let Q be the law of (Z, W, λ, j) for a fresh control and a uniformly sampled pre date, with phase $j \in \{1, \dots, 12\}$. Let P_+ be the corresponding treated-post law. Independent common-law innovation or generator-noise marks may be included. These are laws of oracle inputs, not laws of estimated history-dependent states. Write $p = .35$, $e(z) = \operatorname{expit}(c + z)$, $\rho_{\max} = \sup_z e(z)/\{1 - e(z)\}$, and p_W for the post positive exposure mass, .65 or .50.

Proposition 40 (Oracle-input loss transfer)**.** *For every nonnegative measurable loss L on these inputs,*

$$\mathbb{E}_{P_+} L \le C(p_W)\mathbb{E}_Q L, \qquad C(p_W) = 2\max\{2p_W, 2(1-p_W)\}\frac{1-p}{p}\rho_{\max}. \tag{124}$$

For control-post versus control-pre inputs, the same inequality holds with constant $2\max\{2p_W, 2(1-p_W)\}$.

Proof. The treated/control density ratio of Z is $\{(1-p)/p\}e(Z)/\{1-e(Z)\}$. Relative to the pre exposure density $1/4$, the post density ratios are $2p_W$ on $(0, 2)$ and $2(1-p_W)$ on $(-2, 0)$. The phase ratio is two on phases one through six and zero otherwise. Independence of the primitive streams makes the joint ratio their product; loading and common-law mark densities cancel. Bound that ratio and integrate L. The control/control comparison omits the treatment-composition ratio. □

Here $\rho_{\max} \simeq 2.16579$, giving conservative constants of about 10.46 and 8.04 in the shift and no-shift cells. These are bounds on population loss transfer, not predictions of RMSE ratios. The larger constant also bounds either scoring arm and their mixture. Jensen's inequality transfers a random-post-date squared-error bound to the squared error of the post average. A single specified post date instead has phase ratio twelve, so it requires a different constant.

The nonlinear correction relative to the oracle anchor is $h(Z,W) - a(Z) - b_0 W$. It depends only on $X = (Z,W)$, a direct generator input. A comparison function therefore need not invert the learned bilinear interaction or reconstruct the propensity summaries. This representation observation does not bound approximation by the fixed network, its objective bias, optimization error, or checkpoint selection.

For the same learned integrated predictor evaluated at oracle inputs, put $m_0 = c(Z,W) + f'\lambda$, where $c(Z,W) = .7Z + .5W + h(Z,W) - a(Z)$, and define $a_N = \|m_\phi^o - m_0\|_{L_2(Q)}$. If s_N bounds (123) in target L_2, Proposition 40 and the triangle inequality give

$$\|\widehat{m} - m_0\|_{L_2(P_+)} \leq \sqrt{C(p_W)}\, a_N + s_N. \tag{125}$$

Use the corresponding arm-specific state bounds for the complete ATT score. On the left, P_+ denotes the fresh-unit joint law including the pre-history used to construct $\widehat{m}$; its oracle-input marginal is the law in Proposition 40. The density-ratio argument applies to that marginal, not to the generated histories. The source risk a_N is a fresh oracle-input risk; empirical training MSE on full-panel generated states is not a substitute for it. Section Appendix A.16 retains the objective, generalization, selection and optimization terms needed for that additional step.

Appendix A.24.4 Conditional-mean transport at a fixed pre horizon

For one fixed fitted network, coupled full and pre evaluations obey

$$\|G_\phi(S_0) - T\|_2 \leq \|G_\phi(S_F) - T\|_2 + L_\phi \|\ell_0 - \ell_F\|_2$$

for any common square-integrable reference T and common noise draws. If the references are instead their respective conditional means $m_F(S_F) = \mathbb{E}(Y \mid S_F)$ and $m_0^h(S_0) = \mathbb{E}(Y \mid S_0)$,

the inequality becomes

$$\|G_\phi(S_0) - m_0^h(S_0)\|_2 \le L_\phi\|\ell_0 - \ell_F\|_2 + \|G_\phi(S_F) - m_F(S_F)\|_2 + \|m_F(S_F) - m_0^h(S_0)\|_2. \quad (126)$$

The last term is not eliminated by closeness of the input coordinates. In particular, a full loading contains the target post innovation.

An analytical example establishes the distinction. Take known F, $q = F\lambda + \varepsilon$, independent $\varepsilon \sim N(0, \sigma^2 I)$, and an independent bounded-rectangle loading prior. Gaussian orthogonal projection gives

$$q = F\ell_F + r, \quad r = (I - P_F)\varepsilon, \qquad \mathbb{E}(q_t \mid \ell_F) = f_t'\ell_F.$$

Indeed r is independent of the projected Gaussian component and of λ, hence of ℓ_F, and has mean zero. For a post date, independence of the post innovation from pre history instead gives

$$\mathbb{E}(q_t \mid \ell_0) = f_t'\mathbb{E}(\lambda \mid \ell_0).$$

The posterior mean lies in the loading rectangle, whereas ℓ_0 has unbounded support in every nonzero direction when the pre Gram is invertible. Thus a perfectly learned full-state regression $f_t'\ell$ does not generally equal the pre-state regression at fixed T_0. This example concerns the projection operation, not a new simulation result or the probability limit of the fitted GAN. Correct weighting can still center an ATT score with an imperfect outcome predictor.

Implications for inference. The unchanged state convention admits a conditional joint-growth route. Proposition 37 controls its additional full/pre gap; (125) controls prediction under the target oracle-input law when its source-risk and coordinate-error premises hold. With a conventional product remainder, a separate state-convention term of size L_N/T_0 contributes at most order $r_e L_N/T_0$; its sufficient root-N condition is $\sqrt{N} r_e L_N/T_0 \to 0$. This

does not replace the generally larger anchor-recovery error, the richer-history term, or the factor, rank, optimization, finite-draw and stabilized-weight conditions. If a complete mean-error bound already includes these components, they must not be added a second time as independent biases. Fixed-T_0 conditional-mean sufficiency or a separate nuisance influence expansion would be different routes, not consequences of outer exclusion alone.

Code correspondence. The shared `gscgan/src/gsc_nuisance.py` returns the SVD full-control loadings in `fit_iterated_ife` and assigns full versus own-pre loadings in `fit_stage1_ife` (lines 319–332 of the audited source). In `R/wgan_training.py`, lines 571–604 construct the linear skip, generator state and critic loading features from that same loading array; lines 606–633 construct the pre calibration and selection rows. Lines 166–219 define the generator used in Proposition 39, and lines 906–928 implement its finite-draw post mean. The factor path in `simulations/att_anchor_shift_dgp.py`, lines 122–129, gives (117); the centered response and anchor coefficient are in `simulations/att_table2_anchor_dgp.py`.

The separately versioned external-fold wrapper, preserved at `audit/interface_integration_20261018/source/att_honest_outer.py`, computes own-pre propensity inputs at lines 109–136, retains full-control GAN inputs at lines 232–239, and evaluates scoring histories only after both checkpoints freeze at lines 405–409. Those locations refer to source commit `8bfff7584b2d2c1a1718f235b7ec784d646fd255`. File hashes and the mapping to the released shared core are recorded in `audit/interface_integration_20261018/code_mapping.json`. The deterministic core identities apply without this additional exclusion; using it in conditional inference concerns that separate implementation, not a replacement of the estimator underlying the reported tables.

Appendix A.25 Company bootstrap: targets, fitted maps, and conditional validity

The Geithner bootstrap resamples whole companies within connection-status and original outer-fold strata and refits both branches. It retains the original common-rank, stitched-state estimator, including its weight stabilization and training budgets. This section derives the statistical conditions under which that procedure supports inference. The derivations cover the implemented weight and distribution readouts; regularity of the complete fixed-budget learning algorithm remains an assumption.

Appendix A.25.1 Sampling experiment and estimands

Condition on the event calendar and admitted common environment $\mathcal{C}$. Let F_i denote the original outer-fold label, D_i connection status, $h = (D, F)$ a stratum, and $n_h/n \to \pi_h > 0$. The reference experiment samples independent company paths within strata; dependence across the ten post-event dates of a company is unrestricted. Write E_h for stratum expectation, $p_d = \sum_f \pi_{df}$, and E_d for the corresponding fixed-fold mixture. Take the proportions as fixed along the asymptotic sequence, or require their deviations from the specified proportions to be $o(n^{-1/2})$. The treated-fold weights are $\lambda_f = \pi_{1f}/p_1$, equal to $1/2$ in the application. With $\bar{Y}_i(d) = 10^{-1} \sum_{t \in \mathcal{T}_1} Y_{it}(d)$, the targets are

$$\theta = \sum_f \lambda_f E\{\bar{Y}(1) - \bar{Y}(0) \mid D = 1, F = f, \mathcal{C}\}, \tag{127}$$

$$F_d(y) = \sum_f \lambda_f E\left\{ \frac{1}{10} \sum_{t \in \mathcal{T}_1} \mathbf{1}\{Y_t(d) \le y\} \mid D = 1, F = f, \mathcal{C} \right\},$$

$$\Delta(\tau) = F_1^{-1}(\tau) - F_0^{-1}(\tau). \tag{128}$$

These are company-population targets under the eligibility rule, conditional on the fixed calendar. The QTE compares marginal daily-return quantiles, not quantiles of ten-day returns or individual treatment effects. The 340 treated company-days are not 340 independent

clusters.

This experiment does not automatically give inference for the fixed realized 34-company cohort. Let $\tau_i = \bar{Y}_i(1) - \bar{Y}_i(0)$, $\theta_f = E(\tau_i \mid D_i = 1, F_i = f, \mathcal{C})$, and $\theta_S = n_1^{-1} \sum_{D_i=1} \tau_i$. Under the proportional-stratum sequence,

$$\theta_S - \theta = \frac{1}{n} \sum_i \frac{D_i}{p_1} (\tau_i - \theta_{F_i}) + o_p(n^{-1/2}). \tag{129}$$

Consequently, a population-target influence ψ_P becomes $\psi_P - D(\tau - \theta_F)/p_1$ for error about θ_S. This follows by subtracting (129) from the estimator's expansion. Constant effects eliminate this ATT term; heterogeneous effects generally do not. Realized potential-outcome CDF fluctuations create an analogous distinction for QTE. Company resampling cannot resolve it by relabeling an interval.

Each duplicate inherits its original company's inner and outer roles and ATT evaluation-noise marks. The reference law must therefore treat these as company marks, or establish equivalence to the claimed conditioning scheme. Outer-stratum counts are fixed, but inner-role counts vary across resamples. For finite inner roles R, the variance difference between random and fixed role composition, for a common influence ψ, is $\sum_h \pi_h \operatorname{Var}_{R|h}\{E(\psi \mid h, R)\}$. The law of total variance proves this identity; equality follows if $E(\psi \mid h, R) = E_h \psi$. Neither this equivalence nor negligible remaining cross-company dependence follows merely from fitting factors.

Appendix A.25.2 The stabilized ATT map

Here Y and m denote company averages over the ten post dates. Put

$$R = Y - m, \qquad q = \frac{[e]_{.05}^{.95}}{1 - [e]_{.05}^{.95}}, \qquad M = E_0 q, \qquad a = \min(q, 5M),$$
$$A = E_0 a, \qquad b = \frac{E_0(aR)}{A}, \qquad \Theta = E_1 R - b.$$

The implemented normalize–cap–renormalize policy is exactly

$$w_i = \frac{\min(q_i, 5\bar{q}_0)}{\sum_{D_j=0} \min(q_j, 5\bar{q}_0)}, \qquad \bar{q}_0 = \frac{1}{n_0} \sum_{D_j=0} q_j. \tag{130}$$

Multiplying the capped normalized weights by $\sum_{D_j=0} q_j$ proves the identity. The threshold is estimated, not fixed. Moreover, $q \in [1/19, 19]$ and $A \geq 1/19$, so the denominator is bounded away from zero. This bound is not an identification argument.

Proposition 41 (Derivatives and causal centering)**.** *Suppose the relevant integrals exist and clip/cap boundary perturbations admit the remainder control below. For frozen m, e, the direct sampling influence before within-stratum centering is*

$$\psi_{\text{dir}} = \frac{D}{p_1}(R - E_1 R) - \frac{1 - D}{p_0 A}\{a(R - b) + 5H(q - M)\},$$
$$H = E_0[\mathbf{1}\{q > 5M\}(R - b)]. \tag{131}$$

The derivatives in mean direction v_m and clipped-odds direction v_q are

$$D_m \Theta[v_m] = -E_1 v_m + \frac{E_0(a v_m)}{A}, \tag{132}$$
$$D_q \Theta[v_q] = -\frac{1}{A} E_0 \left[(R - b)\{\mathbf{1}(q < 5M) v_q + 5\mathbf{1}(q > 5M) E_0 v_q\}\right]. \tag{133}$$

Away from clipping boundaries, $v_q = \mathbf{1}\{.05 < e < .95\} v_e/(1 - e)^2$. If m_0 is the untreated conditional mean on the joint information used by the outcome and weighting maps, with conditional-mean comparability between treatment arms on that information, then

$$\Theta - \theta = E_1(m_0 - m) - \frac{E_0\{a(m_0 - m)\}}{A}. \tag{134}$$

Proof. Differentiate $M = E_0 q$ and $a = \min(q, 5M)$. Away from the cap boundary, $da = \mathbf{1}(q < 5M) dq + 5\mathbf{1}(q > 5M) dM$. The ratio derivative of b is $A^{-1} dE_0\{a(R - b)\}$, with b held at its base value on the right. The distribution derivative of M supplies $5H(q - M)/A$,

yielding (131). Substitution of $dR = -v_m$ and $dq = v_q$ gives the two map derivatives. For (134), decompose untreated outcomes as $m_0 + \varepsilon$, whose conditional mean is zero on the joint information, and insert $Y(1) = Y(0) + \tau$ in the treated expectation. Iterated expectation eliminates both residual-error means. □

For explicit remainder control, set $\delta_e = \|\widehat{e} - e\|_\infty$ and $\delta_m = \|\widehat{m} - m\|_{L^2(P_0)}$. Assume the clip and cap boundaries have residual-weighted margins $E_0[(1 + |R|)\mathbf{1}\{\text{dist} \le u\}] \le Cu^{\alpha_e}$ and Cu^{α_a}, respectively, with positive exponents. Use bounded derivative envelopes and finite moments sufficient for the displayed integrals. The population nonlinear remainder is

$$O\{\delta_e^2 + \delta_e^{1+\alpha_e} + \delta_e^{1+\alpha_a} + \delta_e\delta_m\}. \tag{135}$$

To see this, a branch switch requires distance $O(\delta_e)$ from a boundary. The pointwise minimum/clip remainder is $O(\delta_e)$ there; integrating the margin gives the two fractional-power terms. Away from boundaries, bounded odds derivatives give $O(\delta_e^2)$. The mean/weight cross term is bounded by Cauchy–Schwarz. Ratio denominators stay positive. For fixed weights the map is linear in m, so no δ_m^2 term is needed. The bound must be $o(n^{-1/2})$ in both sampling worlds, together with control of empirical cross terms. No-boundary-mass alone gives no rate, and clipping can create atoms.

Equation (134) distinguishes causal centering from orthogonality: $m = m_0$ centers ATT, but (132) need not vanish. Correct raw propensity alone does not restore exact target balance after binding clipping/capping. Thus an ordinary product-of- nuisance-errors argument cannot simply be imposed on the stabilized map.

Appendix A.25.3 A conditional full-refit ATT result

Let W_i^* be independent stratum-wise multinomial bootstrap counts. The conditions below concern target-relevant projections of fitted maps, not root-n estimation of an entire neural function.

Theorem 7 (Projected fitted-map expansion)**.** *Suppose the target is* (127)*, the reference experiment above holds, and the following conditions hold in original and bootstrap samples: (i) causal centering error is* $o(n^{-1/2})$*; (ii) the Taylor remainder in* (135)*, empirical cross terms, selection errors, and generated-state remainders are negligible at that scale; and (iii) the fitted-map contribution*

$$D_m\Theta[\widehat{m} - m] + D_q\Theta[\widehat{q} - q] = n^{-1}\sum_i \xi_i + o_p(n^{-1/2})$$

has a centered bootstrap expansion with the same ξ_i *and counts* $W_i^* - 1$*, where* $E_h\xi = 0$*. This expansion includes every state, history, training, and finite evaluation-noise path, or proves it negligible. Assume stratum laws of large numbers and Lindeberg conditions for* $\psi = \psi_{\text{dir}} + \xi$*, with* $0 < V = \sum_h \pi_h \operatorname{Var}_h(\psi) < \infty$*. Then*

$$\sqrt{n}(\widehat{\theta} - \theta) = n^{-1/2}\sum_i (\psi_i - E_{h_i}\psi) + o_p(1), \tag{136}$$

$$\sqrt{n}(\widehat{\theta}^* - \widehat{\theta}) = n^{-1/2}\sum_i (W_i^* - 1)(\psi_i - \bar{\psi}_{h_i}) + o_{P^*}(1). \tag{137}$$

Both leading limits are $N(0, V)$*, the latter conditionally in probability. Conditional second-moment convergence, for example from an* L^2 *remainder bound and uniform integrability, also gives* $n \operatorname{Var}^*(\widehat{\theta}^*) \to_p V$*. With a growing number of replicates and the corresponding conditional Monte Carlo law of large numbers, the bootstrap-standard-error normal interval has limiting 95% coverage.*

Proof. Insert the projected fitted-map expansion into the Taylor expansion of Θ and add (131). Assumptions (i)–(ii) bound the remaining terms and give (136)–(137). Since $\sum_{i\in h}(W_i^* - 1) = 0$, each bootstrap score is centered within its stratum. Multinomial covariances give leading conditional variance $n^{-1}\sum_h \sum_{i\in h}(\psi_i - \bar{\psi}_h)^2 \to_p V$. The stratum and conditional multinomial central limit theorems give the matching Gaussian laws. Second-moment control transfers the variance; a conditional law of large numbers over

replicates and Slutsky's theorem then justify the standard-error interval. □

Assumption (iii) is the remaining learning interface, rather than a consequence of full refitting. The proposition supplies the exact directions whose expansions are needed; it does not establish those expansions for the implemented fixed-budget GAN.

Appendix A.25.4 Selection and shared states

For a finite rank grid, either of two conditions transfers a valid fixed-branch expansion to a selected estimator. First, a unique limiting risk minimizer may have a gap dominating the uniform selection-risk errors in both the original and bootstrap samples. The probability of any other branch then vanishes in both worlds. Second, all potentially selected near-minimizers may have the same target and influence, with uniformly negligible remainders in both worlds. In the first case, selected and fixed-branch estimators coincide outside an event of vanishing probability. In the second, bound the selected remainder by the largest admitted branch remainder. These arguments prove weak-limit transfer. Variance transfer additionally needs second-moment control on exceptional branches. Similar validation losses or separate normal limits do not imply a common influence. For early stopping, the branch includes stopping/patience decisions and the visited checkpoints, not just a final minimization.

Shared-state dependence can be retained if the entire fitted map satisfies the required expansion. Alternatively, a valid surrogate $\widetilde{\theta}$ transfers its law when

$$\begin{aligned} \sqrt{n}(\widehat{\theta} - \widetilde{\theta}) &= o_p(1), \\ \sqrt{n}\{(\widehat{\theta}^* - \widehat{\theta}) - (\widetilde{\theta}^* - \widetilde{\theta})\} &= o_{P^*}(1), \end{aligned} \tag{138}$$

with a second-moment counterpart for variance consistency. Subtraction and Slutsky prove this assertion. Leave-group-out stability alone does not establish its multinomial-duplication counterpart. Fixed random seeds and finitely many smooth updates also do not imply

convergence of optimization error or of the target-projected learning term. None of these selection or shared-state conditions is established by the 199 application fits.

Appendix A.25.5 The actual QTE readout

The application uses 19 fitted conditional outcome knots, not final Monte Carlo outcome draws. Their levels are

$$.005, .01, .025, .05, .10, .15, .20, .25, .30, .40,$$

$$.50, .60, .70, .80, .90, .95, .975, .99, .995,$$

which differ from the 19 reported QTE levels. For a company-date with ordered knots $q_1, \ldots, q_K$, write

$$H(y;q) = \alpha_1 \mathbf{1}(q_1 \le y) + \sum_{k<K} (\alpha_{k+1} - \alpha_k) \left[\frac{y - q_k}{q_{k+1} - q_k} \right]_0^1 + (1 - \alpha_K)\mathbf{1}(q_K \le y). \tag{139}$$

This right-continuous convention agrees with the implemented interpolation except at the last knot, where the implementation returns α_K. Ties or equality with a numerical-grid point must be negligible or handled explicitly when relating this probability CDF to the actual readout. The mixture averages equally over treated companies and post dates; training propensity weights are not reapplied to this pooling.

Within $q_k < y < q_{k+1}$, put $r_k = (y - q_k)/(q_{k+1} - q_k)$ and $s_k = (\alpha_{k+1} - \alpha_k)/(q_{k+1} - q_k)$. Direct differentiation gives

$$\dot{H} = s_k\{\dot{y} - (1 - r_k)\dot{q}_k - r_k\dot{q}_{k+1}\}. \tag{140}$$

Population differentiation also includes endpoint crossings. Let E_T denote the treated-company and equal-date mixture. The following strong conditions suffice for each endpoint $Q = q_1, q_K$ and its direction $V = \dot{Q}$. Along the parameter path assume $Q_\epsilon = Q + \epsilon V + \rho_\epsilon$, $|V| \le M < \infty$, and $\|\rho_\epsilon\|_\infty = o(|\epsilon|)$. In a neighborhood of y, let the joint law of (Q, V)

have density $k(q, v)$ with respect to $dq\,\nu(dv)$ for a fixed dominating measure ν. Require $k(q, v)$ to be continuous in q at y for ν-almost every v, and bounded by $g(v)$ there, where $\int g(v)\nu(dv) < \infty$. This permits point masses in V. For locally uniform expansions, require uniform continuity and the same envelope and remainder bounds over the target neighborhood. Assume also that internal knot locations have no atoms there and that spacing and derivative envelopes permit differentiation of the clipped ramps under the expectation. These conditions concern the joint endpoint-direction law; continuity of only f_Q and $E(V \mid Q)$ is insufficient. They imply, at fixed y,

$$\begin{aligned} D_\eta F_{0,\eta}(y)[\dot\eta] = & -\alpha_1 f_{q_1}(y) E_T(\dot q_1 \mid q_1 = y) - (1-\alpha_K) f_{q_K}(y) E_T(\dot q_K \mid q_K = y) \\ & - \sum_{k<K} (\alpha_{k+1} - \alpha_k) E_T\left[\frac{\mathbf{1}(q_k < y < q_{k+1})}{q_{k+1} - q_k} \{(1 - r_k)\dot q_k + r_k \dot q_{k+1}\} \right]. \end{aligned} \tag{141}$$

For the endpoint term, bounded V makes every crossing local for small ϵ. Hence

$$\begin{aligned} \frac{P(Q + \epsilon V \le y) - P(Q \le y)}{\epsilon} &= \int \frac{1}{\epsilon} \int_y^{y - \epsilon v} k(q, v)\, dq\, \nu(dv) \\ &= -\int v \int_0^1 k(y - t\epsilon v, v)\, dt\, \nu(dv) \;\longrightarrow\; -\int v k(y, v) \nu(dv). \end{aligned}$$

Dominated convergence uses $|v|g(v)$, which is integrable. The last integral is $-f_Q(y)E(V \mid Q = y)$, interpreted as the joint-density integral if $f_Q(y) = 0$. The same envelope bounds the probability that $Q + \epsilon V$ is within $\|\rho_\epsilon\|_\infty$ of y by $2\|\rho_\epsilon\|_\infty \int g\, d\nu = o(|\epsilon|)$, so the path remainder does not alter this derivative. Apply the result to both endpoint indicators. Dominated convergence for the continuous clipped ramps gives the remaining terms in (141). Differentiation in y similarly gives

$$f_0(y) = \alpha_1 f_{q_1}(y) + (1 - \alpha_K) f_{q_K}(y) + \sum_{k<K} (\alpha_{k+1} - \alpha_k) E_T \frac{\mathbf{1}(q_k < y < q_{k+1})}{q_{k+1} - q_k}. \tag{142}$$

Thus averaging derivatives that freeze all endpoint indicators omits population flux. Central

marginal quantiles do not automatically avoid conditional endpoint crossings.

Every fitted path enters these knots. In the implemented parameterization,

$$q_k = \mu + b_1 + a + \sum_{j<k} d_j \exp(\ell_j), \quad d_j = \max(b_{j+1} - b_j, .001),$$
$$a = 1.25 \tanh(v_1), \quad \ell_j = .55 \tanh(v_{j+1}).$$

Away from the spacing-floor boundary,

$$\dot{q}_k = \dot{\mu} + \dot{b}_1 + \dot{a} + \sum_{j<k} e^{\ell_j}(\dot{d}_j + d_j \dot{\ell}_j), \qquad \dot{d}_j = \mathbf{1}(b_{j+1} - b_j > .001)(\dot{b}_{j+1} - \dot{b}_j).$$

State, network, and standardization paths enter μ, v; residuals and weights enter b. The base knots use weighted midpoint-interpolated order statistics, with positions $c_l = (\sum_{r<l} w_r + w_l/2)/\sum_r w_r$. Their population expansion needs positive residual density, local spacing and maximum-weight control, and treatment of duplicate-company ties. Fixed-branch differentiability is not by itself that expansion. There is no DR-CDF term to cancel (141).

The pooled counterfactual CDF is evaluated at 401 data-dependent grid points g_l. If $A_j = \widehat{F}_0(g_j) < \tau < A_{j+1}$, put $v = (\tau - A_j)/(A_{j+1} - A_j)$. The returned quantile and its derivative are

$$\begin{aligned} Q_{0,g} &= (1-v)g_j + v g_{j+1}, \\ \dot{Q}_{0,g} &= (1-v)\dot{g}_j + v\dot{g}_{j+1} - \frac{g_{j+1} - g_j}{A_{j+1} - A_j}\{(1-v)\dot{A}_j + v\dot{A}_{j+1}\}. \end{aligned} \tag{143}$$

This follows by differentiating the interpolant. The grid endpoints are $L = \min(c) - .75s$, $U = \max(c) + .75s$, where $s = \max\{\mathrm{sd}(c), .25\}$ and c collects observed post outcomes and predicted knots. One must either establish a joint expansion for the needed grid quantities, including these endpoints, or show that actual-grid versus exact-mixture inversion errors, and their centered bootstrap differences, are $o(n^{-1/2})$. Fixed resolution and unique sample

extrema alone do not imply this: the range may grow with n. Exact inversion would not remove the separate 19-knot representation bias.

Appendix A.25.6 A conditional QTE bootstrap result

Theorem 8 (Joint distribution and quantile inference)**.** *For the target* (128)*, suppose the conditional law is causally identified and correctly centered at the inference scale, and the readout regularity conditions above hold. Around each requested marginal quantile, assume that the projected fitted-map term* (141) *has a jointly bootstrappable company-level linear expansion. Require locally uniform $o(n^{-1/2})$ empirical cross terms, Taylor, selection, state, representation, and numerical remainders in the original and centered bootstrap expansions. Suppose the observed company CDF and the counterfactual expansion jointly satisfy stratum functional central limit theorems, with local stochastic equicontinuity; the target CDFs have positive continuous densities at their quantiles. Write their complete influences as $\zeta_1(y), \zeta_0(y)$, including all fitted-map contributions and within-stratum centering. Then the QTE influence is*

$$\psi_\Delta(\tau) = \frac{\zeta_0(q_0(\tau))}{f_0(q_0(\tau))} - \frac{\zeta_1(q_1(\tau))}{f_1(q_1(\tau))}. \tag{144}$$

If its limiting variance is positive, original and bootstrap QTE errors have the same centered Gaussian limit. With a growing number of resamples, the pointwise percentile interval has limiting 95% coverage.

Proof. Writing $H_{\eta,i}(y) = 10^{-1} \sum_t H(y; q_{it}(\eta))$, expand

$$\widehat{F}_0 - F_0 = (P_{T,n} - P_T)H_{\eta_0} + D_\eta[P_T H_{\eta_0}](\widehat{\eta} - \eta_0) + \text{remainder}.$$

The assumptions and (141) turn this into a complete company-level linear expansion jointly with $\widehat{F}_1$. Set $\nu_i(y) = 10^{-1} \sum_t \mathbf{1}(Y_{it} \le y)$. On the full-$n$ scale, its stratum-centered influence is

$$\zeta_{1,i}(y) = \frac{D_i}{p_1} \left[\nu_i(y) - E\{\nu_i(y) \mid D_i = 1, F_i, \mathcal{C}\}\right].$$

The direct pooling contribution for $\widehat{F}_0$ has the same D_i/p_1 scaling and treated-fold centering, with $H_{\eta_0,i}(y)$ in place of $\nu_i(y)$; fitted-map contributions complete ζ_0. The same expansions with multinomial counts hold conditionally. Expand $F_d(\widehat{q}_d)$ at q_d, replace empirical fluctuations at $\widehat{q}_d$ by those at q_d using local equicontinuity, and absorb the stipulated numerical remainder. Solving gives $\widehat{q}_d - q_d = -(\widehat{F}_d - F_d)(q_d)/f_d(q_d) + o_p(n^{-1/2})$. Subtract the two quantile expansions to obtain (144) and repeat conditionally for the bootstrap. The joint Gaussian law retains the covariance of the observed and counterfactual CDFs from the same companies. Its centered symmetry and quantile continuity yield the pointwise percentile result. □

The assumptions isolate what is required from training rather than postulate unconditional validity of its output. Without the fitted-map expansion and negligible causal/model bias, bootstrap regularity around a fitted projection does not establish a causal QTE interval. Nor does pointwise validity establish a simultaneous band.

Appendix A.25.7 Finite Monte Carlo calibration and remaining scope

For the retained $B = 199$, exact 95% calibration is distinct from the growing-B conclusions. Assume the original standardized error and any fixed collection of bootstrap errors converge jointly to independent standard normals. A conditional bootstrap Gaussian limit that is nonrandom implies this joint convergence: conditional independence gives a product of characteristic functions, each converging to the same nonrandom Gaussian characteristic function.

For ATT, the sample variance of B limiting bootstrap normals is $\chi^2_{B-1}/(B-1)$, independent of the original normal. The retained $z_{.975}$ standard-error interval therefore has ideal fixed-B coverage

$$2F_{t_{198}}(1.9599639845) - 1 = 0.9485968608. \tag{145}$$

This argument uses joint weak convergence and continuity of sample standard deviation

at a positive limit; it does not need the stronger moment convergence used for variance consistency in Theorem 7.

For QTE, the implemented linear empirical percentile convention gives, for ordered limiting bootstrap errors $Z_{(1)}, \ldots, Z_{(199)}$,

$$L = .05Z_{(5)} + .95Z_{(6)}, \qquad U = .95Z_{(194)} + .05Z_{(195)}.$$

By centered symmetry, the negative original error has the same law as these independent bootstrap errors. Its rank among 200 draws is uniform. The interval $[Z_{(6)}, Z_{(194)}]$ is contained in $[L, U]$, which is contained in $[Z_{(5)}, Z_{(195)}]$. Hence the ideal fixed-B limiting coverage of the actual interpolated interval lies between $188/200 = .94$ and $190/200 = .95$. These are conditional calibration statements under the stated Gaussian limits, not estimated coverage rates for Geithner.

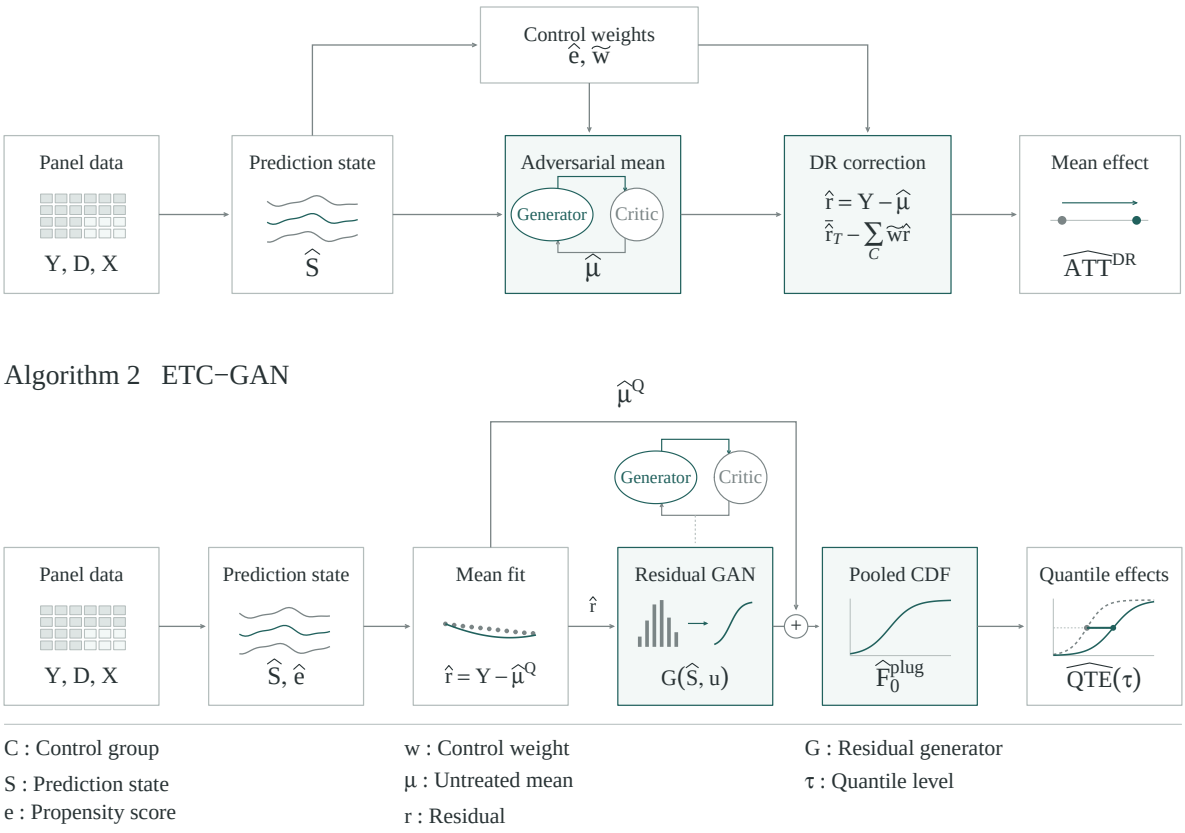
The proofs do not certify the fixed-budget learning expansions, rank and checkpoint regularity, or causal and numerical bias bounds for the fitted application. The 199 resamples assess variability and numerical stability, not repeated-sample coverage. Increasing their number reduces Monte Carlo error but does not repair those premises. The existing point estimates and resamples need not be replaced to state these conditional results; changing shared states, weight rules, rank selection, or the distribution model would define a different estimator and require new corresponding fits.

Code correspondence. The corrected resampling adapters are `att_run.py` and `qte_run.py` in the saved ATT and QTE source snapshots. Both import the common corrected daily-panel adapter `paired.py`. The shared `_global_weights` routine in `common_adjustment.py` implements (130); `wgan_training.py` implements the fixed-budget ATT training and checkpoint decisions. The saved-knot readout in `materialize_geithner_plugin_20261014.py` and the conditional-distribution routines in `run_etcgan_dr_qte_experiment.py` implement (139)–(143), including the endpoint convention noted above.

These correspondences identify the maps covered by the derivatives, not an empirical verification of the theorems' learning conditions.

Appendix B Method details

Appendix B.1 Visual workflow



DR-FGAN: Doubly robust factor-augmented GAN
ETC-GAN: Empirical-transport Cramér GAN
GAN: Generative adversarial network
DR: Doubly robust
ATT: Average treatment effect on the treated
QTE: Quantile treatment effect
CDF: Cumulative distribution function
Y, D, X: Outcome, treatment indicator, observed covariates
S: Observed and factor prediction state
e, w: Propensity score and stabilized control weight
μ, μ^Q: ATT and QTE untreated means
$r, \bar{r}_T$: Outcome residual and treated residual mean
T, C: Treated and control groups
G_θ, θ: Residual generator and its parameters
F_0^{plug}: Plug-in counterfactual CDF
τ: Quantile level

Figure 1: Training and aggregation for ATT and QTE. ATT combines an adversarial untreated-mean fit with residual correction; the QTE branch combines a separate regression mean and residual GAN, averages the conditional counterfactual CDFs over the treated units, and inverts the period-specific or pooled marginal CDF without targeting. Unit, time, and fold indices are suppressed. Arrows show information flow; loops show alternating training updates. Insets illustrate the fitted objects schematically.

We write G for the generator and D for the critic in each branch; the two branches fit separate networks. Parameter subscripts distinguish these networks in the formal expressions below. Fitted inputs are denoted by $\hat{S}_{it}$ and their population counterparts by S_{it}, as defined in Section Appendix A.1.

Appendix B.2 IFE/GSC Three-Step Estimation

Under the linear-anchor conditions, we adapt the IFE/GSC construction of Xu (2017) to whole-unit folds. Within each fold, training controls estimate the common factors. Treated units and held-out controls then use their own pre-treatment outcomes to estimate their loadings.

In the ATT benchmark, both the time-invariant baseline characteristic Z_i and the time-varying external covariate W_{it} are observed, so $X_{it} = (Z_i, W_{it})'$. Writing $\beta_0 = (\beta_Z, \beta_W)'$, the anchored representation is

$$\begin{aligned} Y_{it}(0) &= X_{it}'\beta_0 + \lambda_i' f_t + h_0(X_{it}, \lambda_i, f_t) + \varepsilon_{it} \\ &= \underbrace{\beta_Z Z_i + \beta_W W_{it}}_{\text{Linear covariate effects}} + \underbrace{\lambda_i' f_t}_{\text{Latent factors}} + \underbrace{h_0((Z_i, W_{it})', \lambda_i, f_t)}_{\text{Nonlinear response}} + \varepsilon_{it}. \end{aligned} \tag{146}$$

The factor fit estimates λ_i and f_t. The networks use these estimates and X_{it} to learn the nonlinear response. The linear anchor is fitted in three steps.

Step 1 (Control group factor analysis). For a given rank, estimate $\hat{\beta}$, the factor matrix $\hat{F} = (\hat{f}_1, \ldots, \hat{f}_T)' \in \mathbb{R}^{T \times r}$, and the control loadings $\hat{\Lambda}_{\text{co}} = (\hat{\lambda}_1, \ldots, \hat{\lambda}_{N_{\text{co}}})' \in \mathbb{R}^{N_{\text{co}} \times r}$ from control group data:

$$(\hat{\beta}, \hat{F}, \hat{\Lambda}_{\text{co}}) = \underset{\beta, F, \Lambda_{\text{co}}}{\arg\min} \sum_{i \in \mathcal{C}} \sum_{t=1}^{T} (Y_{it} - X_{it}'\beta - \lambda_i' f_t)^2, \tag{147}$$

subject to $\hat{F}'\hat{F}/T = I_r$ and the usual factor normalization. In the balanced-panel implementation, exact least squares for β and a rank-r truncated SVD of the residual panel alternate

until the objective-change tolerance is reached. Within an outer unit fold only training controls estimate β and the common factor path. Rank selection follows the treated-pre-period leave-one-time-out construction of Xu (2017): for every candidate r and every $s \leq T_0$, estimate each treated loading using the other $T_0 - 1$ periods, predict $Y_{is}(0)$, and average squared errors. The selected fit must converge. Continued improvement at the upper candidate boundary calls for an expanded-grid sensitivity check. Common factors use control outcomes over all T periods; treated post-treatment outcomes are excluded.

Step 2 (Non-training-unit pre-treatment projection). Each treated or scored-control unit's loading is recovered by least squares on its own pre-treatment window:

$$\hat{\lambda}_i = \arg\min_{\lambda} \sum_{s \leq T_0} (Y_{is} - X'_{is}\hat{\beta} - \lambda' \hat{f}_s)^2, \quad i \notin \mathcal{C}_{\text{train}}, \tag{148}$$

The projection excludes the unit's post-treatment outcomes.

Step 3 (Linear counterfactual imputation). The fitted model predicts untreated outcomes as

$$\hat{Y}_{it}^{\text{lfm}}(0) = X'_{it}\hat{\beta} + \hat{\lambda}'_i \hat{f}_t, \quad i \in \mathcal{T},\ t > T_0. \tag{149}$$

Steps 1–2 estimate $\hat{\eta} = (\hat{\beta}, \hat{F}, \hat{\Lambda}, \hat{\lambda}_i, r^*)$, and Step 3 turns these estimates into linear counterfactuals. We keep the factor estimates for adjustment and use them as inputs to the nonlinear mean and distribution learners.

Appendix B.3 Shared propensity scores and control weights

Each unit's treatment probability $\hat{e}_i$ is predicted by a classifier trained on other units, using estimated loadings and pre-treatment summaries of covariates, outcomes, and linear-model residuals. The classifier uses the combined fold-specific states at the selected rank. In the ATT benchmark, its inputs include Z_i and the pre-treatment mean of W, rather than contemporaneous post-treatment W. Both branches share the propensity predictions and

fold assignments. Before score clipping and weight capping, the normalized control odds are

$$\tilde{w}_i = \frac{\hat{e}_i/(1-\hat{e}_i)}{\sum_{j\in\mathcal{C}} \hat{e}_j/(1-\hat{e}_j)}, \qquad i \in \mathcal{C}. \tag{150}$$

The reported estimators apply the study-specific clipping and capping rules and normalize the resulting control weights to sum to one. The shared stabilization map and its inference conditions are given in Section Appendix A.20; each study records its thresholds. These stabilized weights are used in both outcome-learning branches and in the ATT residual correction.

Appendix B.4 WGAN-GP generator and training

Appendix B.4.1 Generator architecture

The generator adds a nonlinear correction to the fold-specific GSC surface:

$$\begin{aligned} \hat{Y}_{it}^{\text{gen}}(0) &= G_\phi(\hat{S}_{it}, \mathbf{z}_{it}) \\ &= X_{it}'\hat{\beta} + \hat{\lambda}_i'\hat{f}_t + \delta_\phi(\hat{S}_{it}, \mathbf{z}_{it}), \qquad \mathbf{z}_{it} \sim \mathcal{N}(0, I_{d_z}). \end{aligned} \tag{151}$$

Here $\hat{Y}_{it}^{\text{gsc}} = X_{it}'\hat{\beta} + \hat{\lambda}_i'\hat{f}_t$. The notation $\delta_\phi(\hat{S}_{it}, \mathbf{z}_{it})$ denotes the implemented correction $\delta_\phi(X_{it}, b_i^\lambda, b_t^f, \mathbf{z}_{it})$, with network parameters ϕ and the fixed buffers defined below. The generator takes the following inputs:

Input	Nature	Source	Role
X_{it}	Observable	Directly observed	Time-varying covariates
$b_i^\lambda \in \mathbb{R}^{d_e}$	Immutable buffer	Zero-padded $\hat{\lambda}_i$	Unit heterogeneity
$b_t^f \in \mathbb{R}^{d_e}$	Immutable buffer	Zero-padded $\hat{f}_t$	Common time trends
$\mathbf{z}_{it} \in \mathbb{R}^{d_z}$	Random noise	$\mathcal{N}(0, I_{d_z})$	Stochastic outputs for mean learning

X_{it}: Observed covariates
b_i^λ, b_t^f: Unit-loading and time-factor buffers
$\hat{\lambda}_i, \hat{f}_t$: Estimated loadings and common factors
$\mathbf{z}_{it}$: Generator noise
d_e, d_z: Buffer and noise dimensions
$\mathcal{N}(0, I_{d_z})$: Standard multivariate normal distribution

The buffers store each fold's Stage-1 estimates. During training, the projection maps and correction network learn how to use them, but their stored coordinates remain fixed.

Changing the noise input $\mathbf{z}_{it}$ generates different outcomes for the same $(X_{it}, b_i^\lambda, b_t^f)$. Averaging a fixed set of these draws gives the untreated mean used for ATT. The training objective combines distributional matching with direct mean calibration and reconstruction, while the QTE estimator learns conditional variances and tails separately.

Appendix B.4.2 State padding and GSC initialization

The buffer dimension d_e may exceed r^*. The first r^* coordinates contain the Stage-1 estimates and the remainder are fixed at zero:

$$b_i^\lambda = (\hat{\lambda}_i', 0')' \in \mathbb{R}^{d_e}, \qquad b_t^f = (\hat{f}_t', 0')' \in \mathbb{R}^{d_e}. \tag{152}$$

The zero blocks are deterministic and remain zero throughout fitting. The GSC surface enters as an additive skip connection, while a squared correction penalty discourages unsupported departures from it.

Appendix B.4.3 Joint subspace interaction

The unit/loading and time/factor buffers are projected to a common dimension d_c and combined via element-wise multiplication:

$$h_{it} = \mathrm{Proj}_{\mathrm{u}}(b_i^\lambda) \odot \mathrm{Proj}_{\mathrm{t}}(b_t^f) \in \mathbb{R}^{d_c}, \tag{153}$$

where $\mathrm{Proj}_{\mathrm{u}} : \mathbb{R}^{d_e} \to \mathbb{R}^{d_c}$ and $\mathrm{Proj}_{\mathrm{t}} : \mathbb{R}^{d_e} \to \mathbb{R}^{d_c}$ are learnable linear maps. Under the linear panel factor assumption, these maps can approximate the identity, giving $h_{it} \approx \hat{\lambda}_i \odot \hat{f}_t$. Summing these element-wise products recovers $\lambda_i' f_t = \sum_k \lambda_{ik} f_{tk}$. Learning the projections

also lets the network represent more general interactions.

The generator concatenates X_{it}, h_{it}, and $\mathbf{z}_{it}$ and outputs the correction:

$$\delta_\phi(X_{it}, b_i^\lambda, b_t^f, \mathbf{z}_{it}) = \mathrm{MLP}_\phi\left([X_{it};\ h_{it};\ \mathbf{z}_{it}]\right). \tag{154}$$

Appendix B.4.4 Critic architecture and gradient penalty

The critic $D_\psi(\cdot)$, with network parameters ψ, is a spectrally normalized multi-layer network, with an additional gradient penalty (Gulrajani et al. 2017):

$$\mathrm{GP}(\psi) = \mathbb{E}\big[(\|\nabla_{\tilde{v}} D_\psi(\tilde{v})\|_2 - 1)^2\big], \tag{155}$$

where $\tilde{v} = \alpha v^{\text{real}} + (1-\alpha)v^{\text{fake}}$, $\alpha \sim \mathrm{Unif}(0,1)$. In implementation the expectation uses the same normalized control weights as the observed and generated observations.

Appendix B.4.5 Training objective

Writing $\mathbb{E}_w$ for the weighted average over training control observations, the critic and generator loss functions are

$$L_D = -\big\{\mathbb{E}_w D_\psi(\hat{S}, Y) - \mathbb{E}_{w,\mathbf{z}} D_\psi(\hat{S}, G_\phi)\big\} + \lambda_{\text{gp}}\mathrm{GP}, \tag{156}$$

$$L_G = -\mathbb{E}_{w,\mathbf{z}} D_\psi(\hat{S}, G_\phi) + \lambda_{\text{reg}}\mathbb{E}_{w,\mathbf{z}}(G_\phi - Y)^2 + \lambda_{\text{cal}}\mathrm{cal} + \lambda_\delta \mathbb{E}_{w,\mathbf{z}}\delta_\phi^2. \tag{157}$$

The adversarial term compares distributions. Reconstruction keeps generated outcomes close to observed controls, while calibration matches treated pre-treatment means and the correction penalty keeps predictions near the linear fit. Reconstruction and the squared correction penalty use IPW weights. Each experiment records its coefficients and training budget.

IPW distributional loss. The critic receives $v_{it}^{\text{real}} = (Y_{it}, X_{it}', \hat{\lambda}_i', \hat{f}_t')'$ and $v_{it}^{\text{fake}} =$

$(\hat{Y}_{it}^{\text{gen}}, X_{it}', \hat{\lambda}_i', \hat{f}_t')'$. The IPW-weighted Wasserstein approximation is

$$W_c^{\text{IPW}}(\psi;\phi) = \mathbb{E}_{\tilde{w}} D_\psi(v_{it}^{\text{real}}) - \mathbb{E}_{\tilde{w}} D_\psi(v_{it}^{\text{fake}}), \tag{158}$$

where $\mathcal{D}_{\text{train}} = \{(i,t) : i \in \mathcal{C},\ t = 1,\ldots,T\}$ is the training set (control group, all periods). The generator minimizes $\mathcal{L}_W^{\text{IPW}}(\phi) = -\mathbb{E}_{\tilde{w}}[D_\psi(v_{it}^{\text{fake}})]$. Thus real, fake, reconstruction, correction, and gradient-penalty batches use the same treated-target control weights.

Transfer calibration loss cal**.** The adversarial loss uses control outcomes, but the generator ultimately predicts outcomes for treated units. We therefore also match the treated units' pre-treatment means. Within every outer fold, training-treated units are split into disjoint calibration and checkpoint-selection subsets. Only the calibration subset enters the gradient updates; the selection subset chooses the final fit.

cal matches the generator's period-specific means on the calibration units to their observations. Training approximates these means with fresh noise draws at each update; fixed draws are used later for checkpoint selection and evaluation. With $\mathcal{T}_{\text{cal}}$ denoting the calibration subset,

$$\text{cal}(\phi) = \frac{1}{T_0}\sum_{t=1}^{T_0}\left\{|\mathcal{T}_{\text{cal}}|^{-1}\sum_{i\in\mathcal{T}_{\text{cal}}}\bar{Y}_{it}^{\text{gen}}(0) - |\mathcal{T}_{\text{cal}}|^{-1}\sum_{i\in\mathcal{T}_{\text{cal}}} Y_{it}\right\}^2. \tag{159}$$

Remark 5 (Source of double robustness)**.** *Weights make the training controls resemble the treated population. Double robustness comes from the Stage-4 AIPW mean moment. The ETC-GAN plug-in QTE requires consistent learning of the conditional distribution.*

Appendix B.4.6 Training details

Training follows the standard WGAN-GP alternating update protocol: within each mini-batch the critic updates K times and the generator updates once, with cosine annealing learning rate decay:

$$\eta_k = \eta_{\min} + \frac{1}{2}(\eta_0 - \eta_{\min})\left(1 + \cos\frac{k\pi}{K_{\max}}\right), \tag{160}$$

where k is the epoch index and $K_{\max}$ is the total number of epochs. Generator gradients are clipped to $\|\nabla_\phi L_G\|_2 \le c_{\max}$.

After every adversarial generator update, checkpoint selection evaluates fixed-draw treated-pre-period MSE on the disjoint selection units. Only post-update adversarial checkpoints are eligible. Patience may stop training, after which the best eligible generator and critic states and the fixed evaluation draw banks are restored.

For each evaluated unit–period, the restored generator's untreated-mean prediction is

$$\hat{\mu}_{it} = \frac{1}{M} \sum_{m=1}^{M} G_\phi(\hat{S}_{it}, \mathbf{z}_m). \tag{161}$$

The evaluation noise bank $\mathbf{z}_1, \ldots, \mathbf{z}_M$ is fixed. Standard post-treatment evaluation uses $M = 50$; study-specific overrides are recorded with the corresponding runs.

The ATT learner trains its deterministic path and main network using the objective above. For QTE, a separate regression first adds a nonlinear correction to the GSC prediction by weighted MSE. We hold that mean fixed, form out-of-fold residuals, and fit ETC-GAN to their conditional quantile function. Both estimators use the same $\hat{U}_i$, $\hat{S}_{it}$, propensity predictions and fold assignments, but their mean predictions come from different networks.

Appendix B.5 ETC-GAN Architecture and Training

The distribution learner uses the observations collected without treatment:

$$\mathcal{O}_0 = \{(i,t) : i \in \mathcal{C},\ 1 \le t \le T\} \cup \{(i,t) : i \in \mathcal{T},\ t \le T_0\}.$$

The QTE-branch regression means are cross-fitted before the residuals $\hat{r}_{it} = Y_{it} - \hat{\mu}^Q_{it}$ are formed. Starting from the GSC surface, a two-hidden-layer SiLU network estimates a nonlinear correction by weighted MSE, without a critic. The regression and ETC-GAN transform the common $\hat{S}_{it}$ into network inputs. The fold-specific $\hat{U}_i$, $\hat{S}_{it}$, GSC surface, propensity vector, and fold map remain fixed during downstream fitting. Pre-treatment outcome and

fold-safe linear-anchor residual summaries enter $\hat{U}_i$; in-sample model-specific residual summaries are excluded. Each coordinate is standardized within the training fold. Controls are sampled with the common normalized propensity odds; treated pre-treatment observations have unit weight. Each optimizer updates only its network parameters. The code class `StageAMeanNet` implements manuscript Stage 3B, despite its name.

The empirical base distribution uses nineteen residual quantiles at

$$(0.005, 0.01, 0.025, 0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.40,$$
$$0.50, 0.60, 0.70, 0.80, 0.90, 0.95, 0.975, 0.99, 0.995).$$

The transport network in equation (162) has two hidden layers of 64 SiLU units. Its last layer is initialized at zero, so training begins at the weighted empirical distribution. The shift is parameterized as $a_\theta(s) = 1.25 \tanh\{a_\theta^{\text{raw}}(s)\}$ and each log-spacing multiplier as $\ell_{\theta k}(s) = 0.55 \tanh\{\ell_{\theta k}^{\text{raw}}(s)\}$. These bounds keep the transport local while the exponential spacing in equation (162) maintains monotonicity.

The global Cramér critic has two spectrally normalized hidden layers with LeakyReLU activation and a sixteen-dimensional normalized output. It receives the conditioning state and the five outcome transformations stated below equation (164). The critic minimizes $-\mathcal{D}_{\text{C}}$, while the generator minimizes equation (165). In the reported specification,

$$(\lambda_E, \lambda_T, \lambda_C, \lambda_\mu, \lambda_R) = (0.35, 0.20, 0.12, 0.15, 0.025).$$

The lower-tail pinball weights equal 1.75 for $\tau \leq 0.30$ and one otherwise, normalized to have mean one. The mean penalty is the squared mini-batch mean of four independent generated residual draws. The transport penalty is the mean squared shift plus the mean squared log-spacing change.

Appendix B.5.1 Monotone transport and objective

Let $b_k = \widehat{Q}^{w}_{\tau_k}(\hat{r})$ be the weighted empirical residual quantiles on a grid $0 < \tau_1 < \cdots < \tau_K < 1$. A two-hidden-layer transport network returns a bounded shift $a_\theta(s)$ and log-spacing multipliers $\ell_{\theta k}(s)$. The conditional quantile generator is

$$\begin{aligned} G_\theta(s, \tau_1) &= b_1 + a_\theta(s), \\ G_\theta(s, \tau_k) &= G_\theta(s, \tau_{k-1}) + (b_k - b_{k-1}) \exp\{\ell_{\theta k}(s)\}, \qquad k = 2, \ldots, K. \end{aligned} \tag{162}$$

The positive increments keep the quantiles in order. To generate a residual $G_\theta(s, u)$, draw $u \sim \mathrm{Unif}(0, 1)$, clamp it to the endpoint levels, and linearly interpolate between adjacent values of $G_\theta(s, \tau_k)$.

The generator is anchored by a tail-weighted pinball loss and two proper energy-score terms. Writing $\rho_\tau(v) = v\{\tau - \mathbf{1}(v < 0)\}$ and $h_-(r) = 0.5 \log\{1 + \exp(-r/0.5)\}$,

$$\begin{aligned} \mathcal{L}_{\mathrm{PB}}(\theta) &= \mathbb{E}_{\hat{w}} \left[\frac{1}{K} \sum_{k=1}^{K} \alpha_k \rho_{\tau_k}\{\hat{r} - G_\theta(s, \tau_k)\} \right], \\ \mathcal{L}_{\mathrm{ES}}(\theta) &= \mathbb{E}_{\hat{w}} \left[|\tilde{r}^a - \hat{r}| - \tfrac{1}{2}|\tilde{r}^a - \tilde{r}^b| \right], \\ \mathcal{L}_{\mathrm{tail}}(\theta) &= \mathbb{E}_{\hat{w}} \left[|h_-(\tilde{r}^a) - h_-(\hat{r})| - \tfrac{1}{2}|h_-(\tilde{r}^a) - h_-(\tilde{r}^b)| \right], \end{aligned} \tag{163}$$

where $\tilde{r}^a$ and $\tilde{r}^b$ are independent generator draws and α_k upweights $\tau_k \le 0.30$.

Adversarial learning uses one global Cramér feature critic, following the energy-distance construction of Bellemare et al. (2017). Let $D_\psi(s, r)$ be a spectrally normalized network applied to s and $(r, |r|, (-r)_+, \tanh r, \min\{r^2, 25\})$. Let (S^r, R) and $(S^{r\prime}, R')$ be independent draws from the weighted real joint distribution, and $(S^g, \widetilde{R})$ and $(S^{g\prime}, \widetilde{R}')$ independent draws from the generated joint distribution, all mutually independent. Both distributions have the

same state marginal. The population feature energy distance is

$$\begin{aligned}\mathcal{D}_{\mathrm{C}}(\psi,\theta) = 2\mathbb{E}\|D_\psi(S^r,R) - D_\psi(S^g,\widetilde{R})\|_2 \\ - \mathbb{E}\|D_\psi(S^r,R) - D_\psi(S^{r\prime},R')\|_2 \\ - \mathbb{E}\|D_\psi(S^g,\widetilde{R}) - D_\psi(S^{g\prime},\widetilde{R}')\|_2. \end{aligned} \tag{164}$$

The implementation averages all real–fake feature pairs and excludes self-pairs from each within-distribution average. Real and generated observations share the state on the cross-pair diagonal. For a batch of size B, this diagonal occupies $1/B$ of the cross pairs and creates an $O(B^{-1})$ criterion discrepancy under bounded feature moments. The finite-batch statistic need not be unbiased or nonnegative; population nonnegativity and equality at matching joint distributions concern (164). The consistency argument includes this discrepancy in the criterion-approximation condition. Unlike the ATT branch's WGAN-GP critic, this critic is controlled by spectral normalization rather than a gradient penalty. With the generator fixed, its loss function is

$$L_D(\psi;\theta) = -\mathcal{D}_{\mathrm{C}}(\psi,\theta).$$

Holding the critic fixed, the ETC-GAN generator minimizes

$$L_G(\theta;\psi) = \mathcal{L}_{\mathrm{PB}} + \lambda_E\mathcal{L}_{\mathrm{ES}} + \lambda_T\mathcal{L}_{\mathrm{tail}} + \lambda_C\mathcal{D}_{\mathrm{C}} + \lambda_\mu\mathcal{L}_\mu + \lambda_R\mathcal{L}_R, \tag{165}$$

where $\mathcal{L}_\mu$ penalizes departures from a zero residual mean and $\mathcal{L}_R$ penalizes excessive shifts and changes in quantile spacing. The theory requires these penalties to preserve the true distribution or to induce vanishing bias.

Appendix B.5.2 Training and run-specific configuration

Training begins with transport-only warm-up and then alternates generator and global critic updates. Checkpoint selection uses held-out control residuals and gives equal weight to

overall and lower-tail pinball loss. A method labelled ETC-GAN can select only a checkpoint reached after an adversarial update. Generated residuals are obtained by uniform-rank interpolation and are added to the fixed QTE-branch regression mean.

Network sizes and training budgets vary by study, as do rank selection, propensity estimation and numerical evaluation. The simulation sections record the settings for each benchmark and supplement. Sections Appendix D.3 and Appendix E describe the applications. In the asymptotic analysis, network complexity can grow with the sample. The political-connections application evaluates its mixture CDF directly from learned quantile knots, while the other studies use their specified generated draws.

Appendix B.6 Marginal targets and plug-in aggregation

The code labels period-specific and pooled population ATT as `ATT-POP-T` and `ATT-POP-POOL`, and their realized treated-sample analogues as `ATT-REAL-T` and `ATT-REAL-POOL`. These labels distinguish the target used for simulation error, score centering and intervals. In the political-connections application, bootstrap intervals concern the eligible firm population conditional on the event calendar and sampling strata. Although population and fixed-cohort estimates have the same sample formula, their sampling errors can differ. The ten-day sum adds the daily ATT estimates without averaging or compounding them.

Explicitly, the population and realized-cohort period targets are

$$ATT_t = \mathbb{E}\{Y_{it}(1) - Y_{it}(0) \mid D_i = 1\}, \qquad ATT_{t,N}^{\text{real}} = \frac{1}{N_{\text{tr}}} \sum_{i \in \mathcal{T}} \{Y_{it}(1) - Y_{it}(0)\}.$$

With uniform post-period weights, their pooled counterparts are

$$ATT = \frac{1}{T - T_0} \sum_{t > T_0} ATT_t, \qquad ATT_N^{\text{real}} = \frac{1}{N_{\text{tr}}(T - T_0)} \sum_{i \in \mathcal{T}} \sum_{t > T_0} \{Y_{it}(1) - Y_{it}(0)\}.$$

The realized finite-sample analogue used as simulation truth is

$$QTE_{t,N}^{\text{real}}(\tau) = Q_\tau\{Y_{it}(1) : i \in \mathcal{T}\} - Q_\tau\{Y_{it}(0) : i \in \mathcal{T}\}.$$

The notation $QTE_t(\tau)$ is reserved for the population target in the theory. Simulation point error must be matched to the declared population or realized treated-sample target.

Let $k(i)$ denote unit i's evaluation fold and $\widehat{F}_\varepsilon^{(-k)}(r \mid s)$ the fitted residual CDF used to evaluate fold k. Shifting each residual CDF by the separate regression mean and averaging over treated units gives

$$\begin{aligned} \widehat{F}_{0t}^{\text{plug}}(y) &= \frac{1}{N_{\text{tr}}} \sum_{i \in \mathcal{T}} \widehat{F}_\varepsilon^{(-k(i))}(y - \hat{\mu}_{it}^{Q} \mid \hat{S}_{it}), \\ \widehat{QTE}_t^{\text{plug}}(\tau) &= \widehat{Q}_{1t}(\tau) - \{\widehat{F}_{0t}^{\text{plug}}\}^{-1}(\tau), \end{aligned} \tag{166}$$

where $\widehat{Q}_{1t}$ is the observed treated-outcome quantile function. Retaining all M ETC-GAN outcome draws gives an empirical approximation to this marginal counterfactual CDF. The corresponding draw-based plug-in estimate is

$$\widehat{QTE}_t^{\text{plug}}(\tau) = Q_\tau\{Y_{it} : i \in \mathcal{T}\} - Q_\tau\left\{\hat{Y}_{it}^{\text{gen},(m)}(0) : i \in \mathcal{T},\ m = 1, \ldots, M\right\}. \tag{167}$$

The first quantile is computed from the observed treated outcomes at period t and the second from all $M \times N_{\text{tr}}$ generated counterfactual draws for that same period.

In the implementation, these draws come from the ETC-GAN residual distribution added to the QTE branch's cross-fitted weighted-MSE regression mean. The empirical transport starts at the weighted residual distribution and learns conditional changes in its location and quantile spacing; the global critic enters through alternating critic and generator updates. This construction preserves monotonicity of the estimated quantile function while allowing its shape to vary with the estimated factor state.

A pooled post-period QTE is a separate aggregated estimand. Let $T^\star$ be drawn from

an explicitly stated weighting distribution ω over $\{T_0 + 1, \ldots, T\}$ (uniform weights are one possible choice). Then

$$QTE_{\text{pool},\omega}(\tau) = Q_\tau\{Y_{T^\star}(1) \mid D = 1\} - Q_\tau\{Y_{T^\star}(0) \mid D = 1\}. \tag{168}$$

Pooling mixes period-specific distributions before taking quantiles, so $QTE_{\text{pool},\omega}(\tau)$ generally differs from both a particular $QTE_t(\tau)$ and their weighted average. The fitted pooled counterfactual CDF is $\sum_{t>T_0} \omega_t \widehat{F}_{0t}^{\text{plug}}(y)$, which is inverted only after this aggregation. The realized analogue of (168) replaces its two population distributions with weighted empirical distributions over treated unit–period observations. We label the four objects `QTE-POP-T`, `QTE-POP-POOL`, `QTE-REAL-T`, and `QTE-REAL-POOL`. The reported pooled comparisons give equal weight to treated units and target periods. In the political-connections application, `QTE-REAL-POOL` labels the sample calculation, while the intervals target `QTE-POP-POOL` under firm sampling and the population-centering conditions in the bootstrap section.

Appendix B.7 Algorithm summary: ATT and plug-in QTE

Algorithm 1 Estimand-specific DR-FGAN and plug-in ETC-GAN algorithm

Require: Panel data; target population or realized cohort; post-period weights; whole-unit fold map; declared rank, training, stabilization, and evaluation settings

Ensure: Period-specific or pooled ATT and marginal plug-in QTE point estimates

1: **Stage 1: shared anchor** Score candidate ranks by treated-pre-period leave-one-period-out prediction when rank selection is used. Within each outer fold, fit the converged iterated IFE model on training controls and project non-training loadings using their own pre-treatment outcomes.

2: **Stage 2: shared adjustment** Construct $\hat{U}_i$ and $\hat{S}_{it}$; fit the common out-of-fold unit propensity and normalized control odds. Store state, fold, weight, and convergence provenance; keep these arrays fixed in downstream fitting.

3: **Stage 3A: ATT mean** Register the Stage-1 factor and loading buffers as immutable; alternate the weighted WGAN-GP critic and generator updates with reconstruction, correction-size, and treated-pre-period mean calibration. Use disjoint calibration and checkpoint-selection units.

4: Restore an eligible adversarial checkpoint and its evaluation draw bank; compute the declared fixed-draw mean $\hat{\mu}_{it}^{ATT}$ for treated and control evaluation units.

5: **Stage 3B: QTE distribution** Cross-fit the separate GSC-anchored weighted-MSE regression mean $\hat{\mu}_{it}^{Q}$ and form untreated out-of-fold residuals.

6: Initialize the monotone transport at weighted empirical residual quantiles. Hold the shared arrays and QTE regression mean fixed; warm up the transport, then alternate the Cramér critic and transport-generator updates.

7: Select an eligible adversarial checkpoint using held-out residual pinball loss.

8: **Stage 4A: ATT aggregation**

$$\widehat{ATT}_t^{\mathrm{DR}} = \frac{1}{N_{\mathrm{tr}}} \sum_{i \in \mathcal{T}} (Y_{it} - \hat{\mu}_{it}^{ATT}) - \sum_{i \in \mathcal{C}} \tilde{w}_i (Y_{it} - \hat{\mu}_{it}^{ATT}).$$

9: **Stage 4B: plug-in QTE aggregation** Shift each fitted residual CDF by $\hat{\mu}_{it}^{Q}$ and average over treated units. For a pooled target, mix period CDFs using the declared ω_t before inversion.

10: Subtract the resulting counterfactual quantile from the matching observed-treated quantile. Use direct knot-based CDF evaluation or the declared generated-draw approximation; do not apply a residual-CDF correction or logistic fluctuation.

11: **Diagnostics and inference** Record rank, overlap, clipping/capping, transfer, checkpoint, and numerical diagnostics. Population ATT unit-score inference uses the joint-growth, nuisance, generated-state, and numerical conditions in Appendix A. For the political-connections application, repeat all estimation stages in the bootstrap to obtain ATT normal and QTE pointwise percentile intervals under the additional bootstrap conditions.

Whole-unit folds prevent direct unit reuse in training. The theory in Appendix A separately accounts for own-pre-period loading error and indirect dependence through the estimated state, and imposes statistical-rate and optimization conditions beyond checkpoint eligibility.

Appendix C Additional simulation results

The first two subsections give the complete no-shift ATT and plug-in QTE benchmarks, retaining their distinct designs and training configurations. The covariate-shift extension then uses paired ATT panels and the same twenty shift panels for QTE. Two small completed pilots connect the branches under the later configuration. A separate historical design assesses pointwise ATT interval coverage.

Appendix C.1 ATT benchmark without covariate shift: complete comparison and design

Table 1 uses twenty no-shift panels under the construction below. We report all methods and retain their paired shift counterparts here; Appendix C.3 interprets the shift extension. This design differs from the archived thirty-replication ATT stress design, originally labelled DGP 1. The QTE comparison in Appendix C.4 uses the same twenty realized shift panels, with pooled quantile effects also identically two. The seed ledger contains forty ATT tasks and twenty QTE tasks, not sixty independent data seeds. The earlier three-seed ATT and eight-seed QTE releases are preserved separately and excluded from these summaries.

Population construction. The untreated mean is

$$\mu_{it}(0) = 0.7Z_i + 0.5W_{it} + 1.5\lambda_{i1}f_{1t} + 1.2\lambda_{i2}f_{2t} + h(Z_i, W_{it}) - a(Z_i),$$

where

$$h(z,w) = 3\tanh(1.2z)\tanh\{1.2(z+w)\},$$
$$a(z) = \frac{3\tanh(1.2z)}{4.8}\left[\log\cosh\{1.2(z+2)\} - \log\cosh\{1.2(z-2)\}\right].$$

The second expression is the analytic expectation of $h(z, W)$ under $W \sim U[-2, 2]$. It removes a persistent nonlinear unit mean from the remainder using population rather than sample centering. The two factor paths before treatment are $f_{1t} = \sqrt{2}\sin(2\pi t/12)$ and $f_{2t} = \sqrt{2}\cos(2\pi t/12)$, for $t = 1, \ldots, 36$. The twelve post-treatment periods repeat phases one through six twice. Thus the factor directions remain present before and after treatment. Loadings are independent $U[-\sqrt{3}, \sqrt{3}]$ variables, with scale coefficients 1.5 and 1.2 in the outcome equation. They are independent of observed covariates, assignment uniforms, and noise.

The assignment probability is $e(Z_i) = \text{expit}(-0.7272164888 + Z_i)$ for $Z_i \sim U[-1.5, 1.5]$. A single Bernoulli draw per unit gives an expected treated share of 0.35, without fixing realized group sizes or redrawing assignments. The latent loadings affect outcomes but do not select treatment conditional on Z. External covariates are independent draws over units and periods, uniform on $[-2, 2]$ before treatment. Their post-treatment law is $0.35U[-2, 0] + 0.65U[0, 2]$ in the shift case and $U[-2, 2]$ in the no-shift case, for both treatment groups. Independent $N(0, 0.3^2)$ disturbances are added to the untreated mean. The treatment effect is always two, giving the common target `ATT-POP-POOL`; realized pooled ATT also equals two.

For the anchor decomposition, define the population control-pre projection

$$b = \frac{\mathbb{E}[W\{h(Z, W) - a(Z)\} \mid D = 0, \text{pre}]}{\mathbb{E}(W^2 \mid D = 0, \text{pre})}, \qquad H_{it} = h(Z_i, W_{it}) - a(Z_i) - bW_{it}.$$

Then the covariate coefficient in the linear anchor is $(0.7, 0.5 + b)'$, with $b \simeq -0.25628$. This decomposes the same outcome law: bW enters the anchor and is subtracted once in the remainder. Population centering gives $\mathbb{E}(H_{it} \mid Z_i, \lambda_i, \text{pre}) = 0$; projection removes the control-pre covariate moment, and independence of the mean-zero loadings removes the population loading moment. As N_{co} and the number of complete pre-treatment factor cycles grow, the bounded pre-treatment remainder is spectrally weak relative to the pervasive rank-

two anchor. The shifted post-treatment block remains spectrally negligible under the stated growth embedding because its length is fixed at twelve. This construction does not extend that argument to proportionally growing shifted post-treatment windows. Finite-panel state diagnostics and the implemented rank choice assess this construction; they are not a proof of selector consistency or generated-state rates for inference.

Estimation protocol. Every panel is fit with three whole-unit folds. Candidate ranks one through four are evaluated using treated-pre-period leave-one-period-out MSPE; the mean across outer folds selects one rank, followed by fold-specific refitting. All forty panels select rank two, as does official GSC under its own selection rule. The iterated IFE solver uses tolerance 10^{-6} and at most 2,000 iterations. The LFM row is computed from the same out-of-fold linear counterfactual surface that initializes the FGAN family. No true loading, population response, propensity, or rotation alignment is passed to a fitted learner.

A common out-of-fold gradient-boosted propensity model uses 100 depth-two trees, learning rate 0.1, minimum leaf size five, and subsampling fraction 0.8. Predictions are clipped to $[0.05, 0.95]$, and normalized control odds are capped at five times their mean and renormalized. The neural methods use a 300-epoch ceiling and patience 100. The strict noADV version sets only the generator adversarial coefficient to zero: it retains trained critics, architecture, initialization, fold map, state, weights, other losses, random-draw schedule, and checkpoint-selection rule. Actual stopping and selected epochs may differ under the shared rule. NN-MSE and histogram gradient boosting (HGB) provide additional mean learners on the same state; they are not objective-only ablations.

The reference implementations use the same observed panels with their own estimation rules. GSC uses `gsynth` 1.2.1, two-way fixed effects, time-varying W, and its own rank search from zero through three; the time-invariant Z is absorbed by unit effects. ASC uses `augsynth` 0.2.0, ridge augmentation, and pre-period cross-validation with the one-standard-error rule. The main-text ASC row uses outcome histories only. ASC-X adds the unit's

pre-treatment mean Z and W through the official covariate interface; it does not adjust on contemporaneous post-treatment exposure. DID/TWFE is a joint regression of observed outcomes on treatment and unit and time effects. TWFE-X additionally includes W linearly. SDID and SDID-X use `synthdid` 0.0.9, with no auxiliary covariate and with time-varying W, respectively. The replication package pins the exact R sources and tuning settings.

SDID reaches the package's default 10,000-iteration limit in nineteen of twenty shift panels and all twenty no-shift panels; SDID-X reaches it in all forty panels. Their saved weights and effect estimates are finite and reproduce the reported numbers. These rows retain the original budget rather than changing the stopping rule after observing results.

Weight stabilization binds in all forty ATT tasks. The average control effective sample size is 60.05 under the shift and 58.88 without it; the respective ranges are 44.91–78.38 and 45.33–72.96. These are results for the implemented stabilized score, not a check of the asymptotically inactive-truncation condition used in the inference theory.

Seed ledger and reporting. Seeds 2026101701 through 2026101720 are paired across the two cases, sharing primitive draws; only the post-treatment exposure law changes. All forty tasks and all seventeen methods are retained. The design arose from earlier exploration; this batch's cases, seeds, methods, and budgets were fixed before fitting. The main-text table retains DR-FGAN, ASC, DID/TWFE, and SDID; the complete ledger here retains the factor-adjusted methods, covariate-adjusted references, and matched ablations. Smoke runs and earlier batches are excluded. Bias, MAE, and RMSE are computed from estimate minus two. SD(error) uses the sample denominator $R - 1$. The package checks $\text{RMSE}^2 = \text{Bias}^2 + (R-1)\text{SD(error)}^2/R$. Paired differences use the same panel in each method comparison; their Monte Carlo SE is the sample SD of the differences divided by $\sqrt{R}$.

Table 1: Complete ATT comparison on the twenty new seeds

Scenario	Method	Bias	MAE	SD(error)	RMSE	MCSE(RMSE)
Shift	LFM	0.20090	0.20090	0.05443	0.20779	0.01185
Shift	FGAN	0.02569	0.03402	0.03139	0.03995	0.00461
Shift	DR-LFM	-0.01109	0.07736	0.09384	0.09214	0.01044

Scenario	Method	Bias	MAE	SD(error)	RMSE	MCSE(RMSE)
Shift	DR-FGAN	-0.00334	0.04230	0.05285	0.05162	0.00818
Shift	NN-MSE	-0.00613	0.02105	0.03043	0.03029	0.00589
Shift	DR-NN	-0.00051	0.04069	0.05157	0.05027	0.00765
Shift	HGB	0.00240	0.02232	0.02868	0.02806	0.00396
Shift	DR-HGB	-0.00842	0.03918	0.05117	0.05058	0.00898
Shift	ASC	0.06660	0.11222	0.12854	0.14189	0.02105
Shift	GSC	0.22342	0.22342	0.06860	0.23321	0.01538
Shift	FGAN (noADV)	0.02078	0.03057	0.02928	0.03530	0.00428
Shift	DR-FGAN (noADV)	-0.00497	0.04219	0.05272	0.05162	0.00821
Shift	DID/TWFE	0.25787	0.25885	0.17460	0.30897	0.04002
Shift	DID/TWFE-X	0.25903	0.25903	0.16917	0.30705	0.03923
Shift	SDID	0.25593	0.25593	0.08126	0.26791	0.01997
Shift	SDID-X	0.25032	0.25032	0.08135	0.26257	0.01958
Shift	ASC-X	-0.03423	0.09857	0.11725	0.11930	0.01312
No shift	LFM	-0.00008	0.04589	0.05397	0.05260	0.00712
No shift	FGAN	-0.00945	0.02558	0.03258	0.03313	0.00695
No shift	DR-LFM	0.00098	0.07862	0.09813	0.09565	0.01279
No shift	DR-FGAN	-0.00378	0.04260	0.05367	0.05245	0.00898
No shift	NN-MSE	-0.02603	0.03326	0.03371	0.04192	0.00704
No shift	DR-NN	-0.00183	0.04385	0.05374	0.05241	0.00832
No shift	HGB	-0.01194	0.02190	0.02701	0.02891	0.00483
No shift	DR-HGB	-0.00780	0.03966	0.05139	0.05069	0.00919
No shift	ASC	0.05363	0.11118	0.12456	0.13273	0.01881
No shift	GSC	-0.00050	0.05303	0.06884	0.06710	0.00981
No shift	FGAN (noADV)	-0.01497	0.02483	0.03139	0.03406	0.00668
No shift	DR-FGAN (noADV)	-0.00526	0.04251	0.05343	0.05234	0.00899
No shift	DID/TWFE	0.03517	0.13259	0.17506	0.17422	0.03127
No shift	DID/TWFE-X	0.03745	0.12950	0.16766	0.16766	0.03049
No shift	SDID	0.00572	0.05442	0.07258	0.07097	0.01153
No shift	SDID-X	0.01035	0.05594	0.07331	0.07220	0.01104
No shift	ASC-X	-0.01583	0.09421	0.10935	0.10775	0.01100

ATT: Average treatment effect on the treated
GAN: Generative adversarial network
DR-FGAN: Doubly robust factor-augmented GAN
LFM: Linear factor model
GSC: Generalized synthetic control
ASC: Augmented synthetic control
DID/TWFE: Difference-in-differences with two-way fixed effects
SDID: Synthetic difference-in-differences
FGAN: Factor-augmented GAN
DR-LFM: LFM with doubly robust correction
NN-MSE: Neural network trained by mean squared error
DR-NN: Neural network with doubly robust correction
HGB: Histogram gradient boosting
DR-HGB: HGB with doubly robust correction
noADV: Generator adversarial term disabled
X suffix: Covariate-adjusted reference specification
MAE: Mean absolute error
RMSE: Root mean squared error
SD(error): Sample standard deviation of estimation errors
MCSE: Monte Carlo standard error
e_s: Estimation error in replication s

Notes: Each cell contains the same twenty new primitive seeds; cells and historical batches are not pooled. MCSE(RMSE) uses the delta-method formula $\mathrm{sd}(e_s^2)/(2\,\mathrm{RMSE}\sqrt{20})$. Within each matched pair, noADV retains the original score and critic training while removing only the generator adversarial coefficient.

No-shift results. Without the exposure shift, LFM has RMSE 0.05260, essentially equal to DR-FGAN's 0.05245. Their biases are −0.00008 and −0.00378; GSC has RMSE 0.06710. The centered nonlinear remainder averages to zero conditional on unit characteristics under the baseline law, so a restricted mean surface can still give an accurate aggregate ATT. Similar ATT errors need not imply similar conditional-mean predictions. DR-FGAN-noADV has RMSE 0.05234; each matched method has smaller absolute error in ten panels. Uncorrected FGAN and FGAN-noADV have RMSEs 0.03313 and 0.03406, respectively, compared with 0.09565 for DR-LFM. NN-MSE, HGB, DR-NN-MSE, and DR-HGB have RMSEs 0.04192, 0.02891, 0.05241, and 0.05069. Outcome-history ASC and ASC-X have RMSEs 0.13273 and 0.10775. These comparisons show similar aggregate accuracy for DR-FGAN and LFM, rather than a separate RMSE gain from adversarial training or residual correction. Residual correction need not reduce finite-sample RMSE, and these results do not isolate a critic gain.

Reproduction. The independent release `att_shared20_20261018` binds executed source commit `7f00c009`, the frozen seed and budget contract, raw predictions, reference outputs, and every method–seed record. The shared-panel manifest verifies the ATT–QTE shift-panel identities and paired pre-treatment blocks. Its materializer recomputes estimates, Monte Carlo summaries, and tables without fitting models. Training checkpoints remain in the original run archive; the portable numerical release inventories omitted files and supplies frozen sources and environment contracts for an independently configured training rerun. The preceding `att_anchored_shift_20261013` package, earlier explorations, and application releases retain their original results; Appendix C.8 records their disposition.

Appendix C.2 QTE benchmark without an imposed covariate-shift break

The thirty-panel benchmark in Table 2 studies nonlinear latent effects and a state-dependent residual mixture. It complements the ATT benchmark rather than extending its seed ledger:

$N = 180$, $T = 40$, and $T_0 = 30$, with seeds 20260825 through 20260854. The covariate and factor generating laws have no treatment-date break. Treatment nevertheless selects on loadings and pre-treatment covariates, so absence of an imposed break does not assert identical realized group–period distributions.

Outcome and assignment laws. The two loading coordinates, two factor coordinates, and two observed covariates are independent standard Gaussian primitive draws. Define $v_{it} = \lambda_i' f_t/\sqrt{2}$ and $L(v) = (1 + e^{-v})^{-1}$. The untreated conditional mean and scale are

$$\mu_{it}(0) = 0.35X_{it,1} - 0.20X_{it,2} + 0.20v_{it} + \tanh(v_{it}) + 0.25X_{it,1}\tanh(v_{it}),$$
$$\sigma_{it} = 0.65 + 0.30L(0.8v_{it}) + 0.15|\lambda_{i1}|.$$

Let $p_{it} = L\{\text{logit}(0.13) - 0.95\tanh(v_{it})\}$. A raw shock is drawn from $N(-3.5, 1.3^2)$ with probability p_{it} and from $N(0,1)$ otherwise. Subtracting its conditional mean gives $\epsilon_{it} =$ raw shock $+ 3.5p_{it}$ and $Y_{it}(0) = \mu_{it}(0) + \sigma_{it}\epsilon_{it}$. For treated post-treatment cells,

$$Y_{it}(1) = \mu_{it}(0) + \sigma_{it}\{\epsilon_{it} + 0.85\max(0, -\epsilon_{it})\}.$$

Treatment cushions negative shocks, producing nonconstant marginal QTEs. The target subtracts quantiles of the two realized potential-outcome samples pooled over the same treated post-treatment cells.

The treatment index is $1.05\lambda_{i1} - 0.65\lambda_{i2} + 0.30\bar{X}_{i,\text{pre},1} - 0.20\bar{X}_{i,\text{pre},2}$. A sample-calibrated intercept sets the average assignment probability to 40/180. The generator accepts a Bernoulli allocation with 28–52 treated units, trying at most 500 allocations; its fallback selects the forty largest propensities. This retained count-restricted design is distinct from the single-draw Bernoulli assignment in the twenty-seed ATT benchmark. Once a panel is saved, a failed fit resumes that panel rather than replacing it.

Frozen estimation configuration. Each replication fits one common nuisance and regression-mean artifact using two whole-unit folds and a prespecified estimation rank of four. The generating latent dimension is two; the fitted rank is neither the true dimension nor the output of the later rank-selection procedure. The regression mean has at most 900 updates and patience 175. The residual transport uses 800 warm-up updates followed by 1,800 distribution-training updates, batches of 256, hidden width 64, two critic updates per adversarial step, learning rate 2×10^{-4}, and 250 evaluation draws per state. All three residual learners share the estimated state, propensities, regression mean, and out-of-fold residuals. The common implementation clips estimated propensities at 0.05 and 0.95 and stabilizes control odds. The source and configuration receipts preserve the full rule.

These settings differ from the twenty-panel shift extension and the two recent pilots, which use three folds, rank search over one through four, at most 300 mean updates, 200 warm-up plus 600 distribution-training updates, batches of 128, and 200 evaluation draws. Accordingly, their errors are reported in separate tables, not pooled or compared as a controlled change in the covariate distribution.

The original architecture comparison contains ETC-GAN, weighted empirical residuals, and non-adversarial ET-Quantile. The empirical alternative uses the weighted residual distribution without learning a conditional transport. ET-Quantile learns the transport without a critic and is not the strict noADV intervention used in the later experiments. All sixty ETC-GAN folds record alternating critic and generator updates, nonzero Cramér checks, and an adversarial-phase checkpoint, giving 7,200 critic updates per panel. CiC and CDF-DiD were subsequently evaluated on all thirty saved panels using only observed group–period outcomes; their definitions are in Appendix C.6.

Results and interpretation. ETC-GAN's overall QTE MAE is 0.1831, compared with 0.1839 for ET-Quantile and 0.1767 for weighted empirical residuals. Its lower-region MAE is 0.2285, compared with 0.2365 and 0.2468, respectively. The empirical baseline has smaller

Table 2: Complete thirty-panel QTE benchmark without an imposed covariate-shift break

Method	Overall	Lower	Central	Upper	CRPS	PIT error
ETC-GAN	0.1831	0.2285	0.1266	0.2036	0.9389	0.0388
ET-Quantile	0.1839	0.2365	0.1222	0.2034	0.9310	0.0347
Weighted empirical residual	0.1767	0.2468	0.1050	0.1903	0.9140	0.0282
CiC (pooled)	0.2025	0.2690	0.1614	0.1840	–	–
CDF-DiD (pooled)	0.2039	0.2727	0.1605	0.1857	–	–

QTE: quantile treatment effect. ETC-GAN: empirical-transport Cramér generative adversarial network. ET-Quantile: empirical-transport quantile learner. noADV: generator adversarial term disabled. CiC: changes-in-changes. CDF-DiD: cumulative distribution function difference-in-differences. MAE: mean absolute error. CRPS: continuous ranked probability score. PIT: probability integral transform.

Notes: All rows use the same thirty archived panels. The first four numerical columns report pooled marginal QTE MAE; regions are $\tau \leq .30$, $.30 < \tau < .70$, and $\tau \geq .70$ on the nineteen-point .05–.95 grid. Lower values indicate better fit. The empirical learner has the lowest overall error, CRPS, and PIT error; ETC-GAN has the lowest lower-region MAE. Conditional scores are not defined for the two pooled marginal procedures (–). ET-Quantile is a non-adversarial comparator, not the strict noADV ablation. Fixed rank four, two folds, and the original budgets are retained.

Table 3: Paired QTE error differences in the thirty-panel benchmark

Method minus ETC-GAN	Overall	Lower	Central	Upper
ETC-GAN	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
ET-Quantile	0.0008 (0.0042)	0.0080 (0.0053)	-0.0044 (0.0051)	-0.0002 (0.0094)
Weighted empirical residual	-0.0064 (0.0073)	0.0183 (0.0060)	-0.0216 (0.0096)	-0.0133 (0.0135)
CiC (pooled)	0.0194 (0.0187)	0.0404 (0.0285)	0.0348 (0.0196)	-0.0196 (0.0255)
CDF-DiD (pooled)	0.0208 (0.0189)	0.0442 (0.0289)	0.0339 (0.0198)	-0.0179 (0.0250)

QTE: quantile treatment effect. ETC-GAN: empirical-transport Cramér generative adversarial network. ET-Quantile: empirical-transport quantile learner. noADV: generator adversarial term disabled. CiC: changes-in-changes. CDF-DiD: cumulative distribution function difference-in-differences. MAE: mean absolute error. CRPS: continuous ranked probability score. PIT: probability integral transform.

Notes: Entries are within-panel MAE differences averaged over thirty panels; parentheses give the sample standard deviation of these differences divided by $\sqrt{30}$. Negative values favor the comparison method. All methods and regions are retained. These Monte Carlo standard errors describe paired accuracy comparisons, not confidence-interval coverage.

central and upper-region errors and better CRPS and PIT calibration. CiC and CDF-DiD have overall MAEs 0.2025 and 0.2039, but smaller upper-region errors than ETC-GAN. The paired comparisons retain this uneven pattern across regions. These results support competitive distributional reconstruction, not uniform dominance of adversarial training.

This design evaluates the complete distribution-learning pipeline with estimated, rather than oracle, adjustment inputs. Its nonlinear latent surface is not constructed to make the nonlinearity a spectrally weak remainder of the specified linear anchor. It therefore supplies a point accuracy benchmark, not verification of the anchor-recovery or inference conditions in Appendix A. The nineteen-point marginal QTE estimates are draw plug-ins without DR-CDF correction; these runs contain no confidence intervals or coverage results.

Reproduction. The unchanged release at `publication_artifacts/etcgan_qte_architecture_point` contains all replication-level metrics, curve summaries, and the source inventory. `publication_artifacts/qte_external_benchmark_allpanels_20261012` contains the complete retrospective pooled benchmarks and paired differences. The SI materializer checks agreement between the two releases before regenerating these tables. Older targeted-CDF and interval releases are separate experiments and do not enter this benchmark.

Appendix C.3 Covariate-shift extension: paired ATT results

The extension changes only the post-treatment law of W in Appendix C.1, from $U[-2,2]$ to $0.35U[-2,0]+0.65U[0,2]$ for both groups. Primitive draws, treatment assignment, nonlinear response, factor paths, and training rules are paired across the twenty seeds. Table 1 retains all results in both cases.

Under the shift, DR-FGAN's RMSE is 0.05162, compared with 0.20779 for LFM and 0.23321 for GSC; their biases are -0.00334, 0.20090, and 0.22342, respectively. The centered nonlinear response $h(Z,W)-a(Z)$ has zero baseline conditional mean. Under the shifted law

its population treated and control means become 0.14071 and −0.07577. Their difference, 0.21648, is the nonlinear component of the untreated group contrast, not the bias of a fitted estimator. This mechanism explains why a restricted response can fit earlier trajectories yet miss the later gap.

DR-FGAN-noADV has RMSE 0.05162, essentially the same as DR-FGAN; each has smaller absolute error in ten panels. Uncorrected FGAN has RMSE 0.03995 and FGAN-noADV has 0.03530. Residual correction reduces FGAN's bias from 0.02569 to −0.00334 but increases SD(error) from 0.03139 to 0.05285, raising RMSE. DR-LFM has RMSE 0.09214. The additional nonlinear learners also reduce shift-induced error: NN-MSE and HGB have RMSEs 0.03029 and 0.02806, while DR-NN-MSE and DR-HGB have 0.05027 and 0.05058. Outcome-history ASC and ASC-X have RMSEs 0.14189 and 0.11930. Thus the gain over LFM is shared by flexible learners; it does not isolate an adversarial-training advantage.

Table 4: Paired ATT error differences on the new seeds

Scenario	Contrast	ΔAE	MCSE	ΔSE	MCSE	Wins
Shift	FGAN − FGAN (noADV)	0.00345	0.00235	0.00035	0.00014	7/20
Shift	DR-FGAN − FGAN	0.00828	0.00712	0.00107	0.00082	9/20
Shift	DR-FGAN − ASC-X	-0.05627	0.01707	-0.01157	0.00321	15/20
Shift	DR-FGAN − SDID-X	-0.20802	0.02160	-0.06628	0.01060	20/20
Shift	DR-FGAN − DID/TWFE-X	-0.21673	0.03914	-0.09162	0.02410	18/20
Shift	DR-FGAN − ASC	-0.06992	0.02266	-0.01747	0.00626	12/20
Shift	DR-FGAN − GSC	-0.18112	0.01930	-0.05172	0.00754	20/20
Shift	DR-FGAN − LFM	-0.15860	0.01583	-0.04051	0.00526	20/20
Shift	DR-FGAN − SDID	-0.21364	0.02168	-0.06911	0.01103	20/20
Shift	DR-FGAN − DR-FGAN (noADV)	0.00010	0.00070	-0.00000	0.00009	10/20
Shift	DR-FGAN − DR-NN	0.00161	0.00239	0.00014	0.00030	6/20
Shift	DR-FGAN − DR-HGB	0.00312	0.00224	0.00011	0.00024	7/20
Shift	DR-FGAN − HGB	0.01998	0.00553	0.00188	0.00071	5/20
Shift	DR-FGAN − DID/TWFE	-0.21656	0.03988	-0.09280	0.02474	18/20
Shift	DR-FGAN − DR-LFM	-0.03506	0.01038	-0.00583	0.00164	15/20
Shift	DR-FGAN − NN-MSE	0.02125	0.00504	0.00175	0.00058	4/20
No shift	FGAN − FGAN (noADV)	0.00075	0.00219	-0.00006	0.00012	7/20
No shift	DR-FGAN − FGAN	0.01701	0.00549	0.00165	0.00070	5/20
No shift	DR-FGAN − ASC-X	-0.05161	0.01242	-0.00886	0.00251	16/20
No shift	DR-FGAN − SDID-X	-0.01334	0.01244	-0.00246	0.00186	10/20
No shift	DR-FGAN − DID/TWFE-X	-0.08691	0.02409	-0.02536	0.01016	16/20
No shift	DR-FGAN − ASC	-0.06859	0.01973	-0.01487	0.00529	16/20
No shift	DR-FGAN − GSC	-0.01043	0.01112	-0.00175	0.00152	12/20
No shift	DR-FGAN − LFM	-0.00330	0.00795	-0.00002	0.00107	13/20
No shift	DR-FGAN − SDID	-0.01182	0.01223	-0.00229	0.00189	12/20
No shift	DR-FGAN − DR-FGAN (noADV)	0.00009	0.00057	0.00001	0.00006	10/20

Scenario	Contrast	ΔAE	MCSE	ΔSE	MCSE	Wins
No shift	DR-FGAN − DR-NN	-0.00125	0.00213	0.00000	0.00022	11/20
No shift	DR-FGAN − DR-HGB	0.00294	0.00211	0.00018	0.00023	6/20
No shift	DR-FGAN − HGB	0.02070	0.00650	0.00192	0.00086	3/20
No shift	DR-FGAN − DID/TWFE	-0.08999	0.02536	-0.02760	0.01082	16/20
No shift	DR-FGAN − DR-LFM	-0.03602	0.01044	-0.00640	0.00221	16/20
No shift	DR-FGAN − NN-MSE	0.00934	0.00678	0.00099	0.00075	6/20

ATT: Average treatment effect on the treated
GAN: Generative adversarial network
DR-FGAN: Doubly robust factor-augmented GAN
LFM: Linear factor model
GSC: Generalized synthetic control
ASC: Augmented synthetic control
DID/TWFE: Difference-in-differences with two-way fixed effects
SDID: Synthetic difference-in-differences
FGAN: Factor-augmented GAN
DR-LFM: LFM with doubly robust correction
NN-MSE: Neural network trained by mean squared error
DR-NN: Neural network with doubly robust correction
HGB: Histogram gradient boosting
DR-HGB: HGB with doubly robust correction
noADV: Generator adversarial term disabled
X suffix: Covariate-adjusted reference specification
AE: Absolute estimation error
SE: Squared estimation error
Δ: First method minus comparator
MCSE: Monte Carlo standard error

Notes: Differences subtract the second method from the first within each seed; negative values favor the first method. Each MCSE is the sample standard deviation of paired differences divided by $\sqrt{20}$. Wins count seeds with lower absolute error. All predeclared contrasts are shown, with DR-FGAN versus DR-LFM and NN-MSE added to complete its comparisons; the package retains all 136 unordered method pairs per scenario.

For DR-FGAN relative to LFM under the shift, the mean paired squared-error difference is −0.04051 (MCSE 0.00526), with smaller absolute error in all twenty panels; the mean paired absolute-error difference is −0.15860 (MCSE 0.01583). Relative to GSC, the corresponding differences are −0.05172 (MCSE 0.00754) and −0.18112 (MCSE 0.01930), again with smaller absolute error in every panel. Relative to outcome-history ASC, the paired squared-error difference is −0.01747 (MCSE 0.00626), with smaller absolute error in twelve panels. The no-shift DR-FGAN–LFM squared-error difference is −0.00002 (MCSE 0.00107), consistent with their nearly equal RMSEs. Table 5 retains the primary seed-level contrasts; all method–seed records are in the release.

Table 5: Seedwise ATT errors for the focal comparisons

Scenario	Seed	DR-FGAN	LFM	GSC	DR-FGAN (noADV)
Shift	2026101701	0.00931	0.28107	0.29562	0.00248
Shift	2026101702	0.05049	0.25590	0.25215	0.05098
Shift	2026101703	0.02277	0.20791	0.28776	0.02263
Shift	2026101704	0.03927	0.13042	0.22041	0.03769
Shift	2026101705	-0.10395	0.11099	0.10643	-0.10939
Shift	2026101706	0.06154	0.21000	0.16548	0.05723
Shift	2026101707	0.08950	0.20874	0.20949	0.08459
Shift	2026101708	0.03351	0.24135	0.24456	0.03190
Shift	2026101709	-0.03947	0.20560	0.17703	-0.04297
Shift	2026101710	-0.04063	0.30694	0.35966	-0.04285
Shift	2026101711	-0.11771	0.16689	0.18239	-0.11617
Shift	2026101712	-0.00457	0.19589	0.21537	-0.00559
Shift	2026101713	0.02835	0.19590	0.16771	0.02656
Shift	2026101714	-0.03375	0.12999	0.15609	-0.03542
Shift	2026101715	-0.02840	0.15881	0.16977	-0.02913
Shift	2026101716	0.01903	0.26045	0.31490	0.01856
Shift	2026101717	-0.00702	0.19751	0.24010	-0.00863
Shift	2026101718	0.03576	0.22371	0.31937	0.03965
Shift	2026101719	-0.04613	0.11006	0.12591	-0.04363
Shift	2026101720	-0.03480	0.21991	0.25816	-0.03780
No shift	2026101701	0.02202	0.01404	0.00851	0.01607
No shift	2026101702	0.04841	0.05249	0.02690	0.04676
No shift	2026101703	0.01684	0.01607	0.06756	0.01582
No shift	2026101704	0.02982	-0.05531	0.03011	0.02913
No shift	2026101705	-0.08701	-0.12160	-0.14849	-0.09057
No shift	2026101706	0.04259	0.03496	-0.03310	0.03931
No shift	2026101707	0.11557	0.00936	0.00028	0.11286
No shift	2026101708	0.04380	-0.02619	-0.05228	0.04482
No shift	2026101709	-0.03779	0.04782	0.00528	-0.04048
No shift	2026101710	-0.03635	0.07849	0.10745	-0.03636
No shift	2026101711	-0.12206	-0.05371	-0.06338	-0.12278
No shift	2026101712	-0.00748	0.02661	0.02968	-0.00989
No shift	2026101713	0.02106	0.03238	-0.00160	0.01952
No shift	2026101714	-0.03857	-0.03349	-0.03563	-0.04053
No shift	2026101715	-0.04400	-0.05021	-0.06867	-0.04354
No shift	2026101716	0.03667	0.06907	0.10842	0.03503
No shift	2026101717	-0.03162	-0.04077	-0.03258	-0.03353
No shift	2026101718	0.01132	0.03517	0.10761	0.01318
No shift	2026101719	-0.04644	-0.07842	-0.09960	-0.04359
No shift	2026101720	-0.01248	0.04168	0.03343	-0.01634

ATT: Average treatment effect on the treated
GAN: Generative adversarial network
DR-FGAN: Doubly robust factor-augmented GAN
LFM: Linear factor model
GSC: Generalized synthetic control
noADV: Generator adversarial term disabled

Notes: Errors equal the estimate minus the true ATT of 2. The reproducibility package retains all 680 method–scenario–seed estimates and errors.

Appendix C.4 Covariate-shift extension: QTE on the shared ATT panels

Table 6 uses the outcome and assignment law in Appendix C.1, with $N = 300$, $T = 48$, $T_0 = 36$, and the shifted exposure distribution in both arms. The nonlinear response, centering function, latent-factor paths, noise, and treatment assignment are unchanged. Since $Y_{it}(1) = Y_{it}(0) + 2$, every period-specific and pooled marginal QTE equals two. The experiment tests recovery of the counterfactual distribution under the changing exposure mixture, not heterogeneous treatment effects.

Shared panels and frozen protocol. The twenty data seeds are 2026101701 through 2026101720. They were fixed before fitting and are disjoint from the earlier ATT and QTE batches. This branch uses the identical realized shift panels as the ATT comparison: observed outcomes, covariates, treatment indicators, and evaluation-only counterfactual arrays are checked against a common pre-fit manifest. Each panel is generated once, including its Bernoulli assignment, with no redraw or failed-seed replacement. All twenty tasks completed; earlier panels are excluded from the averages. This twenty-panel QTE batch is evaluated under the shift design only. A separate two-panel no-shift completion is reported in Appendix C.5; it is not pooled with this batch.

Three whole-unit folds are formed on the original unit ordering using data seed plus 900,000. Each fold evaluates candidate ranks one through four by treated-pre-period leave-one-time-out MSPE; the mean fold score selects a common rank, followed by final fold-specific fits. The true rank is not supplied. All twenty panels select rank two and all candidate and final IFE fits converge within the frozen 2,000-iteration ceiling at tolerance 10^{-6}. Rank selection and state estimation are repeated on every panel.

The recovered state, GBM propensity, clipped and capped normalized control odds, and fold map follow the ATT construction. Propensity seeds equal data seed plus 700,000. The QTE branch separately cross-fits a weighted-MSE regression correction to the linear surface;

it does not reuse the ATT generator mean. This regression uses hidden width 64, at most 300 updates, and patience 75. Its initialization and validation-split seeds equal the data seed plus $500 + k$ and $700 + k$, respectively, for fold $k = 0, 1, 2$. The residual learners use hidden width 64, batches of 128, 200 warm-up generator updates followed by 600 distribution-training updates, two critic updates per adversarial step, and 200 evaluation draws per state. Residual-network initialization is paired across methods and fixed at $81173 + 1009k$ for fold $k = 0, 1, 2$. The new panels therefore test sampling variation, not sensitivity to independent training initializations.

Comparators and scoring. The complete comparison contains ETC-GAN, strict noADV, ET-Quantile, weighted empirical residuals, pooled CiC, and pooled CDF-DiD. The first four share the same upstream state, regression mean, residual inputs, and weights. Strict noADV removes only the generator's adversarial coefficient, preserving critic work, random draws, and eligible checkpoint phase. ET-Quantile has the same 800-update generator budget but no critic. The empirical learner uses the weighted residual distribution without a learned transport. The two pooled comparators use the observed-outcome constructions in Appendix C.6, applied anew to each panel. They do not use the factor state or true counterfactual outcomes.

The predeclared primary readout pools generated untreated outcomes for treated units over the twelve post-treatment periods, then takes quantiles. It is a plug-in estimate before CDF targeting. The overall MAE averages nineteen levels $.05, .10, \ldots, .95$; the lower, central, and upper regions contain six, seven, and six levels, respectively. Conditional CRPS and PIT scores use simulated untreated outcomes for evaluation only. They are not assigned to the marginal CiC and CDF-DiD procedures. Table 6 retains all six learners and scoring dimensions.

ETC-GAN has smaller overall QTE MAE than each pooled comparator in all twenty panels. In contrast, it beats strict noADV in thirteen panels and weighted empirical residuals

Table 6: Complete QTE comparison on the twenty new panels

	Pooled QTE MAE				Distribution fit	
Method	Overall	Lower	Central	Upper	CRPS	PIT error
ETC-GAN	0.0642	0.0710	0.0452	0.0796	0.3797	0.0230
Changes-in-changes (pooled)	0.2907	0.3772	0.2629	0.2367	–	–
CDF-DiD (pooled)	0.2844	0.3792	0.2585	0.2198	–	–
ETC-GAN (ADV off)	0.0645	0.0713	0.0453	0.0800	0.3798	0.0232
ET-Quantile	0.0648	0.0722	0.0435	0.0823	0.3782	0.0221
Weighted empirical residual	0.0656	0.0780	0.0421	0.0805	0.3693	0.0203

QTE: Quantile treatment effect
GAN: Generative adversarial network
ETC-GAN: Empirical-transport Cramér GAN
CDF: Cumulative distribution function
CDF-DiD: CDF difference-in-differences
CiC: Changes-in-changes
MAE: Mean absolute error
ATT: Average treatment effect on the treated
CRPS: Continuous ranked probability score
PIT: Probability integral transform
DR-CDF: Doubly robust CDF correction
ADV off: Generator adversarial term disabled
ET-Quantile: Empirical-transport quantile learner
N: Number of units
T: Number of periods
T_0: Number of pre-treatment periods

Notes: Entries average the same twenty new distribution-shift panels used in the ATT analysis, with $N = 300$, $T = 48$, $T_0 = 36$, and true QTEs equal to 2. The target pools treated unit–period outcomes before quantile inversion. The primary ETC-GAN estimator is the draw plug-in, without DR-CDF correction. Overall averages QTE MAE over $.05, .10, \ldots, .95$; Lower, Central, and Upper average .05–.30, .35–.65, and .70–.95. CRPS and PIT error assess conditional distribution accuracy and calibration; lower is better. These conditional scores are not assigned to marginal CiC or CDF-DiD (–). Earlier seeds are not pooled. ADV off retains the same inputs, random numbers, critic updates, generator budget, and checkpoint eligibility, setting only the generator adversarial coefficient to zero. ET-Quantile has the same generator-step budget; empirical residuals use the shared mean and weights. Residual-transport initialization is fixed across data seeds.

Table 7: Paired QTE-error differences on the twenty new panels

Comparator	Readout	Δ MAE	MCSE	ETC-GAN wins
Changes-in-changes (pooled)	Draw plug-in	-0.22650	0.03247	20/20
CDF-DiD (pooled)	Draw plug-in	-0.22016	0.03497	20/20
ETC-GAN (ADV off)	Draw plug-in	-0.00025	0.00014	13/20
ET-Quantile	Draw plug-in	-0.00061	0.00169	14/20
Weighted empirical residual	Draw plug-in	-0.00134	0.00259	11/20
ETC-GAN (ADV off)	CDF plug-in	-0.00029	0.00014	14/20
ET-Quantile	CDF plug-in	-0.00071	0.00173	14/20
Weighted empirical residual	CDF plug-in	-0.00120	0.00259	11/20
ETC-GAN (ADV off)	CDF targeted	0.00007	0.00007	9/20
ET-Quantile	CDF targeted	0.00177	0.00069	8/20
Weighted empirical residual	CDF targeted	0.00863	0.00330	5/20

QTE: Quantile treatment effect
GAN: Generative adversarial network
ETC-GAN: Empirical-transport Cramér GAN
CDF: Cumulative distribution function
CDF-DiD: CDF difference-in-differences
CiC: Changes-in-changes
MAE: Mean absolute error
ADV off: Generator adversarial term disabled
ET-Quantile: Empirical-transport quantile learner
MCSE: Monte Carlo standard error
Δ: First method minus comparator

Notes: Differences are ETC-GAN minus the comparator's same-panel overall QTE MAE. MCSE is the sample standard deviation of the twenty paired differences divided by $\sqrt{20}$; wins count strictly lower panel-level MAE. Readout refers to ETC-GAN; the pooled comparators use their marginal estimates. Draw plug-in is primary; CDF plug-in and targeted are diagnostic readouts. These are Monte Carlo uncertainties, not effect-estimator standard errors.

in eleven; the average differences are -0.000255 and -0.001337, with Monte Carlo standard errors 0.000145 and 0.002594. Relative to ET-Quantile, it has smaller error in fourteen panels, with a mean difference of -0.000611 (MCSE 0.001688). ETC-GAN's overall MAE is numerically lowest, but its differences from these state-adjusted learners are small. The large gap from the pooled benchmarks is shared across learners rather than specific to adversarial training.

The empirical baseline has the lowest CRPS and PIT error, consistent with the design's independent Gaussian structural disturbance. Its CRPS is 0.36935, compared with ETC-GAN's 0.37967, and it has lower CRPS in all twenty panels. ETC-GAN minus the empirical baseline has mean CRPS difference 0.01033 (MCSE 0.00149). The gain in marginal-QTE accuracy over the pooled comparators does not imply uniformly better conditional-distribution scores.

Definition of the retained logistic correction. This secondary readout is distinct from the primary draw plug-in QTE. Let $\widehat{H}_{it}(y)$ be the out-of-fold conditional CDF obtained by linearly interpolating the nineteen predicted quantiles on the common 401-point outcome grid, with values zero and one beyond the predicted quantile range. Let w_i be the shared control weights: clip propensity estimates to $[.05, .95]$, form control odds, normalize to sum one, cap at $5/N_{\text{co}}$, and renormalize. Write $\widetilde{H}_{it} = \text{clip}(\widehat{H}_{it}, 10^{-5}, 1 - 10^{-5})$. At each threshold with nonconstant control indicators, the frozen routine seeks a common intercept $\delta(y)$ satisfying

$$\frac{\sum_{i \in \mathcal{C}} \sum_{t > T_0} w_i \left[\mathbf{1}\{Y_{it} \leq y\} - L\{\text{logit}\, \widetilde{H}_{it}(y) + \delta(y)\} \right]}{T_{\text{post}} \sum_{i \in \mathcal{C}} w_i} = 0, \tag{169}$$

where L is the logistic function and $T_{\text{post}} = 12$. Starting at zero, its bounded Newton iteration allows at most forty updates, clips each increment to $[-2, 2]$ and each intercept to $[-12, 12]$, and leaves the intercept at zero for thresholds with constant control indicators. Applying the same logistic shift to $\widetilde{H}_{it}(y)$ for treated units, the updated CDFs are averaged equally over treated units and all twelve post-treatment periods. The pooled curve is clipped

to $[0,1]$, isotonic-projected, and given endpoints zero and one with cumulative-maximum repair before linearly interpolated inversion. Subtracting that inverse from the observed pooled treated quantile gives the targeted QTE; no further augmentation is added. Exact grid construction, solver tolerances and repair are preserved in the source capsule in `publication_artifacts/qte_shared20_20261018`: `simulations/run_qte_boundary.py` (`common_grid`, `evaluate`), `gscgan/simulations/run_etcgan_ldml_qte_experiment.py` (`logistic_fluctuation`), and `gscgan/simulations/run_etcgan_dr_qte_experiment.py` (`project_cdf`, `invert_cdf`).

Targeting and finite-sample limits. The original logistic CDF-targeting update is evaluated separately, without selecting a preferred correction strength after inspecting this batch. ETC-GAN's draw-plugin MAE is 0.06422; using its CDF grid before targeting gives 0.06418, while the targeted readout gives 0.09215. Targeting increases error in eighteen panels and reduces it in two. Relative to the draw plug-in, the mean paired increase is 0.02793 (MCSE 0.00463), or about 43 percent. The small draw-to-grid difference does not explain this deterioration. Table 8 preserves the original readouts for every applicable learner, and Table 9 retains each primary seed-level error. These results favor the plug-in estimate over the logistic correction in this design. They assess point accuracy. The fixed-window anchor construction leaves the generated-state rates and cross-fold oracle-equivalence conditions in Appendix A to be established for inference.

Reproduction. The separate release `qte_shared20_20261018` binds executed source commit `e526ec3e`, configuration, panel and fold identities, nuisance predictions, raw distribution draws, and checksummed task receipts. Its materializer recomputes scores and paired summaries before generating the main and SI tables. The portable release inventories omitted training files, retained in the original archive, and provides frozen sources and environment contracts for independently configured training reproduction. Paired Monte Carlo standard errors use the sample standard deviation of within-panel differences divided by $\sqrt{20}$. The pre-

Table 8: Distribution readouts before and after the original CDF correction

Method	Readout	Overall	Lower	Central	Upper	MCSE(Overall)
ETC-GAN	Draw plug-in	0.0642	0.0710	0.0452	0.0796	0.0066
ETC-GAN	CDF plug-in	0.0642	0.0706	0.0449	0.0802	0.0065
ETC-GAN	CDF targeted	0.0922	0.0935	0.0755	0.1103	0.0072
Changes-in-changes (pooled)	Marginal plug-in	0.2907	0.3772	0.2629	0.2367	0.0315
CDF-DiD (pooled)	Marginal plug-in	0.2844	0.3792	0.2585	0.2198	0.0339
ETC-GAN (ADV off)	Draw plug-in	0.0645	0.0713	0.0453	0.0800	0.0066
ETC-GAN (ADV off)	CDF plug-in	0.0645	0.0709	0.0451	0.0806	0.0066
ETC-GAN (ADV off)	CDF targeted	0.0921	0.0933	0.0755	0.1103	0.0072
ET-Quantile	Draw plug-in	0.0648	0.0722	0.0435	0.0823	0.0061
ET-Quantile	CDF plug-in	0.0649	0.0721	0.0433	0.0829	0.0060
ET-Quantile	CDF targeted	0.0904	0.0915	0.0735	0.1090	0.0071
Weighted empirical residual	Draw plug-in	0.0656	0.0780	0.0421	0.0805	0.0050
Weighted empirical residual	CDF plug-in	0.0654	0.0778	0.0418	0.0805	0.0049
Weighted empirical residual	CDF targeted	0.0835	0.0847	0.0685	0.0999	0.0064

QTE: Quantile treatment effect
GAN: Generative adversarial network
ETC-GAN: Empirical-transport Cramér GAN
CDF: Cumulative distribution function
CDF-DiD: CDF difference-in-differences
CiC: Changes-in-changes
MAE: Mean absolute error
ADV off: Generator adversarial term disabled
ET-Quantile: Empirical-transport quantile learner
MCSE: Monte Carlo standard error

Notes: Entries are QTE MAEs, with MCSE for overall MAE. CDF plug-in inverts the common-grid projected CDF without targeting; CDF targeted uses the original full logistic fluctuation and monotonicity projection. For ETC-GAN, targeting increases MAE relative to the primary draw plug-in on 18 of 20 panels and decreases it on 2. Comparing the two CDF readouts isolates targeting from numerical CDF construction. No shrinkage or half-step modification is used. All adverse results are retained.

Table 9: Panel-level QTE MAEs for all twenty registered seeds

Seed	ETC-GAN	ADV off	Empirical	ET-Quantile	CiC	CDF-DiD
2026101701	0.02583	0.02566	0.02765	0.03156	0.26235	0.25556
2026101702	0.10913	0.10928	0.10560	0.11296	0.47825	0.45539
2026101703	0.04930	0.04910	0.04594	0.06742	0.27105	0.28525
2026101704	0.05628	0.05636	0.06203	0.05722	0.26375	0.23367
2026101705	0.03276	0.03272	0.05175	0.03322	0.53357	0.55112
2026101706	0.03357	0.03354	0.04021	0.03194	0.09911	0.07822
2026101707	0.11216	0.11239	0.09864	0.11272	0.16325	0.18498
2026101708	0.05342	0.05520	0.04614	0.05615	0.31006	0.32369
2026101709	0.09284	0.09246	0.08083	0.09339	0.13709	0.12513
2026101710	0.05964	0.05942	0.06689	0.05864	0.39587	0.38346
2026101711	0.14022	0.14093	0.11746	0.12433	0.22355	0.17887
2026101712	0.07758	0.07754	0.07098	0.08071	0.31156	0.32174
2026101713	0.07078	0.07297	0.05574	0.05466	0.25292	0.26611
2026101714	0.06293	0.06295	0.07695	0.06225	0.14885	0.14146
2026101715	0.05038	0.05101	0.06479	0.04238	0.18538	0.15513
2026101716	0.07191	0.07197	0.07108	0.07309	0.66883	0.69639
2026101717	0.05347	0.05371	0.06056	0.06166	0.25202	0.20369
2026101718	0.03874	0.03878	0.05196	0.04461	0.37101	0.38691
2026101719	0.04941	0.04943	0.06425	0.05034	0.25249	0.26309
2026101720	0.04411	0.04413	0.05174	0.04741	0.23341	0.19780

QTE: Quantile treatment effect
GAN: Generative adversarial network
ETC-GAN: Empirical-transport Cramér GAN
CDF: Cumulative distribution function
CDF-DiD: CDF difference-in-differences
CiC: Changes-in-changes
MAE: Mean absolute error
ADV off: Generator adversarial term disabled
ET-Quantile: Empirical-transport quantile learner
Empirical: Weighted empirical residual comparison

Notes: All six methods use the same shifted panel in each row. ETC-GAN and the state-adjusted learners use their primary draw plug-in; pooled benchmarks use their marginal estimates. Historical and smoke panels are excluded.

ceding eight-panel `qte_shared_dgp_20261014` package and earlier checks remain unchanged. The thirty-panel no-shift benchmark in Appendix C.2 has a different DGP and configuration and is not treated as extra replications of this batch.

Appendix C.5 Supplementary no-shift checks on saved panels

Two completed two-seed pilots connect the ATT and QTE analyses under the later three-fold, cross-validated-rank configuration. Both retain the twenty-seed experiment's method definitions and training ceilings. Their small batches remain separate from the twenty-panel and thirty-panel benchmarks, and their primary QTE readout is the draw plug-in without DR-CDF correction.

QTE completion of the original no-shift design (8828). This pilot uses the saved no-shift ATT panels for seeds 2026101701 and 2026101702 in Appendix C.1, with 115 and 105 treated units, respectively. It does not generate replacement panels. All candidate and final factor fits converge, and both panels select rank two. The QTE branch applies its separate regression mean and all six retained residual or pooled-outcome comparators. True pooled QTEs equal two throughout.

ETC-GAN's average QTE MAE is 0.07545, compared with 0.07482 for strict noADV, 0.07720 for ET-Quantile, and 0.06410 for weighted empirical residuals. CiC and CDF-DiD have 0.17702 and 0.16639. The gap from pooled procedures is present without an imposed covariate shift, while the empirical residual baseline is more accurate in both panels. These two panels complete a shared-input illustration; their average cannot serve as a twenty-seed no-shift QTE benchmark.

One panel for ATT and heterogeneous QTE (8829). The second pilot uses seeds 2026102301 and 2026102302, again with $N = 300$, $T = 48$, and $T_0 = 36$. It retains $Z_i \sim U[-1.5, 1.5]$, $\lambda_{ik} \sim U[-\sqrt{3}, \sqrt{3}]$, and $W_{it} \sim U[-2, 2]$ in every period. The two factors are independent stationary Gaussian AR(1) paths, with coefficient 0.7 and innovation variance

Table 10: Complete plug-in QTE comparison in pilot 8828

Method	2026101701	2026101702	Mean MAE	CRPS	PIT error
ETC-GAN	0.06082	0.09009	0.07545	0.38751	0.02650
ETC-GAN (noADV)	0.05954	0.09011	0.07482	0.38777	0.02675
ET-Quantile	0.06061	0.09380	0.07720	0.38634	0.02646
Weighted empirical residual	0.05450	0.07369	0.06410	0.37423	0.01983
CiC (pooled)	0.07977	0.27428	0.17702	–	–
CDF-DiD (pooled)	0.08932	0.24347	0.16639	–	–

QTE: quantile treatment effect. ETC-GAN: empirical-transport Cramér generative adversarial network. ET-Quantile: empirical-transport quantile learner. noADV: generator adversarial term disabled. CiC: changes-in-changes. CDF-DiD: cumulative distribution function difference-in-differences. MAE: mean absolute error. CRPS: continuous ranked probability score. PIT: probability integral transform.

Notes: The seed columns give overall QTE MAE, and the last three columns average the two saved panels. All six methods use the same saved panels. The four state-adjusted learners share nuisance inputs and their frozen budgets; the two pooled procedures use observed group–period distributions. The empirical baseline has the lowest average MAE, CRPS, and PIT error in this pilot; ETC-GAN and strict noADV are close. These two-panel summaries are not combined with any other batch. The primary readout pools generated untreated outcomes before quantile inversion and applies no DR-CDF correction.

0.51. Treatment is a single Bernoulli draw with probability

$$e(Z_i, \lambda_{i1}) = 0.10 + 0.50L(Z_i + 0.8\lambda_{i1}), \qquad L(v) = (1 + e^{-v})^{-1}.$$

Thus a latent loading affects both treatment and the untreated outcome. The untreated mean retains the centered nonlinear response in Appendix C.1, but the disturbance standard deviation is $\sigma_{it} = 0.3\{1 + 0.5\tanh(Z_i + W_{it})\}$. Treatment adds

$$\tau_i = 2 + 0.75\{\tanh(Z_i) - m_{\text{tr}}\}, \qquad m_{\text{tr}} = \mathbb{E}\{\tanh(Z_i) \mid D_i = 1\} = 0.15663462536.$$

This gives population ATT two, but realized treated-unit ATTs 2.035389 and 2.019926. QTE truths are quantile differences between the saved treated post-treatment $Y(1)$ and $Y(0)$ samples, not quantiles of τ_i or a constant vector of two. The two branches use identical observed panels, folds, and weights, with all truth arrays excluded from fitting.

FGAN's treated-post conditional-mean RMSE is 0.51817 and 0.44829, compared with LFM's 1.47109 and 1.41654. This improvement does not translate into lower aggregate ATT error: LFM's two-panel ATT RMSE is 0.01408, versus 0.04329 for DR-FGAN and

Table 11: Complete ATT comparison in the unified no-shift pilot

Method	2026102301	2026102302	Bias	MAE	RMSE
LFM-CV	2.04977	2.03370	0.01407	0.01407	0.01408
FGAN	2.05564	2.01702	0.00867	0.01158	0.01446
DR-LFM-CV	2.09969	2.01104	0.02771	0.03660	0.04590
DR-FGAN	2.06776	2.07189	0.04217	0.04217	0.04329
NN-MSE	2.02436	2.00781	-0.01158	0.01158	0.01159
DR-NN-MSE	2.06491	2.07527	0.04243	0.04243	0.04435
HGB	2.02966	2.00429	-0.01068	0.01068	0.01177
DR-HGB	2.05219	2.01632	0.00660	0.01021	0.01215
ASC-ref	2.14357	2.00322	0.04574	0.06245	0.07741
GSC-ref	2.07714	2.04697	0.03440	0.03440	0.03517
FGAN-noADV	2.03302	2.01330	-0.00450	0.00450	0.00498
DR-FGAN-noADV	2.06833	2.06731	0.04016	0.04016	0.04081
TWFE-ref	2.10771	1.88087	-0.03337	0.10568	0.11083
TWFE-X	2.08440	1.88451	-0.04320	0.09221	0.10183
SDID-ref	2.07884	2.01236	0.01794	0.02551	0.03119
SDID-X	2.07496	2.02613	0.02289	0.02289	0.02832
ASC-X	2.14412	1.91343	0.00112	0.10761	0.10762

ATT: average treatment effect on the treated. DR: doubly robust. FGAN: factor-augmented generative adversarial network. LFM: linear factor model. GSC: generalized synthetic control. ASC: augmented synthetic control. SDID: synthetic difference-in-differences. TWFE: two-way fixed effects. NN-MSE: neural-network mean-squared-error learner. HGB: histogram gradient boosting. noADV: generator adversarial term disabled. MAE: mean absolute error. RMSE: root mean squared error.

Notes: The seed columns are ATT estimates. The realized treated-unit ATT is 2.035389 in the first panel and 2.019926 in the second; errors use these targets rather than the population value two. All seventeen fitted methods are retained. The DR correction increases FGAN's error in both panels; the LFM aggregate ATT error is small despite its less accurate conditional mean. CV denotes the common rank-selection rule; ref denotes the official reference implementation; X adds observed covariates.

Table 12: Complete plug-in QTE comparison in pilot 8829

Method	2026102301	2026102302	Mean MAE	CRPS	PIT error
ETC-GAN	0.07338	0.06164	0.06751	0.42345	0.02988
ETC-GAN (noADV)	0.07355	0.06165	0.06760	0.42346	0.02984
ET-Quantile	0.06782	0.05726	0.06254	0.41813	0.02812
Weighted empirical residual	0.06930	0.03603	0.05267	0.41079	0.02137
CiC (pooled)	0.13492	0.19454	0.16473	–	–
CDF-DiD (pooled)	0.12089	0.18784	0.15436	–	–

QTE: quantile treatment effect. ETC-GAN: empirical-transport Cramér generative adversarial network. ET-Quantile: empirical-transport quantile learner. noADV: generator adversarial term disabled. CiC: changes-in-changes. CDF-DiD: cumulative distribution function difference-in-differences. MAE: mean absolute error. CRPS: continuous ranked probability score. PIT: probability integral transform.

Notes: The seed columns give overall QTE MAE, and the last three columns average the two saved panels. All six methods use the same saved panels. The four state-adjusted learners share nuisance inputs and their frozen budgets; the two pooled procedures use observed group–period distributions. The empirical baseline has the lowest average MAE, CRPS, and PIT error in this pilot; ETC-GAN and strict noADV are close. These two-panel summaries are not combined with any other batch. The primary readout pools generated untreated outcomes before quantile inversion and applies no DR-CDF correction.

0.04081 for DR-FGAN-noADV. The FGAN plug-in errors are 0.02025 and −0.00291; residual correction adds 0.01212 and 0.05487, leaving DR-FGAN errors 0.03237 and 0.05196. The centered response can average out in the ATT even when the linear cell-level predictions are inaccurate.

ETC-GAN's QTE MAE averages 0.06751, close to strict noADV's 0.06760, and above empirical residuals' 0.05267 and ET-Quantile's 0.06254. CiC and CDF-DiD have 0.16473 and 0.15436. The ETC-GAN lower, central, and upper-region MAEs are 0.03959, 0.05138, and 0.11425; the upper-quantile QTEs are the least accurate. The empirical learner also has smaller CRPS and PIT error. No LFM or GSC conditional-distribution estimator is fitted in this QTE comparison.

Table 13: Rank and weight diagnostics in the unified pilot

Seed	N_{tr}	N_{co}	Rank	Clip low	Clip high	Capped	ESS	Max weight
2026102301	101	199	2	19	3	8	64.02	3.19%
2026102302	105	195	2	11	0	4	77.65	2.71%

N_{tr}: treated units. N_{co}: control units. ESS: effective sample size of normalized control weights. ATT: average treatment effect on the treated. QTE: quantile treatment effect.

Notes: Both panels select rank two with all final fits converged. Counts refer to propensity clipping below .05 or above .95 and control-odds capping before the final renormalization. The effective sample sizes are well below the control counts. A final weight can exceed $5/N_{co}$ because the cap precedes renormalization. ATT and QTE share the fold map and stabilized weights.

Both branches select rank two in all folds. Estimated propensity RMSEs relative to the true probabilities are 0.20226 and 0.19604; the fitted weights concentrate on fewer controls than the original counts of 199 and 195. True probabilities lie between approximately 0.13 and 0.57 in these panels, so fitted concentration is not evidence of a structural overlap failure. The saved decomposition reports the residual correction alongside clipping, capping, and effective sample size rather than attributing the ATT error solely to rank choice.

Scope and reproduction. The unified design retains a centered nonlinear remainder and a pervasive linear latent component. Under its selection rule the appropriate control-pre projection coefficient is $b_{\text{co}} = -0.1263145$, not the coefficient from the original Z-only

assignment. Its stationary Gaussian factors have finite moments and geometric mixing, but do not satisfy a uniformly bounded factor-path condition over expanding horizons. These structural checks do not establish generated-state inference rates.

The numerical receipts, configurations, shared-panel audit, and saved weight and mean diagnostics are archived at `publication_artifacts/noshift_pilots_20261023`. The deterministic materializer verifies their checksums and recreates every table from these outputs without training or generating panels. The receipts also retain the secondary CDF-grid and logistic-targeted readouts; neither replaces the primary plug-in result.

Appendix C.6 Pooled distributional benchmark definitions

The observable-outcome comparators in Appendix C.2, Appendix C.4, and Appendix C.5 use pooled group–period distributions, not the estimated factor state. The following constructions are applied separately to every panel, with the same evaluation quantiles as ETC-GAN.

Let $H_{g,s}$ denote the empirical distribution of observed outcomes in group $g \in \{\mathrm{tr}, \mathrm{co}\}$ and period block $s \in \{\mathrm{pre}, \mathrm{post}\}$. Each block pools its unit–period observations uniformly. The changes-in-changes (CiC) benchmark maps every treated pre-treatment observation y to $H^{-1}_{\mathrm{co,post}}\{H_{\mathrm{co,pre}}(y)\}$ and uses the empirical distribution of these transported observations as the untreated post-treatment counterfactual. The transport map uses right-continuous empirical CDFs and a left generalized inverse. Quantiles of the transported sample and the realized target use the same linear interpolation convention. Identification requires a common monotone time transformation, time-stable group-specific latent distributions, and support conditions (Athey and Imbens 2006).

The CDF difference-in-differences (CDF-DiD) benchmark first forms

$$H_0^{\mathrm{raw}}(y) = H_{\mathrm{tr,pre}}(y) + H_{\mathrm{co,post}}(y) - H_{\mathrm{co,pre}}(y).$$

This requires parallel untreated CDF changes at every outcome threshold, a distributional

restriction related to transformation-invariant parallel trends (Roth and Sant'Anna 2023). The raw curve is evaluated on the union of the three observable outcome supports, projected by equal-knot isotonic regression, and bounded to $[0, 1]$ before left-generalized-inverse inversion. The repair produces a valid finite-sample CDF. Multi-period pooling and these interpolation and shape-repair conventions are our implementation choices. Both benchmarks use only observed outcomes for fitting, support selection, and tuning. The observed treated post-treatment quantiles are then differenced from each estimated untreated quantile.

Neither the no-shift mixture benchmark nor the centered-response design imposes either pooled benchmark's identifying restrictions. This comparison assesses state-adjusted reconstruction against pooled distributional procedures; the later strict noADV comparison isolates the adversarial term. The thirty-panel ET-Quantile comparison instead removes critic training and is not a matched objective-only intervention. Conditional CRPS and PIT scores are not assigned to the two marginal procedures because they do not estimate the same conditional distribution object. The complete current results, including the non-adversarial and empirical-residual comparisons, are retained separately for each design in the preceding subsections.

Appendix C.7 Separate-design ATT influence-score calibration

The ATT IF V2 formal50 experiment assesses the ATT influence-score interval under a separate design with $N = 550$, $T_0 = 20$ and $T_{\text{post}} = 10$. Write $a_{it} = \lambda_{i1} f_{t1}$, $b_{it} = \lambda_{i2} f_{t2}$ and $L(v) = (1 + e^{-v})^{-1}$. The untreated law is

$$\begin{aligned} Y_{it}(0) = {} & 0.2 X_{it}'(1.5, -0.8)' + 0.2(a_{it} + b_{it}) + 4 \tanh(a_{it} + 0.5 b_{it}) + 3 L(a_{it}) b_{it} \\ & + 2 X_{it,1} L(a_{it}) + 1.5 X_{it,2} \tanh(b_{it}) + (0.2 + 0.15|a_{it}|)\xi_{it}. \end{aligned} \tag{170}$$

The two loadings are independent uniform draws on $[-\sqrt{3}, \sqrt{3}]$; f_t, X_{it} and ξ_{it} have mutually independent standard Gaussian coordinates. Thus unit paths are iid conditional on the

common factor path. Treatment is drawn once as $D_i \sim \text{Bernoulli}\{L(\alpha + 0.30\lambda_{i1})\}$, where α makes $\mathbb{E}D_i = 0.30$, without treated-count rejection or redraw. At post-treatment event $k = 0, \ldots, 9$, the treatment effect is $(2 + 0.5\lambda_{i1})(1 + 0.2k)$; the population targets average this effect over the treated loading distribution, and then over the ten events for the pooled target. This is separate from both main-text benchmark DGPs.[1]

In each of fifty panels the estimator reruns fold-specific Stage 1 at the registered fixed rank four, the common adjustment state, a whole-unit out-of-fold restricted-loading logistic propensity fit, and the adversarial ATT generator. This historical generator starts from its GSC-based initialization and enters adversarial training directly; it has neither the current ATT generator warm start nor the separate QTE regression-mean fit. The pointwise 95% interval uses only that replication's centered whole-unit out-of-fold Hájek influence score. Known DGP propensities remain diagnostic and never replace estimated propensities in the feasible score.

The pooled pointwise coverage is 0.940 with Monte Carlo standard error 0.034. The mean estimated standard error is 0.0768, SD(error) is 0.0767, and their ratio is 1.001; mean interval length is 0.301. The pooled bias is 0.0059 and its RMSE is 0.0762. Event-time pointwise frequencies range from 0.900 to 0.980, with Monte Carlo standard errors from 0.020 to 0.042. All Stage 1 folds converge, all GAN folds record positive critic updates, and all panels pass the treated-post exclusion gate without propensity clipping, capped controls, or binding stabilization. This check concerns pointwise coverage under the stated design. Coverage for the shared exposure-shift simulations and applications, and simultaneous coverage across events, require separate analysis. The replication archive contains the design, release records, and complete diagnostics.

[1]The release identity and frozen-run inventory are in `simulations/evidence_runs/att_if_coverage/formal50_iid_population_oof_if_v2_release`. They record base source commit `030771c862d9ce30670ae6455a8e9eac594964f9`; the frozen-run file inventory separately binds the executed sources. An archived, byte-identical copy of the frozen DGP module is `publication_artifacts/qte_shared_dgp_20261014/frozen_run/simulations/att_inference_dgp.py`, functions `generate_iid_dgp1_att_v2` and `population_iid_att_truth_v2`; its checksum agrees with the inventoried `frozen_run_spec.json`.

Table 14: Separate-design ATT IF V2 formal50 pointwise calibration. Coverage and its MCSE are computed across fifty iid Bernoulli panels. Event rows target `ATT-POP-T`; the pooled row targets `ATT-POP-POOL`. Mean SE averages the same-replication influence-score SE; SD(error) measures the Monte Carlo dispersion of the point estimates; intervals use the same-replication influence-score SE.

Target	Coverage	MCSE	Mean SE	SD(error)	SE/SD	Mean length
Event 0	0.900	0.042	0.067	0.067	0.992	0.261
Event 1	0.960	0.028	0.072	0.074	0.963	0.281
Event 2	0.960	0.028	0.084	0.087	0.957	0.328
Event 3	0.940	0.034	0.086	0.080	1.077	0.338
Event 4	0.920	0.038	0.092	0.090	1.018	0.359
Event 5	0.940	0.034	0.099	0.090	1.107	0.389
Event 6	0.980	0.020	0.107	0.099	1.082	0.419
Event 7	0.940	0.034	0.114	0.106	1.076	0.446
Event 8	0.940	0.034	0.119	0.118	1.006	0.467
Event 9	0.960	0.028	0.135	0.118	1.142	0.527
Pooled	0.940	0.034	0.077	0.077	1.001	0.301

ATT: Average treatment effect on the treated
IF: Influence function
MCSE: Monte Carlo standard error
SE: Standard error
SD(error): Standard deviation of estimation errors
iid: Independent and identically distributed
V2 formal50: Version-two calibration with fifty replications
ATT-POP-T: Population ATT at a given event time
ATT-POP-POOL: Population ATT averaged over the post-treatment events

Appendix C.8 Exploration history and archived evidence

The centered-response design was selected through exploratory screening and then evaluated on successive new-seed batches. Its current tables use only the twenty-seed expansion: paired shift/no-shift ATT panels and the same twenty shift panels for QTE. The independent thirty-panel QTE benchmark retains its original mixture DGP, fixed rank, and longer training budget; the two recent pilots are reported separately. No estimates are pooled across these designs or configurations. The previous three-seed ATT and eight-seed QTE releases remain at `publication_artifacts/att_anchored_shift_20261013` and `publication_artifacts/qte_shared_dgp_20261014`; their original tables and SI discussions remain inactive manuscript sources in GitHub. The expansion preserves the shift-design gain over LFM and pooled QTE comparators, while changing the no-shift ATT comparison from an earlier LFM advantage to essentially equal DR-FGAN and LFM RMSEs. Earlier ATT nonlinear-factor checks did not reproduce an initial plug-in adversarial advantage; correction sometimes worsened FGAN, and linear or other nonlinear learners sometimes performed better. An observed-covariate precursor helped motivate the exposure-shift mechanism but contained no latent factor component, so its favorable center ranking did not validate factor recovery. Its unfavorable no-shift and weaker-overlap neighbors remain archived with the favorable cells.

Separate QTE mixture and residual-shape explorations found that full CDF correction could worsen point accuracy. A selected weight-shrinkage and half-step correction improved some comparisons with full correction, but not uniformly with the plug-in or non-adversarial alternatives; later seeds reversed early adversarial rankings. Those exploratory corrections are not the primary estimator and do not inherit the original full-score guarantees. The archived inference history also retains the failed unbounded-overlap QTE pilot, the screen-only bounded-overlap pilot, and the separate completed, conservative targeted-QTE calibration. None supplies coverage for the current plug-in QTE, shared-design simulations, or applications.

The other original DGP 1 and DGP 2 comparisons, sensitivity studies, all completed mechanism screens and adverse results, retired trainable-embedding initialization experiments, and the semi-synthetic design without a usable numerical-results archive are preserved in the repository. The pre-cleanup manuscript source is at `historical_archive/manuscript_pre_cleanup_20261016/source/manuscript`; its linked simulation releases preserve the full numerical and selection history. Apart from the explicitly reinstated thirty-panel plug-in QTE benchmark, these archived estimates are unchanged and excluded from the current Monte Carlo summaries.

Appendix D Geithner application: design and additional results

Appendix D.1 Sample and pre-event information

Both estimators use the same 622 firms: 34 connected firms and 588 controls. The market-index record is excluded. The 100 pre-event daily returns run from July 2 through November 20, 2008; the ten post-event daily returns run from November 21 through December 5. Market holidays listed in the source replication program are removed before computing consecutive trading-day return-index ratios. Thus the November 21 return uses the November 20 close, and November 27 is excluded while November 28 is retained. We use close-to-close daily returns throughout; the source study instead uses a one-hour return for the announcement. Returns are expressed in percent and clipped at -60 and 60. Eligibility requires finite return indices and returns throughout the window and nonzero returns on more than half of the pre-event dates. Neither sample inclusion nor adjustment uses annual financial variables. The adjustment variables are estimated factors and loadings together with summaries of pre-event outcomes and linear-factor residuals. No external financing-pressure series is measured in this application.

Annual financial data require a separate timing check because a fiscal period end does not establish when the corresponding figures became public. Thirty-three of the 34 connected firms have FY2008 fiscal period ends after the event. Franklin Resources, Inc. (BEN), the remaining firm, had a September 30 period end and published unaudited preliminary FY2008 results on October 23; its Form 10-K was filed on November 25.[2] The annual file has no point-in-time date or extraction vintage, and we could not verify when BEN's exact four-field Worldscope vector entered the database. The fiscal dates therefore rule out 33 connected firms, while BEN's preliminary release does not establish that all four variables

[2]Franklin Resources, Inc., "Franklin Resources, Inc. Announces Fourth Quarter Results," October 23, 2008; SEC accession 0001193125-08-243375.

were available at the cutoff. We exclude annual 2008 and FY2007 financial fields from estimation and sample selection.

Earlier financial-covariate fits, targeted-CDF results, and their uncertainty diagnostics are retained in the dated manuscript archive at `historical_archive/manuscript_pre_cleanup_20261016/` and its linked replication evidence, separately from the current estimates.

Appendix D.2 Plug-in distributional estimates

ETC-GAN pools the fitted untreated distributions for the 34 connected firms over ten post-event days and compares their quantiles with observed returns. Table 15 reports the full pre-specified nineteen-quantile grid. Saved predictions allow these values to be recomputed directly. The label `QTE-REAL-POOL` denotes the equally weighted sample firm–day calculation. In the bootstrap analysis, that calculation estimates the corresponding firm-population mixture at the fixed event dates. Because pooling precedes quantile inversion, the target differs from both an average of daily QTEs and the distribution of firm-level effects. Point estimates and intervals use the same firm sample and daily-return convention.

The application interpolates conditional CDFs from the monotone quantile knots on a 401-point outcome grid. It then averages the CDFs and interpolates the first crossing of each requested probability, so evaluation requires no additional random draws. The selected rank is six. The weighting rule clips 586 of 622 propensities at its lower bound and caps three control weights; the control effective sample size is 500.919 of 588. These diagnostics measure the extent of stabilization, not population overlap.

Table 15: All plug-in quantile effects in the Geithner application

Quantile	Observed (%)	Counterfactual (%)	QTE (pp)
0.05	-15.952	-15.430	-0.522
0.10	-8.839	-10.924	2.085
0.15	-4.092	-7.737	3.645
0.20	-2.301	-5.496	3.195
0.25	-0.522	-3.865	3.343
0.30	0.000	-2.554	2.554
0.35	0.758	-1.431	2.189
0.40	1.408	-0.429	1.837
0.45	2.319	0.511	1.807
0.50	3.309	1.432	1.876
0.55	4.459	2.382	2.077
0.60	5.545	3.387	2.158
0.65	6.167	4.474	1.693
0.70	6.863	5.695	1.168
0.75	7.935	7.105	0.830
0.80	8.853	8.818	0.035
0.85	10.311	11.109	-0.798
0.90	12.895	14.689	-1.794
0.95	19.629	21.554	-1.925

QTE: Quantile treatment effect pp: Percentage points
CDF: Cumulative distribution function

The sample contains 622 firms, including 34 connected firms. Quantiles pool the 340 connected firm-day observations across ten post-event trading dates. Returns are daily percentages; differences are percentage points. The counterfactual mixes fitted conditional no-connection distributions before quantile inversion. No CDF correction is applied. Table 18 supplies the corresponding pointwise bootstrap intervals.

Appendix D.3 ATT and QTE estimation on the common firm sample

Table 4 reports estimates from separate mean and distribution learners on the same sample of 622 firms, including 34 connected firms. Both use 100 pre-event trading days and ten post-event trading days beginning November 21, 2008. Eligibility depends on the return panel alone. Adjustment uses pre-event outcomes and factors; the simulation's $X_{it} = (Z_i, W_{it})'$ has no measured financing-pressure counterpart in this application.

The outer split uses two stratified whole-firm folds and a fixed seed 20260831. Each fold scores ranks one through ten using treated-pre-period leave-one-time-out prediction.

Averaging those scores chooses a common rank six; the fold-specific minimizers are six and eight. The final IFE fits at rank six. Both estimators use identical factor inputs, propensity predictions and normalized control weights. The replication code verifies this equality before combining the separately fitted ATT and QTE results.

For ATT, WGAN-GP holds the Stage-1 loadings and factors fixed and predicts each firm's mean using a network trained on other firms. The generator and spectrally normalized critic each have three hidden layers of width 128; the factor projection has dimension 16 and the noise dimension is one. Each training step uses a minibatch of 256 and five critic updates. The ceiling is 3,000 generator updates per fold, with patience 500. The two folds stop after 1,247 and 2,879 generator updates; their selected checkpoints follow updates 747 and 2,379. The cosine learning-rate schedule starts at 2×10^{-4} and would reach 10^{-5} at the ceiling; the gradient penalty coefficient is ten. The saved configuration records the remaining objective coefficients and random draws. Treated pre-event calibration and checkpoint selection use separate whole-unit subsets. Selection is evaluated after an adversarial update. Four fixed noise draws enter pre-event evaluation and fifty enter post-event prediction. Because the trainer requires an observed-covariate column, this application passes a zero column with a zero coefficient. It does not change the factor fit or weights and represents no measured covariate.

The propensity model is standardized L2 logistic regression with fixed $C = 0.5$, cross-fitted on the same outer folds. Its probabilities are clipped to $[0.05, 0.95]$, and control odds are normalized, capped at 5/588, and renormalized. The lower probability bound binds for 586 firms, three control weights are capped, and the effective control sample size is 500.919. The QTE estimator uses the same weights. The stabilization analysis treats this clip/cap map as part of the fitted estimator.

The ATT for each day is the connected firms' average outcome residual minus the weighted control residual. Table 16 shows both components. Its ten-day sum adds the daily return contrasts; it is not a compounded portfolio return. QTEs instead invert the

equal-weight mixture of the 340 connected firm–day counterfactual distributions. The conditional mean and distribution learners differ, so the ATT and QTE estimates summarize different fitted distributions. The following section reports bootstrap intervals for the corresponding firm-population target. The source package stores the training histories and daily predictions, including both residual components and the checks of shared inputs. Table 15 reports all nineteen QTEs.

Table 16: Current shared-cohort DR-FGAN daily ATT and residual decomposition

Trading date	Treated residual mean	Weighted control residual	ATT
2008-11-21	2.956883	0.417277	2.539606
2008-11-24	2.433875	-1.415811	3.849687
2008-11-25	0.149177	-0.634479	0.783656
2008-11-26	1.797126	0.343422	1.453704
2008-11-28	-0.191495	0.559767	-0.751262
2008-12-01	-1.329348	0.455290	-1.784638
2008-12-02	-1.245555	-0.241287	-1.004268
2008-12-03	1.593770	0.255653	1.338117
2008-12-04	1.439885	-0.140978	1.580863
2008-12-05	1.228097	0.114048	1.114050

ATT: Average treatment effect on the treated

DR-FGAN: Doubly robust factor-augmented generative adversarial network

Units are percentage points. Each ATT subtracts the normalized treatment-odds-weighted control residual from the treated residual mean. Predictions are out of fold at the company level. The sample has 622 firms and uses consecutive trading-day returns; event day zero is November 21, 2008. Table 17 reports bootstrap intervals.

Appendix D.4 Full-refit firm bootstrap: intervals and diagnostics

Sampling design and target. The bootstrap samples whole 110-date firm records independently within the four connection-status by original-outer-fold strata, whose sizes are

294, 294, 17, and 17. Counts and the event calendar are held fixed. A duplicated firm retains its original outer fold and inner training role, while the counts in each inner role can change. The target is the corresponding eligible firm-population ATT and pooled marginal QTE conditional on the event environment. Resampling whole firms keeps each firm's post-event observations together, rather than treating the 340 connected firm–days as independent observations. It conditions on the common event and calendar, so it does not resample time-series or event uncertainty. Section Appendix A states the conditional firm-sampling distribution and distinguishes this target from the fixed 34-firm potential-outcome target.

Each resample refits the original common-rank, shared-state procedure. The fully outer-isolated estimator is a separate sensitivity analysis. Rank is reselected from one through ten; ten is allowed with a boundary warning. The shared anchor, propensity and weights are then refitted, followed by the two separate outcome learners. ATT retains the 3,000-update ceiling, patience 500, and fifty fixed evaluation draws. QTE retains 500 mean-training steps, 800 warm-up steps and 1,800 adversarial steps, with the same knot-based readout. Training random seeds and inherited firm evaluation-noise marks follow the original protocol.

Each of the 199 ATT replicates is paired with a QTE replicate using the same firm draw, folds and seed. The saved arrays confirm that the two fits use identical inputs, and file hashes verify their source records. The original-sample ATT replay reproduces the daily point estimates exactly. All completed draws, including the eleven selecting rank ten, enter the intervals. The corrected point estimates and both bootstrap batches use the same 622-firm daily-return panel. Earlier 623-record releases and their grouped-deletion diagnostics remain historical outputs, separate from this application's inference.

Interval rules. For ATT, let $\hat{\theta}$ be the original estimate and let s_B be the sample standard deviation of the $B = 199$ full-refit estimates, using denominator $B-1$. The reported normal interval is $[\hat{\theta} - 1.959964 s_B, \hat{\theta} + 1.959964 s_B]$, centered at the original estimate. The ten-day sum and its interval are ten times the daily average and its interval. QTE intervals

are the linearly interpolated 2.5th and 97.5th percentiles of the 199 replicate QTEs, jointly recomputing observed and counterfactual quantiles in each draw. These pointwise intervals do not form a simultaneous band. The theory section gives the validity conditions and finite-B calibration. Coverage of the fitted application procedure has not been evaluated by simulation.

Average and daily effects. The daily-average point estimate is 0.911951 percentage points. Its bootstrap mean is 0.919133, standard deviation 0.519363, and normal interval $[-0.105981, 1.929883]$. Bootstrap standard deviations using the first 50, 100, 150 and 199 draws are 0.450852, 0.479373, 0.536917 and 0.519363. The corresponding ten-day sum is 9.119513, with interval $[-1.059807, 19.298834]$. Both intervals include zero.

The original average treated residual is 0.883241 and the weighted control correction is -0.028710. Their bootstrap means are 0.951299 and 0.032166, with standard deviations 0.477673 and 0.209003. Their difference gives the residual-corrected ATT in each draw.

Table 17 reports all daily normal intervals; each includes zero. The event-day point estimate is 2.539606, compared with bootstrap mean 1.169428 and standard deviation 2.183823. The second day's point estimate, 3.849687, moves in the opposite direction to a bootstrap mean of 5.319127. The daily path thus changes more than its average, as these opposite movements partly offset.

Table 17: Geithner daily ATT: full-refit bootstrap diagnostics

Trading date	ATT	Bootstrap mean	Bootstrap SE	95% interval
2008-11-21	2.540	1.169	2.184	[-1.741, 6.820]
2008-11-24	3.850	5.319	2.722	[-1.485, 9.184]
2008-11-25	0.784	0.991	2.263	[-3.652, 5.220]
2008-11-26	1.454	0.880	1.901	[-2.272, 5.180]
2008-11-28	-0.751	0.016	1.804	[-4.287, 2.785]
2008-12-01	-1.785	-2.119	1.209	[-4.154, 0.585]
2008-12-02	-1.004	-0.530	1.507	[-3.958, 1.950]
2008-12-03	1.338	1.397	1.479	[-1.561, 4.237]
2008-12-04	1.581	1.170	1.200	[-0.770, 3.932]
2008-12-05	1.114	0.898	1.032	[-0.909, 3.137]

ATT: Average treatment effect on the treated SE: Standard error

Units are percentage points. November 21, 2008 is the event date. Each interval is the original daily estimate plus or minus the standard-normal 97.5th percentile times its bootstrap standard deviation. All 199 whole-firm resamples re-estimate rank, factors, weights and networks. The intervals are pointwise and all include zero; they do not form a simultaneous path band. The former grouped-deletion calculations used a different data construction and are not combined with this release.

Distributional effects. Table 18 reports all nineteen quantiles. The lower-quartile, median and upper-quartile point estimates are 3.343002, 1.876418 and 0.829839 percentage points, with pointwise intervals $[1.063713, 3.640307]$, $[0.046579, 2.757063]$ and $[-0.180686, 1.834518]$. The lower-quartile contrast has the clearest separation from zero. The median interval's lower endpoint changes sign across the first 100, 150 and 199 draws: 0.285305, -0.105840 and 0.046579, respectively.

The lower-quartile and median bootstrap means are 2.341590 and 1.347799, respectively 1.001411 and 0.528619 below their original estimates. These differences persist after all 199 refits. The 0.95 QTE interval is much wider, $[-5.184252, 6.218541]$. Weighted post-event control calibration averages 7.932% below the fitted fifth percentile and 90.593% below the

fitted ninety-fifth percentile, rather than 5% and 95%. These are predictive tail-calibration diagnostics, not confidence-interval coverage.

Table 18: Geithner pooled marginal QTEs and pointwise bootstrap intervals

Quantile	Observed (%)	Counterfactual (%)	QTE (pp)	95% interval
0.05	-15.952	-15.430	-0.522	[-3.303, 1.368]
0.10	-8.839	-10.924	2.085	[-2.465, 4.034]
0.15	-4.092	-7.737	3.645	[0.893, 4.243]
0.20	-2.301	-5.496	3.195	[0.756, 3.844]
0.25	-0.522	-3.865	3.343	[1.064, 3.640]
0.30	0.000	-2.554	2.554	[0.980, 3.232]
0.35	0.758	-1.431	2.189	[0.286, 2.835]
0.40	1.408	-0.429	1.837	[0.057, 2.712]
0.45	2.319	0.511	1.807	[-0.006, 2.603]
0.50	3.309	1.432	1.876	[0.047, 2.757]
0.55	4.459	2.382	2.077	[0.347, 3.035]
0.60	5.545	3.387	2.158	[0.637, 2.717]
0.65	6.167	4.474	1.693	[0.417, 2.371]
0.70	6.863	5.695	1.168	[0.032, 2.285]
0.75	7.935	7.105	0.830	[-0.181, 1.835]
0.80	8.853	8.818	0.035	[-0.862, 1.682]
0.85	10.311	11.109	-0.798	[-1.939, 1.844]
0.90	12.895	14.689	-1.794	[-3.204, 2.584]
0.95	19.629	21.554	-1.925	[-5.184, 6.219]

QTE: Quantile treatment effect pp: Percentage points
CDF: Cumulative distribution function

Quantiles pool 34 connected firms over ten post-event trading dates. The counterfactual is the equally weighted mixture of conditional no-connection distributions. QTE is observed minus counterfactual, with no CDF correction. Intervals are the linear-interpolated 2.5th and 97.5th percentiles of 199 full-refit whole-firm bootstrap draws, conditional on the fixed calendar and sampling strata under the stated regularity conditions. All draws, including eleven rank-ten selections, are retained. These are pointwise intervals, not simultaneous bands. The median lower endpoint is 0.047 pp at 199 draws and -0.106 pp at 150 draws; a robust median significance claim is not supported. Extreme-tail endpoints remain imprecise.

Rank and stabilization. Ranks one through ten are selected 7, 21, 24, 27, 29, 23, 27, 15, 15 and 11 times. Original rank six recurs in 23 of 199 draws; its original validation loss margin over rank eight is 0.048024. Because selection is dispersed across the grid, the selection results in the theory require either a two-world selection-gap condition or common first-order behavior across potentially selected ranks.

The lower propensity clip binds for 323–622 rows across resamples; the upper clip binds for 0–21 rows and is active in 26 draws. Weight capping affects 0–31 control rows and is active in 176 draws. After combining duplicate-firm weights, the control effective sample size ranges from 27.171 to 315.608, with median 173.879; the largest weight assigned to one original firm is 10.716%. A bootstrap treated sample contains 34 rows but only 18–26 distinct original firms, with median 22. Effective sample size is therefore computed after combining copies of each firm. The initial cap applies to resampled rows before renormalization; the final mass on an original firm can exceed that threshold after its copies are combined. Daily-average control residual corrections range from −0.886425 to 0.590915. The active clip/cap analysis covers this binding-stabilization regime.

Reproduction and code correspondence. The package `publication_artifacts/geithner_corrected_inference_20261029/` contains the corrected sample and calendar metadata, replicate-level aggregate estimates, source hashes and pairing audit. Its materializer recomputes the intervals and manuscript tables from saved estimates, without refitting models or reading individual financial records. The firm-sample trading-calendar loader is `gscgan/src/geithner_daily_corrected.py`. The ATT score is the saved treated residual mean minus the weighted control residual. The exact stabilization map is `clip -> odds -> normalize -> cap -> renormalize` in `gscgan/src/common_adjustment.py`; the common-rank and inherited-role protocol is recorded with the resampling source snapshot. The QTE readout in `tools/materialize_geithner_plugin_20261014.py` and `gscgan/simulations/run_etcgan_dr_qte_experiment.py` evaluates the nineteen

conditional knots on a 401-point outcome grid, pools firm–day CDFs, and inverts. It adds no evaluation draws and no DR-CDF correction.

The bootstrap theory accounts for the moving cap threshold and numerical quantile map. It requires centered firm-level and target-projected learning expansions, with stable rank and stopping decisions under resampling. Shared-state remainders and causal and numerical bias must be negligible. For the fitted networks, these conditions apply at the stated training budget and fixed optimization seeds. This inference analysis covers the political-connections application only.

Appendix E London dynamic-tariff application: design and diagnostics

We compare three initializations of the full estimator under the original rank grid with a paired fit using a larger grid. We also fit both grids to the same masked-control sample, where untreated outcomes are observed and prediction errors can be checked. The main text uses the first initialization under the original grid.

Appendix E.1 Policy timing, measurement, and sample selection

Residential tariff interventions began on January 4, 2013. The preserved tariff schedule first departs from its normal rate at 14:00 that day, consistent with the implementation report (UK Power Networks 2013a). The panel uses weeks January 2–December 30, 2012 as its 52-week baseline, omits December 31, 2012–January 6, 2013, and uses January 7–April 7, 2013 as its 13-week operating window. Recruitment and incentive offers preceded operation (UK Power Networks 2012). Because household-specific invitation, enrollment, and payment dates are unavailable, identification requires that anticipation did not materially change baseline consumption. Both groups received consumption feedback, with an additional tariff-rate breakdown for dynamic-tariff participants (UK Power Networks 2013b).

Each retained week has at least 330 valid half-hour readings, between 330 and 338 total records, no negative consumption, and no duplicate timestamps. The outcome is mean consumption across the valid readings. The downloaded archive contains 4,443 standard-tariff and 1,123 dynamic-tariff household IDs. Of these, 300 and 54 meet the rule in all baseline weeks; 284 and 52 also meet it in all post weeks. Thus the post rule removes 16 of 300 and two of 54 pre-eligible households. The larger restriction is complete early observation. We count households by their IDs in the data files, since the public landing page and project reports give different inventory totals.

A fixed permutation with seed 2026101561 selects 108 standard-tariff donors and 36

dynamic-tariff targets. The next 36 eligible standard-tariff IDs form the disjoint masked-target sample. This is random subsampling within observed tariff groups, not randomized tariff assignment. Three fixed whole-household folds each contain 36 donors and twelve targets. For the masked sample, the standard-tariff targets' operating-period outcomes are withheld from estimation and used only to score reconstruction.

Z_i is the household's mean over the 52 baseline weekly outcomes. During treated-pre-period leave-one-time-out rank validation, the held-out week is excluded from both the target loading fit and its Z_i; final estimation uses all baseline weeks. W_t is NOAA Heathrow weekly temperature, common to all households (NOAA National Centers for Environmental Information 1999). Twelve of the thirteen post weeks lie within the pre-period temperature range; one is about 0.09 degrees Celsius below it. Only ten baseline weeks fall in the post-period temperature range. Thirty-five of 36 treated baseline-consumption means lie in the donor range. These checks describe marginal support. Identifying the counterfactual also requires overlap in latent characteristics and a common relationship between those characteristics and untreated consumption. The weather change is seasonal and common to both tariff groups.

Appendix E.2 Estimation and equal-budget comparisons

The original factor grid is $\{1, 2, 3\}$, and the expanded grid is $\{1, \ldots, 5\}$. The official GSC reference selects from $\{0, 1, 2, 3\}$ or $\{0, \ldots, 5\}$, respectively, using its own cross-validation protocol. The shared factor model selects rank three or five, while GSC selects two in both real fits. Enlarging the grid changes the factor estimates and, through them, the propensity weights and outcome predictions.

ATT fits have a ceiling of 300 epochs and patience 100. The QTE regression mean has a ceiling of 300 steps and patience 75; residual transport uses 200 pretraining and 600 adversarial steps, followed by 200 draws per household–week. Each matched neural/noADV pair uses the same data, outer folds, corresponding seeds, architectures, and optimization

ceilings. noADV sets only the generator's adversarial coefficient to zero. The critic updates and stopping rules stay the same. The stopping rule can end ATT training at different iterations. QTE estimation uses raw draws without DR-CDF targeting or a nonnegativity transformation.

Initializations 2026101591–2026101593 refer to the same panel and outer folds, while varying propensity fitting, inner splits, and evaluation draws alongside neural-network initialization. The expanded-grid real fit and both masked fits use seed 2026101591. The original and expanded real fits therefore form a paired comparison. The replication files contain each initialization and all nineteen quantiles.

Table 19 reports the paired rank-grid estimates for tariff participants. The remaining tables report the full distributional comparisons and diagnostic fits.

Table 19: London dynamic-tariff application: paired rank-grid sensitivity

Estimator / quantile	Grid 1:3	Grid 1:5	Difference
Panel A. Average effects			
DR-FGAN	-0.0011	-0.0119	-0.0108
DR-FGAN-noADV	-0.0004	-0.0115	-0.0111
LFM	-0.0091	-0.0101	-0.0010
GSC	-0.0132	-0.0132	0.0000
Panel B. ETC-GAN marginal quantile effects			
$\tau = 0.25$	-0.0059	-0.0127	-0.0069
$\tau = 0.50$	0.0043	-0.0013	-0.0056
$\tau = 0.75$	-0.0138	0.0025	0.0163

GAN: Generative adversarial network
DR-FGAN: Doubly robust factor-augmented GAN
ETC-GAN: Empirical-transport Cramér GAN
noADV: Generator adversarial term disabled
LFM: Linear factor model
GSC: Generalized synthetic control
QTE: Quantile treatment effect
CDF: Cumulative distribution function
DR-CDF: Doubly robust CDF correction
τ: Quantile level
kWh: Kilowatt-hour

The paired specifications use the same 36 treated and 108 control households, 52 pre-treatment and 13 post-treatment weeks, and seed 2026101591. Candidate grids are 1:3 and 1:5 (GSC: 0:3 and 0:5); fitted common ranks are 3 and 5, both at their upper boundary. GSC selects 2 and 2. Control-weight effective sample sizes are 40.395 and 28.528. Difference is grid 1:5 minus grid 1:3. DR-FGAN methods apply aggregate weighted residual correction; LFM is the registered plug-in comparator. QTEs invert the equal-weight pooled household-week distributions; there is no DR-CDF update. The expanded grid is sensitivity analysis, not a preferred replacement. All outcome and effect values are weekly mean kWh per valid half-hour reading. Point estimates only; no sampling intervals or causal-accuracy ranking.

Table 20: London: all nineteen marginal QTEs, all distribution estimators

τ	ETC-3	ETC-5	noADV-3	noADV-5	Emp.-3	Emp.-5
0.05	0.0143	-0.0039	0.0143	-0.0040	0.0097	-0.0025
0.10	0.0021	-0.0191	0.0021	-0.0191	-0.0046	-0.0167
0.15	-0.0043	-0.0212	-0.0042	-0.0213	-0.0115	-0.0192
0.20	-0.0103	-0.0224	-0.0103	-0.0225	-0.0174	-0.0212
0.25	-0.0059	-0.0127	-0.0058	-0.0129	-0.0120	-0.0125
0.30	-0.0138	-0.0144	-0.0138	-0.0146	-0.0170	-0.0152
0.35	-0.0223	-0.0204	-0.0223	-0.0205	-0.0236	-0.0225
0.40	-0.0199	-0.0204	-0.0199	-0.0205	-0.0245	-0.0241
0.45	-0.0005	-0.0043	-0.0005	-0.0044	-0.0068	-0.0086
0.50	0.0043	-0.0013	0.0043	-0.0014	-0.0015	-0.0066
0.55	0.0006	-0.0048	0.0006	-0.0049	-0.0039	-0.0106
0.60	0.0001	-0.0016	0.0001	-0.0017	-0.0017	-0.0088
0.65	-0.0059	-0.0035	-0.0059	-0.0035	-0.0078	-0.0128
0.70	-0.0112	0.0012	-0.0112	0.0012	-0.0160	-0.0091
0.75	-0.0138	0.0025	-0.0138	0.0025	-0.0141	-0.0067
0.80	-0.0199	-0.0086	-0.0199	-0.0088	-0.0195	-0.0174
0.85	-0.0098	0.0014	-0.0097	0.0013	-0.0100	-0.0032
0.90	0.0077	0.0179	0.0077	0.0178	0.0071	0.0158
0.95	-0.0227	-0.0084	-0.0227	-0.0081	-0.0257	0.0009

QTE: Quantile treatment effect
GAN: Generative adversarial network
ETC-GAN: Empirical-transport Cramér GAN
ETC: ETC-GAN distribution estimator
noADV: Generator adversarial term disabled
Emp.: Weighted empirical residual comparison
τ: Quantile level
kWh: Kilowatt-hour

Suffix 3 or 5 denotes the candidate-rank ceiling, not a universally valid factor rank. Negative draws are retained. All outcome and effect values are weekly mean kWh per valid half-hour reading. Point estimates only; no sampling intervals or causal-accuracy ranking.

Table 21: London baseline whole-pipeline seed sensitivity

Method	init 01	init 02	init 03			
Panel A. Separate whole-pipeline seed ATT estimates						
DR-FGAN	-0.0011	0.0001	-0.0004	–	–	–
DR-FGAN-noADV	-0.0004	0.0002	-0.0005	–	–	–
LFM	-0.0091	-0.0091	-0.0091	–	–	–
GSC	-0.0132	-0.0132	-0.0132	–	–	–
Panel B. Marginal QTE ranges across the three baseline seeds (not intervals)						
τ	ETC min	ETC max	noADV min	noADV max	Emp. min	Emp. max
0.05	0.0100	0.0192	0.0108	0.0194	0.0017	0.0149
0.10	-0.0043	0.0051	-0.0035	0.0054	-0.0089	0.0027
0.15	-0.0107	-0.0043	-0.0100	-0.0042	-0.0127	-0.0054
0.20	-0.0151	-0.0103	-0.0144	-0.0103	-0.0174	-0.0109
0.25	-0.0068	-0.0053	-0.0063	-0.0051	-0.0120	-0.0036
0.30	-0.0138	-0.0079	-0.0138	-0.0075	-0.0170	-0.0071
0.35	-0.0223	-0.0110	-0.0223	-0.0107	-0.0236	-0.0121
0.40	-0.0199	-0.0079	-0.0199	-0.0077	-0.0245	-0.0111
0.45	-0.0005	0.0100	-0.0005	0.0102	-0.0068	0.0072
0.50	0.0043	0.0100	0.0043	0.0103	-0.0015	0.0121
0.55	0.0006	0.0064	0.0006	0.0065	-0.0039	0.0088
0.60	0.0001	0.0119	0.0001	0.0121	-0.0018	0.0121
0.65	-0.0059	0.0100	-0.0059	0.0100	-0.0084	0.0077
0.70	-0.0112	0.0081	-0.0112	0.0082	-0.0160	0.0043
0.75	-0.0138	0.0023	-0.0138	0.0026	-0.0141	-0.0003
0.80	-0.0199	-0.0038	-0.0199	-0.0037	-0.0215	-0.0165
0.85	-0.0098	0.0058	-0.0097	0.0060	-0.0162	0.0060
0.90	0.0077	0.0287	0.0077	0.0288	0.0039	0.0250
0.95	-0.0227	0.0237	-0.0227	0.0237	-0.0257	-0.0101

ATT: Average treatment effect on the treated
QTE: Quantile treatment effect
GAN: Generative adversarial network
DR-FGAN: Doubly robust factor-augmented GAN
ETC-GAN: Empirical-transport Cramér GAN
ETC: ETC-GAN distribution estimator
noADV: Generator adversarial term disabled
LFM: Linear factor model
GSC: Generalized synthetic control
Emp.: Weighted empirical residual comparison
Min / max: Minimum / maximum across initializations
Init: Whole-pipeline initialization
τ: Quantile level
kWh: Kilowatt-hour

Seeds 2026101591, 2026101592, and 2026101593 use the same panel and outer household folds. The first is predeclared primary; predictions are never averaged into a new estimator. Seeds can change propensity fitting, inner splits, draw banks and neural initialization. Min–max ranges are numerical sensitivity, not confidence intervals. All outcome and effect values are weekly mean kWh per valid half-hour reading. Point estimates only; no sampling intervals or causal-accuracy ranking.

Table 22: London: fixed masked untreated-outcome recovery checks

Method	Grid 3	Grid 5		
Panel A. Absolute ATT point error against zero				
DR-FGAN	0.0227	0.0380	–	–
DR-FGAN-noADV	0.0221	0.0374	–	–
LFM	0.0155	0.0180	–	–
GSC	0.0086	0.0086	–	–
Outcome component	RMSE-3	RMSE-5	MAE-3	MAE-5
DR-FGAN component	0.0813	0.0916	0.0465	0.0542
DR-FGAN-noADV component	0.0813	0.0915	0.0470	0.0540
LFM prediction	0.0811	0.0908	0.0457	0.0533
Distribution estimator	QTE MAE-3	QTE MAE-5	CRPS-3	CRPS-5
ETC-GAN	0.0330	0.0324	0.0402	0.0455
ETC-GAN-noADV	0.0325	0.0339	0.0424	0.0460
Weighted empirical	0.0346	0.0415	0.0407	0.0470

ATT: Average treatment effect on the treated
QTE: Quantile treatment effect
GAN: Generative adversarial network
DR-FGAN: Doubly robust factor-augmented GAN
ETC-GAN: Empirical-transport Cramér GAN
DR: Doubly robust residual correction
noADV: Generator adversarial term disabled
LFM: Linear factor model
GSC: Generalized synthetic control
RMSE: Root mean squared error
MAE: Mean absolute error
CRPS: Continuous ranked probability score
kWh: Kilowatt-hour

The fixed validation panel comprises 36 pseudo-treated and 108 donor standard-tariff households. Their actual untreated post outcomes are masked from fitting and used only for scoring. ATT entries are absolute point errors against zero, not Monte Carlo estimator RMSE. Prediction errors concern the outcome component, not individual DR-corrected potential outcomes; GSC household predictions are unavailable. QTE MAE averages the nineteen absolute marginal quantile errors. CRPS scores saved conditional draws. All outcome and effect values are weekly mean kWh per valid half-hour reading. Point estimates only; no sampling intervals or causal-accuracy ranking.

Table 23: London: common selected ranks and binding stabilization

Run	Rank	GSC	Boundary	ESS	Low clips	Caps
init_01	3	2	yes	40.40	20	4
init_02	3	2	yes	42.26	18	3
init_03	3	2	yes	44.39	20	3
real_r5	5	2	yes	28.53	30	6
masked_r3	3	3	yes	29.53	36	6
masked_r5	5	4	yes	23.69	50	6

GSC: Generalized synthetic control
ESS: Control-weight effective sample size
CV: Cross-validation

The common selected rank is used in final fits; it need not equal each fold's CV winner. Low clips count all households with propensity below 0.05; caps count control weights at the prespecified cap before renormalization. Selected ranks at the improving grid boundary are treated as sensitivity, not rank stability.

Table 24: London: generated-draw support and post-hoc zero-clipping diagnostics

Run	Estimator	Neg. (%)	Max cell (%)	Mean change	QTE changed
init_01	ETC-GAN	1.129	13.5	0.000443	0
init_01	ETC-GAN-noADV	1.129	13.5	0.000443	0
init_01	Weighted empirical	1.068	11.0	0.000572	0
init_02	ETC-GAN	1.166	13.0	0.000454	0
init_02	ETC-GAN-noADV	1.175	13.5	0.000458	0
init_02	Weighted empirical	1.134	10.5	0.000583	0
init_03	ETC-GAN	1.091	8.5	0.000416	0
init_03	ETC-GAN-noADV	1.110	9.0	0.000427	0
init_03	Weighted empirical	0.995	7.0	0.000442	0
real_r5	ETC-GAN	0.543	8.0	0.000150	0
real_r5	ETC-GAN-noADV	0.542	7.5	0.000150	0
real_r5	Weighted empirical	0.610	10.5	0.000304	0
masked_r3	ETC-GAN	2.739	42.0	0.001868	0
masked_r3	ETC-GAN-noADV	2.876	37.5	0.001985	0
masked_r3	Weighted empirical	2.248	25.5	0.002036	0
masked_r5	ETC-GAN	3.580	68.0	0.002232	0
masked_r5	ETC-GAN-noADV	3.637	66.0	0.002245	0
masked_r5	Weighted empirical	3.410	41.0	0.002730	0

GAN: Generative adversarial network
ETC-GAN: Empirical-transport Cramér GAN
noADV: Generator adversarial term disabled
Neg.: Negative generated draws
Max cell: Largest household-week negative-draw share
QTE: Quantile treatment effect
DR: Doubly robust residual correction
ATT: Average treatment effect on the treated

Raw predictions remain unchanged. Mean change is the increase in the counterfactual draw mean under the post-hoc map $y \mapsto \max(y, 0)$, not a new DR ATT. QTE changed counts affected reported levels 0.05 through 0.95. Zero pooled-quantile change does not resolve cell-level support violations. No positive-only conditioning is performed.

Table 25: London: fitted pre-history diagnostics for target households

Run	Saved mean	RMSE	MAE
init_01	Stage-1 IFE	0.0517	0.0293
init_01	QTE Stage-A mean	0.0541	0.0322
real_r5	Stage-1 IFE	0.0454	0.0269
real_r5	QTE Stage-A mean	0.0498	0.0298
masked_r3	Stage-1 IFE	0.0458	0.0276
masked_r3	QTE Stage-A mean	0.0497	0.0319
masked_r5	Stage-1 IFE	0.0433	0.0260
masked_r5	QTE Stage-A mean	0.0478	0.0322

IFE: Interactive fixed effects
QTE: Quantile treatment effect
GAN: Generative adversarial network
FGAN: Factor-augmented GAN
RMSE: Root mean squared error
MAE: Mean absolute error
kWh: Kilowatt-hour

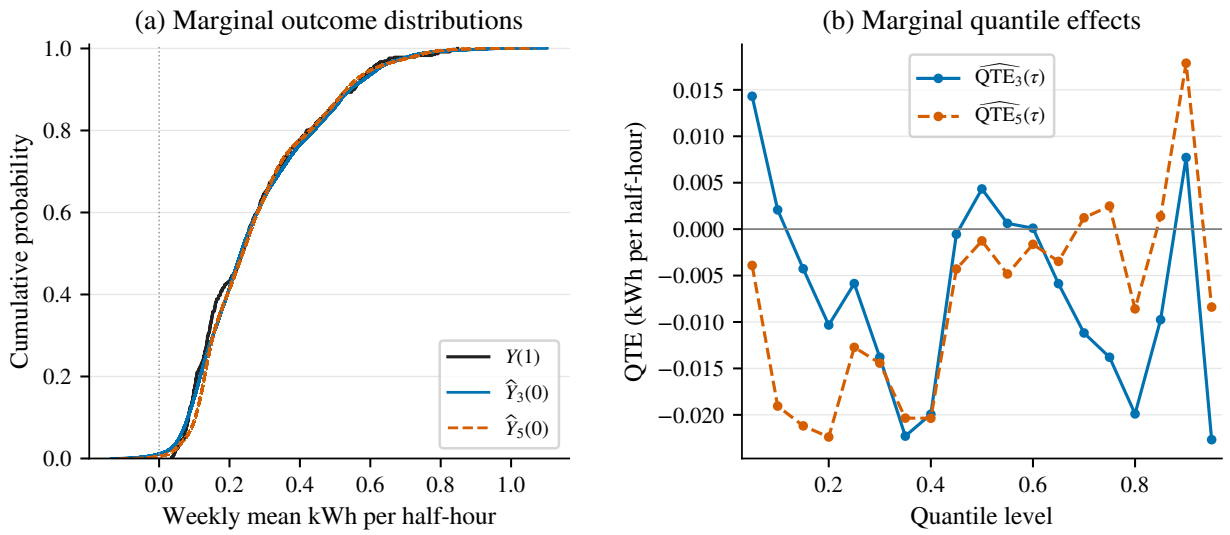
These are final fitted-history reconstructions. Pre outcomes enter target loading estimation and the baseline consumption summary; this is not held-out post-outcome recovery or leave-time-out prediction. Final FGAN pre-period predictions were not saved. All outcome and effect values are weekly mean kWh per valid half-hour reading. Point estimates only; no sampling intervals or causal-accuracy ranking.

Appendix E.3 Rank sensitivity and reconstruction

Figure 2 compares the original and expanded rank grids. Expanding the grid to $\{1, \ldots, 5\}$ selects rank five and moves the ATT to -0.01189; the median and upper-quartile QTEs change sign. In both grids, validation error is lowest at the largest candidate rank. The main-text figure uses the original rank-three specification.

Increasing rank improves the target anchor's pre-period RMSE from 0.05170 to 0.04544 in the real sample and from 0.04584 to 0.04332 in the masked sample. The withheld post-period outcomes provide a separate check of whether better pre-period fit improves prediction. Pre-period predictions were saved for the factor model and the QTE regression mean, but not for the final FGAN mean.

In the masked sample, DR-FGAN's signed ATT error relative to zero changes from -0.02272 to -0.03801, compared with -0.00860 and -0.00858 for GSC on this fixed untreated sample. ETC-GAN's pooled-QTE MAE changes from 0.03305 to 0.03238, while its conditional CRPS worsens from 0.04015 to 0.04550. The noADV ordering also changes with rank. The higher-rank fit therefore improves pre-period reconstruction and pooled-QTE



$Y(1)$: Observed participant consumption
$\widehat{Y}_r(0)$: Estimated consumption without dynamic tariffs at rank r
$\widehat{Q}_1, \widehat{Q}_{0,r}$: Observed and counterfactual quantiles
r: Selected factor rank
τ: Quantile level

ETC-GAN: Empirical-transport Cramér generative adversarial network
QTE: Quantile treatment effect
DR-CDF: Doubly robust cumulative distribution function correction
kWh: Kilowatt-hour

Figure 2: London distributional estimates under the original and expanded rank grids. Panel (a) compares $Y(1)$ with the ETC-GAN estimates $\widehat{Y}_r(0)$. The candidate grids $\{1, 2, 3\}$ and $\{1, \ldots, 5\}$ select $r = 3$ and $r = 5$, respectively. Panel (b) plots $\widehat{\mathrm{QTE}}_r(\tau) = \widehat{Q}_1(\tau) - \widehat{Q}_{0,r}(\tau)$. Both specifications use the same 36 households, 13 operating weeks, initialization, and training budgets. Units are weekly mean kWh per valid half-hour. Both curves use unclipped plug-in draws, without DR-CDF correction or uncertainty bands.

error while worsening ATT error and conditional distribution fit.

The real-sample control-weight effective sample size changes from 40.395 to 28.528 out of 108 donors. DR-FGAN's treated residual mean changes from -0.01003 to -0.01252, while the weighted control residual changes from -0.00896 to -0.00064. The subtraction explains the arithmetic movement in ATT because predictions and weights change together.

Negative ETC-GAN draws account for 1.13% in the original real fit and 0.54% in the expanded fit, versus 2.74% and 3.58% in the masked fits. The largest masked household–week negative share is 68%. Replacing each negative draw by zero, without refitting, leaves all reported pooled quantiles at levels $0.05, 0.10, \ldots, 0.95$ unchanged across the six fits and three distribution learners. Truncation changes conditional distributions and means, however,

because the original generator assigns some probability to negative consumption.

The masked check evaluates untreated households. For tariff participants, causal interpretation requires no material anticipation. It also requires untreated consumption to follow the same conditional relationship as in the donor group, despite selection into participation and record completeness. Initialization ranges and rank differences measure sensitivity, not sampling uncertainty.

Appendix E.4 Replication

The separate London replication archive stores the estimation settings and input hashes alongside saved predictions and completed comparisons. Its table and figure generator reads those predictions without refitting models. All London results are point estimates without a coverage assessment.

The electricity observations are attributed to UK Power Networks and the London Datastore under their Creative Commons Attribution licence. The public replication package separates those data from the Heathrow weather inputs, whose source files must be obtained from NOAA under the applicable upstream terms. The runbook lists source links and input hashes and explains how to prepare the data. Saved predictions reproduce the tables and figures without downloading the raw smart-meter or weather files.

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